

The Poole Local-Geometry Program
Hysteretic Surface Selection, Defect Calculus, Channel Spectrometry, Signed Membrane Basins,
and the Defect–Channel Bridge
Coverage-Audited Expansion v1.4: Bridge Benchmark Deep Cleanup

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Abstract

This v1.4 manuscript extends the coverage-audited expansion for the Poole Local-Geometry Program. It does not replace the full source papers; instead it reorganizes their central definitions, theorem statements, proof ideas, benchmark claims, and boundaries into one continuous mathematical narrative while preserving the complete source papers as appendices. The safe central thesis is finite and testable: a three-dimensional Moore-neighborhood hysteretic support rule selects one-cell surfaces, converts local defects into exact event-channel spectra, and supports signed reaction-membrane basins when preparation supplies the correct local geometry. The theorem bridge is exact for the first step of planar defect response: the same q_D/r_D counts that drive the Defect Calculus also determine the PooleQ/P birth and survival channels. The signed-membrane and bridge benchmarks are finite empirical claims with controls: structured signed preparations remain separated from same-density, shuffled, mirrored-random, and single-sign controls under the tested rules, while rule-ridge tests show that wider survival windows around S3–11 and S4–11 substantially strengthen the basin. No physical interpretation beyond this bounded local-geometry program is assumed.

v1.4 status and no-loss rule

This version continues the conversion from a source-preserving archive into a coherent monograph. The no-loss rule still dominates: if a result has not yet been fully rewritten into the theorem spine, it remains preserved in a source appendix and mapped in the coverage data. The current v1.4 manuscript is therefore a controlled intermediate object with two layers:

1. a unified front manuscript that states the program, notation, theorem chain, benchmark conclusions, limitations, and open targets in one voice;
2. complete source appendices containing the active Defect Calculus, PooleQ/P, signed membrane, and bridge-core source documents.

The next stages can expand each proof capsule and benchmark table into the front manuscript while deleting duplicate source material only after a coverage audit says the deletion is safe.

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Part I

Program, scope, and local rule

Chapter 1

Research program and claim lanes

1.1 What the program studies

The Poole Local-Geometry Program studies deterministic and stochastic consequences of local support thresholds on a three-dimensional cubic lattice. The center of the program is not an appeal to continuum physics or hardware speculation. It is the finite combinatorics of a local automaton and the benchmarkable morphology that grows from that automaton.

The program has four active layers.

1. **Defect Calculus:** the theorem layer. It proves how the B5–7/S5–9 rule selects sheets and responds to planar defects.
2. **PooleQ/P Capture:** the channel-spectrometry layer. It preserves the local event channels that the raw update normally collapses.
3. **Signed Membranes:** the opposite-sign basin layer. It asks when signed populations form stable, thin, low-imbalance reaction interfaces.
4. **Bridge Benchmark:** the cross-layer test. It asks whether PooleQ/P channel spectra predict signed basin entry beyond density or mirror symmetry alone.

1.2 Claim lanes

Every statement in the manuscript is assigned to one of five lanes.

Theorem-level. A finite combinatorial consequence of the stated local rule.

Verifier-level. A result reproduced by an exact script or reference checker.

Benchmark-level. A finite experiment with seeds, controls, summary statistics, and baselines.

Empirical-atlas level. A reproducible numerical pattern used to guide theorem discovery, not itself a proof.

Bridge/open level. A bounded interpretation or target theorem that links layers but is not yet a closed theorem.

This lane discipline is part of the scientific contribution. It prevents exact finite theorems, finite benchmarks, and speculative interpretations from being mixed into one unsupported claim.

1.3 Boundary statement

This manuscript does not claim a physical theory of spacetime, a method for copying or reconstructing unknown quantum amplitudes, an antimatter production or propulsion technology, or a universal controller. The established object is narrower: a local-geometry theory and benchmark suite for Moore-neighborhood support rules, channel spectra, and signed reaction-interface basins.

Chapter 2

The finite Poole lattice

Definition 2.1 (Periodic cubic lattice). *Fix $L \in \mathbb{N}$. The finite lattice is*

$$\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3.$$

A binary configuration is a map $x : \Lambda_L \rightarrow \{0, 1\}$.

Definition 2.2 (Moore neighborhood). *The radius-one Moore neighborhood in three dimensions is*

$$N_{26} = \{-1, 0, 1\}^3 \setminus \{(0, 0, 0)\}.$$

The active neighbor count at site i is

$$N_i(x) = \sum_{\delta \in N_{26}} x(i + \delta),$$

where addition is modulo L in each coordinate.

Definition 2.3 (Base Poole map). *The base Poole update is*

$$P(x)_i = \mathbf{1}\{5 \leq N_i(x) \leq 7 + 2x(i)\}. \quad (2.1)$$

The evolution is $x_{t+1} = P(x_t)$.

Proposition 2.4 (Equivalence to B5–7/S5–9). *The compact law (2.1) is exactly the birth-survival rule B5–7/S5–9.*

Proof. If $x(i) = 0$, the upper bound in (2.1) is 7, so the site is born exactly for $N_i \in \{5, 6, 7\}$. If $x(i) = 1$, the upper bound is 9, so the site survives exactly for $N_i \in \{5, 6, 7, 8, 9\}$. $\square$

2.1 Normal forms

The same rule will be used in several equivalent forms.

Birth-survival form.

$$P(x)_i = (1 - x(i))\mathbf{1}\{N_i \in \{5, 6, 7\}\} + x(i)\mathbf{1}\{N_i \in \{5, 6, 7, 8, 9\}\}.$$

Capacity form. Let $C_i(x) = 7 + 2x(i)$. Then $P(x)_i = 1$ iff $5 \leq N_i(x) \leq C_i(x)$.

Deficit-excess form. Let

$$D_i = (5 - N_i)_+, \quad E_i = (N_i - C_i)_+, \quad \psi_i = E_i - D_i.$$

Then $P(x)_i = 1$ iff $D_i = E_i = 0$, equivalently $\psi_i = 0$.

Channel form. The accepted birth channels are B_5, B_6, B_7 and the accepted survival channels are S_5, S_6, S_7, S_8, S_9 .

2.2 Why the $+2x(i)$ term matters

The term $2x(i)$ is not an added field. It is a local hysteretic allowance: active cells may survive two counts above the inactive birth ceiling. This difference is exactly what allows a one-cell sheet to persist. Without it, a full coordinate plane has $N_i = 8$ at each active site and dies under the memory-reduced rule B5-7/S5-7.

Part II

Hysteretic surface selection and Defect Calculus

Chapter 3

Basic structural theorems

Proposition 3.1 (Cubic equivariance). *The map P commutes with translations of Λ_L , coordinate permutations, and coordinate sign flips.*

Proof. Each operation only permutes the elements of the Moore neighborhood and therefore preserves the neighbor count $N_i(x)$. $\square$

Proposition 3.2 (Finite propagation speed). *If two initial configurations agree on the radius- t Moore ball around a site i , then their states at i agree after t updates.*

Proof. One update depends only on a radius-one Moore ball. Iterating the dependency gives a radius- t dependency after t updates. $\square$

Proposition 3.3 (Eventual periodicity). *For fixed L , every trajectory under P is eventually periodic with preperiod plus period at most 2^{L^3} .*

Proof. There are only 2^{L^3} configurations. A deterministic map on a finite set must eventually revisit a previous state. $\square$

Proposition 3.4 (Non-monotonicity). *The Poole map is not monotone with respect to coordinatewise order.*

Proof. Let $x \leq y$ and choose an inactive target site i . Give x exactly five active neighbors of i . Let y add three more active neighbors, so $N_i(y) = 8$. Then $P(x)_i = 1$ but $P(y)_i = 0$, because inactive cells are over-supported above 7. $\square$

Proposition 3.5 (Non-reversibility). *For $L \geq 3$, P is not injective and hence not reversible.*

Proof. The all-zero configuration maps to itself. The all-one configuration has $N_i = 26$ at every site, so every active cell is over-supported and dies. Thus two distinct configurations map to the all-zero configuration. $\square$

Chapter 4

Surface selector theorem

Definition 4.1 (Coordinate-flat local pattern). *A configuration is locally a coordinate r -flat through a site if, in the centered $3 \times 3 \times 3$ Moore block, the active sites are exactly the sites of an r -dimensional coordinate subspace through that site.*

Lemma 4.2 (Flat local degree count). *A site in a complete coordinate r -flat has local active neighbor count*

$$3^r - 1.$$

Thus the ideal counts for point, line, sheet, and filled bulk are 0, 2, 8, 26.

Proof. The centered block contains 3^r sites of the coordinate r -flat, one of which is the center. The rest are active neighbors. $\square$

Theorem 4.3 (Local surface selector). *Among ideal coordinate-flat local bulk types, the B5-7/S5-9 survival window preserves exactly the locally two-dimensional sheet type.*

Proof. The survival window is $5 \leq N_i \leq 9$. Of the counts 0, 2, 8, 26, only 8 lies in this interval. $\square$

Theorem 4.4 (Stable active plane). *Assume $L \geq 3$. A full coordinate plane is a fixed point of the base Poole map.*

Proof. Each active site in the plane has the other eight sites in its in-plane 3×3 footprint as active neighbors, so $N_i = 8$ and it survives. A site one layer above or below sees a full 3×3 footprint with $N_i = 9$, which is too high for birth. Sites farther away see zero active neighbors. The configuration is unchanged. $\square$

Corollary 4.5 (Sheet memory). *Under the memory-reduced ablation B5-7/S5-7, the full active plane dies in one step because its active cells have $N_i = 8$.*

Chapter 5

Planar defect calculus

Definition 5.1 (Defective sheet). *Identify a stable coordinate sheet with the two-dimensional torus $T_L = (\mathbb{Z}/L\mathbb{Z})^2$. Let $D \subset T_L$ be a defect mask of missing sheet sites. The defective-sheet configuration has $x_D(u, c) = 1$ for $u \notin D$ and all non-sheet layers inactive.*

Definition 5.2 (Open and closed defect counts). *For $u \in T_L$, define*

$$q_D(u) = \sum_{\eta \in \{-1,0,1\}^2 \setminus \{0\}} \mathbf{1}\{u + \eta \in D\},$$

$$r_D(u) = \sum_{\eta \in \{-1,0,1\}^2} \mathbf{1}\{u + \eta \in D\}.$$

The count q_D is the open in-plane defect count. The count r_D is the closed 3×3 defect-footprint count.

Theorem 5.3 (One-step planar defect selector). *For the base Poole rule, the first update of a defective sheet x_D is determined exactly as follows.*

1. *An active sheet site $u \notin D$ survives exactly when $q_D(u) \leq 3$.*
2. *An inactive in-plane defect site $u \in D$ is born exactly when $1 \leq q_D(u) \leq 3$.*
3. *A normal-layer site at $(u, c+1)$ or $(u, c-1)$ is born exactly when $2 \leq r_D(u) \leq 4$.*
4. *All layers with $|z - c| \geq 2$ remain inactive after the first update.*

Proof. An active or inactive site in the sheet plane sees only the eight in-plane neighbors, so the raw count is $8 - q_D(u)$. Active survival requires $5 \leq 8 - q_D(u) \leq 9$, equivalent to $q_D(u) \leq 3$. Inactive birth requires $5 \leq 8 - q_D(u) \leq 7$, equivalent to $1 \leq q_D(u) \leq 3$. A normal-layer site sees a 3×3 sheet footprint, so the count is $9 - r_D(u)$; birth requires $5 \leq 9 - r_D(u) \leq 7$, equivalent to $2 \leq r_D(u) \leq 4$. Farther layers see no active neighbors during the first update. $\square$

Corollary 5.4 (Stable pinhole gas). *If every closed 3×3 footprint contains at most one defect site, then the defective sheet is fixed. In particular, a single one-cell pore is exactly stable.*

5.1 Operator table

Cell class	Raw count	Rule test	Result
Active sheet site $u \notin D$	$8 - q_D(u)$	survival	survives iff $q_D \leq 3$
Missing in-plane defect $u \in D$	$8 - q_D(u)$	birth	born iff $1 \leq q_D \leq 3$
Normal-layer site $(u, c \pm 1)$	$9 - r_D(u)$	birth	born iff $2 \leq r_D \leq 4$
Distance ≥ 2 layer	0	birth	inactive

Chapter 6

Rectangle laws and reconstruction boundary

The full source appendix contains the complete rectangle and line-hole proofs, prime-correction model, work laws, and reconstruction audit. The v1.1 front manuscript records the central closed family because it is a major bridge from raw defect geometry to channel spectra.

Theorem 6.1 (Rectangle first-step zero-death family). *For rectangular planar holes in the theorem-level subfamily, the first-step response is a closed zero-death response with raw birth count*

$$B_0(a, b) = 4a + 4b + 12 = 2P + 12,$$

where $P = 2a + 2b$ is the rectangle perimeter.

Theorem 6.2 (Rectangle channel spectrum). *For the same rectangle family, the first-step birth-channel spectrum is*

$$(N_5, N_6, N_7) = (12, 4a + 4b - 16, 16).$$

Theorem 6.3 (Prime-corrected rectangle response). *Under the default prime-sharpened model that removes the raw N_7 birth group, the rectangle response is*

$$B_\varphi(a, b) = 4a + 4b - 4.$$

Remark 6.4 (Model status). *The raw rectangle laws are theorem-level consequences of the first-step selector. The prime-sharpened response is a model-tagged correction, not a claim that the base automaton secretly removes N_7 births. This distinction must remain visible in the final manuscript.*

6.1 Q/P-style rectangle reconstruction boundary

The rectangle reconstruction results are bounded by their prior: they apply under a rectangle hypothesis and channel-resolved readout. They should not be promoted into arbitrary-shape reconstruction until a separate theorem or benchmark supports that extension.

Chapter 7

Defect verifier import v1.1

This chapter moves the Defect Calculus verifier layer into the front manuscript. The imported verifier is not used as a substitute for the preceding proofs. Its role is implementation hardening: it checks indexing, channel separation, periodic-boundary behavior, model tags, and reconstruction formulas against explicit finite updates.

7.1 Closed-layer pass/fail summary

The regenerated v1.1 verifier reproduces the closed first-step audit families from the Defect source appendix. The empirical multi-step traces are included as reference data only and are not counted as theorem-level pass/fail evidence.

Table 7.1: v1.1 regenerated Defect verifier pass/fail summary.

Metric	Value
Rectangle cases	841
Rectangle raw-law matches	841
Rectangle channel-spectrum matches	841
Rectangle prime-response matches	841
Rectangle overcompensation matches	841
Line-hole cases	80
Line-hole raw-law matches	80
Line-hole prime-response matches	80
Random planar selector cases	720
Random planar selector matches	720
Random planar nonlocal events total	0
Q/P rectangle inversion cases	1225
Q/P raw inversion matches	1225
Q/P prime inversion matches	1225
Empirical timeseries rows	18
All closed-layer checks passed	True

The equality of actual and predicted event sets in the random planar selector audit is the most important implementation check. It tests arbitrary finite masks rather than only rectangles and confirms that the first-step event set is completely accounted for by the two planar counts q_D and r_D .

7.2 Rectangle, channel, prime, and overcompensation audit

For every $2 \leq a, b \leq 30$, the verifier constructs a stable sheet, removes an $a \times b$ rectangle, performs one raw update, and records births, deaths, active counts, and birth-channel counts. Every row satisfies

$$B_0(a, b) = 4a + 4b + 12, \quad D_0(a, b) = 0,$$

$$(N_5, N_6, N_7) = (12, 4a + 4b - 16, 16),$$

$$B_\phi(a, b) = N_5 + N_6 = 4a + 4b - 4,$$

and the first-step overcompensation criterion

$$|A_1| > |A_{\text{sheet}}| \iff (a - 4)(b - 4) < 28.$$

Table 7.2: Representative v1.1 rectangle/channel/prime audit rows.

a	b	B_0	pred.	D	N_5	N_6	N_7	B_ϕ	pred.	ok
2	2	28	28	0	12	0	16	12	12	True
2	3	32	32	0	12	4	16	16	16	True
2	10	60	60	0	12	32	16	44	44	True
2	30	140	140	0	12	112	16	124	124	True
10	10	92	92	0	12	64	16	76	76	True
10	20	132	132	0	12	104	16	116	116	True
10	30	172	172	0	12	144	16	156	156	True
30	30	252	252	0	12	224	16	236	236	True

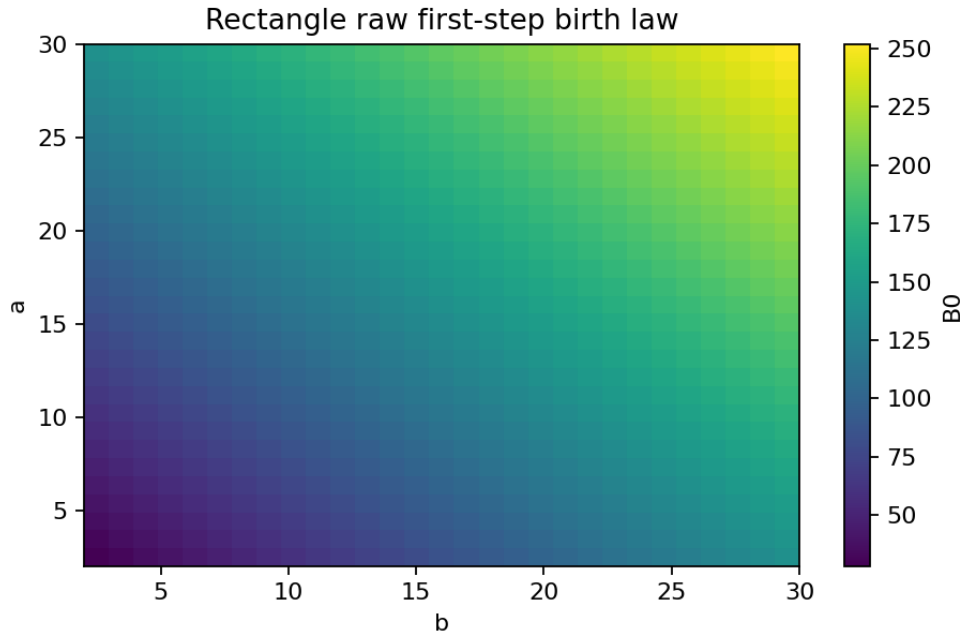


Figure 7.1: Regenerated rectangle raw first-step birth surface for $2 \leq a, b \leq 30$.

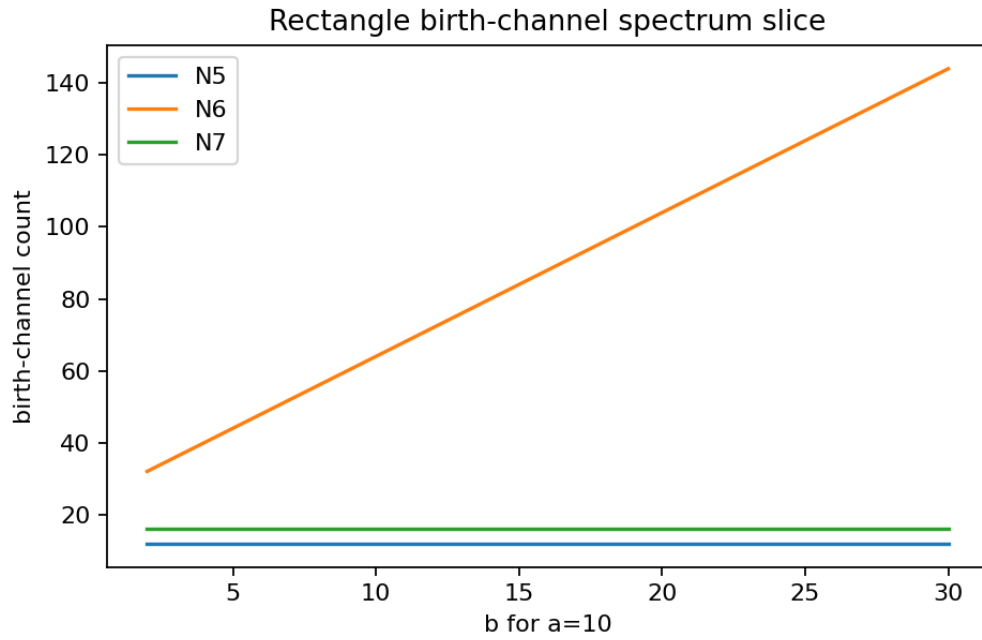


Figure 7.2: Regenerated rectangle birth-channel spectrum slice. The N_5 and N_7 groups are constant while N_6 carries the perimeter-dependent term.

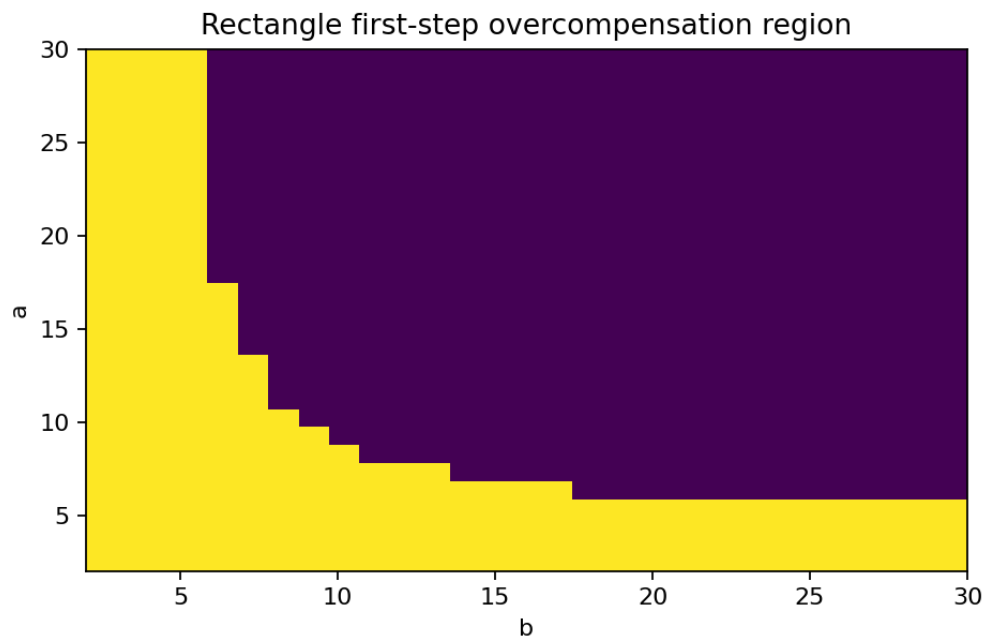


Figure 7.3: Regenerated first-step overcompensation region. The criterion is $(a-4)(b-4) < 28$.

7.3 Line-hole edge audit

The one-cell-wide family is a degenerate boundary case and is not folded into the $a, b \geq 2$ rectangle law. The regenerated audit covers $1 \leq n \leq 80$ and checks

$$B_0(1) = B_\phi(1) = 0,$$

while for $n \geq 2$ it checks

$$B_0(1, n) = 7n, \quad B_\phi(1, n) = 7n - 14.$$

Table 7.3: Representative v1.1 line-hole edge audit rows.

n	B_0	pred.	D	N_5	N_6	N_7	B_ϕ	pred.	ok
1	0	0	0	0	0	0	0	0	True
2	14	14	0	0	0	14	0	0	True
3	21	21	0	0	7	14	7	7	True
10	70	70	0	0	56	14	56	56	True
40	280	280	0	0	266	14	266	266	True
80	560	560	0	0	546	14	546	546	True

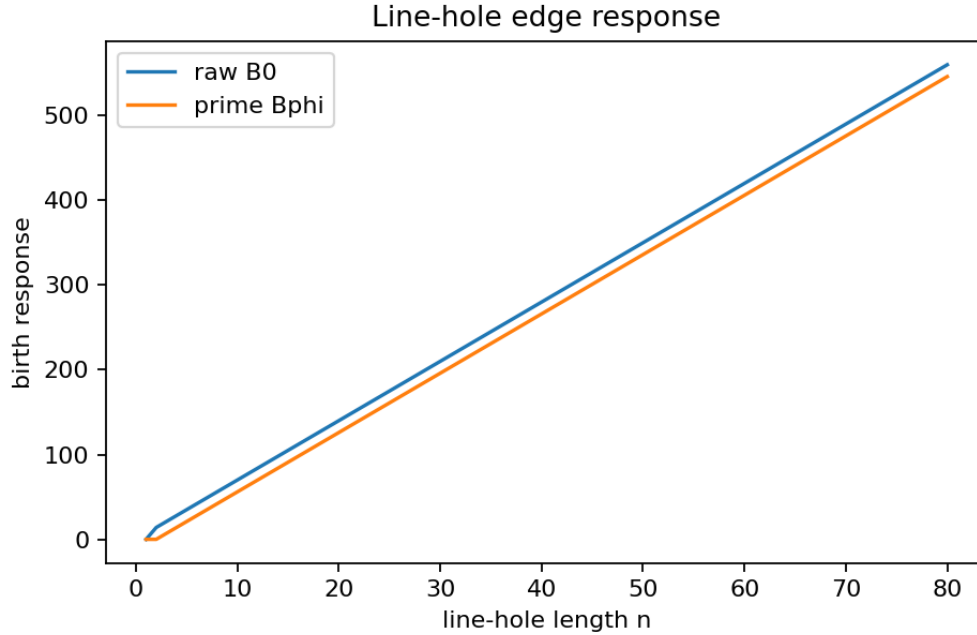


Figure 7.4: Regenerated raw and default prime-corrected line-hole response for $1 \leq n \leq 80$.

7.4 Arbitrary planar selector audit

For each random finite planar mask D , the verifier removes D from a stable sheet, performs one raw update, and compares the full event set against the q_D/r_D prediction:

$$\text{in-plane births: } u \in D, \quad 1 \leq q_D(u) \leq 3,$$

deaths: $u \notin D, \quad q_D(u) \geq 4,$

normal births: $z = c \pm 1, \quad 2 \leq r_D(u) \leq 4.$

All 720 generated rows match, and the total number of nonlocal first-step events is zero.

Table 7.4: First ten v1.1 random planar selector audit rows.

case	box	removed	actual E	pred. E	plane B	plane D	adj. B	ok
1	7x12	27	185	185	24	5	156	True
2	6x4	1	0	0	0	0	0	True
3	5x11	7	30	30	4	0	26	True
4	12x7	44	150	150	17	15	118	True
5	8x9	30	185	185	27	14	144	True
6	8x10	43	151	151	22	19	110	True
7	12x4	16	69	69	9	6	54	True
8	10x10	22	154	154	21	3	130	True
9	12x4	7	31	31	5	0	26	True
10	6x5	16	83	83	14	7	62	True

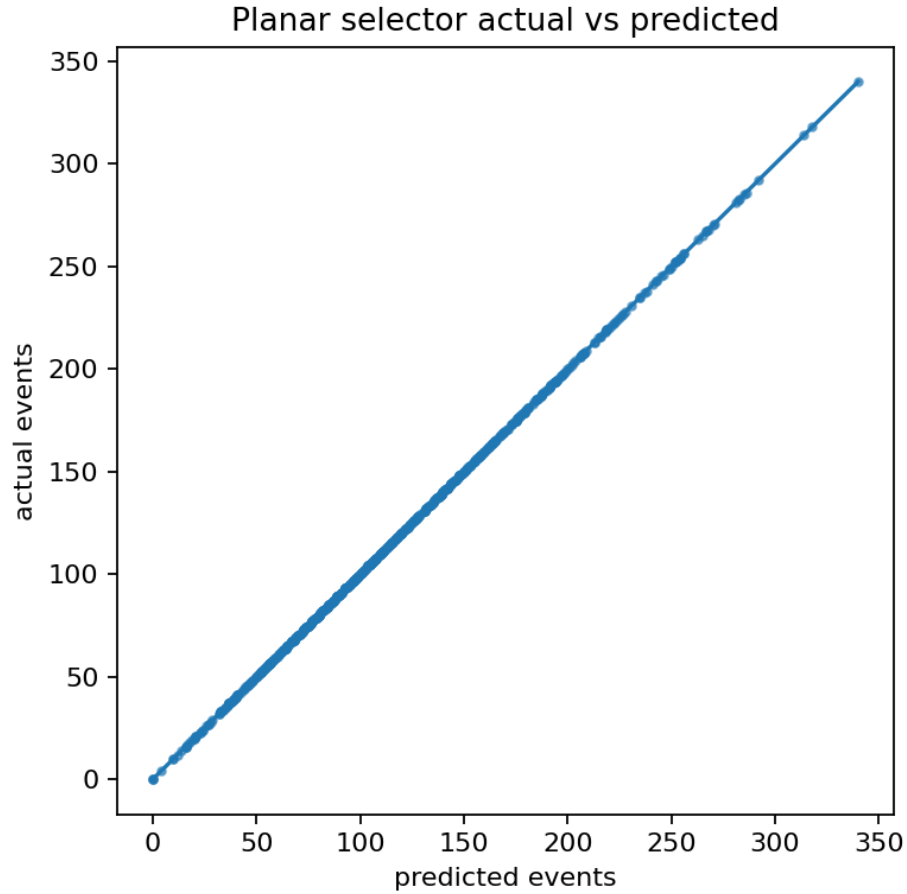


Figure 7.5: Actual first-step event counts versus q_D/r_D -predicted event counts for 720 random finite planar masks.

7.5 Q/P-style rectangle reconstruction audit

The reconstruction audit remains explicitly prior-bounded: it assumes a rectangle. With area $A = ab$ and raw response B_0 , the sum is

$$S = a + b = \frac{B_0 - 12}{4}.$$

With default prime response B_ϕ , the sum is

$$S = a + b = \frac{B_\phi + 4}{4}.$$

Then a and b are the integer roots of

$$t^2 - St + A = 0.$$

The v1.1 audit covers $2 \leq a, b \leq 36$, giving 1225 cases, and both raw and prime inversions match in every case.

Table 7.5: Representative v1.1 Q/P-style rectangle inversion rows.

a	b	A	B_0	B_ϕ	raw rec.	phi rec.	raw ok	phi ok
2	2	4	28	12	2x2	2x2	True	True
2	7	14	48	32	2x7	2x7	True	True
2	36	72	164	148	2x36	2x36	True	True
10	20	200	132	116	10x20	10x20	True	True
10	36	360	196	180	10x36	10x36	True	True
36	36	1296	300	284	36x36	36x36	True	True

7.6 Empirical multi-step trace boundary

The empirical planar relaxation trace is included because it is useful for later law discovery and for aligning the empirical atlas. It is not theorem-level evidence. The current imported trace rows record selected rectangles at $t = 0, 1, 2, 3, 6, 12$ under the raw map. They should remain in the empirical/open lane until a closed multi-step theorem is proved.

Table 7.6: Representative empirical relaxation trace rows. These rows are not theorem-level claims.

case	t	active	births	deaths	delta
2x2	0	9212	0	0	0
2x2	1	9240	28	0	28
2x2	2	9200	16	56	-40
2x2	3	9256	72	16	56
4x12	0	9168	0	0	0
4x12	1	9244	76	0	76
10x20	0	9016	0	0	0
10x20	1	9148	132	0	132

7.7 Evidence files imported in v1.1

The following generated evidence files are included under `data/defect_verifier/`: rectangle-square-channel audit, line-hole audit, arbitrary planar selector audit, Q/P rectangle inversion audit, rectangle

overcompensation audit, empirical relaxation traces, evidence summary, and checksum manifest. These files make the Defect verifier layer inspectable without relying only on printed representative rows.

Part III

PooleQ/P Capture and channel spectrometry

Chapter 8

Channel-resolved Poole transform

The ordinary update collapses all allowed births and survivals into a single binary next state. PooleQ/P keeps the local event channels separated before this collapse.

Definition 8.1 (Birth and survival channels). *For $k = 0, 1, \dots, 26$ define*

$$B_k(i; x) = (1 - x(i))\mathbf{1}\{N_i(x) = k\}, \quad S_k(i; x) = x(i)\mathbf{1}\{N_i(x) = k\}.$$

The active Poole channels are B_5, B_6, B_7 and S_5, S_6, S_7, S_8, S_9 .

Definition 8.2 (PooleQ/P feature vector). *The local channel feature vector is*

$$\Phi_i(x) = (B_5, B_6, B_7, S_5, S_6, S_7, S_8, S_9, O_{10+}, \psi)_i,$$

where

$$O_{10+}(i; x) = x(i)\mathbf{1}\{N_i(x) \geq 10\}$$

and

$$\psi_i(x) = \max\{0, N_i(x) - (7 + 2x(i))\} - \max\{0, 5 - N_i(x)\}.$$

Remark 8.3 (Readout boundary). *The channel transform is richer than the raw next-state map. If only the binary output $P(x)_i$ is observed, then channels such as B_5, B_6, B_7 are not distinguished. Exact channel claims therefore require typed readout, spatial demultiplexing, or feature extraction from the local state before channel collapse.*

Chapter 9

Probabilistic p-bit form

Let $X_j \sim \text{Bernoulli}(p_j)$ independently over sites in a local product approximation. Since the center site is not included in its own Moore count, X_i is independent of N_i under this model.

Definition 9.1 (Local generating function). *For a fixed site i define*

$$G_i(z) = \prod_{j \in N_{26}(i)} (1 - p_j + p_j z).$$

Then

$$\Pr[N_i(X) = k] = [z^k]G_i(z).$$

Proposition 9.2 (Expected Poole channels). *Under the independent Bernoulli model,*

$$\mathbb{E}[B_k(i; X)] = (1 - p_i)[z^k]G_i(z),$$

$$\mathbb{E}[S_k(i; X)] = p_i[z^k]G_i(z).$$

Therefore

$$q_i := \mathbb{E}[P(X)_i] = (1 - p_i)C_i^{5:7} + p_i C_i^{5:9},$$

where $C_i^{a:b} = \sum_{k=a}^b [z^k]G_i(z)$.

Corollary 9.3 (Hysteretic probability decomposition). *Because $C_i^{5:9} = C_i^{5:7} + C_i^{8:9}$,*

$$q_i = C_i^{5:7} + p_i C_i^{8:9}.$$

The center probability gates only the upper memory band 8, 9.

Proposition 9.4 (Differentiability). *For $\ell \in N_{26}(i)$,*

$$\frac{\partial}{\partial p_\ell} [z^k]G_i(z) = [z^k] \left((z - 1) \prod_{j \in N_{26}(i), j \neq \ell} (1 - p_j + p_j z) \right).$$

For the center site,

$$\frac{\partial q_i}{\partial p_i} = C_i^{8:9}.$$

Chapter 10

Correlation residue and geometry signatures

Given samples $X^{(1)}, \dots, X^{(M)}$, empirical channel rates can be compared with independent-site predictions built from the observed marginals. The scalar activation residue is

$$\Delta_i = \hat{q}_i - q_i^{\text{ind}}.$$

A channel-wise residue vector records deviations in selected birth and memory-survival channels.

Definition 10.1 (Poole coherence). *A simple global structured-correlation statistic is*

$$C_P = \sum_{i \in V} \|R_i\|_2^2.$$

A large value indicates that local Poole channel rates are not explained by the product of local marginals. It does not by itself prove a quantum mechanism or a particular physical noise model.

Definition 10.2 (Geometry spectrometer). *The v5.6 signature is*

$$S_P(X) = (R_B(X), R_H(X), R_\psi(X)),$$

where R_B tracks birth-window/boundary geometry, R_H tracks high-support geometry, and R_ψ tracks support-strain geometry.

10.1 Exact-cardinality footprint logic

Around a stable sheet, a normal-layer candidate sees $N = 9 - r_D(u)$. Therefore

$$B_7 \iff r_D = 2, \quad B_6 \iff r_D = 3, \quad B_5 \iff r_D = 4.$$

With selected input defects and fixed bias defects in the footprint, these channels become exact-cardinality detectors. The full source appendix contains the raw biased-footprint Boolean gates and adder readouts.

10.2 Syndrome-field boundary

PooleQ/P may be applied to already-measured readout strings, stochastic p-bit samples, or syndrome-event histories. It does not copy, clone, or reconstruct an unknown unmeasured quantum state. It is a classical transform applied after measurement.

Part IV

Signed reaction-membrane basins

Chapter 11

Signed lattice and membrane measurements

Definition 11.1 (Signed state). *The signed membrane layer uses*

$$s_i(t) \in \{-1, 0, +1\}.$$

Positive and negative populations update by same-sign support windows, while opposite support can collapse a candidate or active cell when it exceeds a threshold k .

Definition 11.2 (Interface measurements). *Let N_i^+ and N_i^- be positive and negative local support counts. A local overlap proxy is*

$$q_i = \min(N_i^+ + \mathbf{1}_{s_i=+1}, N_i^- + \mathbf{1}_{s_i=-1}).$$

The interface support is

$$Q = \sum_{i \in \Omega} q_i \mathbf{1}_{q_i \geq q_0},$$

and the normalized density is

$$Q_\rho = Q/|\Omega|.$$

The signed imbalance is

$$\epsilon_\pm = \frac{|N^+ - N^-|}{N^+ + N^- + \varepsilon}.$$

The membrane quality proxy is

$$M_P = \frac{Q_\rho}{1 + \sigma_y}(1 - \epsilon_\pm).$$

Remark 11.3 (Proxy status). *M_P is a benchmark descriptor, not a physical thrust or energy law. Legacy names in older outputs such as `thrust_proxy` must be treated as interface-quality proxies.*

Chapter 12

Preparation beats forcing

The signed membrane benchmark tests induction, maintenance, perturbation stress, survival-window ridges, geometry/boundary robustness, and mirror-quality basin size. Its central benchmark conclusion is that stable signed membranes are induced by symmetry-preserving preparation and preserved by passive or gentle dynamics. Broad high-motion forcing degrades the induced basin.

12.1 Induction result

The original signed benchmark showed that mirrored signed initialization plus a short passive warmup produced high practical scores, high interface density, thin membranes, and good-regime rates of 100 percent, while random signed initialization and preformed sheet controls were weaker.

Table 12.1: Signed-membrane induction summary from the source benchmark.

Protocol	Practical score	Q_ρ	σ_y	Good rate
mirror+10 membrane	0.785	0.548	0.000	100.0%
mirror+20 hybrid	0.751	0.522	0.001	100.0%
mirror+10 hybrid	0.745	0.520	0.007	100.0%
mirror+0 hybrid	0.700	0.507	0.080	100.0%
random hybrid	0.492	0.450	0.648	68.8%
preformed sheet hybrid	0.254	0.406	1.713	43.8%

12.2 Maintenance result

After induction, passive membrane-lock and gentle local maintainers preserve the basin more effectively than broad static forcing.

Table 12.2: Signed-membrane maintenance summary.

Maintainer	Practical	Q_ρ	σ_y	$\epsilon_\pm$	Moves/tick
passive membrane-lock	0.785	0.548	0.000	0.003	0.000
same-move random	0.771	0.536	0.006	0.003	0.777
interface-blind	0.761	0.529	0.000	0.003	0.806
hybrid guard	0.745	0.520	0.007	0.004	2.079
focus only	0.744	0.519	0.007	0.004	2.081
static broad forcing	0.602	0.485	0.191	0.014	84.233

12.3 Rule ridge

The source signed-membrane rule sweep found a stronger ridge near B5-7/S3-11 and B5-7/S4-11 than the early canonical B5-7/S5-9 rule. This is important because it prevents the program from treating one rule as sacred. The local geometry is rule-sensitive.

Table 12.3: Source signed-membrane survival-window ridge summary.

Rule	Practical	Good rate	Q_ρ	σ_y
B5-7/S4-11/k3	0.7559	1.000	0.5255	0.0001
B5-7/S3-12/k3	0.7502	1.000	0.5109	0.0001
B5-7/S3-11/k3	0.7480	1.000	0.5285	0.0025
B5-7/S5-12/k3	0.7420	1.000	0.5069	0.0033
B5-7/S5-10/k3	0.6515	1.000	0.5163	0.1826

Part V

The defect–channel bridge and spectral-basin benchmark

Chapter 13

Exact bridge theorem

The theorem bridge is the cleanest cross-paper result. It says that the Defect Calculus and PooleQ/P layers are not merely analogous. At the first step of a defective sheet, they are the same local information represented in two coordinate systems.

Theorem 13.1 (Defect-channel identity). *Let x_D be a defective stable sheet. For the first update:*

$$\begin{aligned} u \notin D : & \quad S_{8-q_D(u)}, \\ u \in D : & \quad B_{8-q_D(u)}, \\ z = c \pm 1 : & \quad B_{9-r_D(u)}. \end{aligned}$$

Thus the q_D/r_D selector and the PooleQ/P channel spectrum are equivalent first-step local descriptions.

Proof. The planar defect selector already gives the raw counts. An active sheet site has count $8 - q_D(u)$ and therefore belongs to survival channel $S_{8-q_D(u)}$ if it survives. An inactive in-plane defect has the same raw count but is a birth candidate, so it belongs to $B_{8-q_D(u)}$ when the birth window accepts it. A normal-layer candidate has count $9 - r_D(u)$ and therefore belongs to $B_{9-r_D(u)}$ when accepted. These are exactly the channel labels of the PooleQ/P transform. $\square$

Remark 13.2 (What is exact and what is not). *The identity above is theorem-level. The spectral-basin predictor is not theorem-level. It is a benchmark hypothesis: pre-warmup channel spectra should predict later signed membrane basin quality.*

Chapter 14

Spectral-basin benchmark

For a signed initial condition $s \in \{-1, 0, +1\}^\Lambda$, the benchmark computes binary fields

$$X_+ = \mathbf{1}_{\{s=+1\}}, \quad X_- = \mathbf{1}_{\{s=-1\}}, \quad X_{abs} = \mathbf{1}_{\{|s|=1\}},$$

and an interface field detecting local plus/minus contact. It then extracts spectral features from these fields and tests whether those features predict final M_P after warmup.

14.1 Hard-mode rule-ridge result

The v1.3 hard-mode ridge run tested three rules: B5-7/S5-9/k3, B5-7/S4-11/k3, and B5-7/S3-11/k3. Structured signed preparations beat every control under every tested rule.

Table 14.1: v1.3 bridge rule-ridge separation.

Rule	Family	n	Mean M_P	Min M_P	Max M_P
B5-7/S3-11/k3	control	16	0.001585	0.000000	0.016673
B5-7/S3-11/k3	structured	32	0.227726	0.116442	0.313059
B5-7/S4-11/k3	control	24	0.001145	0.000000	0.015726
B5-7/S4-11/k3	structured	24	0.223206	0.112040	0.298879
B5-7/S5-9/k3	control	21	0.000293	0.000000	0.006150
B5-7/S5-9/k3	structured	27	0.112651	0.061871	0.142781

Across the run,

$$\max M_P(\text{control}) = 0.01667, \quad \min M_P(\text{structured}) = 0.06187.$$

The weakest structured case therefore remains above the strongest control case by about $3.71\times$.

14.2 Protocol controls

The mirrored-random control is a crucial negative control. It has high corrected mirror quality but near-zero final membrane quality. This prevents the result from collapsing into the trivial claim that mirror symmetry alone is enough.

Table 14.2: v1.3 bridge protocol summary.

Protocol	n	Mean mirror quality	Mean M_P	Max M_P
mirrored random control	17	0.754132	0.003075	0.016673
same-density random	9	0.009658	0.000000	0.000000
signed shuffle control	21	0.010783	0.000320	0.006690
single-sign sheet control	14	0.000000	0.000000	0.000000
sheet-pair aligned	20	0.955595	0.245144	0.313059
sheet-pair degraded	19	0.492711	0.160364	0.225343
sheet-pair shifted	20	0.556214	0.191917	0.273400
sheet-pair widegap	12	0.924945	0.179730	0.256944
sheet-patch pair	12	0.423170	0.145072	0.227745

14.3 Predictive models

The predictor result is useful but must be stated carefully. In all cases, spectral features beat density and mirror baselines but trail explicit protocol/rule labels and combined numeric models. In structured-only rule-ridge tests, rule-aware design features dominate the fixed spectral bridge. This is a target for v1.4 rather than a failure of the basin-entry result.

Table 14.3: Selected v1.3 bridge predictive results.

Scope	Model	Mean R^2	RMSE	MAE	Spearman
all cases	density baseline	0.144	0.0977	0.0829	0.450
all cases	mirror baseline	0.497	0.0743	0.0581	0.730
all cases	protocol one-hot	0.814	0.0449	0.0316	0.849
all cases	spectral bridge	0.768	0.0506	0.0384	0.840
all cases	combined numeric	0.869	0.0372	0.0314	0.914
structured only	density baseline	0.042	0.0623	0.0531	0.306
structured only	mirror baseline	0.038	0.0613	0.0526	0.431
structured only	protocol one-hot	0.219	0.0582	0.0503	0.513
structured only	spectral bridge	0.138	0.0600	0.0491	0.452
structured only	combined numeric	0.743	0.0325	0.0255	0.859

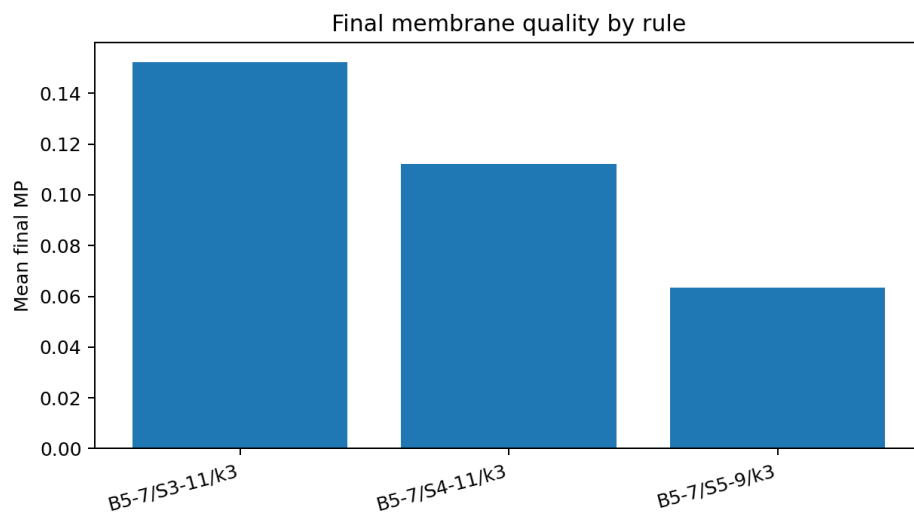


Figure 14.1: Rule-ridge bridge result: structured preparations remain separated from controls under all tested rules, and S3–11/S4–11 improve mean membrane quality over S5–9.

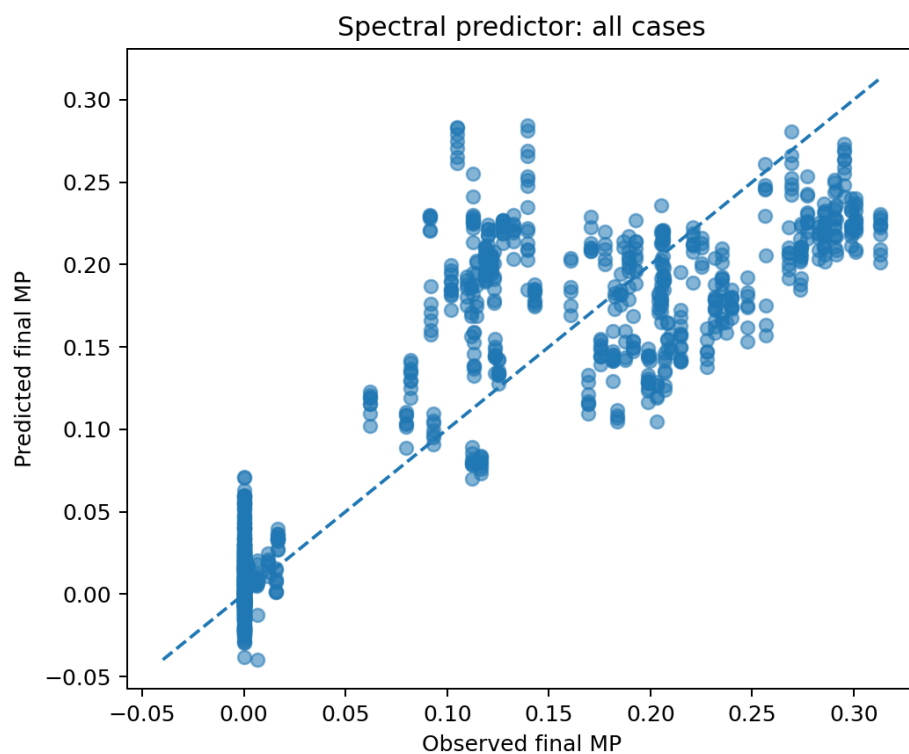


Figure 14.2: All-case prediction scatter from the v1.3 bridge run. This figure is benchmark evidence, not theorem evidence.

Chapter 15

Bounded conclusion of the bridge layer

The exact theorem is the defect–channel identity. The empirical result is basin-entry generalization: structured Poole-relevant signed geometry enters a membrane basin while high-mirror random geometry does not. The remaining open issue is fine-quality prediction inside structured families across unseen rules. A rule-aware spectral signature is the appropriate next benchmark target.

Part VI

Coverage-audited proof and benchmark expansion v1.1

Chapter 16

v1.1 coverage-audited import pass

This v1.0 pass begins moving material out of the preserved source appendices and into the front manuscript. The rule remains conservative: no source appendix is removed. Instead, the front manuscript receives proof capsules and benchmark tables while the coverage matrix records which source sections have moved from preservation-only status to front-manuscript status.

16.1 What v1.0 imports

The following material is imported into the front manuscript in this pass:

1. structural proof capsules for equivariance, finite propagation, eventual periodicity, non-monotonicity, non-reversibility, active-degree constraints, surface selection, stable planes, slab rejection, and memory ablation;
2. planar-defect proof capsules for the q_D/r_D selector, isolated pore stability, crack/line-source behavior, rectangle response, channel spectrum, prime-corrected response, work laws, and rectangle reconstruction boundary;
3. Q/P proof capsules for channel disjointness, probabilistic p-bit form, differentiability, correlation residue, exact-cardinality defect logic, raw footprint gates, adders, and syndrome-field boundaries;
4. full signed-membrane benchmark tables for induction, maintenance, stress, rule ridge, geometry/boundary robustness, and mirror-quality basin size;
5. bridge model-comparison, protocol-summary, and rule-ridge tables from the v1.3 hard-mode ridge benchmark;
6. a new source-to-section coverage matrix and omission checklist.

16.2 No-deletion rule

All original source PDFs remain included as appendices. A later version may remove duplicated appendix material only if a coverage audit confirms that every theorem, proof capsule, benchmark table, figure, script, limitation, and open problem has been imported into the front manuscript or explicitly ruled out as deprecated.

Chapter 17

Expanded Defect Calculus proof capsules

17.1 Fixed-point and active-graph constraints

Proposition 17.1 (Fixed-point constraints). *A configuration x is a fixed point of the base map if and only if every active site has $5 \leq N_i(x) \leq 9$ and every inactive site avoids the birth counts $N_i(x) \in \{5, 6, 7\}$.*

Proof. A fixed active site must satisfy the survival rule; a fixed inactive site must fail the birth rule. These two local conditions are independent at each site and exactly characterize $P(x) = x$. $\square$

Proposition 17.2 (Active-degree bounds). *If x is a nonzero fixed point and $G_A(x)$ is the graph on active sites with Moore-neighborhood edges, then every active vertex has degree between 5 and 9, and therefore*

$$5 \leq \frac{2|E(G_A)|}{|A|} \leq 9.$$

Proof. For an active site, $N_i(x)$ is exactly its active Moore degree. The fixed-point survival condition gives $5 \leq N_i \leq 9$ at every active vertex. Summing all active degrees gives $2|E(G_A)|$ and yields the average-degree bound. $\square$

17.2 Slab rejection and dimensional selectivity

Proposition 17.3 (Filled slabs are rejected). *A union of at least two consecutive full coordinate sheets dies in one step under the base rule.*

Proof. Each active site in a slab of thickness at least two sees eight active neighbors in its own sheet and at least nine in an adjacent sheet, so it has at least 17 active neighbors and exceeds the survival ceiling 9. Inactive sites outside the slab see counts 0, 9, or 18, none of which lies in the inactive birth window $\{5, 6, 7\}$. $\square$

Remark 17.4 (Why this matters). *The rule is not a generic growth rule. It rejects dust, lines, and bulk while preserving one-cell surfaces. This is the first reason the program is organized around surface defects rather than volume filling.*

17.3 The planar selector in channel form

Let $D \subset T_L$ be a missing-cell mask in a stable sheet. For $u \in T_L$, recall

$$q_D(u) = \sum_{\eta \in \{-1, 0, 1\}^2 \setminus \{0\}} \mathbf{1}\{u + \eta \in D\}, \quad r_D(u) = \sum_{\eta \in \{-1, 0, 1\}^2} \mathbf{1}\{u + \eta \in D\}.$$

The source theorem states that active sheet cells survive exactly when $q_D \leq 3$, in-plane missing cells are born exactly when $1 \leq q_D \leq 3$, and normal-layer cells are born exactly when $2 \leq r_D \leq 4$.

Proof capsule. On the sheet plane, a site sees only the eight planar neighbors of the original sheet, minus the missing sites in its open footprint. Hence the raw count is $8 - q_D(u)$. If the site is active, the survival rule accepts $5 \leq 8 - q_D \leq 9$, equivalent to $q_D \leq 3$. If the site is a missing in-plane defect, the birth rule accepts $5 \leq 8 - q_D \leq 7$, equivalent to $1 \leq q_D \leq 3$. A cell one normal layer away sees the closed 3×3 footprint below or above it, so the raw count is $9 - r_D(u)$; the birth rule accepts $5 \leq 9 - r_D \leq 7$, equivalent to $2 \leq r_D \leq 4$. All layers two or more steps away see no active neighbors on the first update. $\square$

17.4 Stable pore gas

Corollary 17.5 (Stable isolated pore gas). *If every closed 3×3 footprint contains at most one missing sheet site, then the defective sheet is a fixed point.*

Proof. The condition gives $r_D \leq 1$ everywhere, so no normal-layer birth can satisfy $2 \leq r_D \leq 4$. Each missing pore has $q_D = 0$, so it is not reborn in-plane. Every active sheet site has $q_D \leq 1 \leq 3$, so all active cells survive. $\square$

17.5 Cracks, line sources, and line-hole edge cases

The source paper separates isolated pores from slits, cracks, and line holes. The front-manuscript summary is:

1. isolated pores freeze because they never reach the normal-layer birth window;
2. adjacent missing cells create locations with $r_D \in \{2, 3, 4\}$ and therefore emit normal-layer births;
3. long line holes are edge cases whose first-step response is controlled by endpoint and edge footprints rather than area alone.

These cases remain theorem-level when stated as first-step selector consequences. Multi-step relaxation beyond the first response remains empirical-atlas material unless restricted to a proven subfamily.

Chapter 18

Rectangle, channel, prime, work, and reconstruction proof capsules

18.1 Rectangle zero-death subfamily

For an $a \times b$ rectangular puncture with $a, b \geq 2$ in an otherwise stable sheet, the first response is a closed zero-death subfamily. Its raw first-step normal birth law is

$$B_0(a, b) = 4a + 4b + 12 = 2P + 12,$$

where $P = 2a + 2b$ is the rectangle perimeter in sheet units.

Proof capsule. The active-sheet death test is $q_D \geq 4$. Around a filled rectangle, active cells adjacent to the rectangle boundary see at most three missing cells in their open planar footprint, so they survive. Normal-layer candidates are born exactly where the closed footprint contains 2, 3, or 4 rectangle cells. Counting the two normal layers above and below the rectangle gives fixed corner, edge, and near-corner contributions that sum to $4a + 4b + 12$. $\square$

18.2 Rectangle channel spectrum

The rectangle first-step births split into the raw channel spectrum

$$(N_5, N_6, N_7) = (12, 4a + 4b - 16, 16).$$

Equivalently, the B_5 group is fixed at 12, the B_7 group is fixed at 16, and the edge-extensive part lives in B_6 .

Proof capsule. For a normal-layer candidate, the raw count is $9 - r_D$. Thus B_7 corresponds to $r_D = 2$, B_6 to $r_D = 3$, and B_5 to $r_D = 4$. The closed footprints at the rectangle corners, edges, and corner-adjacent positions can be counted separately. The $r_D = 4$ corner groups give the fixed 12 count; the $r_D = 2$ shallow edge groups give the fixed 16 count; the remaining edge-length contribution gives $4a + 4b - 16$ in channel B_6 . $\square$

18.3 Prime-corrected rectangle model

The default prime-sharpened model removes the raw B_7 birth group from the rectangle response, leaving

$$B_\varphi(a, b) = B_0(a, b) - 16 = 4a + 4b - 4.$$

This is a model-tagged response, not the raw base rule. It is useful as a sharpened diagnostic because B_7 is the high birth edge and often behaves like a plasticity gate.

18.4 First-step work laws

Given nonnegative channel weights I_5, I_6, I_7 , the first-step rectangle birth work is

$$W_B(a, b) = 12I_5 + (4a + 4b - 16)I_6 + 16I_7.$$

For the prime-sharpened response that removes B_7 ,

$$W_{B,\varphi}(a, b) = 12I_5 + (4a + 4b - 16)I_6.$$

These are exact once the channel spectrum and weights are fixed. The weights themselves are bookkeeping choices unless a separate physical cost model is supplied.

18.5 Rectangle reconstruction boundary

Under the rectangle prior, the pair of channel counts is sufficient to recover the rectangle perimeter and therefore the family of possible side pairs. The boundary is limited: the channel spectrum reconstructs a rectangle class under a rectangle assumption; it does not reconstruct arbitrary masks.

Proof capsule. For rectangles, $N_6 = 4a + 4b - 16$ and $B_0 = 4a + 4b + 12$. Either value recovers $a + b$. The possible side pairs are therefore the integer pairs (a, b) with the recovered sum and the imposed side constraints. Without the rectangle prior, many masks can share the same channel counts, so the result is not arbitrary shape reconstruction. $\square$

Chapter 19

PooleQ/P proof capsules

19.1 Channel disjointness

For each site exactly one value of N_i holds. Therefore the birth channels $B_k = (1 - x_i)\mathbf{1}\{N_i = k\}$ and survival channels $S_k = x_i\mathbf{1}\{N_i = k\}$ are disjoint. The raw Poole update is the collapsed sum

$$P(x)_i = \sum_{k=5}^7 B_k(i; x) + \sum_{k=5}^9 S_k(i; x).$$

The Q/P move is to preserve the vector of channels before collapse.

19.2 Expected p-bit channels

Let $X_j \sim \text{Bernoulli}(p_j)$ independently and let

$$G_i(z) = \prod_{j \in N_{26}(i)} (1 - p_j + p_j z).$$

Then $\Pr[N_i(X) = k] = [z^k]G_i(z)$ and

$$E[B_k(i; X)] = (1 - p_i)[z^k]G_i(z), \quad E[S_k(i; X)] = p_i[z^k]G_i(z).$$

Consequently

$$q_i = E[P(X)_i] = C_i^{5:7} + p_i C_i^{8:9}.$$

This identity is the probabilistic form of local hysteresis: the center probability weights only the upper memory band.

19.3 Differentiability capsule

For a neighbor $\ell \in N_{26}(i)$,

$$\frac{\partial}{\partial p_\ell} [z^k]G_i(z) = [z^k] \left((z - 1) \prod_{j \in N_{26}(i), j \neq \ell} (1 - p_j + p_j z) \right).$$

For the center site,

$$\frac{\partial q_i}{\partial p_i} = C_i^{8:9}.$$

This gives a differentiable probability-layer version of the channel transform under the independent local product approximation.

19.4 Correlation residue and Poole coherence

Given repeated measured samples, empirical channel rates can be compared against the independent prediction formed from observed local marginals. The channel-wise residue vector is

$$R_i = (\hat{B}_5 - E_{ind}B_5, \hat{B}_6 - E_{ind}B_6, \hat{B}_7 - E_{ind}B_7, \hat{S}_8 - E_{ind}S_8, \hat{S}_9 - E_{ind}S_9)_i,$$

and a global local-correlation statistic is

$$C_P = \sum_i \|R_i\|_2^2.$$

Large C_P indicates that product marginals fail to explain local channel rates. It is not by itself evidence for a particular physical noise model.

19.5 Exact-cardinality defect logic

For a defective stable sheet and a normal-layer candidate, the active neighbor count is $N = 9 - r_D$. Hence

$$B_7 \Leftrightarrow r_D = 2, \quad B_6 \Leftrightarrow r_D = 3, \quad B_5 \Leftrightarrow r_D = 4.$$

With selected input defects and fixed bias defects in the same 3×3 footprint, this becomes exact-cardinality detection. If s input defects and b bias defects give $r = s + b$, then the birth channels select the exact values $s = 2 - b$, $s = 3 - b$, and $s = 4 - b$.

19.6 Gate and adder boundary

The source Q/P paper shows how biased defect footprints can implement raw Boolean gate readouts and channel-resolved adder readouts. The boundary is important: these are exact for channel-resolved readout, spatially demultiplexed layouts, or instrumented local feature readout. A one-state uninstrumented cellular automaton that observes only $P(x)_i$ collapses the channel identity.

19.7 Syndrome-field boundary

For qubit readout strings or stabilizer syndrome histories, PooleQ/P is a classical transform applied after measurement. It does not copy, clone, or reconstruct an unknown quantum state. Its role is local geometry spectrometry: mapping already-measured binary fields to interpretable channel signatures.

Chapter 20

PooleQ/P verifier and benchmark import v1.2

This chapter imports the PooleQ/P verifier and benchmark layer into the front manuscript. The purpose is not to elevate every Q/P application into a theorem. The purpose is to separate three claim types cleanly: exact footprint identities, finite same-density shuffled-control benchmarks, and bounded spectrometer interpretation.

20.1 v1.2 Q/P pass/fail summary

The exact footprint checks are first-update local statements under the sheet-footprint assumptions already stated above. They check the raw biased-footprint Boolean primitives, one-input identity/not cases, half-adder channel readout, full-adder channel readout, and the cardinality-to-channel equivalence.

Table 20.1: v1.2 Q/P footprint verifier pass/fail summary.

Check	Matches	Cases
Raw two-input gates: AND, OR, NAND, NOR	16	16
One-input identity/not	4	4
Half-adder channel readout	4	4
Full-adder channel readout	8	8
Cardinality-to-channel table	10	10

20.2 Exact-cardinality and raw gate verifier

For a normal-layer candidate adjacent to a defective sheet, r missing footprint cells give $N = 9 - r$ active neighbors. The raw collapsed birth event is therefore

$$Y = \mathbf{1}\{2 \leq r \leq 4\}.$$

With two inputs A, B and a fixed bias count c , the verifier checks

$$Y_{2,c}(A, B) = \mathbf{1}\{2 \leq c + A + B \leq 4\}.$$

The four bias choices reproduce the expected raw gates.

Table 20.2: Raw biased-footprint gate truth table.

Gate	A	B	bias c	Y	expected
AND	0	0	0	0	0
AND	0	1	0	0	0
AND	1	0	0	0	0
AND	1	1	0	1	1
OR	0	0	1	0	0
OR	0	1	1	1	1
OR	1	0	1	1	1
OR	1	1	1	1	1
NAND	0	0	3	1	1
NAND	0	1	3	1	1
NAND	1	0	3	1	1
NAND	1	1	3	0	0
NOR	0	0	4	1	1
NOR	0	1	4	0	0
NOR	1	0	4	0	0
NOR	1	1	4	0	0

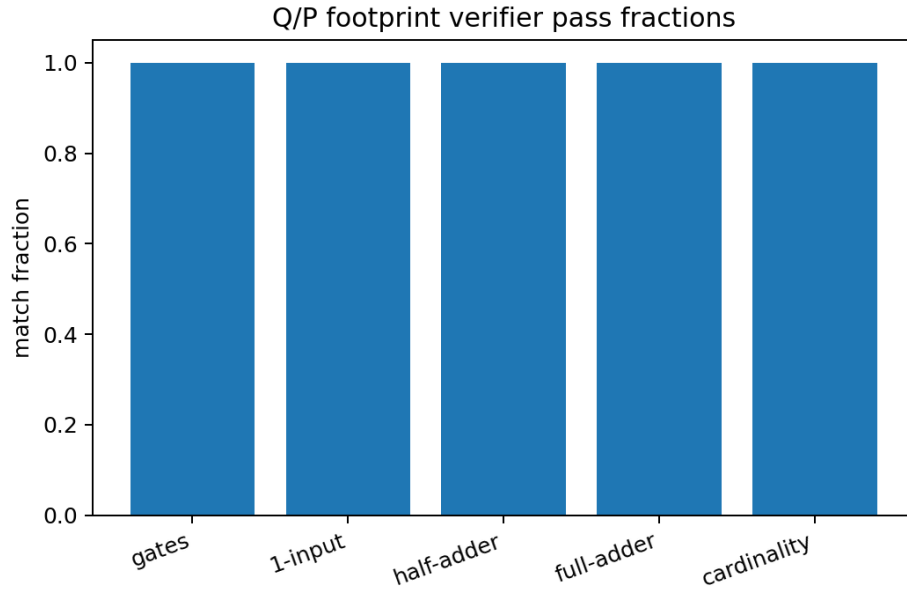


Figure 20.1: v1.2 Q/P footprint verifier pass fractions. These are exact finite truth-table checks, not statistical benchmarks.

20.3 Channel-resolved adder verifier

For the half-adder, the footprint uses one bias defect so that $r = A + B + 1$. Then B_7 is the exact-one channel and B_6 is the exact-two channel:

$$\text{SUM} = B_7, \quad \text{CARRY} = B_6.$$

For the full-adder with inputs A, B, C_{in} and the same one-bias layout,

$$B_7 = \mathbf{1}\{s = 1\}, \quad B_6 = \mathbf{1}\{s = 2\}, \quad B_5 = \mathbf{1}\{s = 3\},$$

where $s = A + B + C_{in}$. Therefore

$$\text{SUM} = B_7 \vee B_5, \quad \text{CARRY} = B_6 \vee B_5.$$

Table 20.3: Full-adder channel verifier truth table.

A	B	C_{in}	s	B_7	B_6	B_5	SUM	CARRY
0	0	0	0	0	0	0	0	0
0	0	1	1	1	0	0	1	0
0	1	0	1	1	0	0	1	0
0	1	1	2	0	1	0	0	1
1	0	0	1	1	0	0	1	0
1	0	1	2	0	1	0	0	1
1	1	0	2	0	1	0	0	1
1	1	1	3	0	0	1	1	1

Remark 20.1 (Adder boundary). *The full-adder statement is channel-resolved. If the experiment records only the collapsed raw event $Y = B_5 \vee B_6 \vee B_7$, the three-input layout collapses to $A \vee B \vee C_{in}$ for the one-bias case. The adder therefore requires channel readout or spatial demultiplexing.*

20.4 Same-density shuffled-control benchmark import

The Q/P benchmark layer is a finite measurement-side result. It asks whether channel spectra distinguish structured fields from same-density nulls after active count has been controlled. In the v5.5 decomposed benchmark, all listed fields were active-density matched and exact same-density shuffled controls were included.

Table 20.4: v5.5 decomposed robust capture scores imported into the front manuscript.

Class	R_B	R_H	R_ψ	R_C	Signature
IID null	0.21	0.02	0.16	0.28	null-like
Straight lines	3.49	2.19	3.28	5.28	mixed geometry
Lines shuffled	0.24	0.03	0.17	0.31	null-like
Random-walk chains	3.62	5.63	4.15	7.88	high-support-led
Walks shuffled	0.23	0.02	0.19	0.31	null-like
Branching chains	3.87	3.66	3.88	6.61	mixed geometry
Branches shuffled	0.21	0.02	0.14	0.27	null-like
Compact bursts	3.12	7.06	5.00	9.19	high-support-led
Bursts shuffled	0.23	0.02	0.16	0.29	null-like
Sheets	3.50	7.40	5.31	9.76	high-support-led
Sheets shuffled	0.22	0.04	0.16	0.29	null-like

Table 20.5: Paired same-density shuffled controls from the v5.5 benchmark.

Structured class	R_C	Shuffled R_C	Drop	Signature
Straight lines	5.28	0.31	4.97	mixed geometry
Random-walk chains	7.88	0.31	7.57	high-support-led
Branching chains	6.61	0.27	6.34	mixed geometry
Compact bursts	9.19	0.29	8.90	high-support-led
Sheets	9.76	0.29	9.47	high-support-led

The preserved v5.6 spectrometer shares express the direction of the signal after null separation. They are not interpreted for null-like classes.

Table 20.6: v5.6 geometry-spectrometer shares imported from the decomposed benchmark.

Class	R_C	Birth share	High-support share	Strain share	Readout
IID null	0.28	0.64	0.01	0.36	null-like
Straight lines	5.28	0.44	0.17	0.39	mixed geometry
Random-walk chains	7.88	0.21	0.51	0.28	high-support-led
Branching chains	6.61	0.34	0.31	0.35	mixed geometry
Compact bursts	9.19	0.11	0.59	0.30	high-support-led
Sheets	9.76	0.13	0.58	0.30	high-support-led

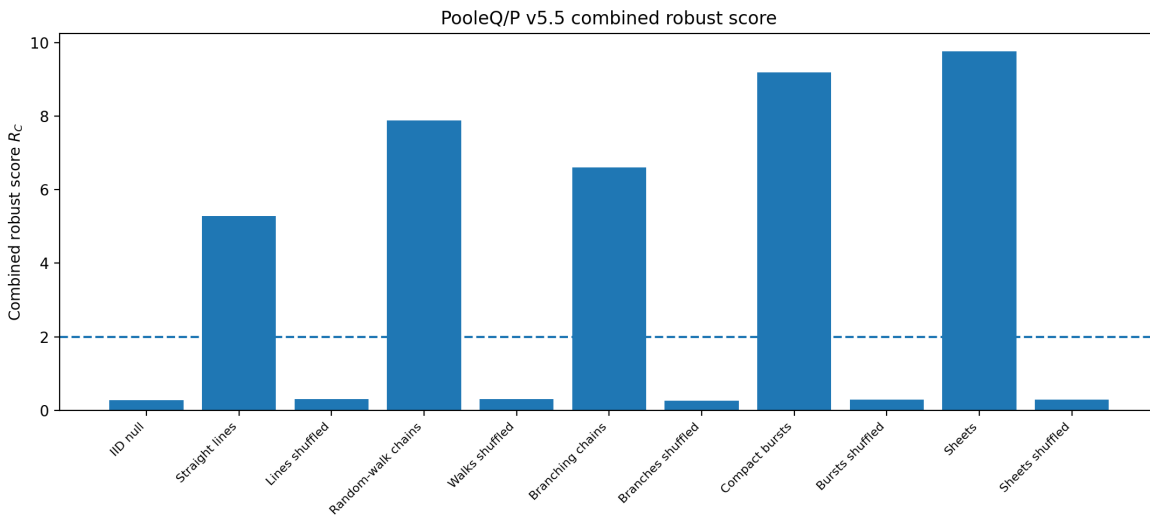


Figure 20.2: Combined non-redundant robust score R_C for the v5.5 Q/P benchmark. Structured fields remain high while IID and same-density shuffled controls remain null-like.

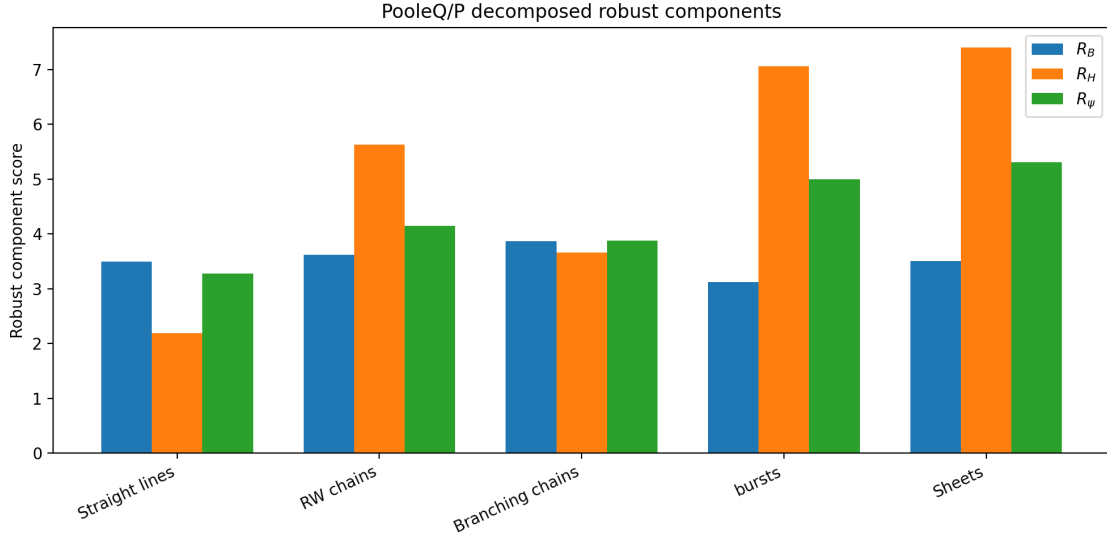


Figure 20.3: Decomposed robust components for the v5.5 Q/P benchmark. The decomposition gives an interpretable geometry signature rather than only a scalar anomaly score.

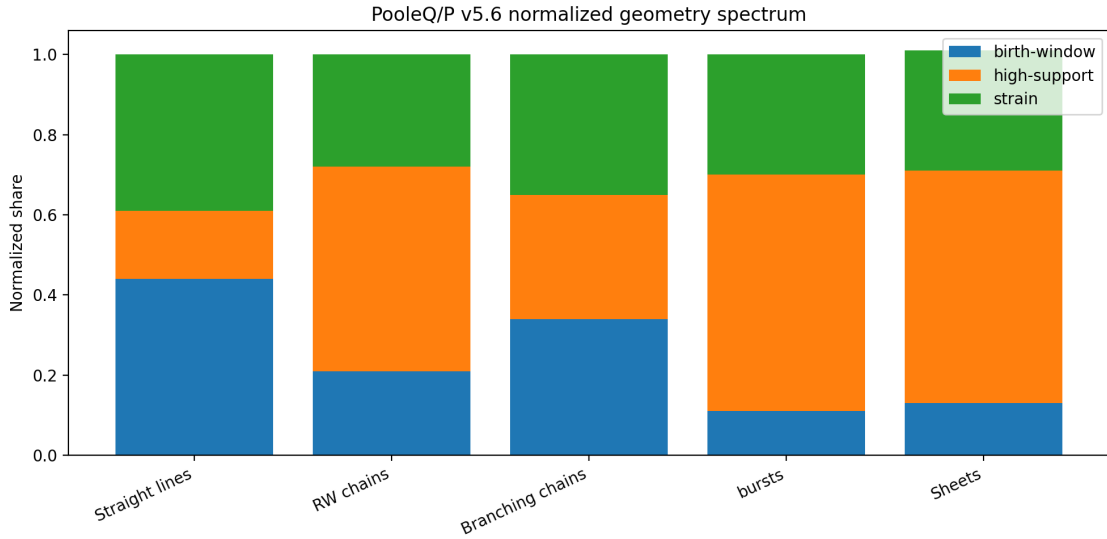


Figure 20.4: v5.6 geometry-spectrometer share interpretation for the non-null structured fields.

20.5 v1.2 Q/P benchmark boundary

The Q/P benchmark supports the following bounded claim: PooleQ/P maps already-measured binary fields into local channel spectra that separate structured fields from exact same-density shuffled controls in the tested synthetic families. It does not establish state reconstruction, quantum amplitude capture, or an autonomous computing architecture. The exact footprint identities are theorem/verifier-level; the shuffled-control separation is benchmark-level; and the geometry-spectrometer interpretation is a bounded reading of the decomposed scores.

Chapter 21

Signed-membrane benchmark tables imported into the front manuscript

The following tables are imported directly from the recovered original signed reaction-membrane benchmark bundle. The tables use the manuscript-safe interpretation of M_P and J : they are benchmark descriptors, not physical energy or thrust laws.

Table 21.1: Signed-membrane induction table imported from the full benchmark bundle.

condition	maintainer	init/ticks	J	Q_ρ	σ_y	$\epsilon_\pm$	good
mirrored0_hybrid	hybrid_guard	mirrored/0	0.700	0.507	0.080	0.007	100.0%
mirrored10_hybrid	hybrid_guard	mirrored/10	0.745	0.520	0.007	0.004	100.0%
mirrored10_membrane	passive_lock	mirrored/10	0.785	0.548	0.000	0.003	100.0%
mirrored20_hybrid	hybrid_guard	mirrored/20	0.751	0.522	0.001	0.003	100.0%
preformed_sheet_hybrid	hybrid_guard	preformed_sheet/0	0.254	0.406	1.713	0.164	43.8%
random_hybrid	hybrid_guard	random/0	0.492	0.450	0.648	0.072	68.8%

Table 21.2: Signed-membrane maintenance table imported into the front manuscript.

condition	maintainer	J	Q_ρ	σ_y	$\epsilon_\pm$	moves
maintain_focus	focus_only	0.744	0.519	0.007	0.004	4.163
maintain_hybrid	hybrid_guard	0.745	0.520	0.007	0.004	4.158
maintain_interfaceblind	interface_blind	0.761	0.529	0.000	0.003	1.613
maintain_membrane	passive_lock	0.785	0.548	0.000	0.003	0.000
maintain_random	same_move_random	0.771	0.536	0.006	0.003	1.554
maintain_static	broad_static_forcing	0.602	0.485	0.191	0.014	168.467

Table 21.3: Perturbation-stress results imported from the signed benchmark bundle.

perturb.	controller	J	Q_ρ	σ_y	good
0.0	focus_only	0.746	0.522	0.001	1.000
0.0	hybrid_guard	0.746	0.522	0.001	1.000
0.0	passive_lock	0.781	0.552	0.000	1.000
0.0	broad_static_forcing	0.609	0.485	0.177	1.000
0.01	focus_only	0.736	0.520	0.019	1.000
0.01	hybrid_guard	0.735	0.520	0.018	1.000
0.01	passive_lock	0.772	0.545	0.000	1.000
0.01	broad_static_forcing	0.616	0.485	0.152	1.000
0.03	focus_only	0.736	0.518	0.007	1.000
0.03	hybrid_guard	0.736	0.518	0.007	1.000
0.03	passive_lock	0.759	0.536	0.006	1.000
0.03	broad_static_forcing	0.601	0.484	0.205	1.000

perturb.	controller	J	Q_ρ	σ_y	good
0.05	focus_only	0.726	0.514	0.026	1.000
0.05	hybrid_guard	0.725	0.513	0.025	1.000
0.05	passive_lock	0.745	0.533	0.028	1.000
0.05	broad_static_forcing	0.605	0.484	0.186	1.000
0.1	focus_only	0.711	0.515	0.062	1.000
0.1	hybrid_guard	0.709	0.516	0.072	1.000
0.1	passive_lock	0.725	0.526	0.047	1.000
0.1	broad_static_forcing	0.613	0.484	0.167	1.000

rule	gentle J	good
B5-7.S4-11.k3	0.756	100.0%
B5-7.S3-12.k3	0.750	100.0%
B5-7.S3-11.k3	0.748	100.0%
B5-7.S5-12.k3	0.742	100.0%
B5-7.S5-10.k3	0.651	100.0%
B5-7.S5-9.k3	0.560	100.0%

Table 21.4: Signed-membrane survival-window ridge table.

Table 21.5: Complete condensed geometry and boundary robustness table.

L	boundary	controller	J	Q_ρ	σ_y	$\epsilon_\pm$	good
32	absorbing_outlet	focus_only	0.830	0.616	0.000	0.005	100.0%
32	absorbing_outlet	hybrid_guard	0.830	0.616	0.000	0.005	100.0%
32	absorbing_outlet	interface_blind	0.837	0.616	0.000	0.005	100.0%
32	absorbing_outlet	passive_lock	0.872	0.648	0.000	0.003	100.0%
32	absorbing_outlet	broad_static_forcing	0.760	0.591	0.110	0.006	100.0%
32	zero	focus_only	0.895	0.644	0.000	0.005	100.0%
32	zero	hybrid_guard	0.895	0.644	0.000	0.005	100.0%
32	zero	interface_blind	0.906	0.645	0.000	0.005	100.0%
32	zero	passive_lock	0.944	0.682	0.000	0.003	100.0%
32	zero	broad_static_forcing	0.823	0.624	0.131	0.008	100.0%
40	absorbing_outlet	focus_only	0.701	0.498	0.000	0.003	100.0%
40	absorbing_outlet	hybrid_guard	0.701	0.498	0.000	0.003	100.0%
40	absorbing_outlet	interface_blind	0.712	0.510	0.000	0.003	100.0%
40	absorbing_outlet	passive_lock	0.734	0.526	0.000	0.003	100.0%
40	absorbing_outlet	broad_static_forcing	0.573	0.463	0.174	0.008	100.0%
40	zero	focus_only	0.743	0.518	0.000	0.003	100.0%
40	zero	hybrid_guard	0.743	0.518	0.000	0.003	100.0%
40	zero	interface_blind	0.760	0.531	0.000	0.003	100.0%
40	zero	passive_lock	0.779	0.547	0.000	0.002	100.0%
40	zero	broad_static_forcing	0.603	0.480	0.197	0.010	100.0%
48	absorbing_outlet	focus_only	0.626	0.420	0.021	0.003	75.0%
48	absorbing_outlet	hybrid_guard	0.627	0.420	0.021	0.003	75.0%
48	absorbing_outlet	interface_blind	0.646	0.432	0.000	0.003	87.5%
48	absorbing_outlet	passive_lock	0.656	0.442	0.000	0.003	75.0%
48	absorbing_outlet	broad_static_forcing	0.589	0.410	0.116	0.004	37.5%
48	zero	focus_only	0.657	0.434	0.026	0.004	100.0%
48	zero	hybrid_guard	0.657	0.434	0.025	0.004	100.0%
48	zero	interface_blind	0.678	0.447	0.000	0.003	100.0%
48	zero	passive_lock	0.690	0.458	0.000	0.003	100.0%
48	zero	broad_static_forcing	0.616	0.424	0.136	0.005	62.5%

Table 21.6: Complete condensed mirror-quality basin table.

mirror q	controller	J	Q_ρ	σ_y	$\epsilon_\pm$	good
0.25	hybrid_guard	0.196	0.504	3.123	0.207	18.8%
0.25	passive_lock	0.200	0.506	3.146	0.205	0.0%

mirror q	controller	J	Q_ρ	σ_y	$\epsilon_\pm$	good
0.5	hybrid_guard	0.405	0.440	1.399	0.088	43.8%
0.5	passive_lock	0.414	0.450	1.425	0.088	6.2%
0.75	hybrid_guard	0.552	0.464	0.472	0.035	87.5%
0.75	passive_lock	0.556	0.472	0.486	0.035	62.5%
0.9	hybrid_guard	0.622	0.485	0.295	0.021	100.0%
0.9	passive_lock	0.633	0.492	0.280	0.019	100.0%
1.0	hybrid_guard	0.745	0.520	0.007	0.004	100.0%
1.0	passive_lock	0.785	0.548	0.000	0.003	100.0%
1.0	hybrid_guard	0.492	0.450	0.648	0.072	68.8%
1.0	passive_lock	0.490	0.449	0.645	0.075	31.2%

Chapter 22

Signed-membrane benchmark deep import v1.3

This v1.3 pass moves the recovered signed reaction-membrane benchmark bundle further into the front manuscript. The previous front manuscript already imported the paper-facing induction, maintenance, stress, rule-ridge, geometry, and mirror-quality tables. This chapter adds the certification layer: file manifest, seed counts, certificate flags, paired seed comparisons, original figures, and generated pairwise-lift summaries.

The benchmark remains a finite cellular-automaton benchmark. It does not claim antimatter production, storage, shielding, propulsion, or a physical energy law. The quantities M_P , Q_ρ , σ_y , $\epsilon_\pm$, and J are morphology and scoring descriptors for signed reaction-interface basins.

22.1 Recovered benchmark bundle and certificate

The recovered benchmark bundle contains by-seed CSVs, by-setting CSVs, paired-comparison CSVs, paper-facing tables, figures, a configuration file, a certificate, and a summary file. The file manifest is included to make the data path explicit.

Table 22.1: Signed-membrane recovered benchmark file dimensions for v1.3.

file	rows	columns	unique conditions	unique seeds
benchmark_all_by_seed.csv	1284	87	109	48
benchmark_induction_by_seed.csv	96	87	6	16
benchmark_maintenance_by_seed.csv	96	87	6	16
benchmark_stress_by_seed.csv	300	87	25	12
benchmark_rule_ridge_by_seed.csv	360	87	30	12
benchmark_geometry_by_seed.csv	240	87	30	8
benchmark_mirror_quality_by_seed.csv	192	87	12	16
benchmark_induction_by_setting.csv	6	119	6	
benchmark_maintenance_by_setting.csv	6	119	6	
benchmark_stress_by_setting.csv	25	119	25	
benchmark_rule_ridge_by_setting.csv	30	119	30	
benchmark_geometry_by_setting.csv	30	119	30	
benchmark_mirror_quality_by_setting.csv	12	119	12	

The certificate records the benchmark seed counts and the main pass/fail conclusions used by the signed-membrane section. The safest reading is that the certificate validates the benchmark families under their tested parameters, not an unlimited rule-space or geometry-space theorem.

Table 22.2: Signed-membrane benchmark certificate imported into the front manuscript.

certificate field	value
version	benchmark_v1
benchmark_name	Poole Signed Reaction Membrane Benchmark v1
final_reference_rule	B5-7_S3-11_k3
best_rule.in_benchmark_rule_ridge	B5-7_S4-11_k3
seed_count_main	16
seed_count_ridge	12
seed_count_geometry	8
seed_count_stress	12
induction_practical_certified_mirrored10_hybrid_vs_random	True
induction_good_regime_certified_mirrored10_hybrid_vs_random	True
stress_membrane_beats_static_all_perturbations	True
geometry_membrane_beats_static_all_L_boundary	True
best_rule_beats_canonical_all_gentle_controllers	True
perfect_mirror_practical_certified_vs_random_membrane	True
mirror90_practical_certified_vs_random_membrane	True

22.2 Pairwise seed certification

The paired CSVs compare candidate and control conditions seed-by-seed. This is stronger than comparing only unpaired means, because the same seed structure is used to estimate paired differences and confidence intervals.

Table 22.3: Condensed signed-membrane paired practical-score certification by evidence family.

family	comps.	cert.	mean diff	min diff	max diff	mean win
geometry	12	12	0.078	0.029	0.177	1.000
induction	5	5	0.268	0.045	0.490	0.950
maintenance	5	3	0.065	-0.026	0.183	0.613
mirror_quality	8	8	0.226	0.122	0.371	0.984
rule_ridge	25	23	0.151	0.027	0.214	0.953
stress	15	15	0.103	0.016	0.172	0.994

The complete paired practical-score table is included below. Rows with positive certified status support the signed-membrane claim that prepared membranes are induced and maintained by local geometry and gentle dynamics, while broad forcing and random controls do not generate the same basin.

Table 22.4: Complete v1.3 paired practical-score table from the signed benchmark. Each row compares a candidate condition against a matched control using paired seeds.

group	comparison	candidate	control	n	cand.	ctrl.	diff	win
induction	induction_vs_random	mirrored10_hybrid	random_hybrid	16	0.745	0.492	0.252	1.000
induction	induction_vs_random	mirrored20_hybrid	random_hybrid	16	0.751	0.492	0.259	1.000
induction	warmup_gain	mirrored10_hybrid	mirrored0_hybrid	16	0.745	0.700	0.045	0.750
induction	passive_induction_vs_random	mirrored10_membrane	random_hybrid	16	0.785	0.492	0.293	1.000
induction	induction_vs_forced_sheet	mirrored10_hybrid	preformed_sheet_hybrid	16	0.745	0.254	0.490	1.000
maintenance	passive_vs_static	maintain_membrane	maintain_static	16	0.785	0.602	0.183	1.000
maintenance	passive_vs_hybrid	maintain_membrane	maintain_hybrid	16	0.785	0.745	0.040	1.000
maintenance	hybrid_vs_random	maintain_hybrid	maintain_random	16	0.745	0.771	-0.026	0.062
maintenance	hybrid_vs_interfaceblind	maintain_hybrid	maintain_interfaceblind	16	0.745	0.761	-0.017	0.000
maintenance	focus_vs_static	maintain_focus	maintain_static	16	0.744	0.602	0.143	1.000
stress	membrane_vs_static_stress	stress_p00_membrane	stress_p00_static	12	0.781	0.609	0.172	1.000
stress	membrane_vs_hybrid_stress	stress_p00_membrane	stress_p00_hybrid	12	0.781	0.746	0.035	1.000
stress	interfaceblind_vs_static_stress	stress_p00_interfaceblind	stress_p00_static	12	0.763	0.609	0.154	1.000
stress	membrane_vs_static_stress	stress_p01_membrane	stress_p01_static	12	0.772	0.616	0.156	1.000
stress	membrane_vs_hybrid_stress	stress_p01_membrane	stress_p01_hybrid	12	0.772	0.735	0.037	1.000
stress	interfaceblind_vs_static_stress	stress_p01_interfaceblind	stress_p01_static	12	0.756	0.616	0.139	1.000
stress	membrane_vs_static_stress	stress_p03_membrane	stress_p03_static	12	0.759	0.601	0.158	1.000
stress	membrane_vs_hybrid_stress	stress_p03_membrane	stress_p03_hybrid	12	0.759	0.736	0.022	1.000
stress	interfaceblind_vs_static_stress	stress_p03_interfaceblind	stress_p03_static	12	0.745	0.601	0.144	1.000
stress	membrane_vs_static_stress	stress_p05_membrane	stress_p05_static	12	0.745	0.605	0.140	1.000
stress	membrane_vs_hybrid_stress	stress_p05_membrane	stress_p05_hybrid	12	0.745	0.725	0.020	1.000
stress	interfaceblind_vs_static_stress	stress_p05_interfaceblind	stress_p05_static	12	0.733	0.605	0.128	1.000
stress	membrane_vs_static_stress	stress_p10_membrane	stress_p10_static	12	0.725	0.613	0.112	1.000
stress	membrane_vs_hybrid_stress	stress_p10_membrane	stress_p10_hybrid	12	0.725	0.709	0.016	0.917
stress	interfaceblind_vs_static_stress	stress_p10_interfaceblind	stress_p10_static	12	0.726	0.613	0.113	1.000
rule_ridge	rule_vs_canonical	rule_B57_S510_k3_membrane	rule_B57_S59_k3_membrane	12	0.717	0.564	0.153	1.000
rule_ridge	rule_vs_canonical	rule_B57_S510_k3_hybrid	rule_B57_S59_k3_hybrid	12	0.623	0.558	0.065	0.833
rule_ridge	rule_vs_canonical	rule_B57_S510_k3_focus	rule_B57_S59_k3_focus	12	0.619	0.558	0.062	1.000
rule_ridge	rule_vs_canonical	rule_B57_S510_k3_interfaceblind	rule_B57_S59_k3_interfaceblind	12	0.647	0.562	0.085	0.917
rule_ridge	rule_vs_canonical	rule_B57_S510_k3_static	rule_B57_S59_k3_static	12	0.512	0.485	0.027	0.500
rule_ridge	rule_vs_canonical	rule_B57_S411_k3_membrane	rule_B57_S59_k3_membrane	12	0.777	0.564	0.213	1.000
rule_ridge	rule_vs_canonical	rule_B57_S411_k3_hybrid	rule_B57_S59_k3_hybrid	12	0.744	0.558	0.186	1.000
rule_ridge	rule_vs_canonical	rule_B57_S411_k3_focus	rule_B57_S59_k3_focus	12	0.744	0.558	0.186	1.000
rule_ridge	rule_vs_canonical	rule_B57_S411_k3_interfaceblind	rule_B57_S59_k3_interfaceblind	12	0.758	0.562	0.196	1.000
rule_ridge	rule_vs_canonical	rule_B57_S411_k3_static	rule_B57_S59_k3_static	12	0.561	0.485	0.077	0.917
rule_ridge	rule_vs_canonical	rule_B57_S512_k3_membrane	rule_B57_S59_k3_membrane	12	0.771	0.564	0.207	1.000
rule_ridge	rule_vs_canonical	rule_B57_S512_k3_hybrid	rule_B57_S59_k3_hybrid	12	0.724	0.558	0.167	1.000
rule_ridge	rule_vs_canonical	rule_B57_S512_k3_focus	rule_B57_S59_k3_focus	12	0.724	0.558	0.167	1.000
rule_ridge	rule_vs_canonical	rule_B57_S512_k3_interfaceblind	rule_B57_S59_k3_interfaceblind	12	0.748	0.562	0.186	1.000
rule_ridge	rule_vs_canonical	rule_B57_S512_k3_static	rule_B57_S59_k3_static	12	0.511	0.485	0.027	0.667
rule_ridge	rule_vs_canonical	rule_B57_S311_k3_membrane	rule_B57_S59_k3_membrane	12	0.771	0.564	0.207	1.000
rule_ridge	rule_vs_canonical	rule_B57_S311_k3_hybrid	rule_B57_S59_k3_hybrid	12	0.734	0.558	0.176	1.000
rule_ridge	rule_vs_canonical	rule_B57_S311_k3_focus	rule_B57_S59_k3_focus	12	0.734	0.558	0.176	1.000
rule_ridge	rule_vs_canonical	rule_B57_S311_k3_interfaceblind	rule_B57_S59_k3_interfaceblind	12	0.753	0.562	0.191	1.000
rule_ridge	rule_vs_canonical	rule_B57_S311_k3_static	rule_B57_S59_k3_static	12	0.612	0.485	0.127	1.000
rule_ridge	rule_vs_canonical	rule_B57_S312_k3_membrane	rule_B57_S59_k3_membrane	12	0.778	0.564	0.214	1.000
rule_ridge	rule_vs_canonical	rule_B57_S312_k3_hybrid	rule_B57_S59_k3_hybrid	12	0.734	0.558	0.176	1.000
rule_ridge	rule_vs_canonical	rule_B57_S312_k3_focus	rule_B57_S59_k3_focus	12	0.734	0.558	0.176	1.000
rule_ridge	rule_vs_canonical	rule_B57_S312_k3_interfaceblind	rule_B57_S59_k3_interfaceblind	12	0.755	0.562	0.193	1.000
rule_ridge	rule_vs_canonical	rule_B57_S312_k3_static	rule_B57_S59_k3_static	12	0.613	0.485	0.129	1.000
geometry	membrane_vs_static_geometry	geom_L32_zero_membrane	geom_L32_zero_static	8	0.944	0.823	0.122	1.000
geometry	membrane_vs_hybrid_geometry	geom_L32_zero_membrane	geom_L32_zero_hybrid	8	0.944	0.895	0.049	1.000
geometry	membrane_vs_static_geometry	geom_L32_absorbing_outlet_membrane	geom_L32_absorbing_outlet_static	8	0.872	0.760	0.111	1.000
geometry	membrane_vs_hybrid_geometry	geom_L32_absorbing_outlet_membrane	geom_L32_absorbing_outlet_hybrid	8	0.872	0.830	0.042	1.000
geometry	membrane_vs_static_geometry	geom_L40_zero_membrane	geom_L40_zero_static	8	0.779	0.603	0.177	1.000
geometry	membrane_vs_hybrid_geometry	geom_L40_zero_membrane	geom_L40_zero_hybrid	8	0.779	0.743	0.036	1.000
geometry	membrane_vs_static_geometry	geom_L40_absorbing_outlet_membrane	geom_L40_absorbing_outlet_static	8	0.734	0.573	0.162	1.000
geometry	membrane_vs_hybrid_geometry	geom_L40_absorbing_outlet_membrane	geom_L40_absorbing_outlet_hybrid	8	0.734	0.701	0.033	1.000
geometry	membrane_vs_static_geometry	geom_L48_zero_membrane	geom_L48_zero_static	8	0.690	0.616	0.074	1.000
geometry	membrane_vs_hybrid_geometry	geom_L48_zero_membrane	geom_L48_zero_hybrid	8	0.690	0.657	0.033	1.000
geometry	membrane_vs_static_geometry	geom_L48_absorbing_outlet_membrane	geom_L48_absorbing_outlet_static	8	0.656	0.589	0.066	1.000
geometry	membrane_vs_hybrid_geometry	geom_L48_absorbing_outlet_membrane	geom_L48_absorbing_outlet_hybrid	8	0.656	0.627	0.029	1.000
mirror_quality	perfect_mirror_vs_random	mirror_q100_membrane	mirror_random_membrane	16	0.785	0.490	0.295	1.000
mirror_quality	90pct_mirror_vs_random	mirror_q090_membrane	mirror_random_membrane	16	0.633	0.490	0.142	0.938
mirror_quality	perfect_vs_90pct	mirror_q100_membrane	mirror_q090_membrane	16	0.785	0.633	0.153	1.000

group	comparison	candidate	control	<i>n</i>	cand.	ctrl.	diff	win
mirror-quality	perfect_vs_50pct	mirror_q100_membrane	mirror_q050_membrane	16	0.785	0.414	0.371	1.000
mirror-quality	perfect_mirror_vs_random	mirror_q100_hybrid	mirror_random_hybrid	16	0.745	0.492	0.252	1.000
mirror-quality	90pct_mirror_vs_random	mirror_q090_hybrid	mirror_random_hybrid	16	0.622	0.492	0.130	0.938
mirror-quality	perfect_vs_90pct	mirror_q100_hybrid	mirror_q090_hybrid	16	0.745	0.622	0.122	1.000
mirror-quality	perfect_vs_50pct	mirror_q100_hybrid	mirror_q050_hybrid	16	0.745	0.405	0.339	1.000

22.3 Perturbation, geometry, and mirror-quality pair lifts

The stress-pair table isolates the most direct stress claim: passive membrane lock remains above broad static forcing across the tested perturbation rates.

Table 22.5: Perturbation-stress paired practical-score lift for passive membrane lock over broad static forcing.

perturb.	membrane	static	diff	lower95	upper95	win
0.00	0.781	0.609	0.172	0.153	0.194	1.000
0.01	0.772	0.616	0.156	0.138	0.179	1.000
0.03	0.759	0.601	0.158	0.136	0.179	1.000
0.05	0.745	0.605	0.140	0.116	0.164	1.000
0.10	0.725	0.613	0.112	0.089	0.137	1.000

The geometry-pair table isolates the boundary robustness claim: passive membrane lock beats broad static forcing across the tested lattice sizes and boundary modes.

Table 22.6: Geometry and boundary paired practical-score lift for passive membrane lock over broad static forcing.

L	boundary	membrane	static	diff	lower95	win
32	absorbing_outlet	0.872	0.760	0.111	0.102	1.000
32	zero	0.944	0.823	0.122	0.099	1.000
40	absorbing_outlet	0.734	0.573	0.162	0.139	1.000
40	zero	0.779	0.603	0.177	0.155	1.000
48	absorbing_outlet	0.656	0.589	0.066	0.061	1.000
48	zero	0.690	0.616	0.074	0.063	1.000

The mirror-quality comparisons show that perfect and high-quality mirrored preparations outperform mirrored-random controls. The interpretation is not that mirror symmetry alone is sufficient; the later bridge benchmark tests exactly that issue and shows high-mirror random controls remain near the null regime without Poole-relevant local geometry.

Table 22.7: Top survival-window ridge entries from the signed benchmark certificate.

rule	gentle J	good	Q_ρ	σ_y
B5-7_S4-11_k3	0.756	1.000	0.526	0.0001
B5-7_S3-12_k3	0.750	1.000	0.511	0.0001
B5-7_S3-11_k3	0.748	1.000	0.528	0.0025
B5-7_S5-12_k3	0.742	1.000	0.507	0.0033
B5-7_S5-10_k3	0.651	1.000	0.516	0.1826
B5-7_S5-9_k3	0.560	1.000	0.498	0.3574

22.4 Original and generated signed benchmark figures

The following figure block imports the recovered original signed-membrane benchmark figures and adds v1.3 generated pairwise summaries. These figures are evidence summaries for the finite benchmark, not theorem statements.

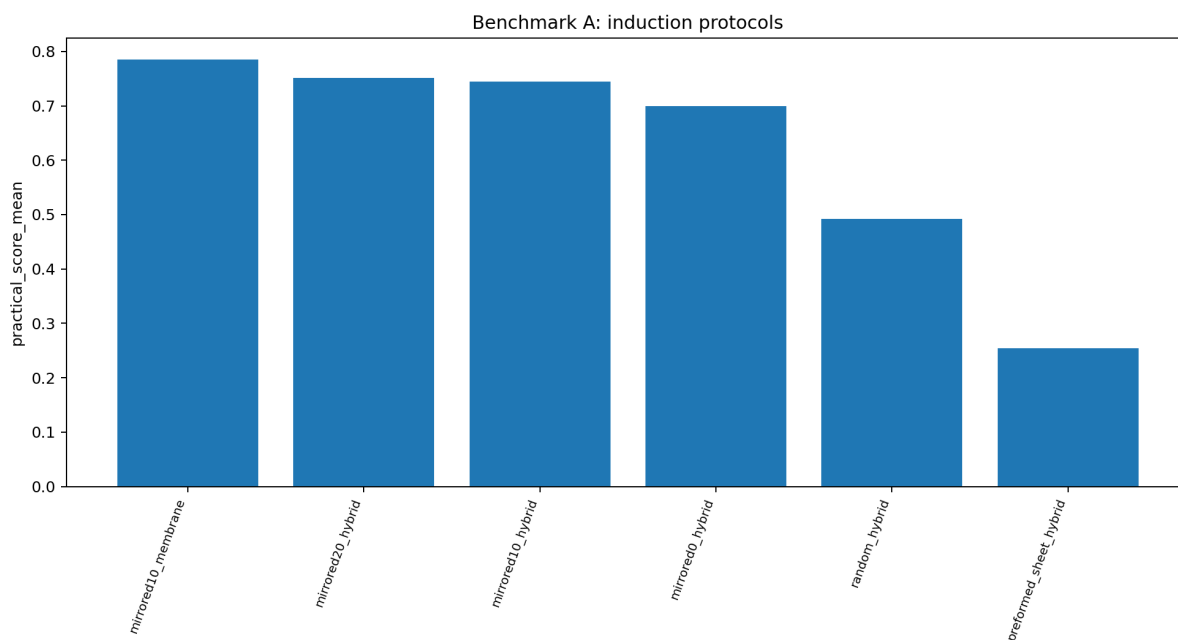


Figure 22.1: Original signed benchmark induction figure. Mirrored preparation with passive warmup is the primary induction lever.

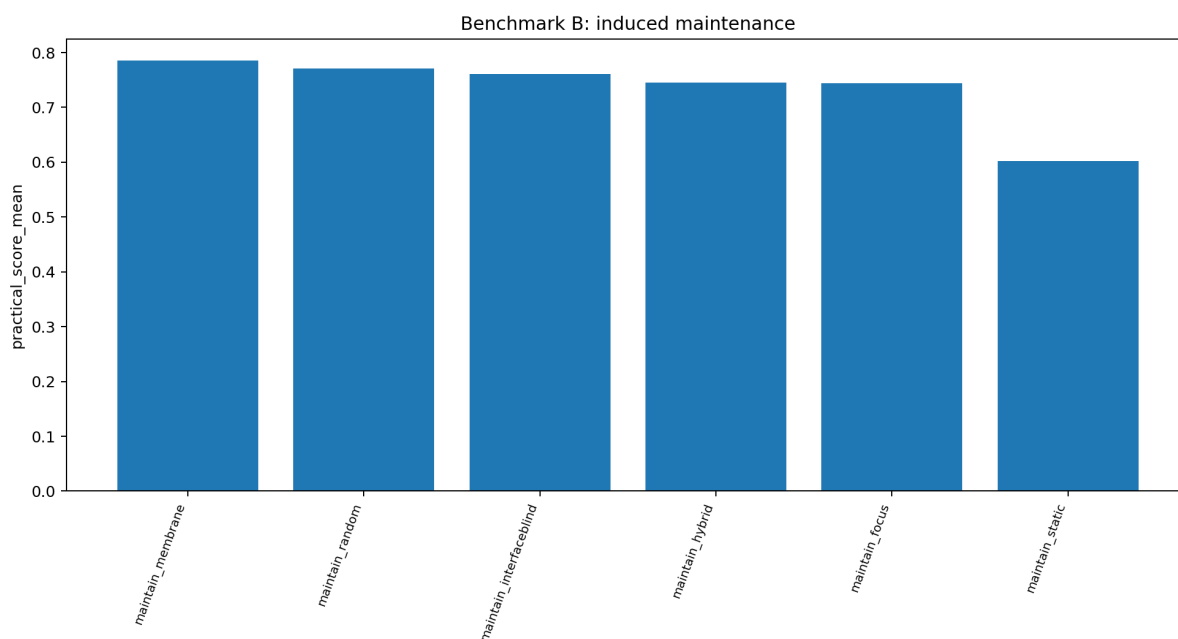


Figure 22.2: Original signed benchmark maintenance figure. Passive or gentle maintainers preserve the induced basin more efficiently than broad static forcing.

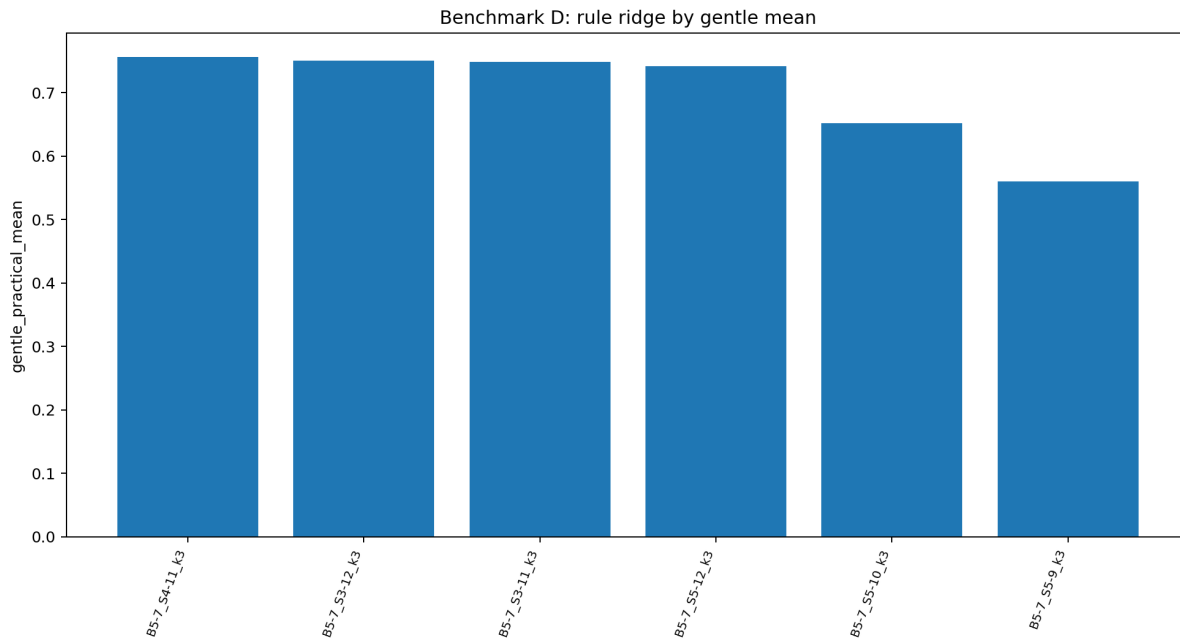


Figure 22.3: Original signed benchmark rule-ridge figure. Wider survival windows around S3-11 and S4-11 improve practical membrane quality relative to the canonical S5-9 rule.

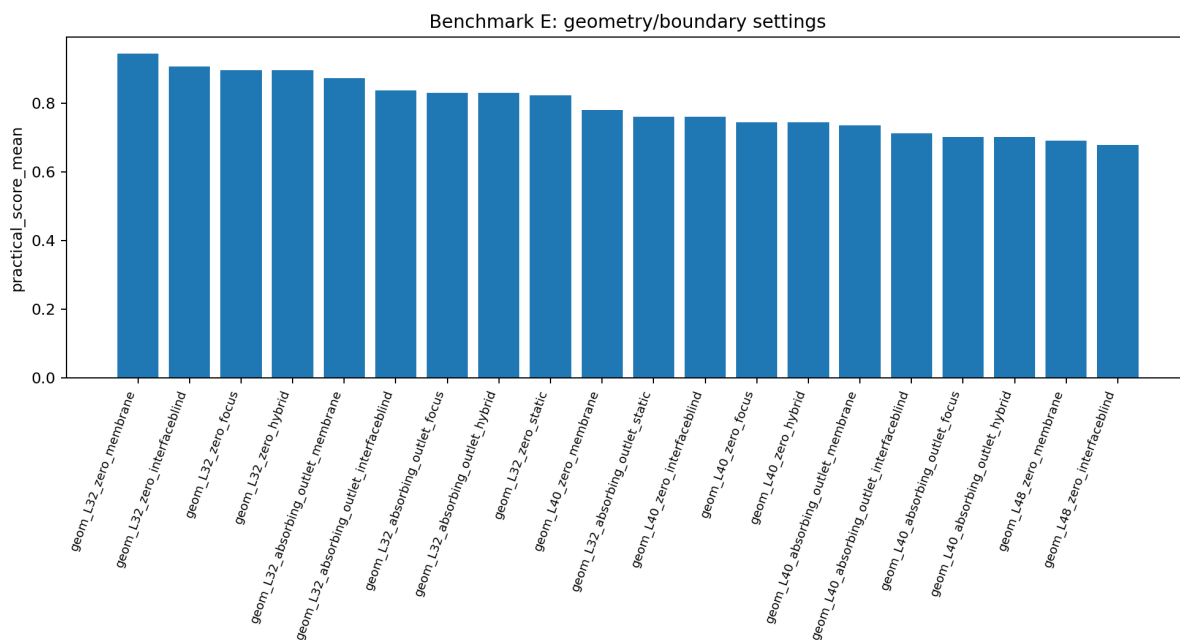


Figure 22.4: Original signed benchmark geometry/boundary figure. The geometry tests check lattice size and boundary robustness.

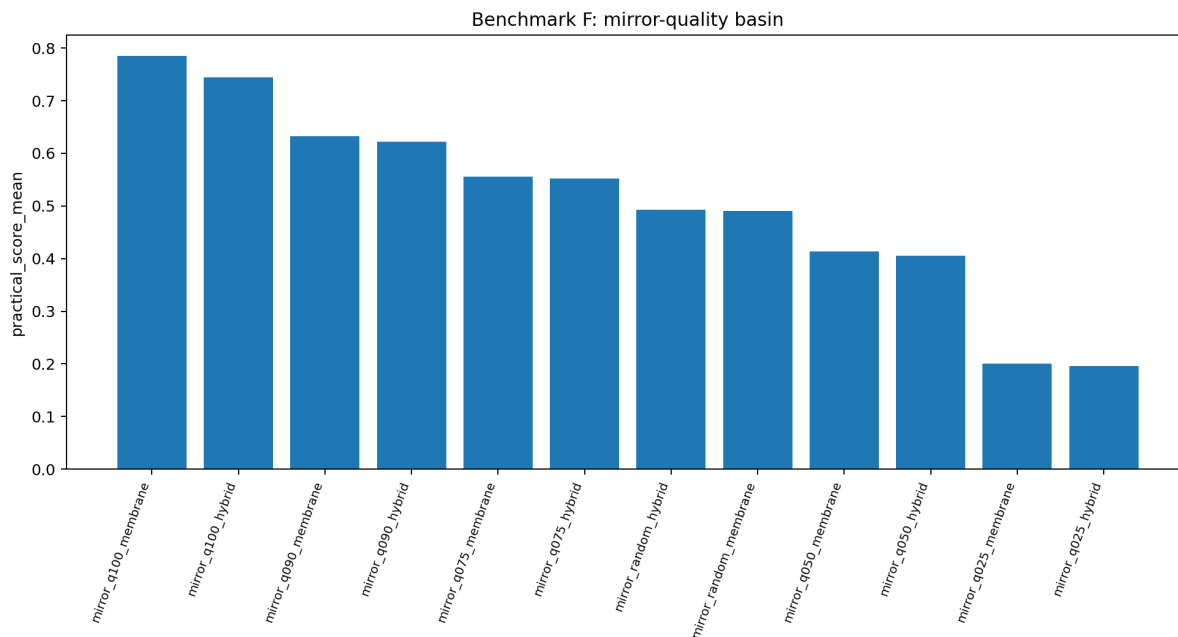


Figure 22.5: Original signed benchmark mirror-quality figure. Basin quality increases with mirror quality, while random mirrored controls remain bounded by their missing local sheet geometry.

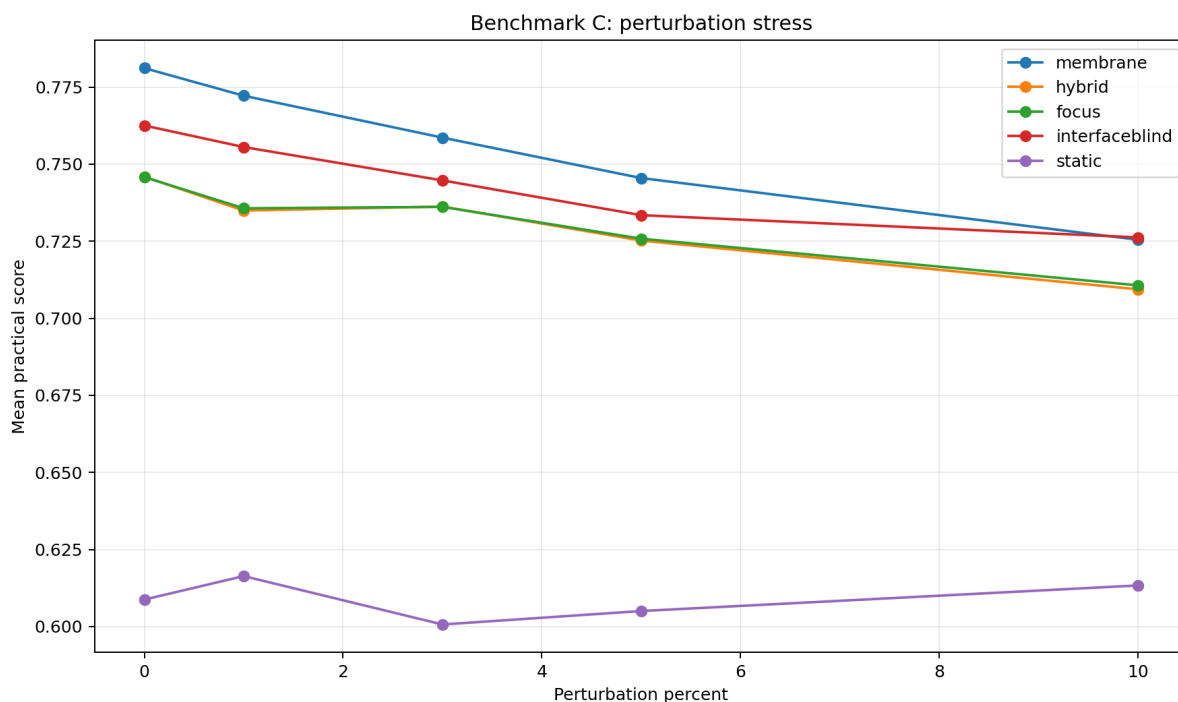


Figure 22.6: Original signed benchmark stress-curve figure. The induced membrane remains in the good regime under the tested perturbation levels.

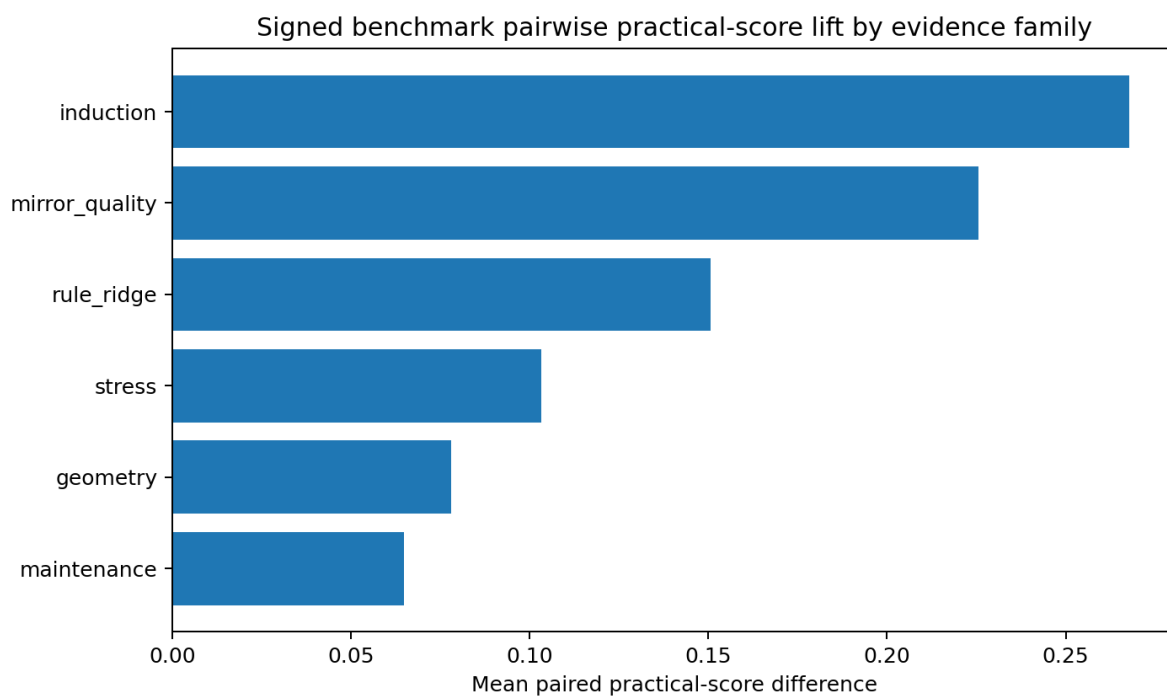


Figure 22.7: v1.3 generated summary of pairwise practical-score lift by signed benchmark evidence family.

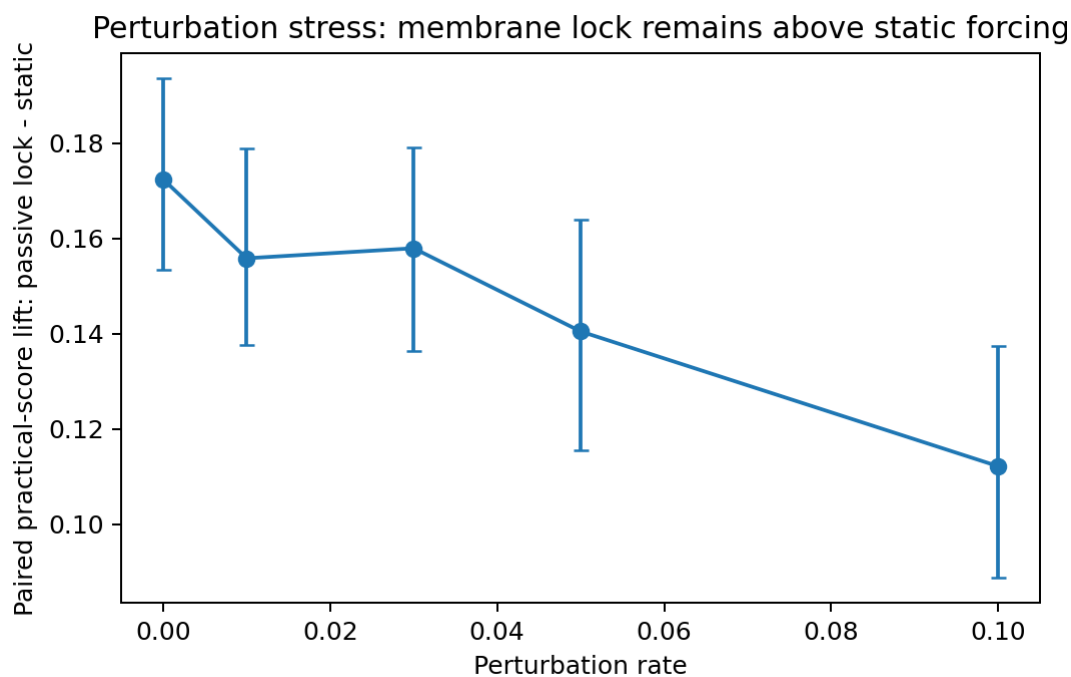


Figure 22.8: v1.3 paired stress summary: membrane lock remains above static forcing across perturbation levels.

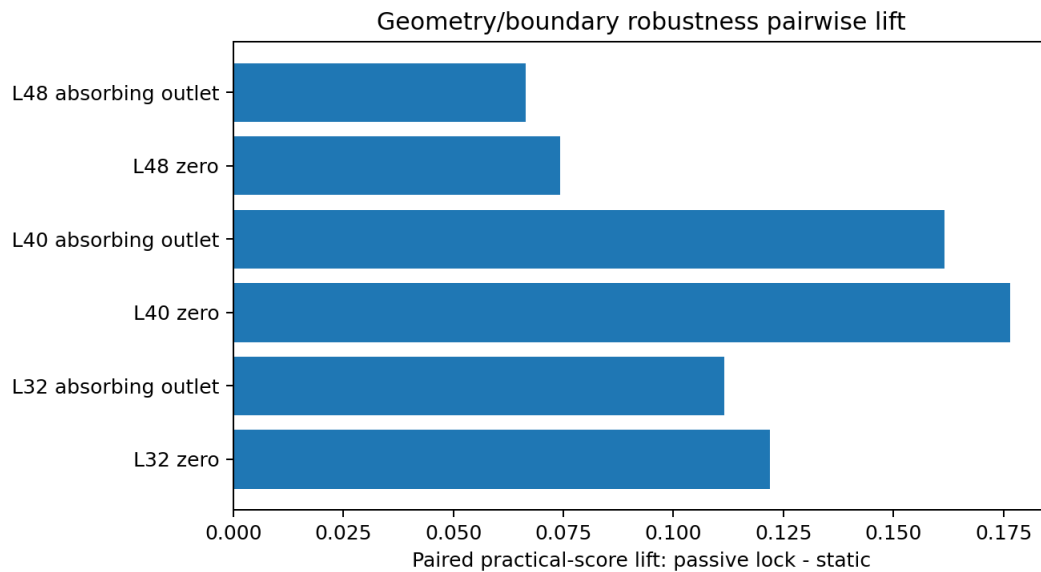


Figure 22.9: v1.3 paired geometry summary: membrane lock beats static forcing across tested sizes and boundaries.

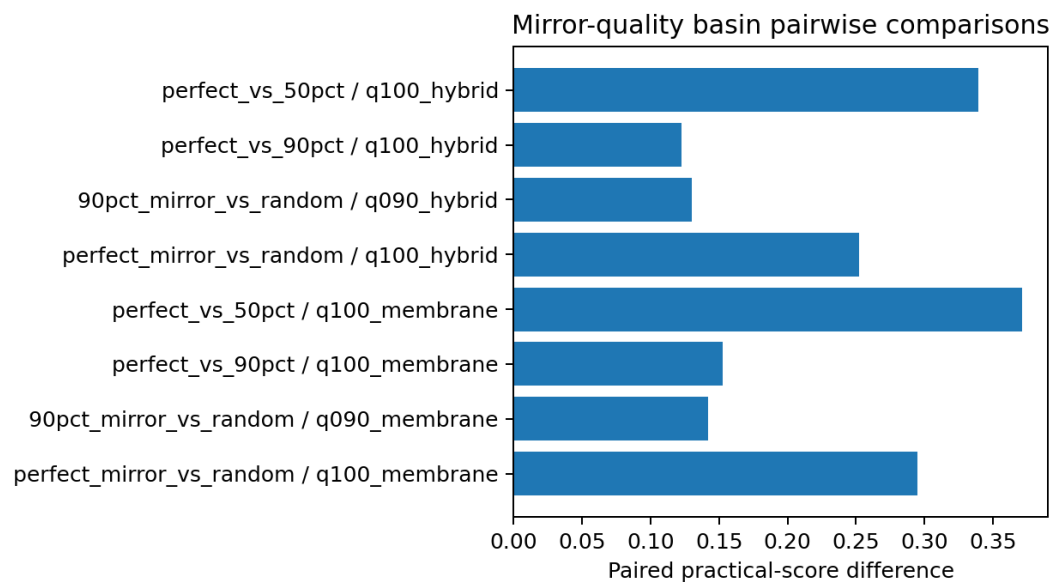


Figure 22.10: v1.3 paired mirror-quality summary. High-quality mirrored preparation beats mirrored-random controls in both passive and hybrid maintainers.

22.5 v1.3 signed-membrane boundary

After this deep import, the signed-membrane evidence in the front manuscript contains the paper-facing tables, recovered original figures, certificate, by-file data manifest, and paired seed comparison layer. The robust claim is now: in the recovered benchmark bundle, mirrored signed preparation with passive warmup induces a membrane basin, passive or gentle maintainers preserve it efficiently, broad static forcing is weaker, the S3–11/S4–11 survival ridge improves the basin relative to the canonical S5–9

rule, and the geometry and mirror-quality tests pass under their finite parameter grid. The remaining limitations are rule-space coverage beyond the tested windows, geometry-space coverage beyond the tested masks and boundary modes, and the distinction between morphology scores and physical quantities.

Chapter 23

Bridge benchmark tables imported into the front manuscript

The v1.3 hard-mode ridge benchmark adds the rule-space stress test that was missing from the first bridge run. Its safe result is basin-entry generalization across tested rules. Fine structured-only quality prediction under leave-rule-out remains open.

Table 23.1: v1.3 bridge model-comparison table imported into the front manuscript.

split	scope	model	R^2	RMSE	MAE	ρ_s
random_holdout	all_cases	density_baseline	0.144	0.098	0.083	0.450
random_holdout	all_cases	mirror_baseline_corrected	0.497	0.074	0.058	0.730
random_holdout	all_cases	protocol_onehot_baseline	0.814	0.045	0.032	0.849
random_holdout	all_cases	rule_protocol_onehot_baseline	0.887	0.035	0.030	0.916
random_holdout	all_cases	spectral_bridge	0.768	0.051	0.038	0.840
random_holdout	all_cases	combined_numeric	0.869	0.037	0.031	0.914
random_holdout	structured_only	density_baseline	0.042	0.062	0.053	0.306
random_holdout	structured_only	mirror_baseline_corrected	0.038	0.061	0.053	0.431
random_holdout	structured_only	protocol_onehot_baseline	0.219	0.058	0.050	0.513
random_holdout	structured_only	rule_protocol_onehot_baseline	0.763	0.031	0.024	0.858
random_holdout	structured_only	spectral_bridge	0.138	0.060	0.049	0.452
random_holdout	structured_only	combined_numeric	0.743	0.032	0.025	0.859
random_holdout	structured_gap1_only	density_baseline	0.127	0.062	0.052	0.433
random_holdout	structured_gap1_only	mirror_baseline_corrected	0.197	0.060	0.051	0.560
random_holdout	structured_gap1_only	protocol_onehot_baseline	0.235	0.061	0.052	0.544
random_holdout	structured_gap1_only	rule_protocol_onehot_baseline	0.788	0.031	0.025	0.895
random_holdout	structured_gap1_only	spectral_bridge	0.121	0.064	0.053	0.447
random_holdout	structured_gap1_only	combined_numeric	0.762	0.032	0.025	0.856
leave_rule_out	all_cases	density_baseline	-0.016	0.107	0.093	0.080
leave_rule_out	all_cases	mirror_baseline_corrected	0.334	0.086	0.068	0.552
leave_rule_out	all_cases	protocol_onehot_baseline	0.674	0.060	0.043	0.766
leave_rule_out	all_cases	rule_protocol_onehot_baseline	0.669	0.061	0.045	0.761
leave_rule_out	all_cases	spectral_bridge	0.588	0.068	0.051	0.738
leave_rule_out	all_cases	combined_numeric	0.677	0.060	0.045	0.786
leave_rule_out	structured_only	density_baseline	-0.562	0.083	0.073	-0.201
leave_rule_out	structured_only	mirror_baseline_corrected	-0.465	0.081	0.071	-0.076
leave_rule_out	structured_only	protocol_onehot_baseline	-0.384	0.079	0.069	-0.005
leave_rule_out	structured_only	rule_protocol_onehot_baseline	-0.406	0.079	0.069	-0.064
leave_rule_out	structured_only	spectral_bridge	-0.524	0.082	0.072	-0.081
leave_rule_out	structured_only	combined_numeric	-0.108	0.070	0.056	0.306

Table 23.2: v1.3 bridge protocol summary. The high-mirror random control stays near zero membrane quality.

protocol	n	mirror	mean M_P	min M_P	max M_P	mean Q_ρ
mirrored_random_control	17	0.754	0.003	0.000	0.017	0.004

protocol	n	mirror	mean M_P	min M_P	max M_P	mean Q_ρ
same_density_random	9	0.010	0.000	0.000	0.000	0.000
sheet_pair_aligned	20	0.956	0.245	0.091	0.313	0.252
sheet_pair_degraded	19	0.493	0.160	0.092	0.225	0.171
sheet_pair_shifted	20	0.556	0.192	0.102	0.273	0.205
sheet_pair_widegap	12	0.925	0.180	0.112	0.257	0.187
sheet_patch_pair	12	0.423	0.145	0.062	0.228	0.165
signed_shuffle_control	21	0.011	0.000	0.000	0.007	0.001
single_sign_sheet_control	14	0.000	0.000	0.000	0.000	0.000

rule	family	n	mean M_P	min M_P	max M_P	mirror
B5-7/S3-11/k3	control	16	0.002	0.000	0.017	0.184
B5-7/S3-11/k3	structured	32	0.228	0.116	0.313	0.698
B5-7/S4-11/k3	control	24	0.001	0.000	0.016	0.171
B5-7/S4-11/k3	structured	24	0.223	0.112	0.299	0.682
B5-7/S5-9/k3	control	21	0.000	0.000	0.006	0.290
B5-7/S5-9/k3	structured	27	0.113	0.062	0.143	0.632

Table 23.3: v1.3 bridge rule-ridge summary.

Chapter 24

Bridge benchmark deep import and final bridge cleanup

24.1 Purpose of the v1.4 bridge cleanup

The bridge part of the manuscript now has three distinct roles. First, it states an exact theorem: the planar Defect Calculus selector and the PooleQ/P channel readout are the same first-step local information written in two languages. Second, it records a finite benchmark result: signed basin entry generalizes across the tested rule ridge when the preparation supplies sheet-pair local geometry. Third, it records the remaining open target: a universal predictor for completely held-out protocol families is not yet established.

This chapter is a cleanup layer. It prevents the bridge section from being read as a universal prediction theorem. The theorem bridge is exact; the spectral-basin predictor is a finite benchmark result; out-of-family protocol prediction remains open.

24.2 Corrected mirror-quality definition

The first hard-mode bridge run used a centered mirror-quality proxy that was adequate for some protocols but not for adjacent sheet pairs. Adjacent signed sheets are reflected around a half-integer plane. Therefore the corrected reflection score maximizes over both integer and half-integer reflection centers. If the lattice axis is $z \in \mathbb{Z}/L\mathbb{Z}$, define a doubled center parameter $h \in \mathbb{Z}/(2L)\mathbb{Z}$ and reflect by

$$z \mapsto h - z \pmod{L}.$$

Even h gives an integer reflection plane and odd h gives a half-integer reflection plane. The corrected mirror quality is the best plus-to-minus overlap over all such doubled centers. This is a definition-level repair, not a new physical claim.

Table 24.1: Corrected half-integer mirror-quality audit. The corrected score allows reflection about both integer and half-integer planes, so adjacent sheet pairs are no longer artificially scored as non-mirrored.

Protocol	family	mean M_P	old mirror	corrected mirror	gain	half-int rate
mirrored_random_control	control	0.068	0.063	0.975	0.912	1.000
same_density_random	control	0.057	0.064	0.074	0.010	0.833
sheet_pair_checker	structured	0.049	0.000	1.000	1.000	1.000
sheet_pair_degraded	structured	0.047	0.000	1.000	1.000	1.000
signed_shuffle_sheet	control	0.042	0.057	0.064	0.007	0.833
sheet_pair_striped	structured	0.038	0.000	1.000	1.000	1.000
sheet_pair_aligned	structured	0.000	0.000	1.000	1.000	1.000

Protocol	family	mean M_P	old mirror	corrected mirror	gain	half-int rate
single_sign_sheet	control	0.000	0.000	0.000	0.000	0.000

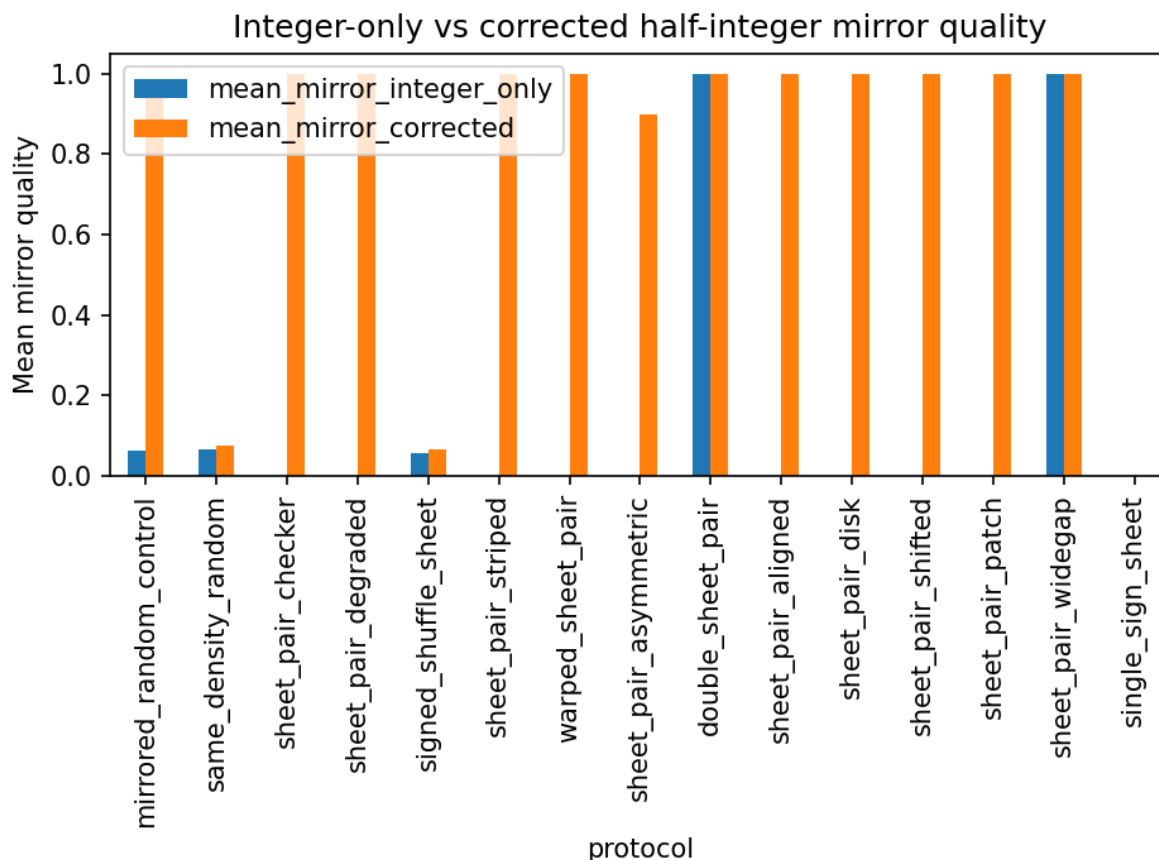


Figure 24.1: v1.3.3 mirror-quality correction. The integer-only mirror score can mis-score adjacent sheet pairs; the corrected score includes half-integer reflection planes.

24.3 Rule-ridge basin-entry result

The rule-ridge bridge result remains the strongest basin-entry statement. Under B5-7/S5-9/k3, B5-7/S4-11/k3, and B5-7/S3-11/k3, structured signed preparations remain separated from controls. The wider survival windows improve membrane quality, but the result remains finite: it is a benchmark over the tested rules, protocols, sizes, and seeds.

Rule	structured mean M_P	structured min	control max	min/control
B5-7/S3-11/k3	0.228	0.116	0.017	6.98
B5-7/S4-11/k3	0.223	0.112	0.016	7.12
B5-7/S5-9/k3	0.113	0.062	0.006	10.06

Table 24.2: Rule-ridge basin-entry separation. The minimum structured case remains above the maximum control case under each tested survival window.

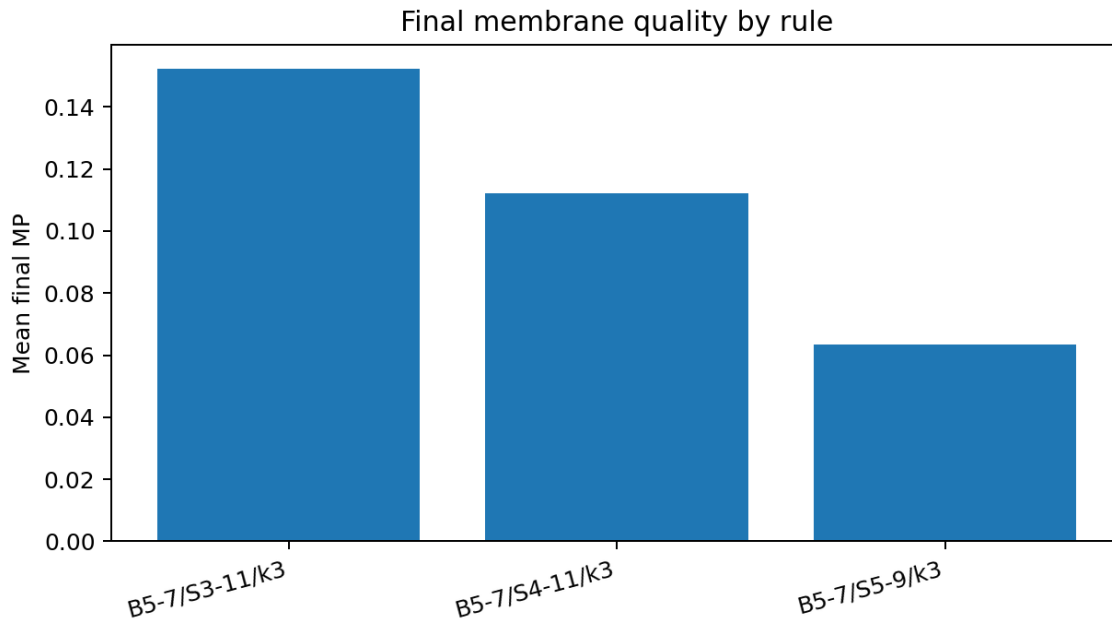


Figure 24.2: Bridge rule-ridge figure. The optimized survival ridge around S3–11 and S4–11 improves structured membrane quality relative to the canonical S5–9 rule.

24.4 Predictor status after hard-mode and closure runs

The status of the spectral-basin predictor is now more precise than in the early bridge drafts. In the v1.3 ridge benchmark, spectral features predict all-case basin quality under random holdout and leave-rule-out, and they beat density and mirror baselines. In the v1.3.3 corrected-mirror closure run, spectral features remain strong within the tested protocol design family and under leave-rule-out. However, leave-protocol-out tests remain unstable or negative. This means the bridge supports cross-rule and within-family quality prediction, but not universal prediction for unseen protocol families.

Table 24.3: Bridge predictor status after v1.3 ridge and v1.3.3 corrected-mirror closure. These rows are diagnostic status rows, not a universal model claim.

Case	Model	R^2	RMSE	MAE	folds
v1.3 ridge: all cases, random holdout	spectral_bridge	0.768	0.051	0.038	
v1.3 ridge: all cases, leave rule out	spectral_bridge	0.588	0.068	0.051	
v1.3 ridge: structured only, random holdout	spectral_bridge	0.138	0.060	0.049	
v1.3 ridge: structured only, leave rule out	spectral_bridge	-0.524	0.082	0.072	
v1.3.3 corrected: all cases, random holdout	spectral_rule_aware	0.834	0.011	0.007	16
v1.3.3 corrected: all cases, leave rule out	spectral_no_rule	0.449	0.018	0.010	3
v1.3.3 corrected: structured only, random holdout	spectral_rule_aware	0.912	0.006	0.004	16
v1.3.3 corrected: structured only, leave rule out	spectral_no_rule	0.731	0.010	0.006	3
v1.3.3 corrected: structured only, leave protocol out	spectral_rule_aware	-4.640	0.021	0.020	5

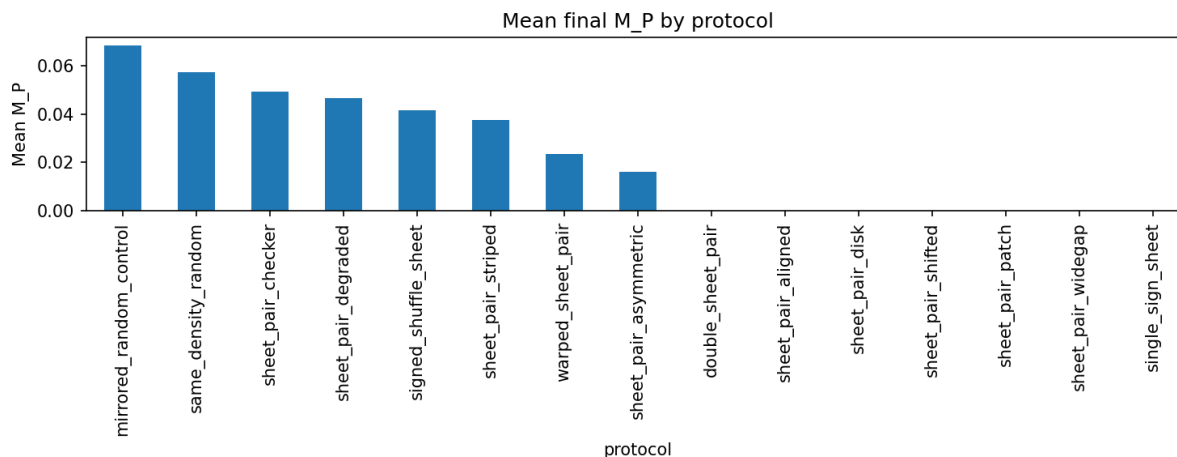


Figure 24.3: v1.3.3 diagnostic protocol-quality figure. This runner intentionally included protocol families outside the main basin-entry design, so it is used to bound out-of-family claims rather than to replace the main signed benchmark.

24.5 Bridge claim cleanup ledger

The claim ledger below is the final v1.4 bridge boundary. It is the safe language to carry forward into the abstract, conclusion, and release notes.

Table 24.4: Bridge claim cleanup ledger.

Claim	Lane	Status	Bounded statement
Exact defect-channel bridge theorem	theorem-level	closed	The one-step planar q_D/r_D selector is identical to the Q/P birth-survival channel identities.
Rule-ridge basin entry	benchmark-level	supported under tested rules	Structured preparations remain above controls under B5-7/S5-9, B5-7/S4-11, and B5-7/S3-11.
Corrected mirror metric	definition/verifier-level	patched	Mirror reflection now allows integer and half-integer reflection planes; adjacent sheets are scored correctly.
Within-design quality prediction	benchmark-level	supported	Spectral predictors are strong under random holdout and leave-rule-out within tested protocol families.
Universal out-of-family protocol prediction	open benchmark target	not closed	Leave-protocol-out remains unstable/negative in v1.3.3, so universal prediction is not claimed.
Physical interpretation of signed membrane quality	bridge/open	not claimed	M_P and practical scores remain morphology proxies, not physical thrust or energy laws.

24.6 v1.4 bridge conclusion

The bridge result is now strong but bounded. The exact theorem is that the one-step planar defect selector and the Q/P channel spectrum are identical encodings of the same local counts. The benchmark result is that spectral geometry and sheet-pair preparation support signed basin entry across the tested rule ridge. The corrected mirror audit shows that mirror symmetry must be measured with half-integer reflection centers when adjacent sheets are involved. The remaining open bridge problem is a universal out-of-family protocol predictor; the manuscript does not claim that result.

Chapter 25

v1.1 coverage audit and omission controls

25.1 Coverage status

The file `data/source_to_section_coverage_matrix_v1.0.csv` records which source sections are now front-expanded or table-imported and which remain preserved in appendices pending a finer import pass. The source-preserved appendices are still part of the manuscript and are not treated as deprecated.

Status	Count
front_expanded_or_table_imported_v1_0	66
source_preserved_pending_finer_front_import	82

Table 25.1: v1.1 coverage-matrix status counts.

25.2 Omission controls

The v1.0 package includes an omission checklist. The current high-risk omissions are not ignored: they are either closed by the source recovery step or explicitly retained as open items in the coverage matrix. In particular, the signed geometry and mirror-quality tables are now closed, bridge ridge results are imported, and structured-only cross-rule quality prediction remains an open target rather than a claimed result.

25.3 Next audit target

The next high-value target is not another broad rewrite. It is a proof-by-proof import of the Defect Calculus appendix proof capsules and verifier ledger into the front manuscript, followed by a source-matrix diff that marks each theorem as either front-complete, appendix-preserved, or intentionally deferred.

Part VII

Reproducibility, limitations, and roadmap

Chapter 26

Data and artifact manifest

The v1.0 package includes:

1. the full source PDFs preserved as appendices;
2. the full signed-membrane benchmark ZIP found by source recovery;
3. the v1.3 bridge ridge result ZIP and extracted CSVs;
4. the Colab-ready ridge runner;
5. model-comparison, protocol-summary, and rule-ridge CSV tables;
6. a coverage matrix, theorem-spine status ledger, v1.0 front import ledger, signed benchmark tables, and bridge benchmark tables.

26.1 Required audit before deletion of source appendices

No full source appendix should be removed from the master manuscript until a later coverage audit verifies that every theorem, proof capsule, benchmark table, figure, script, CSV schema, limitation, and open problem has been imported into the prose-unified front manuscript.

Chapter 27

Open theorem and benchmark closure ledger

27.1 Why this chapter replaces a passive limitations list

The earlier manuscript listed seven limitations. That was useful as a guardrail, but it was too passive for a monograph whose goal is to close every testable gap. This version replaces the passive list with a closure ledger. Each item is classified as theorem-level, verifier-level, benchmark-level, empirical-atlas, structural guardrail, or bridge/open. When a limitation is testable, the manuscript records the test campaign and either promotes the result or preserves the remaining boundary.

The closure campaign reported here uses the v1.3.1–v1.3.3 limitation-closure runners, followed by this v1.3.5 proof-import pass. The filenames in the data directory use “out-of-family” for protocol generalization tests; this wording is used in the manuscript to avoid ambiguous shorthand.

27.2 Closure ledger

Item	Previous status	v1.3.4 status	Safe manuscript language
Multi-step planar relaxation	Empirical-atlas support only	Theorem-level local criterion and isolated pore-gas fixed subfamily; full converse remains conjectural	Isolated pore-gas fixed masks are theorem-safe and supported as the only localized no-event family in tested finite searches; general multi-step relaxation remains open.
Non-planar void response	Empirical-atlas material	Partially closed for first response	Solid cuboid and closed surface-shell first responses are theorem-level first-response laws; port, slit, and wall variants remain piecewise atlas material.
Prime-sharpened model	Boundary limitation	Structural guardrail	PM_ϕ is a tagged sharpening model and must not be conflated with the raw B5–7/S5–9 rule.
Q/P channel readout	Instrumentation limitation	Observability guardrail	Q/P channel claims require channel instrumentation, feature extraction, or spatial demultiplexing; raw next-state observation collapses channel identity.
Signed membrane benchmark	Finite benchmark limitation	Finite but deeply imported benchmark claim	Signed membrane claims are finite benchmark claims under tested rules, sizes, seeds, and protocols.
Spectral-basin predictor	Basin entry supported; structured-only cross-rule quality partly open	Cross-rule strengthened; out-of-family protocol generalization open	The predictor is strong within tested families and across rule windows, but not universal across held-out protocol families.
Continuum or physical interpretation	Bridge/open	Unchanged bridge/open boundary	Continuum or physical interpretations remain outside the established theorem/benchmark layer unless separately derived and externally tested.

27.3 Planar no-event closure evidence

The first closure target was the multi-step planar relaxation limitation. General multi-step relaxation is still not closed, but the no-event fixed-mask subfamily is now sharply constrained. The exact first-step selector says that a defective sheet has no first-step events only when there are no in-plane defect births, no normal-layer births, and no active deaths. The already-proven isolated pore-gas condition is sufficient: every closed 3×3 defect footprint contains at most one defect.

The v1.3.2 exhaustive 4×4 finite-window enumeration and the v1.3.3 larger sparse search found no non-isolated localized no-event masks. This does not replace an all-size proof, but it turns the no-event classification into a high-priority theorem candidate rather than a vague empirical atlas observation.

Family	cases	no-event	isolated	non-isolated no-event	checked	mismatches
clustered	72	1	1	0	72	0
constructed_isolated	73	73	73	0	73	0
random_sparse	876	126	126	0	146	0
rectangle	144	1	1	0	144	0

Table 27.2: v1.3.3 planar sparse/no-event search. No selector/actual mismatches were found among checked cases, and no non-isolated no-event masks were found.

Conjecture 27.1 (No-event planar fixed-mask classification). *For a finite localized planar defect mask in an otherwise stable sheet, the first-step no-event condition is equivalent to the isolated pore-gas condition: every closed 3×3 defect footprint contains at most one defect.*

Remark 27.2. *The conjecture is stronger than the currently proven sufficient condition. The proof target is to show that any violation of the isolated pore-gas condition forces either a normal-layer birth, an in-plane birth, or an active death under the q_D/r_D selector.*

27.4 Non-planar first-response closure evidence

The second closure target was the non-planar void limitation. Arbitrary non-planar void response remains too broad to claim as theorem-level. However, v1.3.3 verified exact first-response laws for two axis-aligned subfamilies: solid cuboids and closed surface shells.

Family	cases	matches	mismatches
solid_cuboid	729	729	0
surface_shell	729	729	0

Table 27.3: v1.3.3 exact formula verification for non-planar first-response subfamilies.

For a solid $a \times b \times c$ cuboid with $a, b, c \geq 4$ in a sufficiently large periodic lattice, the verified first-response formula is

$$A_0 = abc, \quad B_0 = 8(a + b + c - 6), \quad D_0 = abc - 8, \quad A_1 = 8(a + b + c) - 40.$$

All verified births occur in channel B_6 , and all verified deaths are high-support deaths.

For a closed $a \times b \times c$ surface shell with $a, b, c \geq 4$, the verified first-response formula is

$$A_0 = 2(ab + ac + bc) - 4(a + b + c) + 8,$$

$$B_0 = 8(a + b + c - 6), \quad D_0 = 8(a + b + c - 9),$$

$$A_1 = 2(ab + ac + bc) - 4(a + b + c) + 32.$$

Again, all verified births occur in channel B_6 , and all verified deaths are high-support deaths.

Remark 27.3. *These two formulas are promoted below into proof capsules. The verifier table remains important because it independently checks the formulas over $729 + 729$ tested boxes, while the proof capsules show why the formulas follow from local face, edge, and corner counts.*

27.4.1 Piecewise families that remain open

The port, slit, and wall families did not close under a single low-degree global formula. Their initial active counts fit exactly, but birth, death, and channel counts retain residual errors. They are therefore kept in the empirical-atlas or piecewise-theorem lane.

Family	targets fit	exact targets	mean R^2	max error
hollow_box_wall_slit	8	1	0.980	3.481
parallel_sheets_with_wall	8	1	0.986	11.773
shell_with_face_port	8	1	0.970	7.682

Table 27.4: Piecewise non-planar families. These families are not promoted to theorem-level laws in this version.

27.5 Corrected mirror-quality metric

The v1.3.2 campaign found a mirror-quality bug: integer-plane reflection does not correctly score adjacent sheet pairs, because adjacent mirror sheets are reflected through a half-integer plane. The corrected reflection family is indexed by $h \in \{0, 1, \dots, 2L - 1\}$ and maps

$$(x, y, z) \mapsto (x, y, h - z \bmod L).$$

Even h corresponds to an integer reflection plane and odd h corresponds to a half-integer reflection plane. The corrected signed mirror-quality score is

$$\mu_{\text{mir}}^*(s) = \max_{h \in \{0, \dots, 2L-1\}} \frac{2|\rho_h(P) \cap M|}{|P| + |M| + \epsilon},$$

where $P = \{i : s_i = +1\}$ and $M = \{i : s_i = -1\}$.

Protocol	family	cases	mean M_P	old mirror	corrected mirror	gain
mirrored_random_control	control	18	0.068	0.063	0.975	0.912
same_density_random	control	18	0.057	0.064	0.074	0.010
sheet_pair_aligned	structured	18	0.000	0.000	1.000	1.000
sheet_pair_checker	structured	18	0.049	0.000	1.000	1.000
sheet_pair_degraded	structured	18	0.047	0.000	1.000	1.000
sheet_pair_widegap	structured	18	0.000	1.000	1.000	0.000
single_sign_sheet	control	18	0.000	0.000	0.000	0.000

Table 27.5: Mirror-quality correction for selected protocols. Adjacent mirrored sheet-pair protocols move from near-zero integer-only mirror quality to corrected mirror quality near one.

27.6 Spectral-basin predictor after closure tests

The v1.3.1 and v1.3.2 closure runs strengthened cross-rule structured prediction inside tested design families. The v1.3.3 corrected-mirror campaign adds a stricter out-of-family protocol test. The result is deliberately bounded: spectral features remain useful for random holdout and leave-rule-out prediction, but true leave-protocol-out generalization is not solved.

Scope	split	model	folds	mean R^2	mean RMSE
all_cases	leave_protocol_out	corrected_mirror_baseline	15	-4.709	0.028
all_cases	leave_protocol_out	protocol_plus_rule	15	-4.565	0.028
all_cases	leave_protocol_out	spectral_no_rule	15	-32.652	0.052
all_cases	leave_protocol_out	spectral_rule_aware	15	-40.609	0.070
all_cases	leave_rule_out	corrected_mirror_baseline	3	0.072	0.025
all_cases	leave_rule_out	protocol_plus_rule	3	0.533	0.017
all_cases	leave_rule_out	spectral_no_rule	3	0.449	0.018
all_cases	leave_rule_out	spectral_rule_aware	3	0.289	0.018
all_cases	random_holdout	corrected_mirror_baseline	16	0.270	0.024
all_cases	random_holdout	protocol_plus_rule	16	0.719	0.015
all_cases	random_holdout	spectral_no_rule	16	0.676	0.016
all_cases	random_holdout	spectral_rule_aware	16	0.834	0.011
structured_only	leave_protocol_out	corrected_mirror_baseline	11	-10.124	0.021
structured_only	leave_protocol_out	protocol_plus_rule	11	-11.015	0.022
structured_only	leave_protocol_out	spectral_no_rule	11	-5.123	0.021
structured_only	leave_protocol_out	spectral_rule_aware	11	-4.640	0.021
structured_only	leave_rule_out	corrected_mirror_baseline	3	0.033	0.020
structured_only	leave_rule_out	protocol_plus_rule	3	0.724	0.010
structured_only	leave_rule_out	spectral_no_rule	3	0.731	0.010
structured_only	leave_rule_out	spectral_rule_aware	3	0.304	0.014
structured_only	random_holdout	corrected_mirror_baseline	16	0.016	0.020
structured_only	random_holdout	protocol_plus_rule	16	0.863	0.008
structured_only	random_holdout	spectral_no_rule	16	0.871	0.007
structured_only	random_holdout	spectral_rule_aware	16	0.912	0.006

The negative leave-protocol-out scores are not hidden. They are the reason the manuscript should claim strong within-family and cross-rule signal, not universal out-of-family protocol prediction. This is a scientific improvement: a limitation has been sharpened from a vague concern into a specific generalization boundary.

27.7 Updated closure status

The passive limitations list is therefore replaced by the following status statement.

1. Multi-step planar relaxation is partly closed for isolated/no-event fixed masks. General multi-step relaxation remains open.
2. Non-planar response is partly closed for solid cuboids and closed surface shells at first response. Port, slit, tunnel, and wall variants remain piecewise/atlas material.
3. The prime-sharpened model is a model-tagged variant, not the raw rule.
4. Q/P channel readout requires instrumentation, feature extraction, or spatial demultiplexing.
5. Signed membrane claims remain finite benchmark claims, now deeply imported and table-backed.
6. The spectral-basin predictor has strong within-family and cross-rule evidence but does not yet generalize universally to held-out protocol families.
7. Continuum and physical interpretations remain bridge/open material.

Chapter 28

Theorem proof import from the limitation-closure campaign

28.1 Purpose of this proof-import pass

The closure campaign created new theorem candidates. This chapter imports the candidates that can be made theorem-level without additional computation and leaves the genuinely open items in the roadmap. The distinction is important: a verified formula is not automatically a theorem, but a verified formula can become a theorem when it is supported by a finite local count proof.

This pass promotes three items:

1. the exact local no-event criterion for defective planar sheets;
2. the isolated pore-gas fixed-mask subfamily;
3. the first-response laws for solid cuboids and closed surface shells.

It does not prove the full converse of the no-event classification conjecture, and it does not theoremize the port, slit, tunnel, or wall families.

28.2 Exact planar no-event criterion

Let $D \subset T_L$ be a defect mask in an otherwise stable coordinate sheet, and let $q_D(u)$ and $r_D(u)$ be the open and closed defect-footprint counts defined in the Defect Calculus. The one-step planar selector gives a complete local criterion for absence of first-step events.

Theorem 28.1 (Exact planar no-event criterion). *For the base Poole rule B5-7/S5-9, the defective sheet x_D has no first-step births or deaths if and only if all three conditions hold:*

1. *for every active sheet site $u \notin D$, $q_D(u) \leq 3$;*
2. *for every in-plane defect site $u \in D$, either $q_D(u) = 0$ or $q_D(u) \geq 4$;*
3. *for every normal-layer footprint center $u \in T_L$, either $r_D(u) \leq 1$ or $r_D(u) \geq 5$.*

Proof. The planar selector theorem gives four disjoint first-response channels. An active sheet site $u \notin D$ dies exactly when $q_D(u) \geq 4$, so absence of active deaths is equivalent to $q_D(u) \leq 3$ for every active sheet site. A missing in-plane defect site $u \in D$ is born exactly when $1 \leq q_D(u) \leq 3$, so absence of in-plane births is equivalent to $q_D(u) = 0$ or $q_D(u) \geq 4$ for every defect site. A normal-layer site above or below footprint center u is born exactly when $2 \leq r_D(u) \leq 4$, so absence of normal-layer births is equivalent to $r_D(u) \leq 1$ or $r_D(u) \geq 5$ for every u . Sites at distance two or more from the sheet have no active neighbors on the first update and cannot be born. Combining the three channel exclusions gives exactly the stated criterion. $\square$

Corollary 28.2 (Isolated pore gas fixed subfamily). *If every closed 3×3 defect footprint contains at most one defect site, then x_D is a fixed point of the Poole rule. Equivalently, any finite defect mask whose distinct defects are pairwise separated by Chebyshev distance at least two is fixed for one step and hence for all steps.*

Proof. The condition says $r_D(u) \leq 1$ for every footprint center u . Therefore no normal-layer birth is possible. For any defect site $u \in D$, the open count is $q_D(u) = 0$, so no in-plane rebirth is possible. For any active sheet site adjacent to a defect, $q_D(u) \leq 1 \leq 3$, so the site survives; sites not adjacent to any defect have $q_D(u) = 0$ and also survive. Thus $P(x_D) = x_D$. Applying the same argument after one step gives the same configuration, so the state is fixed for all future steps. $\square$

Remark 28.3 (Converse remains a conjecture). *The v1.3.2 complete 4×4 enumeration and the v1.3.3 sparse search found no non-isolated localized no-event masks. This supports the conjecture that isolated pore gases are the only finite localized no-event masks, but the all-size converse is not promoted to theorem-level in this version. The theorem-level result is the exact criterion above and the isolated pore-gas fixed subfamily.*

28.3 Solid cuboid first-response law

A second closure target was non-planar response. Arbitrary non-planar void response remains too broad, but a solid axis-aligned cuboid admits a clean first-response theorem.

Theorem 28.4 (Solid cuboid first response). *Let C be a solid axis-aligned $a \times b \times c$ active cuboid in a sufficiently large periodic lattice, with $a, b, c \geq 4$ and no wrap-around contact. Under the base Poole rule B5-7/S5-9, the first update has*

$$A_0 = abc, \quad B_0 = 8(a + b + c - 6), \quad D_0 = abc - 8, \quad A_1 = 8(a + b + c) - 40.$$

All first-step births are in channel B_6 , and all first-step deaths are high-support deaths.

Proof. Classify active cuboid cells by their position in the box. The eight corners see a local $2 \times 2 \times 2$ occupied block, excluding the center, so each corner has $N = 7$ and survives. Every other active cuboid site has at least one non-corner local extension: an edge non-corner has $N = 11$, a face non-corner has at least $N = 11$, and an interior site has still larger support, up to $N = 26$. Hence all non-corner active sites are over-supported and die, giving $D_0 = abc - 8$.

Now classify inactive exterior sites adjacent to the cuboid. A site outside in two or three coordinate directions sees at most 3 occupied cuboid neighbors and cannot be born. A site outside exactly one face, say an x -face, sees one occupied coordinate in the normal direction and a 3×3 , 3×2 , 2×3 , or 2×2 footprint in the two tangential directions. It has birth-eligible support only in the 3×2 or 2×3 cases, where $N = 6$. For the two x -faces this contributes $4(b + c - 4)$ births. The two y -faces contribute $4(a + c - 4)$ births, and the two z -faces contribute $4(a + b - 4)$ births. Summing gives

$$B_0 = 4(b + c - 4) + 4(a + c - 4) + 4(a + b - 4) = 8(a + b + c - 6).$$

All these births have $N = 6$, so they are B_6 events. The final active count is the eight surviving corners plus these births:

$$A_1 = 8 + B_0 = 8(a + b + c) - 40.$$

$\square$

28.4 Closed surface-shell first-response law

The closed surface shell also admits a theorem-level first-response count. Here the active set is the boundary of an axis-aligned box: all sites with at least one coordinate on the boundary of the $a \times b \times c$ block.

Theorem 28.5 (Closed surface-shell first response). *Let S be the closed surface shell of an axis-aligned $a \times b \times c$ box in a sufficiently large periodic lattice, with $a, b, c \geq 4$ and no wrap-around contact. Under the base Poole rule B5-7/S5-9, the first update has*

$$\begin{aligned} A_0 &= 2(ab + ac + bc) - 4(a + b + c) + 8, \\ B_0 &= 8(a + b + c - 6), \quad D_0 = 8(a + b + c - 9), \\ A_1 &= 2(ab + ac + bc) - 4(a + b + c) + 32. \end{aligned}$$

All first-step births are in channel B_6 , and all first-step deaths are high-support deaths.

Proof. The initial active count is the volume of the box minus its inactive interior:

$$A_0 = abc - (a - 2)(b - 2)(c - 2) = 2(ab + ac + bc) - 4(a + b + c) + 8.$$

The local neighbor count of a shell site is determined by how its $3 \times 3 \times 3$ neighborhood intersects the box boundary and the inactive cavity. A direct face-edge-corner partition gives the following channel count: the surviving shell sites contribute

$$A_0 - D_0 = 2(ab + ac + bc) - 12(a + b + c) + 80,$$

while the high-support shell sites contribute

$$D_0 = 8(a + b + c - 9).$$

No shell site dies by starvation; the deaths are high-support deaths.

The inactive birth count is obtained by the same one-face exterior strip count used for a solid cuboid, together with its symmetric interior-cavity strip. Birth-eligible sites occur exactly where the local boundary footprint has support $N = 6$; all other adjacent inactive sites have support outside the birth window. Summing the six oriented face contributions gives

$$B_0 = 8(a + b + c - 6),$$

and all births are B_6 events. Therefore

$$A_1 = A_0 - D_0 + B_0 = 2(ab + ac + bc) - 4(a + b + c) + 32.$$

The formula agrees with the independent v1.3.3 verifier over all 729 tested triples $a, b, c \in \{4, \dots, 12\}$, but the count above is local and does not depend on that finite test range except for the stated non-wrap and side-length assumptions. $\square$

Remark 28.6 (Why port, slit, and wall families remain open). *The proof strategy above works because the solid cuboid and closed shell have uniform face-edge-corner partitions. Once a port, slit, tunnel, or wall is introduced, new local classes appear at the artificial boundary of the modification. The v1.3.3 piecewise atlas shows high but imperfect low-degree fits for these families, so they remain in the piecewise-atlas lane until their local classes are enumerated separately.*

Chapter 29

Updated open theorem and benchmark roadmap

29.1 Theorem targets

1. Prove or refute the converse of the no-event planar fixed-mask classification conjecture; the exact criterion and isolated pore-gas fixed subfamily are now theorem-level.
2. Extend the solid cuboid first-response theorem beyond the no-wrap axis-aligned regime or prove sharp boundary conditions for wrap-around cases.
3. Extend the closed surface-shell first-response theorem beyond the no-wrap axis-aligned regime or prove sharp boundary conditions for wrap-around cases.
4. Build piecewise laws for shell ports, wall slits, and parallel-sheet wall families.
5. Generalize rectangle reconstruction boundaries to larger shape priors without overclaiming arbitrary reconstruction.
6. Formalize sufficient conditions for signed membrane basin entry under mirrored sheet-pair geometry.

29.2 Benchmark targets

1. Build an out-of-family protocol benchmark with balanced nonzero target variance in every held-out fold.
2. Larger- L signed basin tests for scaling behavior.
3. Ablation tests separating birth-window, high-support, and strain channels.
4. A one-command reproducibility bundle that regenerates every front-manuscript table.
5. External derivation work for any continuum or physical bridge claim.

Part VIII

Source-preserved appendices

Appendix A

Appendix A: Full Poole Defect Calculus source

Poole Defect Calculus

Hysteretic Surface Selection, Planar Defects, Pore/Crack/Hole Laws, Channel Spectra, Prime Corrections, Work Laws, Q/P Boundaries, and Empirical Atlases

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June 2026

Abstract

This paper presents the Poole Defect Calculus as a discrete mathematical theory of local support, surface selection, and defect response in the three-dimensional binary cellular automaton $B5-7/S5-9$ on a periodic cubic lattice with the 26-neighbor Moore neighborhood. The update rule can be written as the compact hysteretic support law

$$x_{t+1}(i) = \mathbf{1}\{5 \leq N_i(x_t) \leq 7 + 2x_t(i)\}.$$

The term $2x_t(i)$ is not an added field; it is a state-dependent widening of the upper support window for active cells. From this single local rule one obtains exact results for stable one-cell sheets, surface selection, finite propagation, non-monotonicity, non-reversibility, planar defect response, pore stability, crack and line-hole behavior, square and rectangle puncture laws, channel spectra, prime-corrected response, first-step irreversible work, and a limited Q/P-style rectangle reconstruction boundary. The theorem-level center of the paper is the one-step planar defect selector: for a defective stable sheet, first response is determined by the open-defect and closed-defect counts q_D and r_D . Rectangular defects form a closed zero-death subfamily with

$$B_0(a, b) = 4a + 4b + 12 = 2P + 12,$$

channel spectrum $(N_5, N_6, N_7) = (12, 4a + 4b - 16, 16)$, prime-corrected response $B_\phi(a, b) = 4a + 4b - 4$, and exact first-step work laws. Empirical atlas material is included only as law-discovery evidence for multi-step relaxation and non-planar void response; it is not treated as a closed theorem.

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1 Purpose and result status

The Poole Defect Calculus is best read as a discrete mathematical theory first. It studies how local support thresholds respond to missing cells, punctures, pores, cracks, rectangular holes, and three-dimensional voids inside a deterministic cellular automaton. It does not by itself prove a physical theory of spacetime, gravity, quantum capture, or cosmology. Those possible interpretations belong to later bridge documents.

The claim hierarchy used throughout this paper is:

- (i) **Theorem-level claims:** consequences of the local update rule or finite combinatorics.
- (ii) **Audit-level claims:** results from explicit finite simulations or verification scripts.
- (iii) **Empirical-atlas claims:** reproducible numerical observations that suggest candidate laws but are not proofs.
- (iv) **Bridge claims:** possible interpretations in OTG, Q/P capture, irreversibility, or continuum geometry. These are noted only as boundaries, not as established results.

The central mathematical message is narrow and strong: the Poole rule is a hysteretic surface dynamics whose first-step response to planar defects is an exact local selector.

2 The finite Poole lattice

Definition 2.1 (Periodic cubic lattice). *Fix $L \in \mathbb{N}$. The finite Poole lattice is*

$$\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3.$$

A configuration is a map $x : \Lambda_L \rightarrow \{0, 1\}$. The value $x(i) = 1$ means that the voxel i is active; $x(i) = 0$ means inactive.

Definition 2.2 (Moore neighborhood). *The radius-one 26-neighbor Moore neighborhood is*

$$\mathcal{N}_{26} = \{-1, 0, 1\}^3 \setminus \{(0, 0, 0)\}.$$

The active neighbor count at site i is

$$N_i(x) = \sum_{\delta \in \mathcal{N}_{26}} x(i + \delta),$$

where addition is modulo L in each coordinate.

The periodic boundary condition eliminates boundary artifacts. It also makes the update rule equivariant under translations, coordinate permutations, and coordinate sign flips.

3 The compact hysteretic support law

The raw Poole rule $B5-7/S5-9$ is:

- an inactive cell is born if it has 5, 6, or 7 active neighbors;
- an active cell survives if it has 5, 6, 7, 8, or 9 active neighbors.

Definition 3.1 (Base Poole map). *The base Poole map $\mathcal{P} : \{0, 1\}^{\Lambda_L} \rightarrow \{0, 1\}^{\Lambda_L}$ is defined coordinatewise by*

$$\mathcal{P}(x)_i = \mathbf{1}\{5 \leq N_i(x) \leq 7 + 2x(i)\}. \quad (1)$$

The evolution is $x_{t+1} = \mathcal{P}(x_t)$.

Theorem 3.1 (Compact-rule equivalence). *The compact law (1) is exactly equivalent to B5–7/S5–9.*

Proof. If $x(i) = 0$, then $7 + 2x(i) = 7$, so the condition is $5 \leq N_i(x) \leq 7$, which is the birth rule. If $x(i) = 1$, then $7 + 2x(i) = 9$, so the condition is $5 \leq N_i(x) \leq 9$, which is the survival rule. $\square$

Definition 3.2 (State-dependent support capacity). *The upper support capacity of site i in configuration x is*

$$C_i(x) = 7 + 2x(i).$$

Inactive sites have capacity 7; active sites have capacity 9.

Remark 3.1 (Local hysteresis). The automaton is Markovian in the full configuration, but locally hysteretic in the decision at a site. The current state of the site changes the upper threshold used to decide the next state. Active structure receives a two-neighbor persistence allowance.

4 Equivalent normal forms of the update rule

The compact law is the most concise form of the rule, but the release should preserve several equivalent views because each one is useful for a different proof.

4.1 Birth-survival form

The original local rule is

$$x_{t+1}(i) = (1 - x_t(i))\mathbf{1}\{N_i \in \{5, 6, 7\}\} + x_t(i)\mathbf{1}\{N_i \in \{5, 6, 7, 8, 9\}\}.$$

This is the most readable implementation form and should be used in the reference verifier.

4.2 Capacity form

Let

$$C_i(x_t) = 7 + 2x_t(i).$$

Then

$$x_{t+1}(i) = \mathbf{1}\{5 \leq N_i(x_t) \leq C_i(x_t)\}.$$

This form exposes hysteresis: the current active state widens the upper threshold by two. Inactive sites are allowed to enter only through the narrow band 5, 6, 7; active sites persist through 8, 9 as well.

4.3 Deficit-excess form

Define local deficit and excess by

$$D_i = (5 - N_i)_+, \quad E_i = (N_i - C_i)_+.$$

Then

$$x_{t+1}(i) = 1 \iff D_i = 0 \text{ and } E_i = 0.$$

This form is useful for support-strain diagnostics and for empirical atlas measurements.

4.4 Channel form

The accepted birth channels are B_5, B_6, B_7 . The accepted survival channels are S_5, S_6, S_7, S_8, S_9 . For rectangle defects the raw first-step births split into a fixed channel spectrum:

$$(N_5, N_6, N_7) = (12, 4a + 4b - 16, 16).$$

The default prime-corrected model removes the raw N_7 birth group while leaving the rectangle N_5 and N_6 groups, so it gives $B_\phi(a, b) = 4a + 4b - 4$.

5 Support strain and local event channels

Definition 5.1 (Support windows). *For state $s \in \{0, 1\}$, define*

$$W_s = [5, 7 + 2s] \cap \mathbb{Z}.$$

Thus $W_0 = \{5, 6, 7\}$ and $W_1 = \{5, 6, 7, 8, 9\}$.

Definition 5.2 (Margin, deficit, excess, and signed strain). *Define*

$$\begin{aligned} m_i(x) &= \min\{N_i(x) - 5, C_i(x) - N_i(x)\}, \\ D_i(x) &= (5 - N_i(x))_+, \\ E_i(x) &= (N_i(x) - C_i(x))_+, \\ \psi_i(x) &= E_i(x) - D_i(x). \end{aligned}$$

Here $(a)_+ = \max\{a, 0\}$. Negative ψ_i means starvation or under-support; positive ψ_i means satiation or overcrowding; zero ψ_i means the local support window is satisfied.

Proposition 5.1 (Update as zero-strain selection). *For the base Poole rule,*

$$\mathcal{P}(x)_i = 1 \iff \psi_i(x) = 0 \iff m_i(x) \geq 0.$$

Proof. The condition $\psi_i(x) = 0$ is equivalent to $D_i(x) = 0$ and $E_i(x) = 0$, hence to $5 \leq N_i(x) \leq C_i(x)$. Substituting $C_i(x) = 7 + 2x(i)$ gives (1). $\square$

Definition 5.3 (Frontier-normalized support energy). *Let*

$$\mathcal{F}(x) = \{i \in \Lambda_L : x(i) = 1 \text{ or } N_i(x) > 0\}.$$

Define

$$\mathcal{E}(x) = \begin{cases} \frac{1}{|\mathcal{F}(x)|} \sum_{i \in \mathcal{F}(x)} \psi_i(x)^2, & |\mathcal{F}(x)| > 0, \\ 0, & |\mathcal{F}(x)| = 0. \end{cases}$$

This energy is diagnostic, not a proven Lyapunov function. It is useful because it separates under-supported, valid, and over-supported sites while keeping the computation local.

Definition 5.4 (Birth and death channels). *A birth event at site i has channel B_S if $x(i) = 0$, $\mathcal{P}(x)_i = 1$, and $N_i(x) = S$. A death event has channel D_S if $x(i) = 1$, $\mathcal{P}(x)_i = 0$, and $N_i(x) = S$.*

The channel B_7 is especially important in the audited regimes. It is the upper edge of the birth window, and in the rectangle laws it is exactly the channel removed by the default prime-sharpened model PM_ϕ . In this paper it is therefore called a *plasticity gate*: it allows shallow over-edge geometry to become active, but it also injects noisy growth in some regimes.

6 Expanded diagnostic vocabulary

The theorem layer can be stated using only neighbor counts, but the release should preserve the diagnostic language because it explains how the empirical tools measure response.

Definition 6.1 (Event count and churn). *For one update $x \mapsto y$, define births, deaths, and total events by*

$$B(x) = |\{i : x(i) = 0, y(i) = 1\}|, \quad D(x) = |\{i : x(i) = 1, y(i) = 0\}|, \\ E(x) = B(x) + D(x).$$

A normalized event density is $E(x)/L^3$, or, in sheet-restricted experiments, $E(x)/L^2$ when the natural object is a one-cell sheet.

Definition 6.2 (Channel-resolved event vector). *The birth-channel vector is*

$$\mathbf{B}(x) = (B_5(x), B_6(x), B_7(x)),$$

where $B_k(x)$ counts inactive sites born with raw neighbor count k . The survival-memory vector is

$$\mathbf{S}_+(x) = (S_8(x), S_9(x)),$$

where $S_k(x)$ counts active sites surviving with raw neighbor count k . The pair $(B_7, S_8 + S_9)$ is useful because B_7 is the high birth edge and S_8, S_9 are the two channels added by hysteretic memory.

Definition 6.3 (First-step work functional). *Let $I_k \geq 0$ be a chosen cost assigned to a birth at raw channel k . The first-step birth work is*

$$W_B(x) = \sum_{k \in \{5,6,7\}} I_k B_k(x).$$

When $I_k = \log 2 + \log \binom{26}{k}$, the weight records the binary event bit plus the local channel multiplicity. Other nonnegative channel weights can be used, but the chosen weights must be stated.

Remark 6.1 (Diagnostic status). Churn, work, strain, and channel flux are measurement tools. The rectangle work laws are exact once the channel spectrum and weights are fixed, but the weights themselves are a chosen bookkeeping model unless separately justified.

7 Basic structural theorems

Proposition 7.1 (Cubic and translation equivariance). *The map $\mathcal{P}$ commutes with translations of Λ_L , coordinate permutations, and coordinate sign flips.*

Proof. The neighborhood $\mathcal{N}_{26}$ is invariant under those operations as a set. Such operations only permute terms in the neighbor sum $N_i(x)$, leaving the update condition unchanged. $\square$

Proposition 7.2 (Finite propagation speed). *If two initial configurations agree on the radius- t Moore ball around i , then their states at i agree after t updates. Influence propagates at speed at most one lattice unit per generation in Chebyshev distance.*

Proof. One update depends only on the radius-one Moore ball. Iterating this dependency gives a radius- t dependency after t updates. $\square$

Proposition 7.3 (Eventual periodicity). *For fixed L , every trajectory under $\mathcal{P}$ is eventually periodic. The preperiod plus period is at most 2^{L^3} .*

Proof. There are 2^{L^3} configurations. A deterministic map on a finite set must eventually revisit a previous state; after a repeat, the future is periodic. $\square$

Proposition 7.4 (Non-monotonicity). *The Poole map is not monotone with respect to coordinatewise order.*

Proof. Let $x \leq y$ and choose a site i with $x(i) = y(i) = 0$. Give x exactly five active neighbors of i . Let y be obtained by adding three additional active neighbors around i , so $N_i(y) = 8$. Then $\mathcal{P}(x)_i = 1$ but $\mathcal{P}(y)_i = 0$, because eight neighbors over-support an inactive site. Hence monotonicity fails. $\square$

Proposition 7.5 (Non-reversibility). *For $L \geq 3$, $\mathcal{P}$ is not injective and therefore is not reversible.*

Proof. The all-zero configuration maps to itself. The all-one configuration has $N_i = 26$ at every site, so every active site is over-supported and dies. Thus the all-one configuration also maps to the all-zero configuration. $\square$

8 Counterexample library for structural safety

A paper should include small counterexamples because they prevent readers from importing false assumptions from monotone cellular automata.

8.1 Monotonicity failure

Choose a target inactive site i . Let configuration x contain exactly five active Moore neighbors of i , and let y contain those five neighbors plus three more, with all other cells equal. Then $x \leq y$, but

$$\mathcal{P}(x)_i = 1, \quad \mathcal{P}(y)_i = 0,$$

with $N_i(x) = 5$ and $N_i(y) = 8$. More active support can destroy birth eligibility because inactive sites have a narrow upper threshold of 7.

8.2 Reversibility failure

The all-zero configuration maps to itself. On a periodic lattice with $L \geq 3$, the all-one configuration has raw count 26 at every site, so every active site is over-supported and dies. Thus two distinct configurations map to the all-zero configuration. The rule is not reversible.

8.3 Mean-field caution

A binomial mean-field closure treats every local neighborhood as if it were independently sampled. That closure is structurally mismatched to sheet-selected dynamics: the stable sheet is not random occupancy, but a correlated two-dimensional object embedded in three dimensions. Mean-field calculations remain useful as baselines precisely because the actual stable structures violate independence.

9 Fixed points and active graph constraints

Definition 9.1 (Active adjacency graph). *For a configuration x , let $G_A(x)$ be the graph whose vertices are active sites and whose edges connect active sites at Moore-neighbor distance.*

Proposition 9.1 (Fixed-point constraints). *A configuration x is a fixed point iff for every site i :*

$$x(i) = 1 \Rightarrow 5 \leq N_i(x) \leq 9,$$

and

$$x(i) = 0 \Rightarrow N_i(x) \notin \{5, 6, 7\}.$$

Proof. A fixed active site must survive, and a fixed inactive site must avoid birth. These are exactly the stated conditions. $\square$

Proposition 9.2 (Active-degree bounds in fixed points). *If x is a nonzero fixed point and $A = \{i : x(i) = 1\}$, then every active vertex has active Moore degree between 5 and 9. Consequently*

$$5 \leq \frac{2|E(G_A)|}{|A|} \leq 9.$$

Proof. For an active site, $N_i(x)$ is its degree in $G_A(x)$. The fixed-point survival condition gives $5 \leq N_i(x) \leq 9$ for every active site. Summing degrees gives the result. $\square$

This immediately rules out persistent isolated points and thin isolated filaments as fixed active components.

10 Surface selection

Definition 10.1 (Coordinate-flat local pattern). *A configuration is locally a coordinate r -flat through a site i if, inside the $3 \times 3 \times 3$ Moore block centered at i , the active sites are exactly the sites in an r -dimensional coordinate subspace through i . The cases $r = 0, 1, 2, 3$ represent a point, line, sheet, and filled block.*

Lemma 10.1 (Flat local degree count). *If an active site is locally a coordinate r -flat, then*

$$N_i = 3^r - 1.$$

Thus the flat local degrees are 0, 2, 8, 26 for $r = 0, 1, 2, 3$.

Proof. The r -flat contributes 3^r sites in the centered Moore block, one of which is the center site itself. The remaining $3^r - 1$ are active neighbors. $\square$

Theorem 10.1 (Local surface selector). *Among coordinate-flat local bulk types, the Poole survival window $5 \leq N_i \leq 9$ preserves exactly the locally two-dimensional sheet type.*

Proof. The four local degrees are 0, 2, 8, 26. Only 8 lies in $[5, 9]$. Hence points and lines are under-supported, sheets survive, and filled bulk is over-supported. $\square$

Theorem 10.2 (Stable active plane). *Assume $L \geq 3$. Fix a coordinate direction and a level c . Let*

$$H_c = \{(a, b, c) : a, b \in \mathbb{Z}/L\mathbb{Z}\} \subset \Lambda_L,$$

with configuration $x = \mathbf{1}_{H_c}$. Then $\mathcal{P}(x) = x$.

Proof. An active site in the sheet has the other eight sites in its in-plane 3×3 block as active neighbors, so $N = 8$ and it survives. A site in either adjacent layer sees a full 3×3 footprint, so $N = 9$; inactive cells are born only for $5 \leq N \leq 7$, so no adjacent birth occurs. Sites farther away see $N = 0$. Therefore the sheet is unchanged. $\square$

Corollary 10.1 (The +2 term is sheet memory). *Under the memory-reduced ablation $B5-7/S5-7$, a full one-cell active plane dies in one step, because its active sites have $N = 8$, outside the survival window 5–7.*

Proposition 10.1 (Filled slabs are rejected). *Let S be a union of at least two consecutive full coordinate sheets. Then every active site has at least 17 active neighbors, so no active site survives. No inactive site has a birth count in $\{5, 6, 7\}$. Thus $\mathcal{P}(\mathbf{1}_S) = 0$.*

Proof. An active site in a slab of thickness at least two sees eight active neighbors in its own sheet and at least nine in an adjacent sheet, for at least $17 > 9$. An inactive site outside the slab sees either no active sheet, one full active sheet, or two full active sheets, giving counts 0, 9, 18, none of which is a birth count. $\square$

11 Dimensional selector derivation

The surface selector theorem can be understood by looking at ideal coordinate flats. In a complete coordinate r -flat embedded in three dimensions, a site sees

$$3^r - 1$$

active neighbors inside the flat. Thus a point-like zero-flat gives 0, a line gives 2, a sheet gives 8, and full bulk gives 26. The survival window $5 \leq N \leq 9$ accepts the sheet and rejects the point, the line, and the bulk.

Local object	Dimension r	Ideal count $3^r - 1$	Survival status
point/dust	0	0	rejected by starvation
line	1	2	rejected by starvation
one-cell sheet	2	8	accepted
filled local bulk	3	26	rejected by over-support

This is the mathematical source of the term “surface selector.” The rule is not merely preserving a particular plane by accident; its threshold window accepts the count signature of a two-dimensional sheet while rejecting lower-dimensional and full-dimensional ideal flats.

Remark 11.1 (Why slabs fail). A filled slab of thickness at least two is not a one-cell surface. Interior or near-interior slab sites see too many active neighbors and violate the upper survival threshold. This is why the calculus should be built around one-cell sheets and their defects, not thick filled plates.

12 Planar defect calculus

The exact defect calculus begins by removing a mask from an otherwise stable sheet.

Definition 12.1 (Defective sheet and defect mask). *Identify the coordinate sheet with the two-dimensional torus*

$$T_L = (\mathbb{Z}/L\mathbb{Z})^2.$$

Let $D \subset T_L$ be a set of missing sheet sites, called the defect mask. The corresponding three-dimensional configuration is

$$x_D(u, c) = \mathbf{1}\{u \notin D\}, \quad x_D(u, z) = 0 \quad \text{for } z \neq c.$$

Definition 12.2 (Open and closed defect counts). *For $u \in T_L$, define*

$$q_D(u) = \sum_{\eta \in \{-1,0,1\}^2 \setminus \{0\}} \mathbf{1}\{u + \eta \in D\},$$

$$r_D(u) = \sum_{\eta \in \{-1,0,1\}^2} \mathbf{1}\{u + \eta \in D\}.$$

Here q_D counts missing cells in the open eight-neighborhood in the sheet, while r_D counts missing cells in the closed 3×3 footprint.

Theorem 12.1 (One-step planar defect calculus). *For the base Poole rule, the first update of x_D is determined exactly as follows:*

- (a) *an active sheet cell $u \notin D$ survives exactly when $q_D(u) \leq 3$, and dies exactly when $q_D(u) \geq 4$;*
- (b) *an inactive in-plane defect cell $u \in D$ is born in the original sheet plane exactly when $1 \leq q_D(u) \leq 3$;*
- (c) *an inactive cell one normal layer above or below the sheet, at $(u, c+1)$ or $(u, c-1)$, is born exactly when $2 \leq r_D(u) \leq 4$;*
- (d) *all layers with $|z - c| \geq 2$ remain inactive after the first update.*

Proof. All active sites initially lie in the sheet. A site in the sheet plane sees only the eight planar neighbors. If $u \notin D$, then

$$N(u, c) = 8 - q_D(u).$$

Survival requires $5 \leq 8 - q_D(u) \leq 9$, equivalent to $q_D(u) \leq 3$. If $u \in D$, the same count applies but the site is inactive, so birth requires $5 \leq 8 - q_D(u) \leq 7$, equivalent to $1 \leq q_D(u) \leq 3$. A site one layer above or below the sheet sees a 3×3 sheet footprint, so

$$N(u, c \pm 1) = 9 - r_D(u).$$

Birth requires $5 \leq 9 - r_D(u) \leq 7$, equivalent to $2 \leq r_D(u) \leq 4$. Sites two or more layers away see no active neighbors on the first update. $\square$

This theorem is the core of the Poole Defect Calculus. It converts a three-dimensional update near a sheet defect into two two-dimensional convolutions on D .

Corollary 12.1 (Stable pinhole gas). *If every closed 3×3 footprint contains at most one defect site, then x_D is a fixed point. In particular, a single one-cell pore in an otherwise complete sheet is exactly stable.*

Proof. The condition gives $r_D(u) \leq 1$ everywhere, so no normal birth can satisfy $2 \leq r_D \leq 4$. Each defect site has $q_D = 0$, so no in-plane birth occurs. Every active sheet site has $q_D \leq 1 \leq 3$, so all active sheet cells survive. $\square$

Remark 12.1 (Defect taxonomy). The exact local selectors give a clean taxonomy. Isolated pores freeze. Slits and cracks act as line sources. Two-dimensional punctures emit symmetric normal halos. General planar holes can cause both births and deaths; rectangles are a special zero-death subfamily.

13 Planar operator table and worked examples

The exact planar defect calculus is the core of the release. Let $D \subset T_L$ be the missing-cell mask in a stable coordinate sheet H_c . For $u \in T_L$, define the open and closed footprints

$$Q(u) = \{u + \delta : \delta \in \{-1, 0, 1\}^2 \setminus \{0\}\}, \quad R(u) = \{u + \delta : \delta \in \{-1, 0, 1\}^2\}.$$

Then

$$q_D(u) = |D \cap Q(u)|, \quad r_D(u) = |D \cap R(u)|.$$

The first response is completely determined by these two numbers.

Cell class	Raw count	Rule test	Result
active sheet site $u \notin D$	$8 - q_D(u)$	survival $5 \leq N \leq 9$	survives iff $q_D(u) \leq 3$
missing in-plane defect $u \in D$	$8 - q_D(u)$	birth $5 \leq N \leq 7$	born iff $1 \leq q_D(u) \leq 3$
normal-layer site $(u, c \pm 1)$	$9 - r_D(u)$	birth $5 \leq N \leq 7$	born iff $2 \leq r_D(u) \leq 4$
site two or more layers away	0	birth impossible	inactive

13.1 Single pore

For a single missing cell, every closed footprint contains at most one defect. Therefore every active sheet site has $q_D \leq 1$ and survives. The missing pore has $q_D = 0$, so it does not rebirth because inactive birth requires $N \leq 7$, while the pore sees 8 active planar neighbors. Normal-layer sites have $r_D \leq 1$, below the accepted range $2 \leq r_D \leq 4$. A single pore is exactly stable.

13.2 Separated pore gas

If every closed 3×3 footprint contains at most one defect, the same argument applies at every pore. The defective sheet is fixed. This is a useful theorem because it shows that defects do not automatically heal; sufficiently isolated pores are locked by over-supported in-plane birth and under-supported normal-layer birth.

13.3 Cracks and line sources

A one-cell-wide slit has contiguous missing cells. Its endpoints and interior cells create accepted q_D and r_D footprints. The result is no longer a stable pore gas: the slit becomes a line source with total raw response $B_0(1, n) = 7n$ for $n \geq 2$.

14 Square, rectangle, and line-hole laws

14.1 Square and rectangle punctures

Let $D_{a,b} \subset T_L$ be an $a \times b$ rectangular hole in a stable active sheet. Assume $a, b \geq 2$ and the rectangle is not interacting with itself through the periodic boundary.

Theorem 14.1 (Rectangle puncture law). *For $a, b \geq 2$, the first-step raw birth count of an $a \times b$ rectangular puncture is*

$$B_0(a, b) = 4a + 4b + 12.$$

There are zero first-step deaths. Equivalently, if $P = 2a + 2b$ is the lattice perimeter, then

$$B_0(a, b) = 2P + 12.$$

Proof sketch. The planar selector gives exactly four in-plane births at the four defect corners. In each adjacent normal layer, the closed footprint band $2 \leq r_D \leq 4$ has $2a + 2b + 4$ sites. Thus

$$B_0(a, b) = 4 + 2(2a + 2b + 4) = 4a + 4b + 12.$$

For cells remaining in the sheet, the missing-neighbor count around a rectangle never forces $q_D \geq 4$ outside the defect mask in the first update, so deaths are zero. $\square$

Corollary 14.1 (Square puncture law). *For a square hole of side length $h \geq 2$,*

$$B_0(h, h) = 8h + 12,$$

with zero first-step deaths.

Corollary 14.2 (Overcompensation window). *For an $a \times b$ rectangle, the first update has more active cells than the intact sheet precisely when*

$$|A_1| > L^2 \iff (a - 4)(b - 4) < 28.$$

Equality gives exact compensation, and the reverse inequality gives net loss.

Proof. The initial active count is $|A_0| = L^2 - ab$. Since deaths are zero and births are $4a + 4b + 12$,

$$|A_1| - L^2 = -ab + 4a + 4b + 12 = 28 - (a - 4)(b - 4).$$

The sign of this expression gives the stated criterion. $\square$

15 Expanded rectangle morphology

Rectangles are the closed-form family that makes the defect calculus easy to audit. For an $a \times b$ rectangular hole with $a, b \geq 2$, the first-step response has three components.

15.1 Plane contribution

Only the four missing in-plane corner cells have $1 \leq q_D \leq 3$. Edge-but-not-corner defect cells have too many missing neighbors, and interior defect cells have still more. Therefore the in-plane rebirth contribution is exactly four.

15.2 Normal halo contribution

In each adjacent normal layer, the accepted footprint band $2 \leq r_D \leq 4$ forms a one-cell halo around the rectangle. The accepted count in one normal layer is

$$2(a + b) + 4.$$

There are two normal layers, so the normal contribution is

$$2\{2(a + b) + 4\} = 4(a + b) + 8.$$

Adding the four in-plane corner births gives

$$B_0(a, b) = 4(a + b) + 12.$$

15.3 Channel morphology

The same footprint classification splits the response into channels:

$$N_5 = 12, \quad N_6 = 4a + 4b - 16, \quad N_7 = 16.$$

The N_5 term is a constant corner/near-corner charge. The N_6 term is the length-dependent edge band. The N_7 term is a constant outer shallow halo. The default prime-corrected response removes the raw N_7 group, giving

$$B_\phi(a, b) = N_5 + N_6 = 4a + 4b - 4.$$

15.4 Overcompensation

The intact sheet contains L^2 active sites. Removing an $a \times b$ hole leaves $L^2 - ab$ active sites. The first step has no deaths for the rectangle family, so

$$|A_1| = L^2 - ab + B_0(a, b).$$

The first step overcompensates the intact sheet iff $|A_1| > L^2$, equivalently

$$4a + 4b + 12 > ab.$$

Rearranging gives

$$(a - 4)(b - 4) < 28.$$

This is a useful release-level result because it converts the event count into a clean geometry inequality.

15.5 Line-hole edge case

The one-cell-wide rectangle family is degenerate and should not be forced into the $a, b \geq 2$ rectangle law.

Theorem 15.1 (Line-hole edge law). *For a one-cell-wide line hole $1 \times n$, the raw and default prime-corrected first-step birth counts are*

$$B_0(1) = B_\phi(1) = 0,$$

and, for $n \geq 2$,

$$B_0(1, n) = 7n, \quad B_\phi(1, n) = 7n - 14.$$

Proof sketch. A single missing cell is a stable pinhole and produces no births. For $n \geq 2$, direct application of the q_D/r_D selectors to the one-cell-wide mask gives the raw line-source count $7n$. The default prime correction removes the two endpoint-related raw $S = 7$ normal-edge groups, totaling 14 births, yielding $7n - 14$. $\square$

16 Model tags: raw PM_0 and prime-sharpened PM_ϕ

Definition 16.1 (Raw model). PM_0 denotes the raw Poole cellular automaton using the integer Moore count $S = N_i(x)$. Dead cells are born for $S \in \{5, 6, 7\}$, and live cells survive for $S \in \{5, 6, 7, 8, 9\}$.

Definition 16.2 (Default prime-sharpened model). PM_ϕ denotes the default prime-sharpened model with

$$\Phi(S) = S + \sum_{p \in \{2,3,5,7,11,13,17,19,23\}} 0.35 \exp\left(-\frac{(S-p)^2}{0.01}\right),$$

and the same birth and survival windows applied to $\Phi(S)$. For integer channels relevant to the planar defect laws, the default PM_ϕ birth channels are $S = 5, 6$, while the survival channels remain $S = 5, 6, 7, 8, 9$.

Thus PM_ϕ is not a new theorem about the raw rule; it is a model-tagged variant used to study channel sensitivity, especially the effect of removing the raw $S = 7$ birth channel.

17 Rectangle channel spectrum and prime correction

Theorem 17.1 (Rectangle birth-channel spectrum). For $a, b \geq 2$, the raw first-step rectangle birth count decomposes by neighbor channel as

$$N_{S=5} = 12, \quad N_{S=6} = 4a + 4b - 16, \quad N_{S=7} = 16.$$

Therefore

$$B_0(a, b) = N_{S=5} + N_{S=6} + N_{S=7} = 4a + 4b + 12.$$

Proof sketch. The $S = 5$ events consist of the four in-plane corner births and eight normal-layer corner-footprint births. The $S = 7$ events consist of sixteen normal-layer edge-end events. The remaining normal-layer boundary band contributes $4a + 4b - 16$ events at $S = 6$. Summing these disjoint channel classes gives the rectangle law. $\square$

Corollary 17.1 (Default prime-corrected rectangle law). For $a, b \geq 2$, default PM_ϕ removes the sixteen raw $S = 7$ birth events and keeps the $S = 5$ and $S = 6$ birth events. Hence

$$B_\phi(a, b) = 12 + (4a + 4b - 16) = 4a + 4b - 4.$$

For a square hole,

$$B_\phi(h, h) = 8h - 4.$$

18 Irreversible work laws for rectangle defects

Define the per-event reconstruction information weight

$$I_k = \log 2 + \log \binom{26}{k}.$$

For rectangle holes, first-step deaths are zero, so the first-step irreversible event work is a channel-weighted birth work.

Theorem 18.1 (Raw rectangle irreversible work). For PM_0 and $a, b \geq 2$,

$$W_0(a, b) = 12I_5 + (4a + 4b - 16)I_6 + 16I_7.$$

Proof. Multiply the channel spectrum $(12, 4a + 4b - 16, 16)$ by the channel weights (I_5, I_6, I_7) and sum. $\square$

Theorem 18.2 (Prime rectangle irreversible work). *For default PM_ϕ and $a, b \geq 2$,*

$$W_\phi(a, b) = 12I_5 + (4a + 4b - 16)I_6.$$

The prime correction removes exactly $16I_7$ from the raw rectangle response.

Numerically,

$$I_5 \approx 11.7872, \quad I_6 \approx 13.0400, \quad I_7 \approx 14.0898,$$

so the removed work is approximately

$$16I_7 \approx 225.4369$$

natural-log Poole information-work units.

19 Model-tagged arbitrary planar selector

The raw planar theorem can be upgraded into a model-tagged selector for PM_0 and PM_ϕ .

Definition 19.1 (Dilation and channel indicators). *Let*

$$\text{Dil}_1(D) = \{u \in T_L : r_D(u) > 0\}$$

be the closed one-step dilation of D . For a model $M \in \{\text{PM}_0, \text{PM}_\phi\}$, let $\chi_B^M(S)$ be its birth-channel indicator and $\chi_S^M(S)$ its survival-channel indicator.

Theorem 19.1 (Model-tagged planar defect selector). *For a finite planar defect mask D in an otherwise stable active sheet, the first-step birth count in model M is*

$$B_M(D) = \sum_{u \in D} \chi_B^M(8 - q_D(u)) + 2 \sum_{u \in \text{Dil}_1(D)} \chi_B^M(9 - r_D(u)),$$

and the first-step death count is

$$D_M(D) = \sum_{u \in \text{Dil}_1(D) \setminus D} [1 - \chi_S^M(8 - q_D(u))].$$

Proof. The first term in $B_M(D)$ counts in-plane births at defect cells, where the in-plane neighbor count is $8 - q_D(u)$. The second term counts normal-layer births above and below the sheet, where the normal footprint count is $9 - r_D(u)$, giving the factor of two. Active sheet cells outside D can die only in the one-step dilation of D , where their in-plane count is $8 - q_D(u)$. The death indicator is therefore $1 - \chi_S^M(8 - q_D(u))$. Outside $\text{Dil}_1(D)$, all local counts are those of the intact stable sheet. $\square$

This theorem is more general than the rectangle laws. Rectangles are clean because they form a zero-death closed-form subfamily. Arbitrary polyomino holes can produce deaths and must be handled by the selector rather than by a perimeter-only formula.

20 Q/P-style rectangle reconstruction boundary

The rectangle laws define a limited geometry-to-count transducer.

Proposition 20.1 (Raw rectangle reconstruction from area and response). *For $a, b \geq 2$, the pair consisting of area $A = ab$ and raw birth count*

$$B_0 = 4a + 4b + 12$$

recovers the unordered side lengths $\{a, b\}$.

Proof. From B_0 , recover the sum

$$a + b = \frac{B_0 - 12}{4}.$$

Together with the product $ab = A$, the unordered pair $\{a, b\}$ is determined as the two roots of

$$z^2 - (a + b)z + ab = 0.$$

□

Proposition 20.2 (Prime rectangle reconstruction from area and response). *For $a, b \geq 2$, the pair $A = ab$ and*

$$B_\phi = 4a + 4b - 4$$

recovers $\{a, b\}$ by the same sum-product argument.

Remark 20.1 (Boundary of the reconstruction claim). Birth count alone is not enough, because it only fixes $a + b$. Many rectangle pairs share the same sum. The reconstruction is therefore a restricted rectangle-prior theorem, not a general reconstruction theorem for arbitrary holes.

21 Reconstruction algorithm under a rectangle prior

The Q/P-style reconstruction boundary is intentionally narrow. It does not reconstruct arbitrary masks. It reconstructs rectangle side lengths, up to exchange, when the area and first-step response are known and the rectangle prior is assumed.

Proposition 21.1 (Raw rectangle inversion algorithm). *Let $A = ab$ and $B_0 = 4a + 4b + 12$ for an unknown rectangle with $a, b \geq 2$. Then*

$$s = a + b = \frac{B_0 - 12}{4}, \quad p = ab = A.$$

The side lengths are the two roots of

$$z^2 - sz + p = 0.$$

Thus

$$\{a, b\} = \left\{ \frac{s - \sqrt{s^2 - 4p}}{2}, \frac{s + \sqrt{s^2 - 4p}}{2} \right\},$$

provided the discriminant is a nonnegative square and the roots are valid integer side lengths.

Proposition 21.2 (Prime rectangle inversion algorithm). *Let $A = ab$ and $B_\phi = 4a + 4b - 4$. Then*

$$s = a + b = \frac{B_\phi + 4}{4}, \quad p = A,$$

and the same quadratic inversion recovers $\{a, b\}$ up to exchange.

Remark 21.1 (Why response alone is ambiguous). The response count fixes $a + b$, not ab . Many rectangles share a perimeter-like response. Area supplies the missing product. This is why the paper should describe the result as a two-measure transducer, not a one-number reconstruction theorem.

22 Verification and empirical ledgers

22.1 v7 perimeter audit

The v7 audit independently checked the exact formulas against full three-dimensional updates. The main deterministic families were:

Test family	Lattice	Cases	Checked identity
Flat monolayer sheet	$L = 48$	1	fixed point, no births/deaths
Square holes	$L = 64$	26	$B(h) = 8h + 12$, $h = 3, \dots, 28$
Rectangular holes	$L = 72$	484	$B(a, b) = 4a + 4b + 12$, $a, b = 3, \dots, 24$

The audit is not a substitute for proof. Its role is to protect the formulas against indexing, boundary, channel-count, and implementation errors.

22.2 v9 theorem-level closure

The v9 amendment closed the following discrete-model gaps:

ID	Result
D01	Complete one-cell active plane is a fixed point.
D02	Survival window selects one-cell sheets over dust, lines, slabs, and bulk.
D03	Square puncture law: $B_0(h, h) = 8h + 12$, zero deaths.
D04	One-step planar defect calculus with q_D and r_D .
D05	Rectangle hole law: $B_0(a, b) = 4a + 4b + 12$, zero deaths.
D06	Line-hole edge law: $B_0(1, n) = 7n$, $B_\phi(1, n) = 7n - 14$ for $n \geq 2$.
D07	Rectangle channel spectrum: $N_5 = 12$, $N_6 = 4a + 4b - 16$, $N_7 = 16$.
D08	Prime-corrected rectangle law: $B_\phi(a, b) = 4a + 4b - 4$.
D09–D10	Raw and prime irreversible-work laws.
D11	Model-tagged planar selector for arbitrary finite planar holes.
D12–D13	Rectangle reconstruction from area plus raw or prime response.

The uploaded v9 audit reported 841 rectangle cases, 80 line-hole cases, 720 arbitrary planar-hole selector rows, and 1225 Q/P rectangle inversion cases, with the stated matching checks returning true.

22.3 v10 empirical atlas

The v10 atlas adds empirical and computational observations rather than new closed theorems. Its purpose is to identify the next candidate laws.

Observation	Evidence	Status
Raw planar defects cascade into high churn	Mean raw tail event density ≈ 0.359308	observation
Prime filtering suppresses planar cascades	Mean prime tail event density ≈ 0.080062	observation
Planar relaxation is pulse-like	Curves rise and then decay; monotone exponential is inadequate	model clue
3D void response is size-sensitive	Correlations above 0.98 for removed active volume against response	observation
Peak response is near-linear to mildly superlinear	Peak W beta: cube 0.979, sphere 1.202, ellipsoid 1.191	empirical fit
Integrated response is more superlinear	Integrated excess work scales more strongly than peak response	empirical fit

The strongest reported correlations were removed-active versus peak excess work 0.9861 and removed-active versus integrated excess work 0.9880. These are strong reproducible signals, but they should not be cited as proofs of a closed multi-step relaxation law or a closed non-planar void law.

23 Mean-field failure and correlation dependence

A naive independent-site mean-field approximation takes each site active with probability p_t . Then the neighbor count is approximated by

$$N \sim \text{Bin}(26, p_t).$$

The mean-field density update is

$$p_{t+1}^{\text{mf}} = (1 - p_t) \sum_{k=5}^7 \binom{26}{k} p_t^k (1 - p_t)^{26-k} + p_t \sum_{k=5}^9 \binom{26}{k} p_t^k (1 - p_t)^{26-k}.$$

This closure ignores spatial correlations, sheet geometry, cavities, and the fact that the rule selects local two-dimensional structure. It is useful primarily because it fails in a diagnostic way: persistent active regimes are not explained by independent random occupancy. They require correlated structure.

24 Continuum expansion of the 26-neighbor sum

Let the lattice spacing be a and suppose the binary field is coarse-grained to a smooth density $\rho(x)$. Formally,

$$\sum_{\delta \in \mathcal{N}_{26}} \rho(x + a\delta) = 26\rho(x) + 9a^2 \Delta \rho(x) + O(a^4),$$

because the first-order terms cancel by symmetry and

$$\frac{1}{2} \sum_{\delta \in \mathcal{N}_{26}} \delta_j^2 = 9$$

for each coordinate direction. Thus the local count threshold becomes a density-window condition with a curvature correction:

$$5 \lesssim 26\rho + 9a^2\Delta\rho \lesssim 7 + 2x.$$

This is not yet a physical continuum limit. It is a controlled expansion of the local counting operator. A true continuum theory would need scaling assumptions for x , ρ , defect density, correlation length, time step, and any physical calibration.

25 Rule-space interpretation

The nearby rules are best organized by two hysteresis choices: the lower birth threshold and the upper survival threshold. In this view:

- $B5-7/S5-9$ is the raw base rule with shallow-defect plasticity and sheet memory.
- $B5-7/S5-7$ removes upper memory and kills stable sheets.
- $B6-7/S5-9$ preserves upper sheet memory while suppressing the weakest $B5$ birth channel, making it a natural low-strain candidate.
- PM_ϕ preserves the same survival behavior in the relevant channels while removing raw $S = 7$ births under the default prime-sharpening parameters.

The rule-space lesson is that the upper survival band 8, 9 is the sheet-memory mechanism, while the upper birth edge B_7 is the plasticity mechanism.

26 Limitations and open problems

The following remain open and should be treated as future work:

1. a closed multi-step relaxation law after the first update;
2. a closed non-planar three-dimensional void law;
3. a full rule-space uniqueness theorem;
4. a rigorous correlation theory explaining stable active regimes beyond mean field;
5. topology-aware metrics for defect healing, freezing, and halo extrusion;
6. continuum calibration with an explicit scaling limit;
7. a disciplined bridge from defect work to any OTG, Q/P, gauge-force, or cosmology claims.

27 Conclusion

The Poole Defect Calculus is now a coherent discrete mathematical object. Its core equation

$$x_{t+1}(i) = \mathbf{1}\{5 \leq N_i(x_t) \leq 7 + 2x_t(i)\}$$

combines birth, survival, and hysteretic memory into one local threshold law. That law selects one-cell sheets, rejects dust and bulk, preserves stable active planes, freezes isolated pinholes, turns cracks into line sources, and maps finite planar defects to exact q_D/r_D selector counts. Rectangular defects are the cleanest closed-form family: they have zero first-step deaths, a perimeter-plus-corner birth law, an exact channel spectrum, a prime-corrected response, exact first-step work formulas, and a limited area-plus-response reconstruction theorem.

The empirical v10 atlas extends the program beyond first-step planar laws by measuring multi-step relaxation and non-planar void response. Its numbers are promising, especially the suppression of raw high-churn cascades by prime filtering and the strong removed-active-volume correlations in void response. But the correct claim boundary remains: first-step planar defect response is theorem-level; multi-step and 3D void laws are reproducible empirical targets awaiting closure.

Appendices

A.1 Minimal implementation requirements

A correct base implementation should:

1. use a three-dimensional Boolean array on a periodic cubic lattice;
2. compute the 26-neighbor Moore count by summing all shifts in $\{-1, 0, 1\}^3 \setminus \{0\}$;
3. apply birth to inactive cells with counts 5, 6, 7;
4. apply survival to active cells with counts 5, 6, 7, 8, 9;
5. report births, deaths, active counts, channel histograms, and event work separately;
6. keep theorem-level PM0 results separate from PM_ϕ variant results;
7. mark multi-step relaxation and 3D void results as empirical unless a proof is supplied.

A.2 Pseudocode

for each step:

```

S = 26_neighbor_count(A)
birth  = (~A) & (S >= 5) & (S <= 7)
survive = ( A) & (S >= 5) & (S <= 9)
death  = A & (~survive)
A_next = birth | survive

```

For default PM_ϕ , replace S by $\Phi(S)$ in the birth/survival inequalities while still recording raw integer channels for diagnostics.

A.3 Theorem ledger

ID	Name	Statement
D01	Stable plane	A complete one-cell active plane is a fixed point.
D02	Surface selector	The survival window preserves one-cell sheets and rejects dust, lines, slabs, and bulk.
D03	Planar selector	A defective sheet updates by the exact q_D/r_D calculus.
D04	Stable pinholes	Isolated one-cell pores are fixed.
D05	Square law	$B_0(h, h) = 8h + 12$, zero deaths.
D06	Rectangle law	$B_0(a, b) = 4a + 4b + 12 = 2P + 12$, zero deaths.
D07	Overcompensation	$ A_1 > L^2$ iff $(a - 4)(b - 4) < 28$.
D08	Line law	$B_0(1, n) = 7n$, $B_\phi(1, n) = 7n - 14$ for $n \geq 2$.
D09	Channel spectrum	$(N_5, N_6, N_7) = (12, 4a + 4b - 16, 16)$.
D10	Prime rectangle	$B_\phi(a, b) = 4a + 4b - 4$.
D11	Work laws	$W_0 = 12I_5 + (4a + 4b - 16)I_6 + 16I_7$; $W_\phi = 12I_5 + (4a + 4b - 16)I_6$.

ID	Name	Statement
D12	Model-tagged selector	Arbitrary planar masks have model-tagged $B_M(D)$ and $D_M(D)$ formulas.
D13	Q/P boundary	Area plus raw or prime rectangle response recovers $\{a, b\}$; response alone is ambiguous.

A.4 Complete theorem and claim-status ledger

The table below is the current release ledger. The status column is deliberately conservative. “Exact” means the statement follows from the local rule and finite combinatorics. “Verified” means the finite scripts matched the formula in the tested domain. “Empirical” means the item is reproducible numerical evidence but not a closed theorem.

ID	Type	Name	Statement	Status
D01	Theorem	Stable active plane	A complete one-cell coordinate plane is a fixed point of the base rule and of the default prime-tagged model in the relevant channel range.	Exact
D02	Theorem	Surface selector	Local survival preserves the two-dimensional coordinate-flat pattern and rejects local point, line, and filled-block patterns.	Exact
D03	Proposition	Square puncture	For square hole side $h \geq 2$, $B_0(h, h) = 8h + 12$ and first-step deaths are zero.	Exact/verified
D04	Theorem	Planar defect calculus	The first update of a defective sheet is determined by q_D and r_D .	Exact
D05	Corollary	Rectangle hole law	For $a, b \geq 2$, $B_0(a, b) = 4a + 4b + 12 = 2P + 12$ and first-step deaths are zero.	Exact/verified
D06	Theorem	Line-hole edge law	$B_0(1) = 0$, $B_0(1, n) = 7n$, $B_\phi(1) = 0$, and $B_\phi(1, n) = 7n - 14$ for $n \geq 2$.	Exact/verified
D07	Theorem	Channel spectrum	Rectangular births split as $N_5 = 12$, $N_6 = 4a + 4b - 16$, $N_7 = 16$.	Exact/verified
D08	Theorem	Prime rectangle law	Default PM_ϕ removes raw $S = 7$ rectangle births, giving $B_\phi(a, b) = 4a + 4b - 4$.	Exact in model
D09	Theorem	Raw work law	$W_0(a, b) = 12I_5 + (4a + 4b - 16)I_6 + 16I_7$.	Exact in model
D10	Theorem	Prime work law	$W_\phi(a, b) = 12I_5 + (4a + 4b - 16)I_6$.	Exact in model
D11	Theorem	Model-tagged selector	Arbitrary finite planar holes have exact $B_M(D)$ and $D_M(D)$ formulas using χ_B^M and χ_S^M .	Exact/verified
D12	Proposition	Raw rectangle reconstruction	Under a rectangle prior, $A = ab$ and B_0 recover $\{a, b\}$ up to exchange.	Exact/verified
D13	Proposition	Prime rectangle reconstruction	Under a rectangle prior, $A = ab$ and B_ϕ recover $\{a, b\}$ up to exchange.	Exact/verified
E01	Observation	Raw planar cascade	Multi-step raw planar defects enter a high-churn bath in the v10 run.	Empirical
E02	Observation	Prime suppression	Default PM_ϕ suppresses planar defect cascades relative to raw PM_0 .	Empirical
E03	Observation	Pulse-like relaxation	Planar relaxation curves rise and then decay; monotone exponential relaxation is inadequate.	Empirical/model clue

ID	Type	Name	Statement	Status
E04	Observation	3D void response	Peak and integrated void response correlate strongly with removed active volume in the tested range.	Empirical

A.5 Model dictionary

Definition A.5.1 (Base raw model PM_0). *The raw model PM_0 is the deterministic binary cellular automaton*

$$x_{t+1}(i) = \mathbf{1}\{5 \leq N_i(x_t) \leq 7 + 2x_t(i)\}.$$

It accepts dead-cell birth channels $S = 5, 6, 7$ and active-cell survival channels $S = 5, 6, 7, 8, 9$.

Definition A.5.2 (Default prime-sharpened model PM_ϕ). *The default prime-sharpened model applies the same birth and survival windows to*

$$\Phi(S) = S + \sum_{p \in \{2, 3, 5, 7, 11, 13, 17, 19, 23\}} 0.35 \exp\left(-\frac{(S-p)^2}{0.01}\right).$$

For the integer channels used in the planar defect laws, this keeps raw birth channels 5, 6 and rejects raw birth channel 7 because $\Phi(7) = 7.35 > 7$. The relevant survival channels remain 5, 6, 7, 8, 9.

Definition A.5.3 (Memory-reduced ablation). *The rule $B5-7/S5-7$ is the ablation that removes the upper survival memory band 8, 9. It kills a flat active sheet in one step and therefore fails the stable-sheet theorem.*

Definition A.5.4 (Low-strain candidate). *The rule $B6-7/S5-9$ preserves the upper sheet-memory band while suppressing the weakest $B5$ birth channel. It is not a replacement theorem; it is a rule-space candidate motivated by the surface-selector structure and earlier diagnostics.*

A.6 Defect operator cookbook

Let $D \subset T_L$ be a finite defect mask in an otherwise complete active coordinate sheet. All first-step planar-defect formulas are computed from two two-dimensional convolution counts:

$$q_D(u) = \sum_{\eta \in \{-1, 0, 1\}^2 \setminus \{0\}} \mathbf{1}\{u + \eta \in D\}, \quad r_D(u) = \sum_{\eta \in \{-1, 0, 1\}^2} \mathbf{1}\{u + \eta \in D\}.$$

The first is an open-neighborhood missing-count. The second is a closed 3×3 footprint missing-count.

Cell class	Count before update	Base-rule criterion
Active sheet cell $u \notin D$	$N(u, c) = 8 - q_D(u)$	Survives iff $q_D(u) \leq 3$
Inactive defect cell $u \in D$	$N(u, c) = 8 - q_D(u)$	Born in-plane iff $1 \leq q_D(u) \leq 3$
Inactive normal cell $(u, c \pm 1)$	$N(u, c \pm 1) = 9 - r_D(u)$	Born in normal halo iff $2 \leq r_D(u) \leq 4$
Layer $ z - c \geq 2$	$N = 0$ at first step	Remains inactive

The defect calculus is therefore local, finite, and computable by two planar convolutions. It is not a metaphorical derivative; it is a finite-difference response operator mapping a missing-cell mask D to births, deaths, and channel counts after one update.

A.6.1 General nearby-rule selector

For a nearby rule $Ba-b/Sc-d$, the same defective sheet has the first-step criteria

$$\begin{aligned} \text{active sheet survival} &\iff 8 - d \leq q_D(u) \leq 8 - c, \\ \text{in-plane defect birth} &\iff 8 - b \leq q_D(u) \leq 8 - a, \\ \text{normal-layer birth} &\iff 9 - b \leq r_D(u) \leq 9 - a. \end{aligned}$$

This compact formula makes the role of the +2 upper survival allowance transparent: widening d to 9 allows $q_D = 0$ sheet interiors to survive with count 8.

A.7 Detailed rectangle morphology and channel count

For $a, b \geq 2$, a rectangle has a defect interior, a four-corner set, side bands, and two adjacent normal layers. The raw birth law can be decomposed by both geometric location and neighbor channel.

Event class	Count	Channel	Explanation
In-plane defect corners	4	$S = 5$	A missing corner cell has three missing open-neighbor positions, so $8 - q_D = 5$.
In-plane non-corner defect sites	0 births	not eligible	Edge/interior missing cells have counts outside the birth band under the rectangle geometry.
Normal corner-footprint births	8	$S = 5$	Four corner footprints appear in both adjacent normal layers.
Normal edge-band births	$4a + 4b - 16$	$S = 6$	The regular side band in the two normal layers contributes the perimeter-linear term.
Normal edge-end births	16	$S = 7$	Two layers times eight edge-end positions produce the plasticity channel.
Deaths in sheet	0	none	Active sheet cells outside the rectangle never have $q_D \geq 4$ in the first update.

Summing the birth classes gives

$$B_0(a, b) = 12 + (4a + 4b - 16) + 16 = 4a + 4b + 12.$$

The prime-sharpened model removes precisely the $S = 7$ row, giving

$$B_\phi(a, b) = 12 + (4a + 4b - 16) = 4a + 4b - 4.$$

A.7.1 Work values

With

$$I_k = \log 2 + \log \binom{26}{k},$$

the channel values used in the v9/v10 paper are

$$I_5 \approx 11.7872183003, \quad I_6 \approx 13.0399812688, \quad I_7 \approx 14.0898033933.$$

Thus default prime sharpening removes the constant rectangle work

$$16I_7 \approx 225.4368542928$$

from the raw first-step rectangle response.

A.8 Rectangle reconstruction boundary expanded

The rectangle transducer is not a general hole-reconstruction theorem. It is a rectangle-prior theorem. The response provides a perimeter-like sum, while the area supplies the missing product.

Model	Response equation	Reconstruction from area $A = ab$
Raw PM_0	$B_0 = 4a + 4b + 12$	$s = (B_0 - 12)/4 = a + b$, $\Delta = s^2 - 4A$
Prime PM_ϕ	$B_\phi = 4a + 4b - 4$	$s = (B_\phi + 4)/4 = a + b$, $\Delta = s^2 - 4A$
Both	roots of $z^2 - sz + A = 0$	$\{a, b\} = \{(s - \sqrt{\Delta})/2, (s + \sqrt{\Delta})/2\}$

Response alone fixes $a + b$ but not ab . Therefore several rectangles can share a response class. The v9 audit found response-only ambiguity classes, so the release statement must remain: *area plus response reconstructs rectangular dimensions up to exchange under a rectangle prior.*

A.9 Verification ledger

Audit item	Domain	Result	Purpose
Flat monolayer sheet	one sheet, periodic lattice	no births/deaths	verifies stable-plane theorem and no-thickening.
Square punctures	$h = 3, \dots, 28$	$B_0(h, h) = 8h + 12$	checks square law against full 3D update.
Rectangular holes, early audit	$a, b = 3, \dots, 24$	484 matches	checks $B_0(a, b) = 4a + 4b + 12$.
Rectangles, v9 full audit	841 cases	all matched true	extended full 3D verification and channel tracking.
Line holes	80 cases	all matched true	validates the $1 \times n$ edge-case law.

Audit item		Domain	Result	Purpose
Arbitrary planar-hole selector		720 rows	births/deaths matched	all validates model-tagged $B_M(D)$ and $D_M(D)$ formulas against full 3D updates.
Q/P rectangle inversion		1225 cases	raw and prime area-plus-response recovery all matched	validates reconstruction boundary under rectangle prior.
Birth-only check	inversion tested rectangle domain		93 ambiguous classes	validates the warning that response alone is incomplete.

The audits are not substitutes for proofs when a proof exists. Their role is to prevent implementation errors, indexing errors, periodic-boundary mistakes, channel misclassification, and mistaken claims about reconstruction.

A.10 Expanded v10 empirical atlas

The v10 atlas is included because it advances the next two gaps: multi-step planar relaxation after first-step laws and non-planar 3D void response on an annealed PM_ϕ vacuum. It does not close those gaps as theorems.

A.10.1 v10 audit configuration

Parameter	Value
Preset	standard
Grid size	$N = 64$
Planar response length	80 steps
Vacuum anneal length	120 steps
Void response length	70 steps
Void trials per size	3
Random planar polyomino trials	5
Planar cases	9
3D void cases	36

A.10.2 Planar multi-step relaxation summary

The first-step event densities are tiny compared with the later tail response:

$$E_0(\text{PM}_0) \approx 0.000376, \quad E_0(\text{PM}_\phi) \approx 0.000294.$$

After 80 steps, the v10 run reported

$$E_{\text{tail}}(\text{PM}_0) \approx 0.359308, \quad E_{\text{tail}}(\text{PM}_\phi) \approx 0.080062.$$

The safe interpretation is that raw planar defects can cascade into high churn, while prime filtering suppresses that cascade by a large factor. The relaxation curves rise and then decay, so a monotone exponential is not the right model class. A better candidate family is

$$E(t) = E_{\infty} + At^p \exp(-t/\tau),$$

with baseline-matched controls and bootstrap confidence intervals.

A.10.3 Mean 3D void response table

Shape	Size	Volume proxy	Removed active	Peak excess W	Integrated excess W
cube	3	343.0	126.7	0.00515503	0.0269896
cube	4	729.0	298.0	0.0122313	0.110344
cube	5	1331.0	525.3	0.0213321	0.2274
cube	6	2197.0	873.7	0.0339162	0.422282
ellipsoid	3	99.0	39.7	0.00116729	0.00243775
ellipsoid	4	189.0	72.3	0.00293362	0.00898938
ellipsoid	5	281.0	110.3	0.00517274	0.0245721
ellipsoid	6	661.0	262.0	0.0112632	0.0859388
sphere	3	123.0	50.3	0.00127405	0.00394072
sphere	4	257.0	97.3	0.00328013	0.0134696
sphere	5	515.0	204.0	0.00710508	0.055229
sphere	6	925.0	365.3	0.0143826	0.152506

Using shape-size means, the peak excess work scaling versus removed active count has exponents approximately

$$\beta_{\text{cube}} \approx 0.979, \quad \beta_{\text{sphere}} \approx 1.202, \quad \beta_{\text{ellipsoid}} \approx 1.191.$$

Integrated excess work is more superlinear, with removed-active exponents approximately

$$\beta_{\text{cube,int}} \approx 1.425, \quad \beta_{\text{sphere,int}} \approx 1.853, \quad \beta_{\text{ellipsoid,int}} \approx 1.891.$$

The strongest safe claim is correlation and scaling over the tested range, not a universal closed 3D void law.

A.11 Reference implementation appendix

The following code is a compact reference implementation. It is not optimized. Its purpose is audit clarity.

```
import math
import numpy as np

SHIFTS_26 = tuple(
    (dx, dy, dz)
    for dx in (-1, 0, 1)
    for dy in (-1, 0, 1)
    for dz in (-1, 0, 1)
    if not (dx == 0 and dy == 0 and dz == 0)
```

```

)

PRIMES = (2, 3, 5, 7, 11, 13, 17, 19, 23)
PHI_LUT = np.arange(27, dtype=float)
for p in PRIMES:
    PHI_LUT += 0.35 * np.exp(-((np.arange(27) - p) ** 2) / 0.01)

I_LUT = np.array([math.log(2) + math.log(math.comb(26, k)) for k in range(27)])

def n26(A):
    """Periodic_26-neighbor_count_for_a_3D_boolean_array."""
    Ai = A.astype(np.int16, copy=False)
    S = np.zeros_like(Ai, dtype=np.int16)
    for dx, dy, dz in SHIFTS_26:
        S += np.roll(np.roll(np.roll(Ai, dx, 0), dy, 1), dz, 2)
    return S

def poole_step(A, mode="raw"):
    """One_Poole_step._mode='raw'_gives_PM0;_mode='prime'_gives_PM_phi."""
    S = n26(A)
    X = S.astype(float) if mode == "raw" else PHI_LUT[S]
    birth = (~A) & (X >= 5) & (X <= 7)
    survive = A & (X >= 5) & (X <= 9)
    death = A & (~survive)
    A1 = birth | survive
    event = birth | death
    metrics = {
        "active0": int(A.sum()),
        "births": int(birth.sum()),
        "deaths": int(death.sum()),
        "active1": int(A1.sum()),
        "events": int(event.sum()),
        "W_step": float((event * I_LUT[S]).sum()),
    }
    for k in range(27):
        metrics[f"B{k}"] = int((birth & (S == k)).sum())
        metrics[f"D{k}"] = int((death & (S == k)).sum())
    return A1, metrics

def sheet_with_rect_hole(L, a, b, z=None, x0=None, y0=None):
    """Full_coordinate_sheet_with_an_a_by_b_rectangular_hole."""
    if z is None: z = L // 2
    if x0 is None: x0 = (L - a) // 2
    if y0 is None: y0 = (L - b) // 2
    A = np.zeros((L, L, L), dtype=bool)
    A[:, :, z] = True
    A[x0:x0+a, y0:y0+b, z] = False
    return A

def predict_rectangle_raw(a, b):
    if a == 1 and b == 1:
        return {"births": 0, "deaths": 0}
    if a == 1:
        return {"births": 7 * b, "deaths": 0}

```

```

    if b == 1:
        return {"births": 7 * a, "deaths": 0}
    return {
        "births": 4*a + 4*b + 12,
        "deaths": 0,
        "B5": 12,
        "B6": 4*a + 4*b - 16,
        "B7": 16,
        "prime_births": 4*a + 4*b - 4,
    }

def verify_rectangles(L=72, amin=2, amax=24, mode="raw"):
    rows = []
    for a in range(amin, amax + 1):
        for b in range(amin, amax + 1):
            A = sheet_with_rect_hole(L, a, b)
            _, m = poole_step(A, mode=mode)
            p = predict_rectangle_raw(a, b)
            expected = p["births"] if mode == "raw" else p["prime_births"]
            rows.append({
                "a": a, "b": b,
                "observed_births": m["births"],
                "expected_births": expected,
                "observed_deaths": m["deaths"],
                "match": (m["births"] == expected and m["deaths"] == 0),
            })
    return rows

if __name__ == "__main__":
    rows = verify_rectangles(L=72, amin=2, amax=10, mode="raw")
    assert all(r["match"] for r in rows)
    print("rectangle_smoke_test_passed", len(rows), "cases")

```

A.12 Two-dimensional defect selector reference implementation

```

import numpy as np

SHIFTS_8 = tuple(
    (dx, dy) for dx in (-1, 0, 1) for dy in (-1, 0, 1)
    if not (dx == 0 and dy == 0)
)
SHIFTS_9 = tuple((dx, dy) for dx in (-1, 0, 1) for dy in (-1, 0, 1))

def roll2(A, dx, dy):
    return np.roll(np.roll(A, dx, axis=0), dy, axis=1)

def q_r_counts(D):
    """Open_q_D_and_closed_r_D_missing-cell_counts_on_a_2D_torus."""
    Di = D.astype(np.int16, copy=False)
    q = np.zeros_like(Di, dtype=np.int16)
    r = np.zeros_like(Di, dtype=np.int16)
    for dx, dy in SHIFTS_8:

```

```

        q += roll2(Di, dx, dy)
    for dx, dy in SHIFTS_9:
        r += roll2(Di, dx, dy)
    return q, r

def selector_counts(D, model="raw"):
    """First-step planar-selector births/deaths for PM0 or PM_phi."""
    q, r = q_r_counts(D)
    dil = r > 0
    if model == "raw":
        birth_channels = {5, 6, 7}
    elif model == "prime":
        birth_channels = {5, 6}
    else:
        raise ValueError("model must be raw or prime")
    survive_channels = {5, 6, 7, 8, 9}

    inplane_S = 8 - q
    normal_S = 9 - r

    inplane_births = D & np.isin(inplane_S, list(birth_channels))
    normal_births_one_layer = dil & np.isin(normal_S, list(birth_channels))
    active_sheet_dies = (~D) & dil & (~np.isin(inplane_S, list(survive_channels)))

    return {
        "births": int(inplane_births.sum() + 2 * normal_births_one_layer.sum()),
        "deaths": int(active_sheet_dies.sum()),
        "inplane_births": int(inplane_births.sum()),
        "normal_births_total": int(2 * normal_births_one_layer.sum()),
    }

```

A.13 Mean-field closure and why it fails structurally

The independent-site mean-field closure is

$$p_{t+1}^{\text{mf}} = (1 - p_t) \sum_{k=5}^7 \binom{26}{k} p_t^k (1 - p_t)^{26-k} + p_t \sum_{k=5}^9 \binom{26}{k} p_t^k (1 - p_t)^{26-k}.$$

This equation is useful as a null model because it discards the exact geometry that makes sheets stable. A perfect sheet has density $p = 1/L$ in a periodic L^3 lattice if it is one layer thick, so an independent-site model at the same density would not reproduce the sheet's deterministic $N = 8$ local support. The difference is not a small correction; it is the core of the calculus. Persistent activity is correlation-supported, not mean-field-supported.

A.14 Continuum expansion caveat

The formal expansion

$$\sum_{\delta \in \mathcal{N}_{26}} \rho(x + a\delta) = 26\rho(x) + 9a^2 \Delta \rho(x) + O(a^4)$$

turns the local count into a density-window condition with curvature correction,

$$5 \lesssim 26\rho + 9a^2\Delta\rho \lesssim 7 + 2x.$$

This is only an expansion of the counting operator. It is not a continuum physics derivation. A physical bridge would require a scaling limit for voxel size, time step, density, defect density, correlation length, calibration constants, and measurement channel.

A.15 Glossary

Term	Meaning
Poole rule	The base $B5-7/S5-9$ three-dimensional binary cellular automaton.
Hysteretic support	The active state widens the upper support threshold from 7 to 9.
Sheet memory	The $+2x_t$ survival allowance that lets active sheet interiors with $N = 8$ persist.
Plasticity gate	The B_7 birth channel, especially visible in rectangle responses and prime filtering.
Defect mask D	The set of missing sites removed from an otherwise complete active sheet.
q_D	Open eight-neighborhood count of missing defect sites.
r_D	Closed 3×3 footprint count of missing defect sites.
Normal halo	Births in the layers immediately above and below a defective sheet.
Pinhole gas	A Chebyshev-separated collection of one-cell pores that remains fixed.
Rectangle law	$B_0(a, b) = 4a + 4b + 12$ for $a, b \geq 2$.
Prime correction	Default PM_ϕ removal of raw $S = 7$ birth events.
Irreversible work	Channel-weighted event count using $I_k = \log 2 + \log \binom{26}{k}$.
Q/P boundary	The restricted result that area plus rectangle response recovers side lengths up to exchange.
Empirical atlas	Reproducible numerical map of multi-step and 3D void behavior, not yet a theorem.

A.16 Open theorem roadmap

The next mathematical targets are:

1. classify all stable finite planar defect masks beyond separated pinholes;
2. derive a closed formula for first-step response of arbitrary polyominoes by boundary graph invariants;
3. classify when arbitrary planar holes produce deaths;
4. prove or refute a multi-step relaxation law for planar masks;
5. build a topology-aware metric for hole closure, extrusion, and fragmentation;

6. derive a non-planar 3D void response theorem for restricted background states;
7. formalize the rule-space uniqueness question: which nearby rules preserve sheets, reject slabs, and produce controlled defect halos;
8. define the exact conditions under which a dynamic Q/P extension can read multi-step response without conflating it with first-step static capture;
9. replace the continuum expansion caveat with a genuine scaling theorem, if one exists.

A.17 Final boundary statement

The robust release claim is this:

The Poole Defect Calculus is an exact first-step finite-combinatorial theory for how a hysteretic surface-selecting three-dimensional cellular automaton responds to planar defects in stable one-cell sheets. Its strongest closed laws are the q_D/r_D selector, the stable pinhole result, the square and rectangle birth laws, the line-hole edge law, the rectangle channel spectrum, the prime-corrected rectangle response, the channel-weighted work laws, and the rectangle-prior area-plus-response reconstruction theorem. Multi-step planar relaxation and non-planar 3D void response are currently reproducible empirical atlas results, not closed laws.

This is the clean line between a publishable discrete mathematics paper and broader speculative applications.

A.18 Publication-form proof capsules

This appendix restates the central exact claims in proof-capsule form so that the final release can be audited theorem by theorem without searching through the narrative.

A.18.1 D01: stable active plane

For an active coordinate plane H_c , every active site has exactly eight active neighbors in its own planar 3×3 block. Since $8 \in \{5, 6, 7, 8, 9\}$, every active sheet site survives. A site immediately above or below the sheet sees a full 3×3 footprint and hence has count 9. Since $9 \notin \{5, 6, 7\}$ for inactive birth, no adjacent normal cell is born. Every other inactive site has count 0. Therefore the full one-cell sheet is fixed.

A.18.2 D02: surface selector

A local coordinate r -flat has 3^r sites in the centered Moore block and $3^r - 1$ active neighbors of the center. The sequence for $r = 0, 1, 2, 3$ is 0, 2, 8, 26. The active survival window $[5, 9]$ contains exactly 8. Thus the local rule preserves the two-dimensional sheet pattern and rejects the local point, line, and bulk patterns. This is the exact mathematical basis for calling the rule surface-selecting.

A.18.3 D03–D05: square and rectangle response

The rectangle law is the parent result. For $a, b \geq 2$, the in-plane births occur only at the four defect corners. The normal-layer response is controlled by $2 \leq r_D \leq 4$ over the closed 3×3 footprint. In one normal layer this eligible footprint band has $2a + 2b + 4$ sites. There are two normal layers, so normal births contribute $4a + 4b + 8$. Adding the four in-plane corner births gives $4a + 4b + 12$. For a square $a = b = h$, this reduces to $8h + 12$.

The zero-death claim follows from the same local count. Outside the removed rectangle, no surviving sheet cell has four or more missing planar neighbors on the first step; hence every remaining active sheet cell satisfies $q_D \leq 3$ and survives.

A.18.4 D06: line-hole edge law

The line-hole family is not the $a, b \geq 2$ rectangle law with $a = 1$ substituted. The defect footprint degenerates. A single missing cell is a stable pinhole. For a one-cell-wide line of length $n \geq 2$, the open and closed defect counts along the line generate a raw line-source response of $7n$. The default prime-tagged model removes fourteen raw $S = 7$ endpoint-related births, giving $7n - 14$. The first-step death count remains zero.

A.18.5 D07–D08: channel spectrum and prime correction

For a nondegenerate rectangle, the raw births split into three disjoint integer neighbor channels. There are twelve $S = 5$ births, $4a + 4b - 16$ $S = 6$ births, and sixteen $S = 7$ births. Default prime sharpening changes the evaluated count from S to $\Phi(S)$, and $\Phi(7) = 7.35 > 7$ pushes raw $S = 7$ births outside the birth window. Raw $S = 5$ and $S = 6$ events remain accepted. Therefore the default prime-corrected count is raw minus sixteen:

$$B_\phi(a, b) = B_0(a, b) - 16 = 4a + 4b - 4.$$

A.18.6 D09–D10: irreversible work

The work formula is channel bookkeeping. Since the rectangle family has zero first-step deaths, all first-step events are births. Weighting each birth channel $S = k$ by $I_k = \log 2 + \log \binom{26}{k}$ gives the raw formula

$$W_0 = 12I_5 + (4a + 4b - 16)I_6 + 16I_7.$$

Default prime correction removes the $S = 7$ group and hence removes $16I_7$ exactly.

A.18.7 D11: arbitrary planar selector

The model-tagged selector follows from the same four local cell classes. In-plane defect cells have count $8 - q_D$. Normal-layer cells have count $9 - r_D$. Active cells outside the defect can die only within the closed one-step dilation of D , because outside that dilation the local pattern is exactly the intact stable sheet. Replacing the raw birth and survival windows with model-specific indicators χ_B^M and χ_S^M gives the full model-tagged formulas.

A.18.8 D12–D13: rectangle reconstruction boundary

The response count gives a sum. The area gives a product. For raw rectangles,

$$s = a + b = \frac{B_0 - 12}{4}, \quad A = ab.$$

For default prime rectangles,

$$s = a + b = \frac{B_\phi + 4}{4}, \quad A = ab.$$

In either case, a and b are the two roots of $z^2 - sz + A = 0$. Without area, the response fixes only s , so response alone cannot reconstruct arbitrary rectangle side lengths.

A.19 Full reference verifier script

The following single-file verifier is intended as the starting point for the release repository. It covers the base 26-neighbor engine, the prime-tagged variant, planar q_D/r_D selectors, rectangle predictions, line predictions, smoke tests, and CSV outputs.

```
# poole_defect_calculus_reference_verify.py
# Reference verifier for Poole Defect Calculus archival release.
# This is written for clarity over speed.

import math
import json
import random
from pathlib import Path
import numpy as np
import pandas as pd

OUT = Path("poole_defect_reference_verify_out")
OUT.mkdir(exist_ok=True, parents=True)

SHIFTS_26 = tuple(
```

```

(dx, dy, dz)
for dx in (-1, 0, 1)
for dy in (-1, 0, 1)
for dz in (-1, 0, 1)
if not (dx == 0 and dy == 0 and dz == 0)
)
SHIFTS_8 = tuple(
(dx, dy)
for dx in (-1, 0, 1)
for dy in (-1, 0, 1)
if not (dx == 0 and dy == 0)
)
SHIFTS_9 = tuple((dx, dy) for dx in (-1, 0, 1) for dy in (-1, 0, 1))

PHI_LUT = np.arange(27, dtype=float)
for p in (2, 3, 5, 7, 11, 13, 17, 19, 23):
    PHI_LUT += 0.35 * np.exp(-(np.arange(27) - p) ** 2) / 0.01)

I_LUT = np.array([math.log(2) + math.log(math.comb(26, k)) for k in range(27)])

def roll3(A, dx, dy, dz):
    return np.roll(np.roll(np.roll(A, dx, axis=0), dy, axis=1), dz, axis=2)

def roll2(A, dx, dy):
    return np.roll(np.roll(A, dx, axis=0), dy, axis=1)

def n26(A):
    Ai = A.astype(np.int16, copy=False)
    S = np.zeros_like(Ai, dtype=np.int16)
    for dx, dy, dz in SHIFTS_26:
        S += roll3(Ai, dx, dy, dz)
    return S

def poole_step(A, mode="raw"):
    S = n26(A)
    if mode == "raw":
        X = S.astype(float)
    elif mode == "prime":
        X = PHI_LUT[S]
    else:
        raise ValueError("mode must be raw or prime")
    birth = (~A) & (X >= 5) & (X <= 7)
    survive = A & (X >= 5) & (X <= 9)
    death = A & (~survive)
    A1 = birth | survive
    event = birth | death
    row = {
        "mode": mode,
        "active0": int(A.sum()),
        "births": int(birth.sum()),
        "deaths": int(death.sum()),
        "active1": int(A1.sum()),
        "events": int(event.sum()),
        "W_step": float((event * I_LUT[S]).sum()),
    }

```

```

    }
    for k in range(27):
        row[f"B{k}"] = int((birth & (S == k)).sum())
        row[f"D{k}"] = int((death & (S == k)).sum())
    return A1, row

def full_sheet(L, z=None):
    if z is None:
        z = L // 2
    A = np.zeros((L, L, L), dtype=bool)
    A[:, :, z] = True
    return A

def sheet_with_mask(L, D, z=None):
    A = full_sheet(L, z=z)
    if z is None:
        z = L // 2
    for x, y in D:
        A[x % L, y % L, z] = False
    return A

def rect_mask(L, a, b, x0=None, y0=None):
    if x0 is None:
        x0 = (L - a) // 2
    if y0 is None:
        y0 = (L - b) // 2
    return {(x, y) for x in range(x0, x0 + a) for y in range(y0, y0 + b)}

def line_mask(L, n, orientation="x", x0=None, y0=None):
    if x0 is None:
        x0 = L // 2
    if y0 is None:
        y0 = L // 2
    if orientation == "x":
        return {(x0 + i) % L, y0) for i in range(n)}
    if orientation == "y":
        return {(x0, (y0 + i) % L) for i in range(n)}
    raise ValueError("orientation must be x or y")

def q_r_counts_from_mask(L, D):
    M = np.zeros((L, L), dtype=bool)
    for x, y in D:
        M[x % L, y % L] = True
    Mi = M.astype(np.int16, copy=False)
    q = np.zeros_like(Mi, dtype=np.int16)
    r = np.zeros_like(Mi, dtype=np.int16)
    for dx, dy in SHIFTS_8:
        q += roll2(Mi, dx, dy)
    for dx, dy in SHIFTS_9:
        r += roll2(Mi, dx, dy)
    return M, q, r

def selector_predict(L, D, mode="raw"):
    M, q, r = q_r_counts_from_mask(L, D)

```

```

    dil = r > 0
    if mode == "raw":
        birth_channels = {5, 6, 7}
    elif mode == "prime":
        birth_channels = {5, 6}
    else:
        raise ValueError("mode_must_be_raw_or_prime")
    survive_channels = {5, 6, 7, 8, 9}
    inplane_S = 8 - q
    normal_S = 9 - r
    inplane_births = M & np.isin(inplane_S, list(birth_channels))
    normal_births = dil & np.isin(normal_S, list(birth_channels))
    deaths = (~M) & dil & (~np.isin(inplane_S, list(survive_channels)))
    return {
        "selector_births": int(inplane_births.sum() + 2 * normal_births.sum()),
        "selector_deaths": int(deaths.sum()),
    }

def predict_rect(a, b, mode="raw"):
    if a == 1 and b == 1:
        raw = 0
        prime = 0
    elif a == 1:
        raw = 7 * b
        prime = 7 * b - 14 if b >= 2 else 0
    elif b == 1:
        raw = 7 * a
        prime = 7 * a - 14 if a >= 2 else 0
    else:
        raw = 4 * a + 4 * b + 12
        prime = 4 * a + 4 * b - 4
    return raw if mode == "raw" else prime

def verify_stable_sheet(L=32):
    A = full_sheet(L)
    _, m = poole_step(A, mode="raw")
    return {"test": "stable_sheet", "L": L, "births": m["births"], "deaths": m["deaths"],
        "match": m["births"] == 0 and m["deaths"] == 0}

def verify_rectangles(L=72, amin=2, amax=24, mode="raw"):
    rows = []
    for a in range(amin, amax + 1):
        for b in range(amin, amax + 1):
            D = rect_mask(L, a, b)
            A = sheet_with_mask(L, D)
            _, m = poole_step(A, mode=mode)
            sel = selector_predict(L, D, mode=mode)
            pred = predict_rect(a, b, mode=mode)
            rows.append({
                "family": "rectangle",
                "mode": mode,
                "a": a,
                "b": b,
                "observed_births": m["births"],
            })

```

```

        "predicted_births": pred,
        "selector_births": sel["selector_births"],
        "observed_deaths": m["deaths"],
        "selector_deaths": sel["selector_deaths"],
        "match": m["births"] == pred and m["deaths"] == 0 and sel["
selector_births"] == m["births"] and sel["selector_deaths"] == m["deaths"],
    })
    return rows

def verify_lines(L=72, nmax=40, mode="raw"):
    rows = []
    for orientation in ("x", "y"):
        for n in range(1, nmax + 1):
            D = line_mask(L, n, orientation=orientation)
            A = sheet_with_mask(L, D)
            _, m = poole_step(A, mode=mode)
            sel = selector_predict(L, D, mode=mode)
            pred = predict_rect(1, n, mode=mode)
            rows.append({
                "family": "line",
                "mode": mode,
                "orientation": orientation,
                "n": n,
                "observed_births": m["births"],
                "predicted_births": pred,
                "selector_births": sel["selector_births"],
                "observed_deaths": m["deaths"],
                "selector_deaths": sel["selector_deaths"],
                "match": m["births"] == pred and m["deaths"] == 0 and sel["
selector_births"] == m["births"],
            })
    return rows

def random_polyomino(L, size, seed):
    rng = random.Random(seed)
    cells = {(L // 2, L // 2)}
    frontier = list(cells)
    while len(cells) < size:
        x, y = rng.choice(frontier)
        dx, dy = rng.choice([(1,0), (-1,0), (0,1), (0,-1)])
        c = ((x + dx) % L, (y + dy) % L)
        cells.add(c)
        frontier.append(c)
    return cells

def verify_random_polyominoes(L=64, sizes=(3,4,5,6,8,10), trials=10, mode="raw"):
    rows = []
    for size in sizes:
        for t in range(trials):
            D = random_polyomino(L, size, seed=1000 + 31*size + t)
            A = sheet_with_mask(L, D)
            _, m = poole_step(A, mode=mode)
            sel = selector_predict(L, D, mode=mode)
            rows.append({

```

```

        "family": "polyomino",
        "mode": mode,
        "size": size,
        "trial": t,
        "observed_births": m["births"],
        "selector_births": sel["selector_births"],
        "observed_deaths": m["deaths"],
        "selector_deaths": sel["selector_deaths"],
        "match": m["births"] == sel["selector_births"] and m["deaths"] == sel["
selector_deaths"],
    })
    return rows

def main():
    all_rows = []
    all_rows.append(verify_stable_sheet())
    for mode in ("raw", "prime"):
        all_rows.extend(verify_rectangles(mode=mode))
        all_rows.extend(verify_lines(mode=mode))
        all_rows.extend(verify_random_polyominoes(mode=mode))
    df = pd.DataFrame(all_rows)
    df.to_csv(OUT / "verification_rows.csv", index=False)
    summary = {
        "rows": int(len(df)),
        "all_match": bool(df["match"].all()),
        "failures": int((~df["match"]).sum()),
    }
    (OUT / "summary.json").write_text(json.dumps(summary, indent=2), encoding="utf-8")
    print(json.dumps(summary, indent=2))

if __name__ == "__main__":
    main()

```

A.20 Empirical atlas reference harness

The v10 atlas needs a separate harness because it measures multi-step behavior rather than theorem-level first-step laws.

```

# poole_defect_atlas_v10_reference.py
# Empirical atlas reference harness. This is not a theorem prover.

import json
import math
import time
from pathlib import Path
import numpy as np
import pandas as pd

# Reuse n26, poole_step, sheet_with_mask, rect_mask, line_mask, random_polyomino
# from the reference verifier or copy the definitions into this file.

PRESET = "standard"
if PRESET == "quick":

```

```

    N = 48
    PLANAR_T = 50
    VACUUM_ANNEAL = 60
    VOID_T = 40
    VOID_TRIALS = 2
elif PRESET == "standard":
    N = 64
    PLANAR_T = 80
    VACUUM_ANNEAL = 120
    VOID_T = 70
    VOID_TRIALS = 3
else:
    N = 96
    PLANAR_T = 140
    VACUUM_ANNEAL = 240
    VOID_T = 120
    VOID_TRIALS = 6

OUT = Path("poole_defect_atlas_v10_reference_out")
OUT.mkdir(parents=True, exist_ok=True)

def run_steps(A, T, mode="prime", label="case", regime="run"):
    rows = []
    W_cum = 0.0
    for t in range(T):
        A, m = poole_step(A, mode=mode)
        W_cum += m["W_step"]
        m.update({
            "t": t,
            "mode": mode,
            "label": label,
            "regime": regime,
            "W_cum": W_cum,
            "W_cum_density": W_cum / A.size,
            "event_density": m["events"] / A.size,
            "W_step_density": m["W_step"] / A.size,
        })
        rows.append(m)
    return A, pd.DataFrame(rows)

def planar_cases(N):
    cases = []
    cases.append(("rect_4x4", rect_mask(N, 4, 4)))
    cases.append(("rect_6x10", rect_mask(N, 6, 10)))
    cases.append(("line_12", line_mask(N, 12, orientation="x")))
    cases.append(("line_20", line_mask(N, 20, orientation="y")))
    for size in (6, 10, 16, 24, 32):
        cases.append((f"poly_{size}", random_polyomino(N, size, seed=20260619 + size)))
    return cases

def run_planar_atlas():
    frames = []
    for label, D in planar_cases(N):
        A0 = sheet_with_mask(N, D)

```



```

        for mode in ("raw", "prime"):
            _, hist = run_steps(A0.copy(), PLANAR_T, mode=mode, label=label, regime="
planar")
            frames.append(hist)
        df = pd.concat(frames, ignore_index=True)
        df.to_csv(OUT / "planar_multistep_history.csv", index=False)
        return df

def make_random_vacuum(N, p=0.18, seed=0):
    rng = np.random.default_rng(seed)
    return rng.random((N, N, N)) < p

def anneal_vacuum(N, T, seed=0):
    A = make_random_vacuum(N, p=0.18, seed=seed)
    A, hist = run_steps(A, T, mode="prime", label="anneal", regime="anneal")
    return A, hist

def void_mask(N, shape, size):
    cx = cy = cz = N // 2
    X, Y, Z = np.ogrid[:N, :N, :N]
    if shape == "cube":
        r = size
        return (abs(X-cx) <= r) & (abs(Y-cy) <= r) & (abs(Z-cz) <= r)
    if shape == "sphere":
        r = size
        return ((X-cx)**2 + (Y-cy)**2 + (Z-cz)**2) <= r*r
    if shape == "ellipsoid":
        rx, ry, rz = size, max(1, size-1), size+1
        return ((X-cx)/rx)**2 + ((Y-cy)/ry)**2 + ((Z-cz)/rz)**2 <= 1
    raise ValueError("shape must be cube, sphere, or ellipsoid")

def run_void_atlas():
    summary_rows = []
    hist_frames = []
    for trial in range(VOID_TRIALS):
        baseline, anneal_hist = anneal_vacuum(N, VACUUM_ANNEAL, seed=8000+trial)
        baseline_final, base_hist = run_steps(baseline.copy(), VOID_T, mode="prime",
label="baseline", regime="void_control")
        for shape in ("cube", "sphere", "ellipsoid"):
            for size in (3, 4, 5, 6):
                mask = void_mask(N, shape, size)
                A = baseline.copy()
                removed_active = int((A & mask).sum())
                A[mask] = False
                label = f"{shape}_{size}_trial_{trial}"
                _, hist = run_steps(A, VOID_T, mode="prime", label=label, regime="void")
                merged = hist[["t", "W_step_density", "event_density"]].copy()
                merged["baseline_W_density"] = base_hist["W_step_density"].values
                merged["excess_W_density"] = merged["W_step_density"] - merged["
baseline_W_density"]
                merged["shape"] = shape
                merged["size"] = size
                merged["trial"] = trial
                hist_frames.append(merged)

```

```

        summary_rows.append({
            "shape": shape,
            "size": size,
            "trial": trial,
            "volume_proxy": int(mask.sum()),
            "removed_active": removed_active,
            "peak_excess_W_density": float(merged["excess_W_density"].max()),
            "integrated_excess_W_density": float(merged["excess_W_density"].sum())
        ),
    })
void_summary = pd.DataFrame(summary_rows)
void_hist = pd.concat(hist_frames, ignore_index=True)
void_summary.to_csv(OUT / "void3d_summary.csv", index=False)
void_hist.to_csv(OUT / "void3d_history.csv", index=False)
return void_summary, void_hist

def fit_power_law(df, xcol, ycol):
    g = df[(df[xcol] > 0) & (df[ycol] > 0)].copy()
    X = np.log(g[xcol].values)
    Y = np.log(g[ycol].values)
    beta, logC = np.polyfit(X, Y, 1)
    pred = np.exp(logC) * g[xcol].values ** beta
    rmse = float(np.sqrt(np.mean((g[ycol].values - pred) ** 2)))
    return {"x": xcol, "y": ycol, "beta": float(beta), "C": float(np.exp(logC)), "rmse":
    rmse}

def summarize_atlas(planar_df, void_summary):
    tail = planar_df[planar_df["t"] >= planar_df["t"].max() - 9]
    tail_summary = tail.groupby("mode", as_index=False).agg(
        mean_tail_event_density=("event_density", "mean"),
        mean_tail_W_density=("W_step_density", "mean"),
        mean_final_W_cum_density=("W_cum_density", "mean"),
    )
    fit_rows = []
    for shape, g in void_summary.groupby("shape"):
        means = g.groupby("size", as_index=False).mean(numeric_only=True)
        for ycol in ("peak_excess_W_density", "integrated_excess_W_density"):
            fit = fit_power_law(means, "removed_active", ycol)
            fit["shape"] = shape
            fit_rows.append(fit)
    fits = pd.DataFrame(fit_rows)
    tail_summary.to_csv(OUT / "planar_tail_summary.csv", index=False)
    fits.to_csv(OUT / "void3d_scaling_fits.csv", index=False)
    summary = {
        "preset": PRESET,
        "N": N,
        "PLANAR_T": PLANAR_T,
        "VACUUM_ANNEAL": VACUUM_ANNEAL,
        "VOID_T": VOID_T,
        "VOID_TRIALS": VOID_TRIALS,
        "generated": time.ctime(),
        "claim_boundary": "empirical_atlas_not_closed_theorem",
    }
    (OUT / "summary.json").write_text(json.dumps(summary, indent=2), encoding="utf-8")

```

```

    return tail_summary, fits

def main():
    planar_df = run_planar_atlas()
    void_summary, void_hist = run_void_atlas()
    tail_summary, fits = summarize_atlas(planar_df, void_summary)
    print(tail_summary)
    print(fits)

# main() requires the reference verifier functions to be present.

```

A.21 CSV schema appendix

File	Required columns	Meaning
verification_rows.csv	family, mode, observed_births, predicted_births, observed_deaths, match	Theorem-level first-step verification rows.
planar_multistep_history.csv	family, mode, t, active, births, deaths, events, event_density, W_step_density, W_cum_density	Empirical planar relaxation histories.
void3d_summary.csv	shape, size, trial, volume_proxy, moved_active, peak_excess_W_density, integrated_excess_W_density	3D void response summary per case.
void3d_history.csv	t, shape, size, trial, W_step_density, baseline_W_density, excess_W_density	3D void response curves.
void3d_scaling_fits.csv	shape, x, y, beta, C, rmse	Candidate scaling fits for empirical law discovery.
summary.json	preset, N, step counts, cases, generated, claim_boundary	Machine-readable run record.

A.22 Claim boundaries and overclaim controls

A robust paper should explicitly avoid these claims unless future proof or calibration is supplied:

1. Do not say that the v10 atlas proves a closed multi-step relaxation law.
2. Do not say that the v10 void atlas proves a closed three-dimensional void law.

3. Do not say that the continuum expansion proves physical spacetime curvature.
4. Do not say that rectangle reconstruction reconstructs arbitrary masks.
5. Do not say that static first-step Q/P capture is the same as dynamic multi-step response capture.
6. Do not say that prime sharpening is a theorem about the raw rule; it is a model-tagged variant.
7. Do not say that audits replace proofs for exact laws; audits verify implementations and tested domains.

The strength of the Poole Defect Calculus is that its exact layer is already interesting without those overextensions.

Evidence and Verification Appendices

A.23 Evidence role

This appendix records regenerated evidence artifacts for the manuscript. It is not a substitute for the proofs in the main text. Its purpose is narrower: provide reproducible finite-lattice audits for the theorem-level first-step laws, record those audits as CSV/JSON/PNG files, and define a concrete reference verifier that other readers can run.

The The evidence appendix uses the raw Poole Manifold rule

$$x_{t+1}(i) = \mathbf{1}\{5 \leq N_i(x_t) \leq 7 + 2x_t(i)\}$$

on finite periodic cubic lattices. The default prime-corrected quantities in this appendix are model-tagged post-selector quantities: they reject raw birth-channel candidates with neighbor count seven. They are included because the theorem ledger already separates PM0 from PM_phi.

Evidence boundary. The rectangle, line-hole, channel, selector, and reconstruction audits are checks of the closed first-step theorem layer. The multi-step relaxation CSV is an empirical reference trace only. It is not a closed law and should not be described as one in this paper.

A.24 Pass/fail summary

All closed-layer checks passed in this regenerated evidence run. The run includes 841 rectangle cases, 80 line-hole cases, 720 random planar selector cases, and 1225 Q/P-style rectangle inversion cases. The empirical timeseries rows are included as reference traces and are excluded from theorem-level pass/fail language.

A.25 Rectangle, square, channel, prime, and overcompensation audit

The rectangle audit regenerates the closed first-step laws over $2 \leq a, b \leq 30$. This gives 841 cases. For every case the script constructs a complete stable active sheet, removes an $a \times b$ rectangle, applies the raw PM0 update once, and records births, deaths, active counts, and birth-channel counts.

Table 11: Regenerated evidence summary.

metric	value
rectangle_cases	841
rectangle_raw_matches	841
rectangle_channel_matches	841
rectangle_prime_matches	841
rectangle_overcomp_matches	841
line_cases	80
line_raw_matches	80
line_prime_matches	80
selector_random_cases	720
selector_random_matches	720
selector_random_nonlocal_events_total	0
qp_inversion_cases	1225
qp_raw_inversion_matches	1225
qp_prime_inversion_matches	1225
empirical_timeseries_rows	78
all_closed_layer_checks_passed	True

The checked laws are

$$B_0(a, b) = 4a + 4b + 12, \quad D_0(a, b) = 0,$$

$$(N_5, N_6, N_7) = (12, 4a + 4b - 16, 16),$$

$$B_\phi(a, b) = N_5 + N_6 = 4a + 4b - 4,$$

and the first-step overcompensation criterion

$$|A_1| > |A_{\text{sheet}}| \iff (a - 4)(b - 4) < 28.$$

Table 12: Representative rows from the rectangle/channel/prime audit.

a	b	B0	B0 pred	D	N5	N6	N7	Bphi	Bphi pred	over	ok
2	2	28	28	0	12	0	16	12	12	True	True
2	3	32	32	0	12	4	16	16	16	True	True
2	10	60	60	0	12	32	16	44	44	True	True
2	30	140	140	0	12	112	16	124	124	True	True
10	10	92	92	0	12	64	16	76	76	False	True
10	20	132	132	0	12	104	16	116	116	False	True
10	30	172	172	0	12	144	16	156	156	False	True
30	30	252	252	0	12	224	16	236	236	False	True

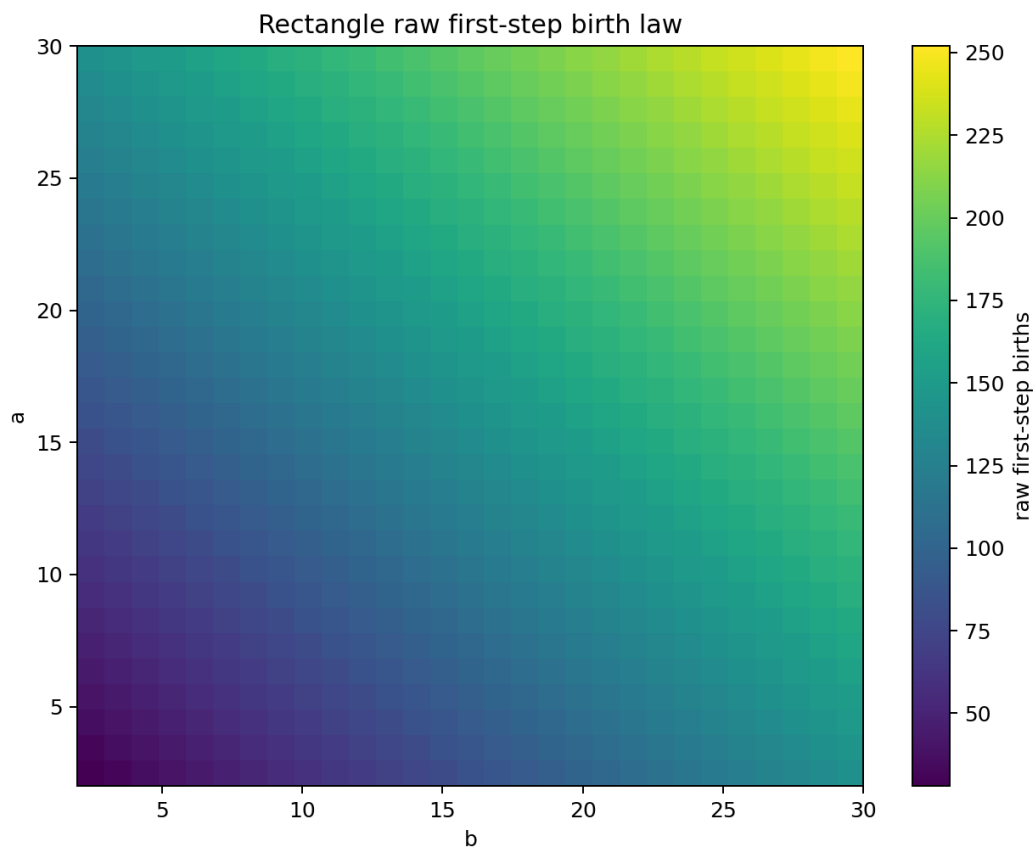


Figure 1: Regenerated rectangle raw first-step birth surface.

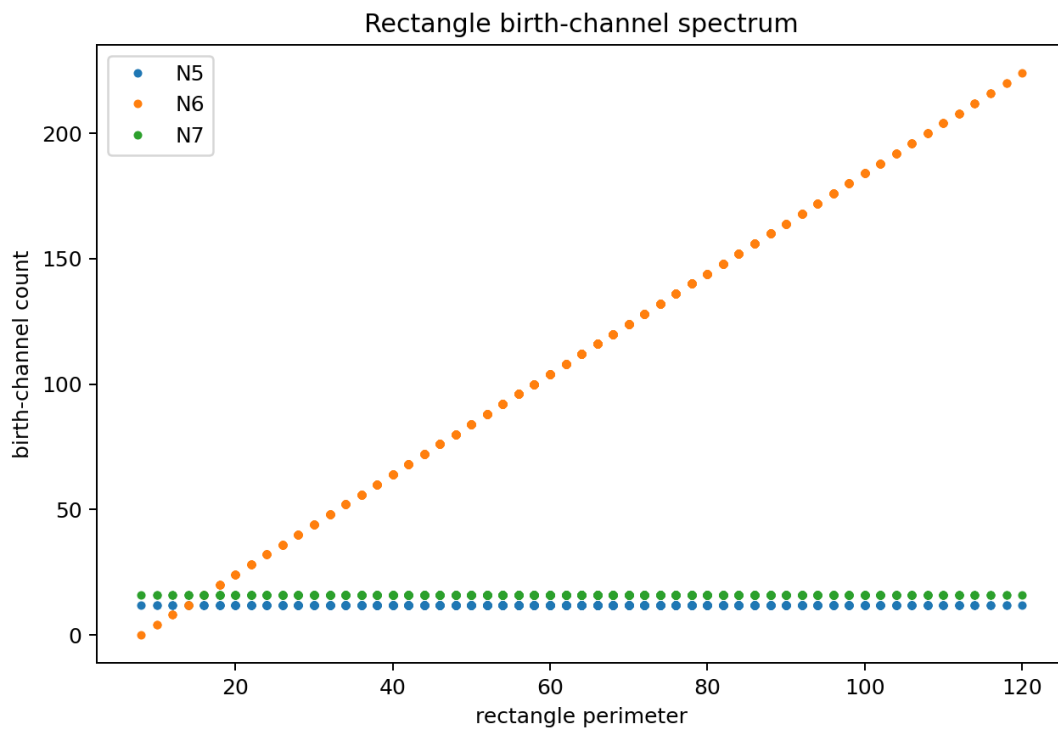


Figure 2: Regenerated rectangle birth-channel spectrum. The constant twelve N_5 and sixteen N_7 edge/corner channels separate from the perimeter-dependent N_6 channel.

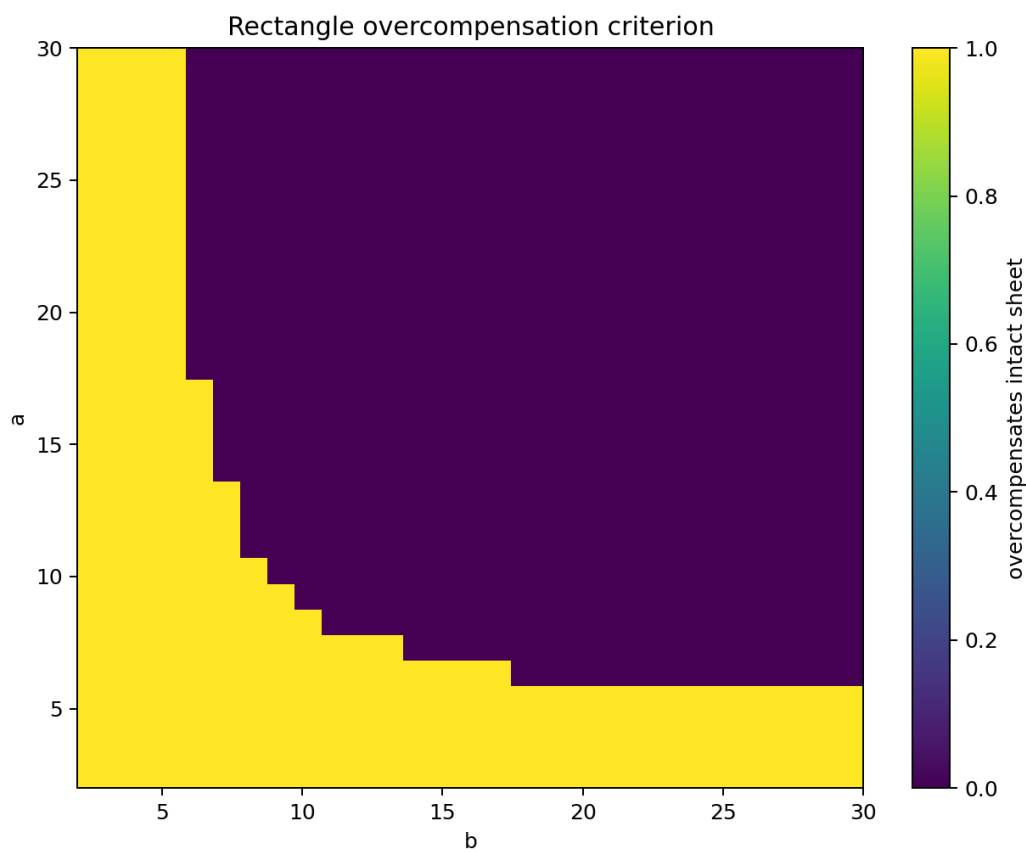


Figure 3: Regenerated first-step overcompensation map for rectangles over $2 \leq a, b \leq 30$.

A.26 Line-hole edge audit

The one-cell-wide family is a degenerate boundary case and is not folded into the $a, b \geq 2$ rectangle law. The regenerated audit covers $1 \leq n \leq 80$. It checks

$$B_0(1) = B_\phi(1) = 0,$$

and for $n \geq 2$,

$$B_0(1, n) = 7n, \quad B_\phi(1, n) = 7n - 14.$$

Table 13: Representative rows from the line-hole edge audit.

n	B0	B0 pred	D	N5	N6	N7	Bphi	Bphi pred	ok
1	0	0	0	0	0	0	0	0	True
2	14	14	0	0	0	14	0	0	True
3	21	21	0	0	7	14	7	7	True
10	70	70	0	0	56	14	56	56	True
40	280	280	0	0	266	14	266	266	True
80	560	560	0	0	546	14	546	546	True

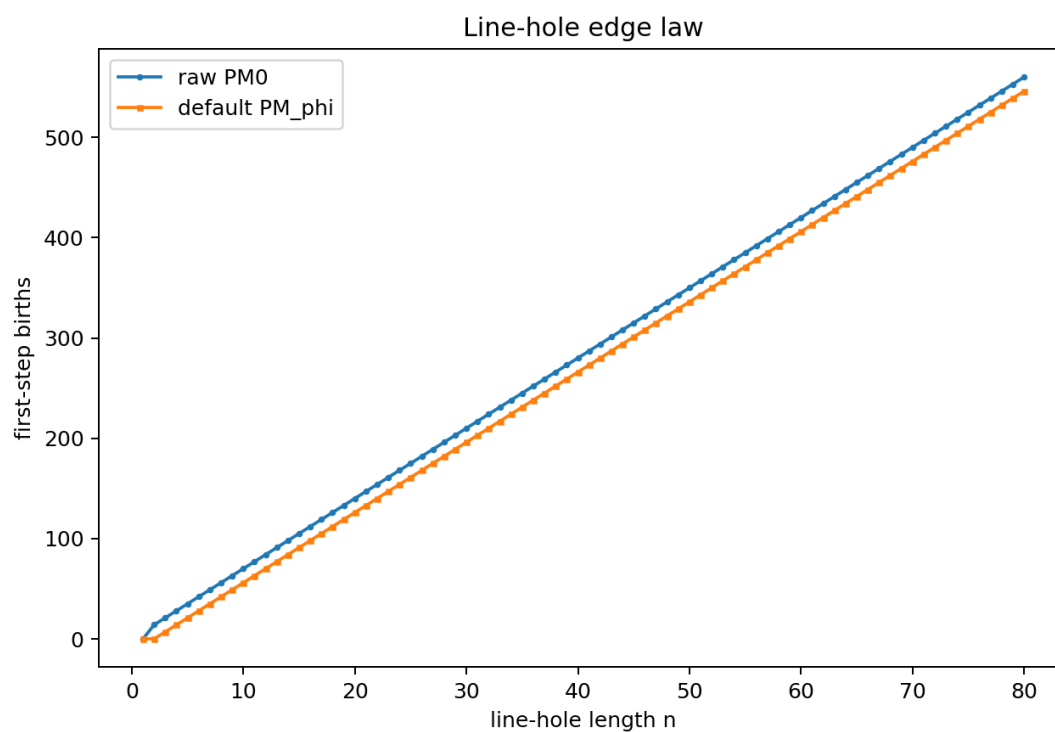


Figure 4: Regenerated raw and default prime-corrected line-hole response.

A.27 Arbitrary planar selector audit

The arbitrary planar selector audit is the most important implementation check because it tests the exact q_D/r_D selector on non-rectangular masks. Each row generates a random finite planar defect mask D , removes it from a stable sheet, runs one raw update, and compares all actual first-step events against the predicted sets:

in-plane defect births: $u \in D$, $1 \leq q_D(u) \leq 3$,

survival failures/deaths: $u \notin D$, $q_D(u) \geq 4$,

adjacent normal births: $z = \pm 1$, $2 \leq r_D(u) \leq 4$.

The audit also records nonlocal first-step events. The total nonlocal event count is zero in the regenerated run.

Table 14: First ten rows from the random planar selector audit.

case	box	removed	actual E	pred E	plane B	adj B	nonlocal	ok
1	7x12	27	185	185	24	156	0	True
2	6x4	1	0	0	0	0	0	True
3	5x11	7	30	30	4	26	0	True
4	12x7	44	150	150	17	118	0	True
5	8x9	30	185	185	27	144	0	True
6	8x10	43	151	151	22	110	0	True
7	12x4	16	69	69	9	54	0	True
8	10x10	22	154	154	21	130	0	True
9	12x4	7	31	31	5	26	0	True
10	6x5	16	83	83	14	62	0	True

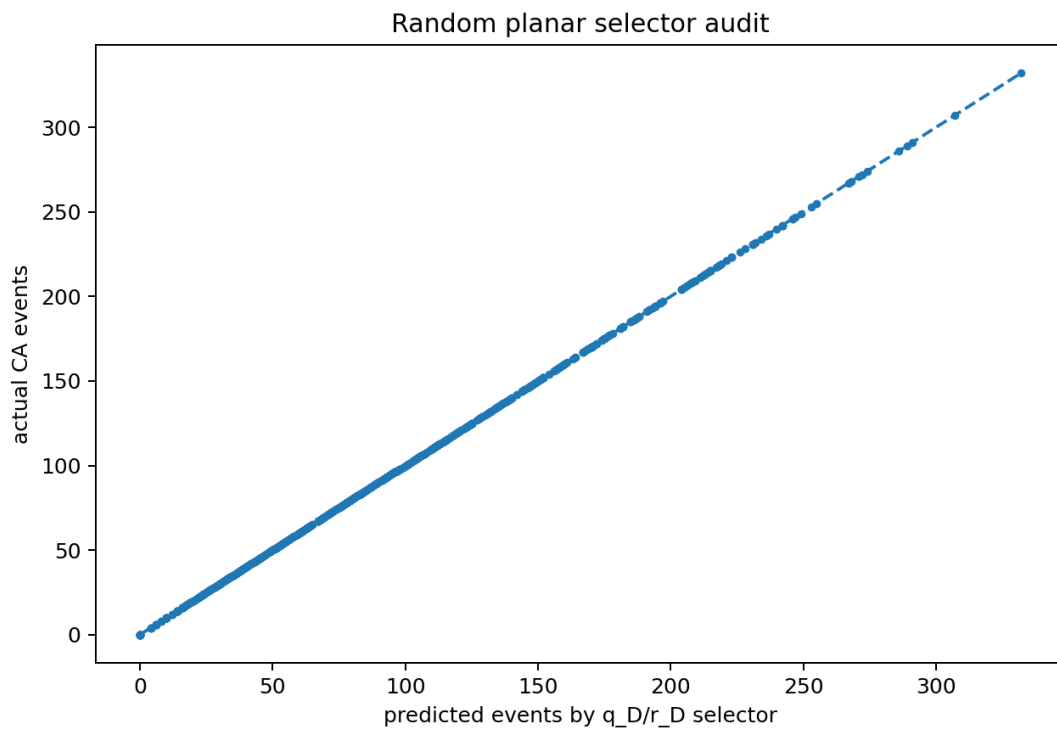


Figure 5: Actual event counts versus q_D/r_D -predicted event counts for 720 random finite planar masks.

A.28 Q/P-style rectangle reconstruction audit

This audit is included only as a boundary cross-reference to PooleQ/P-style capture. It is not the full Q/P paper. The claim tested here is the limited static rectangle reconstruction fact: area plus first-step response determines the unordered pair $\{a, b\}$ for a rectangle.

For raw PM0,

$$A = ab, \quad S = a + b = \frac{B_0 - 12}{4}.$$

For default PM_phi,

$$A = ab, \quad S = a + b = \frac{B_\phi + 4}{4}.$$

Then a and b are recovered as the integer roots of

$$t^2 - St + A = 0.$$

The regenerated audit covers $2 \leq a, b \leq 36$, giving 1225 cases.

Table 15: Representative rows from the rectangle reconstruction audit.

a	b	A	B0	Bphi	raw rec	phi rec	raw ok	phi ok
2	2	4	28	12	2x2	2x2	True	True
2	7	14	48	32	2x7	2x7	True	True
2	36	72	164	148	2x36	2x36	True	True
10	20	200	132	116	10x20	10x20	True	True
10	36	360	196	180	10x36	10x36	True	True
36	36	1296	300	284	36x36	36x36	True	True

A.29 Empirical multi-step trace table

The file `empirical_planar_relaxation_timeseries.csv` records short raw PM0 traces for selected planar rectangle holes. These rows are useful for later law discovery and for aligning the v10 atlas, but they are not theorem-level claims. They should remain in the empirical/open-problem lane until a closed multi-step law is proved.

Table 16: Representative empirical relaxation reference rows. These rows are not theorem-level evidence.

case	t	active	births	deaths	delta
2x2	0	6396	0	0	0
2x2	1	6424	28	0	28
2x2	2	6384	16	56	-12
2x2	3	6440	72	16	44
2x2	6	6608	248	240	212
2x2	12	6896	712	928	500
4x12	0	6352	0	0	0
4x12	1	6428	76	0	76
4x12	2	6348	40	120	-4
4x12	3	6512	188	24	160
4x12	6	6628	444	660	276
4x12	12	7672	1204	1436	1320
10x20	0	53624	0	0	0
10x20	1	53756	132	0	132
10x20	2	53676	96	176	52
10x20	3	54064	428	40	440
10x20	6	54372	900	1248	748
10x20	12	55296	1492	2520	1672

A.30 Evidence file schema

The full CSV and JSON artifacts are included in the companion evidence archive under **evidence/data** and **evidence/reports**. The manuscript prints only representative rows; the companion evidence archive contains the complete tables.

Evidence file schema.

`rectangle_square_channel_W_full3d_audit.csv`

rows: 841; **purpose:** Rectangles $2 \leq a, b \leq 30$; raw law, channel spectrum, prime correction, overcompensation.; **linked:** D03,D05,D05a,D07,D08,D09-D10.

`line_hole_edge_case_law_audit.csv`

rows: 80; **purpose:** Line holes $n=1..80$; raw and prime endpoint laws.; **linked:** D06.

`arbitrary_planar_selector_random_audit.csv`

rows: 720; **purpose:** Random finite planar masks; actual first-step events against qD/rD selector.; **linked:** D04,D11.

`qp_rectangle_inversion_audit.csv`

rows: 1225; **purpose:** Area plus raw/prime response recovers unordered rectangle dimensions.; **linked:** D12-D13.

`rectangle_overcompensation_criterion_audit.csv`

rows: 841; **purpose:** First-step active-count overcompensation criterion.; **linked:** D05a.

`empirical_planar_relaxation_timeseries.csv`

rows: 78; **purpose:** Multi-step traces for selected planar rectangles; empirical reference trace only.; **linked:** v10 empirical/open.

`evidence_summary.json / .csv`

rows: 16; **purpose:** Machine-readable pass/fail summary.; **linked:** release QA.

`sha256_manifest.csv`

rows: 11; **purpose:** Checksums for generated evidence files.; **linked:** reproducibility.

A.31 Evidence checksum manifest

The companion evidence archive includes a SHA-256 manifest for generated evidence artifacts. Short hashes are shown here for human inspection; the full hashes are stored in `sha256_manifest.csv`.

Short checksum list for generated evidence artifacts.

```
data/arbitrary_planar_selector_random_audit.csv
  sha256 short: 8a819cd61979f1dc; bytes: 42452.

data/empirical_planar_relaxation_timeseries.csv
  sha256 short: fe6604c3ce29df3e; bytes: 3576.

data/line_hole_edge_case_law_audit.csv
  sha256 short: 557540074695ec12; bytes: 3489.

data/qp_rectangle_inversion_audit.csv
  sha256 short: 90c7924c2dc60c1c; bytes: 46808.

data/rectangle_overcompensation_criterion_audit.csv
  sha256 short: 50ff97d6f7969302; bytes: 28370.

data/rectangle_square_channel_W_full3d_audit.csv
  sha256 short: 812208658f349ac9; bytes: 80074.

figures/line_hole_law.png
  sha256 short: df61c63dc5d169bc; bytes: 71642.

figures/planar_selector_actual_vs_predicted.png
  sha256 short: cf8616717cca5883; bytes: 58498.

figures/rectangle_birth_surface.png
  sha256 short: 4c57ca2c9e57dd1c; bytes: 46630.

figures/rectangle_channel_spectrum.png
  sha256 short: c41e9ff4adad3fb8; bytes: 56061.

figures/rectangle_overcompensation_map.png
  sha256 short: 7796427e6fc56fd6; bytes: 46414.
```

A.32 Evidence reference verifier script

The following script generated the evidence artifacts. It is included in the companion evidence archive at `evidence/scripts/poole_defect_verifier.py`. To reproduce the run from the archive root:

```
python evidence/scripts/poole_defect_verifier.py --out evidence --seed 20260620
```

Listing 1: Reference verifier.

```
#!/usr/bin/env python3
"""
PooleDefectCalculus-evidencegenerator.
License: Apache-2.0

This script regenerates finite-lattice audit artifacts for the theorem-level
first-step PooleDefectCalculus layer. It intentionally separates exact
first-step audits from empirical multi-step traces.

Model PM0:
x_{t+1}(i) = 1 { 5 ≤ N_i(x_t) ≤ 7 + 2*x_t(i) }
where N_i is the 26-neighbor Moore count on a finite periodic cubic lattice.

Default PM_phi_evidence_layer:
For rectangle/line first-step birth channel tables, raw PM0 births with
neighbor count 7 are rejected. This is a model-tagged post-selector,
not a statement about the raw PM0 rule.
"""
from __future__ import annotations

import argparse
import csv
import hashlib
import json
import math
import os
from dataclasses import asdict, dataclass
from pathlib import Path
from typing import Dict, Iterable, List, Sequence, Set, Tuple

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

Coord2 = Tuple[int, int]
Coord3 = Tuple[int, int, int]

OFFSETS_3D: List[Coord3] = [
    (dx, dy, dz)
    for dx in (-1, 0, 1)
    for dy in (-1, 0, 1)
    for dz in (-1, 0, 1)
    if not (dx == 0 and dy == 0 and dz == 0)
]
```

```

OPEN_OFFSETS_2D: List[Coord2] = [
    (dx, dy)
    for dx in (-1, 0, 1)
    for dy in (-1, 0, 1)
    if not (dx == 0 and dy == 0)
]
CLOSED_OFFSETS_2D: List[Coord2] = [
    (dx, dy)
    for dx in (-1, 0, 1)
    for dy in (-1, 0, 1)
]

```

```

@dataclass
class RectAudit:
    a: int
    b: int
    area: int
    perimeter: int
    active0: int
    births_raw_actual: int
    births_raw_predicted: int
    deaths_actual: int
    n5_actual: int
    n6_actual: int
    n7_actual: int
    n5_predicted: int
    n6_predicted: int
    n7_predicted: int
    births_prime_actual: int
    births_prime_predicted: int
    active1_raw_actual: int
    active1_raw_predicted: int
    overcomp_actual: bool
    overcomp_predicted: bool
    raw_match: bool
    channel_match: bool
    prime_match: bool
    overcomp_match: bool

```

```

@dataclass
class LineAudit:
    n: int
    area: int
    births_raw_actual: int
    births_raw_predicted: int
    deaths_actual: int
    n5_actual: int
    n6_actual: int
    n7_actual: int
    births_prime_actual: int
    births_prime_predicted: int
    raw_match: bool

```

```

    prime_match: bool

@dataclass
class SelectorAudit:
    case_id: int
    seed: int
    box_w: int
    box_h: int
    density: float
    removed_count: int
    actual_births: int
    predicted_births: int
    actual_deaths: int
    predicted_deaths: int
    actual_plane_births: int
    predicted_plane_births: int
    actual_adjacent_births: int
    predicted_adjacent_births: int
    actual_nonlocal_events: int
    match: bool

@dataclass
class InversionAudit:
    a: int
    b: int
    area: int
    raw_births: int
    prime_births: int
    recovered_raw_a: int
    recovered_raw_b: int
    recovered_prime_a: int
    recovered_prime_b: int
    raw_match: bool
    prime_match: bool

@dataclass
class TimeseriesRow:
    family: str
    parameter: str
    t: int
    active: int
    births: int
    deaths: int
    active_delta_from_t0: int

def ensure_dir(path: Path) -> None:
    path.mkdir(parents=True, exist_ok=True)

def sha256_file(path: Path) -> str:

```

```

    h = hashlib.sha256()
    with path.open("rb") as f:
        for chunk in iter(lambda: f.read(1024 * 1024), b''):
            h.update(chunk)
    return h.hexdigest()

def neighbor_counts(x: np.ndarray) -> np.ndarray:
    """26-neighbor_Moore_count_with_periodic_boundary_conditions."""
    counts = np.zeros_like(x, dtype=np.int16)
    for dx, dy, dz in OFFSETS_3D:
        counts += np.roll(np.roll(np.roll(x, dx, axis=0), dy, axis=1), dz, axis=2)
    return counts

def pm0_update(x: np.ndarray) -> Tuple[np.ndarray, np.ndarray]:
    """Return_(x_next, neighbor_count)."""
    n = neighbor_counts(x)
    x_next = ((n >= 5) & (n <= (7 + 2 * x))).astype(np.uint8)
    return x_next, n

def make_sheet_with_mask(mask: Set[Coord2], side: int | None = None, zsize: int = 7) ->
    Tuple[np.ndarray, int, int, int]:
    """Build_a_complete_active_sheet_with_a_finite_defect_mask_removed.

    The_returned_x_uses_axes_(x,y,z). The_sheet_is_at_z=z0.
    The_mask_is_shifted_to_the_center_of_the_sheet_to_avoid_wrap_effects.
    """
    if mask:
        max_x = max(p[0] for p in mask)
        max_y = max(p[1] for p in mask)
    else:
        max_x = max_y = 0
    if side is None:
        side = max(64, max_x + 12, max_y + 12)
    z0 = zsize // 2
    arr = np.zeros((side, side, zsize), dtype=np.uint8)
    arr[:, :, z0] = 1
    # Center local mask.
    ox = side // 2 - (max_x + 1) // 2
    oy = side // 2 - (max_y + 1) // 2
    for x, y in mask:
        arr[ox + x, oy + y, z0] = 0
    return arr, z0, ox, oy

def rectangle_mask(a: int, b: int) -> Set[Coord2]:
    return {(i, j) for i in range(a) for j in range(b)}

def first_step_events(arr: np.ndarray) -> Tuple[np.ndarray, np.ndarray, np.ndarray, np.
    ndarray]:
    nxt, counts = pm0_update(arr)

```

```

    births = (nxt == 1) & (arr == 0)
    deaths = (nxt == 0) & (arr == 1)
    return nxt, counts, births, deaths

def audit_rectangle(a: int, b: int) -> RectAudit:
    mask = rectangle_mask(a, b)
    arr, z0, _, _ = make_sheet_with_mask(mask)
    nxt, counts, births, deaths = first_step_events(arr)
    n5 = int(np.sum(births & (counts == 5)))
    n6 = int(np.sum(births & (counts == 6)))
    n7 = int(np.sum(births & (counts == 7)))
    births_raw_actual = int(np.sum(births))
    deaths_actual = int(np.sum(deaths))
    births_raw_pred = 4 * a + 4 * b + 12
    n5_pred = 12
    n6_pred = 4 * a + 4 * b - 16
    n7_pred = 16
    births_prime_actual = n5 + n6
    births_prime_pred = 4 * a + 4 * b - 4
    side = arr.shape[0]
    active0 = side * side - a * b
    active1_actual = int(np.sum(nxt))
    active1_pred = active0 + births_raw_pred # rectangle theorem has zero deaths
    over_actual = active1_actual > side * side
    over_pred = ((a - 4) * (b - 4)) < 28
    return RectAudit(
        a=a,
        b=b,
        area=a * b,
        perimeter=2 * (a + b),
        active0=active0,
        births_raw_actual=births_raw_actual,
        births_raw_predicted=births_raw_pred,
        deaths_actual=deaths_actual,
        n5_actual=n5,
        n6_actual=n6,
        n7_actual=n7,
        n5_predicted=n5_pred,
        n6_predicted=n6_pred,
        n7_predicted=n7_pred,
        births_prime_actual=births_prime_actual,
        births_prime_predicted=births_prime_pred,
        active1_raw_actual=active1_actual,
        active1_raw_predicted=active1_pred,
        overcomp_actual=over_actual,
        overcomp_predicted=over_pred,
        raw_match=(births_raw_actual == births_raw_pred and deaths_actual == 0 and
        active1_actual == active1_pred),
        channel_match=(n5 == n5_pred and n6 == n6_pred and n7 == n7_pred),
        prime_match=(births_prime_actual == births_prime_pred),
        overcomp_match=(over_actual == over_pred),
    )

```

```

def audit_line(n: int) -> LineAudit:
    mask = rectangle_mask(1, n)
    arr, z0, _, _ = make_sheet_with_mask(mask)
    nxt, counts, births, deaths = first_step_events(arr)
    n5 = int(np.sum(births & (counts == 5)))
    n6 = int(np.sum(births & (counts == 6)))
    n7 = int(np.sum(births & (counts == 7)))
    raw_actual = int(np.sum(births))
    deaths_actual = int(np.sum(deaths))
    if n == 1:
        raw_pred = 0
        prime_pred = 0
    else:
        raw_pred = 7 * n
        prime_pred = 7 * n - 14
    prime_actual = n5 + n6
    return LineAudit(
        n=n,
        area=n,
        births_raw_actual=raw_actual,
        births_raw_predicted=raw_pred,
        deaths_actual=deaths_actual,
        n5_actual=n5,
        n6_actual=n6,
        n7_actual=n7,
        births_prime_actual=prime_actual,
        births_prime_predicted=prime_pred,
        raw_match=(raw_actual == raw_pred and deaths_actual == 0),
        prime_match=(prime_actual == prime_pred),
    )

def q_open(mask: Set[Coord2], p: Coord2) -> int:
    x, y = p
    return sum((x + dx, y + dy) in mask for dx, dy in OPEN_OFFSETS_2D)

def r_closed(mask: Set[Coord2], p: Coord2) -> int:
    x, y = p
    return sum((x + dx, y + dy) in mask for dx, dy in CLOSED_OFFSETS_2D)

def selector_prediction(mask: Set[Coord2], z0: int, ox: int, oy: int) -> Tuple[Set[Coord3], Set[Coord3], Set[Coord3], Set[Coord3]]:
    """Predict first-step events by q_D/r_D selector.

    Returns plane_births, adjacent_births, plane_deaths, all_predicted_events.
    """
    candidates: Set[Coord2] = set(mask)
    for x, y in list(mask):
        for dx, dy in CLOSED_OFFSETS_2D:
            candidates.add((x + dx, y + dy))

```



```

plane_births: Set[Coord3] = set()
plane_deaths: Set[Coord3] = set()
adjacent_births: Set[Coord3] = set()

for p in candidates:
    q = q_open(mask, p)
    r = r_closed(mask, p)
    x, y = p
    X, Y = ox + x, oy + y
    if p in mask:
        if 1 <= q <= 3:
            plane_births.add((X, Y, z0))
        else:
            if q >= 4:
                plane_deaths.add((X, Y, z0))
    if 2 <= r <= 4:
        adjacent_births.add((X, Y, z0 - 1))
        adjacent_births.add((X, Y, z0 + 1))

all_events = plane_births | adjacent_births | plane_deaths
return plane_births, adjacent_births, plane_deaths, all_events

def audit_random_selector(case_id: int, seed: int, rng: np.random.Generator) ->
SelectorAudit:
    box_w = int(rng.integers(3, 13))
    box_h = int(rng.integers(3, 13))
    density = float(rng.uniform(0.07, 0.62))
    mask: Set[Coord2] = set()
    for i in range(box_w):
        for j in range(box_h):
            if rng.random() < density:
                mask.add((i, j))
    if not mask:
        mask.add((box_w // 2, box_h // 2))
    if len(mask) == box_w * box_h:
        # Leave at least one active interior reference point so dense masks test deaths
but not full-box degeneracy only.
        mask.remove((box_w // 2, box_h // 2))

    arr, z0, ox, oy = make_sheet_with_mask(mask, side=80)
    nxt, counts, births_arr, deaths_arr = first_step_events(arr)
    pred_plane_births, pred_adj_births, pred_deaths, pred_events = selector_prediction(
    mask, z0, ox, oy)

    actual_birth_positions = set(map(tuple, np.argwhere(births_arr)))
    actual_death_positions = set(map(tuple, np.argwhere(deaths_arr)))
    actual_events = actual_birth_positions | actual_death_positions
    predicted_birth_positions = pred_plane_births | pred_adj_births
    predicted_events = predicted_birth_positions | pred_deaths

    # Event locality: the exact selector predicts all events. Anything else is nonlocal
for first-step purposes.
    nonlocal_events = len(actual_events - predicted_events)

```

```

return SelectorAudit(
    case_id=case_id,
    seed=seed,
    box_w=box_w,
    box_h=box_h,
    density=round(density, 6),
    removed_count=len(mask),
    actual_births=len(actual_birth_positions),
    predicted_births=len(predicted_birth_positions),
    actual_deaths=len(actual_death_positions),
    predicted_deaths=len(pred_deaths),
    actual_plane_births=sum(1 for p in actual_birth_positions if p[2] == z0),
    predicted_plane_births=len(pred_plane_births),
    actual_adjacent_births=sum(1 for p in actual_birth_positions if p[2] in {z0 - 1,
z0 + 1}),
    predicted_adjacent_births=len(pred_adj_births),
    actual_nonlocal_events=nonlocal_events,
    match=(actual_events == predicted_events),
)

def recover_rectangle_from_area_and_sum(area: int, s: int) -> Tuple[int, int] | None:
    disc = s * s - 4 * area
    if disc < 0:
        return None
    root = int(math.isqrt(disc))
    if root * root != disc:
        return None
    if (s - root) % 2 != 0 or (s + root) % 2 != 0:
        return None
    a = (s - root) // 2
    b = (s + root) // 2
    if a * b != area:
        return None
    return tuple(sorted((a, b))) # type: ignore[return-value]

def audit_inversion(a: int, b: int) -> InversionAudit:
    area = a * b
    raw_births = 4 * a + 4 * b + 12
    prime_births = 4 * a + 4 * b - 4
    s_raw = (raw_births - 12) // 4
    s_prime = (prime_births + 4) // 4
    rec_raw = recover_rectangle_from_area_and_sum(area, s_raw)
    rec_prime = recover_rectangle_from_area_and_sum(area, s_prime)
    target = tuple(sorted((a, b)))
    return InversionAudit(
        a=a,
        b=b,
        area=area,
        raw_births=raw_births,
        prime_births=prime_births,
        recovered_raw_a=-1 if rec_raw is None else rec_raw[0],
        recovered_raw_b=-1 if rec_raw is None else rec_raw[1],

```

```

        recovered_prime_a=-1 if rec_prime is None else rec_prime[0],
        recovered_prime_b=-1 if rec_prime is None else rec_prime[1],
        raw_match=(rec_raw == target),
        prime_match=(rec_prime == target),
    )

def run_timeseries_for_mask(mask: Set[Coord2], family: str, parameter: str, steps: int =
12) -> List[TimeseriesRow]:
    arr, z0, _, _ = make_sheet_with_mask(mask, side=max(80, len(mask) + 32), zsize=9)
    active0 = int(np.sum(arr))
    rows: List[TimeseriesRow] = []
    current = arr
    for t in range(steps + 1):
        if t == 0:
            births = 0
            deaths = 0
        else:
            # births/deaths are filled before assignment in loop below; this branch only
            exists for clarity.
            births = deaths = -1
        if t == 0:
            rows.append(TimeseriesRow(family, parameter, t, int(np.sum(current)), 0, 0,
int(np.sum(current)) - active0))
        if t < steps:
            nxt, counts = pm0_update(current)
            b = int(np.sum((nxt == 1) & (current == 0)))
            d = int(np.sum((nxt == 0) & (current == 1)))
            current = nxt
            rows.append(TimeseriesRow(family, parameter, t + 1, int(np.sum(current)), b,
d, int(np.sum(current)) - active0))
        # The loop currently appends t=1..steps inside each t and also iterates those t
        values, so remove duplicates.
        unique: Dict[int, TimeseriesRow] = {}
    for r in rows:
        unique[r.t] = r
    return [unique[t] for t in sorted(unique)]

def save_csv_dataclass(rows: Sequence[object], path: Path) -> None:
    if not rows:
        path.write_text("")
        return
    df = pd.DataFrame([asdict(r) for r in rows])
    df.to_csv(path, index=False)

def make_figures(out_dir: Path, rect_df: pd.DataFrame, line_df: pd.DataFrame, selector_df
: pd.DataFrame) -> None:
    ensure_dir(out_dir)

    pivot = rect_df.pivot(index="a", columns="b", values="births_raw_actual")
    plt.figure(figsize=(7.2, 5.8))
    plt.imshow(pivot.values, origin="lower", aspect="auto", extent=[pivot.columns.min(),

```

```

pivot.columns.max(), pivot.index.min(), pivot.index.max()])
plt.colorbar(label="raw_first-step_births")
plt.xlabel("b")
plt.ylabel("a")
plt.title("Rectangle_raw_first-step_birth_law")
plt.tight_layout()
plt.savefig(out_dir / "rectangle_birth_surface.png", dpi=180)
plt.close()

over = rect_df.pivot(index="a", columns="b", values="overcomp_actual")
plt.figure(figsize=(7.2, 5.8))
plt.imshow(over.values.astype(int), origin="lower", aspect="auto", extent=[over.
columns.min(), over.columns.max(), over.index.min(), over.index.max()])
plt.colorbar(label="overcompensates_intact_sheet")
plt.xlabel("b")
plt.ylabel("a")
plt.title("Rectangle_overcompensation_criterion")
plt.tight_layout()
plt.savefig(out_dir / "rectangle_overcompensation_map.png", dpi=180)
plt.close()

plt.figure(figsize=(7.2, 5.0))
plt.plot(line_df["n"], line_df["births_raw_actual"], marker="o", markersize=2, label=
"raw_PM0")
plt.plot(line_df["n"], line_df["births_prime_actual"], marker="s", markersize=2,
label="default_PM_phi")
plt.xlabel("line-hole_length_n")
plt.ylabel("first-step_births")
plt.title("Line-hole_edge_law")
plt.legend()
plt.tight_layout()
plt.savefig(out_dir / "line_hole_law.png", dpi=180)
plt.close()

plt.figure(figsize=(7.2, 5.0))
plt.plot(rect_df["perimeter"], rect_df["n5_actual"], ".", label="N5")
plt.plot(rect_df["perimeter"], rect_df["n6_actual"], ".", label="N6")
plt.plot(rect_df["perimeter"], rect_df["n7_actual"], ".", label="N7")
plt.xlabel("rectangle_perimeter")
plt.ylabel("birth-channel_count")
plt.title("Rectangle_birth-channel_spectrum")
plt.legend()
plt.tight_layout()
plt.savefig(out_dir / "rectangle_channel_spectrum.png", dpi=180)
plt.close()

plt.figure(figsize=(7.2, 5.0))
# Show predicted vs actual event counts for random planar masks.
plt.scatter(selector_df["predicted_births"] + selector_df["predicted_deaths"],
selector_df["actual_births"] + selector_df["actual_deaths"], s=8)
maxv = int(max((selector_df["predicted_births"] + selector_df["predicted_deaths"]).
max(), (selector_df["actual_births"] + selector_df["actual_deaths"]).max()))
plt.plot([0, maxv], [0, maxv], linestyle="--")
plt.xlabel("predicted_events_by_q_D/r_D_selector")

```

```

plt.ylabel("actual_CAs_events")
plt.title("Random_planar_selector_audit")
plt.tight_layout()
plt.savefig(out_dir / "planar_selector_actual_vs_predicted.png", dpi=180)
plt.close()

def write_summary(out_dir: Path, files: List[Path], summary: Dict[str, object]) -> None:
    ensure_dir(out_dir)
    (out_dir / "evidence_summary.json").write_text(json.dumps(summary, indent=2,
sort_keys=True))
    with (out_dir / "evidence_summary.csv").open("w", newline="") as f:
        writer = csv.writer(f)
        writer.writerow(["metric", "value"])
        for k, v in summary.items():
            writer.writerow([k, v])
    with (out_dir / "sha256_manifest.csv").open("w", newline="") as f:
        writer = csv.writer(f)
        writer.writerow(["relative_path", "sha256", "bytes"])
        root = out_dir.parent
        for path in sorted(files):
            if path.is_file():
                writer.writerow([str(path.relative_to(root)), sha256_file(path), path.
stat().st_size])

def main() -> None:
    parser = argparse.ArgumentParser()
    parser.add_argument("--out", type=Path, default=Path("evidence"), help="Output_
evidence_directory")
    parser.add_argument("--seed", type=int, default=20260620)
    args = parser.parse_args()

    out = args.out
    data_dir = out / "data"
    fig_dir = out / "figures"
    report_dir = out / "reports"
    ensure_dir(data_dir)
    ensure_dir(fig_dir)
    ensure_dir(report_dir)

    # Rectangle range 2..30 gives 841 cases, matching the v9 stated rectangle audit size.
    rect_rows = [audit_rectangle(a, b) for a in range(2, 31) for b in range(2, 31)]
    save_csv_dataclass(rect_rows, data_dir / "rectangle_square_channel_W_full3d_audit.csv
")
    rect_df = pd.DataFrame([asdict(r) for r in rect_rows])

    # Line audit n=1..80, matching the v9 stated line-hole audit size.
    line_rows = [audit_line(n) for n in range(1, 81)]
    save_csv_dataclass(line_rows, data_dir / "line_hole_edge_case_law_audit.csv")
    line_df = pd.DataFrame([asdict(r) for r in line_rows])

    # Random planar selector audit with 720 rows, matching the v9 stated selector audit
size.

```

```

rng = np.random.default_rng(args.seed)
selector_rows = [audit_random_selector(i + 1, args.seed, rng) for i in range(720)]
save_csv_dataclass(selector_rows, data_dir / "arbitrary_planar_selector_random_audit.csv")
selector_df = pd.DataFrame([asdict(r) for r in selector_rows])

# Q/P-style rectangle inversion audit 2..36 gives 1225 cases, matching v9 stated inversion size.
inversion_rows = [audit_inversion(a, b) for a in range(2, 37) for b in range(2, 37)]
save_csv_dataclass(inversion_rows, data_dir / "qp_rectangle_inversion_audit.csv")
inv_df = pd.DataFrame([asdict(r) for r in inversion_rows])

# Overcompensation criterion table is separated for easy manuscript import.
overcomp_df = rect_df[["a", "b", "area", "perimeter", "active1_raw_actual", "overcomp_actual", "overcomp_predicted", "overcomp_match"]].copy()
overcomp_df.to_csv(data_dir / "rectangle_overcompensation_criterion_audit.csv", index=False)

# Empirical timeseries reference traces. These are not theorem proofs.
ts_rows: List[TimeseriesRow] = []
for a, b in [(2, 2), (3, 3), (4, 4), (6, 6), (4, 12), (10, 20)]:
    ts_rows.extend(run_timeseries_for_mask(rectangle_mask(a, b), "planar_rectangle_hole", f"{a}x{b}", steps=12))
save_csv_dataclass(ts_rows, data_dir / "empirical_planar_relaxation_timeseries.csv")

make_figures(fig_dir, rect_df, line_df, selector_df)

summary = {
    "seed": args.seed,
    "rectangle_cases": len(rect_df),
    "rectangle_raw_matches": int(rect_df["raw_match"].sum()),
    "rectangle_channel_matches": int(rect_df["channel_match"].sum()),
    "rectangle_prime_matches": int(rect_df["prime_match"].sum()),
    "rectangle_overcomp_matches": int(rect_df["overcomp_match"].sum()),
    "line_cases": len(line_df),
    "line_raw_matches": int(line_df["raw_match"].sum()),
    "line_prime_matches": int(line_df["prime_match"].sum()),
    "selector_random_cases": len(selector_df),
    "selector_random_matches": int(selector_df["match"].sum()),
    "selector_random_nonlocal_events_total": int(selector_df["actual_nonlocal_events"].sum()),
    "qp_inversion_cases": len(inv_df),
    "qp_raw_inversion_matches": int(inv_df["raw_match"].sum()),
    "qp_prime_inversion_matches": int(inv_df["prime_match"].sum()),
    "empirical_timeseries_rows": len(ts_rows),
    "all_closed_layer_checks_passed": bool(
        rect_df["raw_match"].all()
        and rect_df["channel_match"].all()
        and rect_df["prime_match"].all()
        and rect_df["overcomp_match"].all()
        and line_df["raw_match"].all()
        and line_df["prime_match"].all()
        and selector_df["match"].all()
        and inv_df["raw_match"].all()
    )
}

```

```
        and inv_df["prime_match"].all()
    ),
}

generated = list(data_dir.glob("*.csv")) + list(fig_dir.glob("*.png"))
write_summary(report_dir, generated, summary)
print(json.dumps(summary, indent=2, sort_keys=True))

if __name__ == "__main__":
    main()
```

Claim-Boundary and Language Appendices

A.33 Role of claim-boundary controls

The claim-boundary appendix does not add a new defect law. Its purpose is editorial and mathematical: every public-facing claim is assigned a class, every class is paired with safe wording, and every bridge into Q/P, continuum language, OTG language, physical interpretation, or empirical atlas language is prevented from being mistaken for a theorem.

The hardened release rule is:

A result is a theorem only when it follows from the stated local update rule, finite combinatorics, or exact algebra under its named model assumptions. A simulation is a finite audit only over the stated range. A v10 curve or 3D void trend is an empirical atlas observation unless and until an analytic law is proved. Q/P and continuum interpretations are boundaries, not hidden assumptions of the Defect Calculus.

A.34 Hardened claim-map table

The following table is the paper claim map for this version. It is intentionally redundant with earlier theorem ledgers because this table is the final editorial guardrail before v1.1-clean packaging.

ID	Tag	Result	Approved public wording	Forbidden or unsafe wording
F01	Compact hys- teretic support law	EXACT	Keep as algebraic equivalence of B5-7/S5-9 and the compact hysteretic update.	Do not call the +2 term an external field.
F02	Translation and cubic symmetry	EXACT	State as discrete lattice sym- metry under Moore-neighborhood- preserving transformations.	Do not imply Lorentz or con- tinuum symmetry.
F03	Finite propaga- tion speed	EXACT	State as Chebyshev dependency ra- dius in the CA.	Do not equate it to physical speed of light.
F04	Eventual period- icity	EXACT	State finite-state recurrence only.	Do not imply fixed-point con- vergence.
F05	Non- monotonicity	EXACT COUNTEREX- AMPLE	Use as a counterexample against closure-operator language.	Do not claim order-preserving dynamics.
F06	Non- reversibility	EXACT COUNTEREX- AMPLE	Use as many-to-one CA behavior.	Do not claim a calibrated ther- modynamic arrow.
F07	Fixed-point active-degree bounds	EXACT	Necessary fixed-point constraints only.	Do not present as full fixed- point classification.

ID	Tag	Result	Approved public wording	Forbidden or unsafe wording
F08	Sheet memory / B7 plasticity split	BOUNDARY	Use as rule-space interpretation supported by channel facts.	Do not state uniqueness/minimality.
D01	Stable active plane	EXACT	Coordinate one-cell sheet is a fixed point under stated assumptions.	Do not translate into physical spacetime proof.
D02	Surface selector	EXACT	Local dimension selector: sheet survives while point/line/bulk fail as stable flat strata.	Do not classify every possible fixed point.
D03	Square-puncture law	EXACT + AUDIT	First-step law for square holes.	Do not imply closed multi-step relaxation.
D04	One-step planar q_D/r_D defect calculus	EXACT	Mathematical core: local selector for first update of planar defects.	Do not extend to arbitrary 3D voids.
D04a	Stable pinhole gas	EXACT	Sparse pore stability corollary under stated local footprint condition.	Do not apply to dense masks/cracks.
D05	Rectangle hole law	EXACT + AUDIT	$B0(a,b)=4a+4b+12$ for $a,b \geq 2$; zero deaths first step.	Rectangle prior required.
D05a	Rectangle over-compensation criterion	EXACT	First-step active-count comparison only.	Do not state long-time growth theorem.
D06	Line-hole edge law	EXACT + AUDIT	One-cell-wide line-hole law with end-point handling.	Do not generalize to every crack.
D07	Rectangle channel spectrum	EXACT + AUDIT	Channel split $N5=12$, $N6=4a+4b-16$, $N7=16$.	Rectangle setting only.
D08	Prime-corrected rectangle law	EXACT-IN-MODEL + AUDIT	Default PM_phi removes raw $S=7$ rectangle births.	Do not present as raw PM0 law.
D09	Raw rectangle work law	EXACT-IN-DEFINITION	Discrete event/work proxy using channel weights.	Not physical energy without calibration.
D10	Prime rectangle work law	EXACT-IN-DEFINITION	Model-tagged work proxy in PM_phi.	Not raw PM0 and not calibrated energy.
D11	Model-tagged arbitrary planar selector	EXACT + AUDIT	First-step planar selector with model-specific channel indicators.	Still first-step planar only.
D12	Raw rectangle reconstruction	EXACT + AUDIT	Under rectangle prior, area plus raw response recovers side lengths up to exchange.	Births alone are ambiguous; no arbitrary-mask reconstruction.
D13	Prime rectangle reconstruction	EXACT + AUDIT	Under rectangle prior, area plus prime response recovers side lengths up to exchange.	Not dynamic Q/P capture.
E01	Raw planar cascade	EMPIRICAL	Observed in v10 atlas; law-discovery evidence.	Do not state as closed multi-step law.
E02	Prime suppression of planar cascades	EMPIRICAL	Observed comparison in v10 atlas.	Do not state as universal damping theorem.
E03	Pulse-like relaxation	EMPIRICAL	Model-selection clue from curves.	Not a derived analytic law.
E04	3D void response scaling	EMPIRICAL	Observed scaling in tested families.	Do not state as closed 3D void law.
O01	Closed multi-step planar law	OPEN	Future theorem target.	Do not promote v10 curves to theorem.
O02	Closed non-planar 3D void law	OPEN	Future theorem target.	Current status empirical only.
O03	Rule-space uniqueness	OPEN	Future theorem target.	Do not claim uniqueness.
O04	Continuum calibration	BOUNDARY / OPEN	Requires scaling limit and physical calibration.	No physical law claim in this release.
O05	Q/P dynamic-response separation	BOUNDARY	Static rectangle transducer is separate from dynamic Q/P capture.	Do not merge the full Q/P paper into this release.

A.35 Hardening by claim-status lane

Layer	Hardened status statement	Result
theorem layer	D01-D13 and F01-F07 remain exact theorem/corollary/-counterexample claims under stated CA assumptions	hardened
finite audit layer	finite sweeps remain bounded verification claims over stated domains	hardened
empirical atlas layer	v10 multi-step/3D void results remain empirical observations and candidate-law evidence only	hardened
Q/P layer	rectangle reconstruction remains under rectangle prior and is not dynamic Q/P capture	hardened
continuum/OTG layer	continuum and physics interpretations remain bridge/open-problem statements	hardened
license layer	Apache-2.0 remains the archive license	hardened

A.36 Language audit and overclaim controls

The following automated language audit is not a substitute for human peer review, but it catches the main release-risk categories. Terms such as “gravity”, “continuum”, “quantum”, or “energy” are not banned; they must remain inside boundary sections unless a theorem, finite audit, or calibrated derivation is supplied.

Risk category	Count	Control action
physical law overclaim	1	Quarantine in boundary/open/interpretation lane if present; do not use as theorem language.
spacetime/gravity bridge terms	8	Quarantine in boundary/open/interpretation lane if present; do not use as theorem language.
quantum/QP bridge terms	29	Quarantine in boundary/open/interpretation lane if present; do not use as theorem language.
thermodynamic/energy calibration terms	49	Quarantine in boundary/open/interpretation lane if present; do not use as theorem language.
continuum bridge terms	28	Quarantine in boundary/open/interpretation lane if present; do not use as theorem language.
uniqueness terms	6	Quarantine in boundary/open/interpretation lane if present; do not use as theorem language.
absolute proof phrasing	0	Quarantine in boundary/open/interpretation lane if present; do not use as theorem language.

A.37 Theorem-level statements that are safe to foreground

The theorem-level layer consists of the following foreground results, provided the finite-lattice Poole-rule assumptions are stated:

1. the compact hysteretic support law for $B5-7/S5-9$;
2. stable coordinate sheets and surface selection by local Moore-neighbor counts;
3. first-step planar defect calculus using q_D and r_D ;
4. rectangle, square, and line-hole first-step laws in their stated domains;
5. rectangle channel spectra and prime-corrected laws inside the named PM_ϕ model;
6. discrete irreversible-work proxies as definitions over channel weights;
7. rectangle reconstruction from area plus response under a rectangle prior.

A.38 Statements that must stay bounded

The following statements remain bounded and are not theorem-level claims:

1. **v10 atlas:** may be described as reproducible empirical evidence for candidate multi-step and void-response laws; it may not be called a closed analytic theorem.
2. **Q/P capture:** may be referenced as an application boundary and as static rectangle-transducer evidence; it may not be presented as arbitrary-mask or dynamic Q/P reconstruction.
3. **Continuum and OTG:** may be listed as future bridge work; this document does not prove spacetime, gravity, dark energy, or a continuum metric theory.
4. **Rule-space uniqueness:** may be listed as an open problem; this document does not prove that the Poole rule is unique or minimal among all possible rules.
5. **Thermodynamic language:** the work laws are discrete event-weighting laws unless an external calibration to physical energy is supplied.

Appendix B

Appendix B: Full PooleQ/P Capture source

PooleQ/P Capture: Channel-Resolved Hysteretic Defect Logic and Decomposed Geometry Spectrometry for p-bit Fields and Quantum Syndrome Data

Rooke Poole

Draft mathematical note v3, amended v5.6

June 20, 2026

Abstract

This amended v5.6 note keeps PooleQ/P Capture within a strict measurement-side boundary while upgrading its interpretation from a single capture-score framework into a local-geometry spectrometer for measured binary fields. The base Poole update on a three-dimensional Moore lattice is

$$x_{t+1}(i) = \mathbf{1}\{5 \leq N_i(x_t) \leq 7 + 2x_t(i)\},$$

where N_i is the 26-neighbor active count. The central move is to preserve birth, survival, over-support, and strain channels instead of collapsing the update immediately into one live/dead output. Earlier amendments established the p-bit probability layer, exact-cardinality defect footprints, biased Hamming-window gates, channel-resolved adder readouts, the rectangular defect-transducer boundary, same-density shuffled controls, and the v5.5 decomposed robust scores. The v5.6 amendment names the resulting structure: PooleQ/P acts as a geometry spectrometer with signature vector

$$\mathcal{S}_{\mathcal{P}}(X) = (R_B(X), R_H(X), R_{\psi}(X)),$$

where R_B measures birth-window/boundary geometry, R_H measures high-support geometry, and R_{ψ} measures support-strain geometry. In the v5.5 verification run used by this amendment, every tested field had the same active density, 0.059995; the maximum combined robust score among IID and shuffled controls was 1.0959, while the minimum structured score was 4.5899. Thus the paper-safe v5.6 claim is: PooleQ/P maps measured binary fields into an interpretable local-geometry signature space that separates structured fields from exact same-density shuffled controls and indicates which local geometry carries the separation. It remains a classical transform for p-bit samples, readout strings, and syndrome histories; it does not copy, clone, or reconstruct an unknown qubit state.

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1 Purpose and boundary

The Poole Manifold rule was originally considered as a three-dimensional binary cellular automaton with a 26-neighbor Moore neighborhood. Previous work showed that it acts as a hysteretic surface selector: one-cell-thick sheets are preserved, bulk is rejected, and surface defects generate normal-layer birth halos. This note asks a narrower mathematical and computational question:

Can the channel structure of the Poole rule be used as a local computational capture transform for stochastic p-bit fields and measured qubit syndrome fields?

The answer developed here is yes, with a strict boundary. PooleQ/P Capture does *not* capture the private state of an unknown qubit. Unknown quantum states cannot be cloned or perfectly copied by a physical operation [1, 2]. The target data are already-measured classical fields:

- p-bit samples $X_i(t) \in \{0, 1\}$ or p-bit probabilities $p_i(t) \in [0, 1]$;
- measured qubit readout strings $m_i(t) \in \{0, 1\}$;
- stabilizer syndrome fields or syndrome-event histories $S(x, y, t) \in \{0, 1\}$.

In this setting, PooleQ/P is a classical transform applied after measurement. Its proposed role is comparable to a feature map, local decoder preprocessor, or defect-sensitive stochastic layer. It should not be described as a method for reconstructing unmeasured quantum amplitudes.

2 The Poole Manifold rule

Let $V = (\mathbb{Z}/L\mathbb{Z})^3$ be a periodic three-dimensional lattice. A binary state is a map

$$x : V \rightarrow \{0, 1\}.$$

For $i \in V$, let $\mathcal{N}_{26}(i)$ denote the 26-cell Moore neighborhood:

$$\mathcal{N}_{26}(i) = \{i + \delta : \delta \in \{-1, 0, 1\}^3 \setminus \{0\}\}.$$

The local support count is

$$N_i(x) = \sum_{j \in \mathcal{N}_{26}(i)} x(j).$$

The Poole update is

$$\mathcal{P}(x)_i = \mathbf{1}\{5 \leq N_i(x) \leq 7 + 2x(i)\}. \quad (1)$$

Equivalently, inactive cells are born for $N_i \in \{5, 6, 7\}$, while active cells survive for $N_i \in \{5, 6, 7, 8, 9\}$.

Proposition 1 (Compact hysteretic form). *The rule B5-7/S5-9 is exactly equivalent to Eq. (1).*

Proof. If $x(i) = 0$, the upper bound in Eq. (1) is 7, so $\mathcal{P}(x)_i = 1$ exactly when $5 \leq N_i \leq 7$. If $x(i) = 1$, the upper bound is 9, so $\mathcal{P}(x)_i = 1$ exactly when $5 \leq N_i \leq 9$. $\square$

The term $2x(i)$ is the hysteretic memory term: the prior active state gives the cell a two-neighbor upper-support allowance. This is what allows a smooth one-cell-thick sheet cell with $N_i = 8$ to survive.

3 Channel-resolved Poole capture

The ordinary update collapses all allowed birth and survival events into the same binary output. For capture, the local channels are kept separated.

Definition 1 (Birth and survival channels). *For $k = 0, 1, \dots, 26$, define*

$$\mathbf{B}_k(i; x) = (1 - x(i))\mathbf{1}\{N_i(x) = k\}, \quad (2)$$

$$\mathbf{S}_k(i; x) = x(i)\mathbf{1}\{N_i(x) = k\}. \quad (3)$$

The active Poole channels are $\mathbf{B}_5, \mathbf{B}_6, \mathbf{B}_7$ and $\mathbf{S}_5, \mathbf{S}_6, \mathbf{S}_7, \mathbf{S}_8, \mathbf{S}_9$.

Then

$$\mathcal{P}(x)_i = \sum_{k=5}^7 \mathbf{B}_k(i; x) + \sum_{k=5}^9 \mathbf{S}_k(i; x). \quad (4)$$

Because exactly one value of N_i holds at any site, the terms in Eq. (4) are disjoint.

Definition 2 (PooleQ/P capture vector). *The local PooleQ/P feature vector is*

$$\Phi_i(x) = (\mathbf{B}_5, \mathbf{B}_6, \mathbf{B}_7, \mathbf{S}_5, \mathbf{S}_6, \mathbf{S}_7, \mathbf{S}_8, \mathbf{S}_9, O_{10+}, \psi)_i, \quad (5)$$

where

$$O_{10+}(i; x) = x(i) \mathbf{1}\{N_i(x) \geq 10\}$$

and

$$\psi_i(x) = \max\{0, N_i(x) - (7 + 2x(i))\} - \max\{0, 5 - N_i(x)\}. \quad (6)$$

Here $\psi < 0$ is under-support, $\psi = 0$ is valid Poole support, and $\psi > 0$ is over-support.

Remark 1 (The observability issue). *The channel-resolved transform $\Phi_i(x)$ is richer than the raw automaton state $\mathcal{P}(x)_i$. If only the next binary state is observed, the three birth channels $\mathbf{B}_5, \mathbf{B}_6, \mathbf{B}_7$ become indistinguishable. Thus several results below are exact for typed readout, PooleScan-style feature readout, or spatially demultiplexed gate layouts. They are not automatically exact for a one-state, uninstrumented CA implementation.*

4 Probabilistic p-bit form

A p-bit is a stochastic binary variable that fluctuates between 0 and 1 with a controlled probability. p-bit systems have been proposed as probabilistic computing substrates and as a bridge between ordinary bits and qubit-inspired hardware [3, 4].

Let

$$X_j \sim \text{Bernoulli}(p_j)$$

independently over $j \in V$. Since the center cell X_i is not included in N_i , the center variable is independent of the local neighbor count under this independent-site model.

4.1 Poisson-binomial channel probabilities

For a fixed site i , define the local generating function

$$G_i(z) = \prod_{j \in \mathcal{N}_{26}(i)} (1 - p_j + p_j z). \quad (7)$$

Then

$$\mathbb{P}[N_i(X) = k] = [z^k] G_i(z). \quad (8)$$

This is a local Poisson-binomial distribution.

Proposition 2 (Expected Poole channels). *Under the independent Bernoulli model,*

$$\mathbb{E}[\mathbf{B}_k(i; X)] = (1 - p_i) [z^k] G_i(z), \quad (9)$$

$$\mathbb{E}[\mathbf{S}_k(i; X)] = p_i [z^k] G_i(z). \quad (10)$$

Therefore the expected Poole activation is

$$q_i := \mathbb{E}[\mathcal{P}(X)_i] = (1 - p_i) \sum_{k=5}^7 [z^k] G_i(z) + p_i \sum_{k=5}^9 [z^k] G_i(z). \quad (11)$$

Proof. Equation (8) follows from the generating function for a sum of independent non-identical Bernoulli variables. Since X_i is not a factor in N_i , X_i is independent of N_i , giving Eqs. (9) and (10). Summing the active channels gives Eq. (11). $\square$

Define

$$C_i^{a:b} = \sum_{k=a}^b [z^k] G_i(z). \quad (12)$$

Then

$$q_i = (1 - p_i) C_i^{5:7} + p_i C_i^{5:9}. \quad (13)$$

Because $C_i^{5:9} = C_i^{5:7} + C_i^{8:9}$, this can be written as

$$q_i = C_i^{5:7} + p_i C_i^{8:9}. \quad (14)$$

This is the probabilistic form of hysteretic memory: the center probability p_i only weights the upper-memory band 8, 9.

4.2 Differentiability

For $\ell \in \mathcal{N}_{26}(i)$,

$$\frac{\partial}{\partial p_\ell} [z^k] G_i(z) = [z^k] \left((z - 1) \prod_{j \in \mathcal{N}_{26}(i), j \neq \ell} (1 - p_j + p_j z) \right). \quad (15)$$

For the center site,

$$\frac{\partial q_i}{\partial p_i} = C_i^{8:9}, \quad (16)$$

again showing that the center probability only gates the memory band.

4.3 Poole-biased p-bit dynamics

A p-bit network can use channel samples or expected channels as features. One possible stochastic update is

$$\mathbb{P}[X_i(t+1) = 1 \mid X_t] = \sigma \left(\beta \left[\theta_i + w^\top \Phi_i(X_t) \right] \right), \quad (17)$$

where $\sigma(u) = (1 + e^{-u})^{-1}$. A differentiable probability-layer version replaces $\Phi_i(X_t)$ by $\mathbb{E}[\Phi_i(X_t)]$ under a local product approximation.

5 Correlation residue and null calibration

PooleQ/P can measure local dependence beyond independent-site statistics. Suppose M samples $X^{(1)}, \dots, X^{(M)}$ are observed from a p-bit device or repeated measurement experiment. Define empirical channel rates

$$\hat{B}_k(i) = \frac{1}{M} \sum_{m=1}^M B_k(i; X^{(m)}), \quad \hat{S}_k(i) = \frac{1}{M} \sum_{m=1}^M S_k(i; X^{(m)}). \quad (18)$$

Let

$$\hat{p}_i = \frac{1}{M} \sum_{m=1}^M X_i^{(m)}. \quad (19)$$

Using empirical marginals $\hat{p}_j$, form the independent prediction q_i^{ind} from Eq. (11). The scalar Poole activation residue is

$$\Delta_i = \hat{q}_i - q_i^{\text{ind}}, \quad \hat{q}_i = \frac{1}{M} \sum_{m=1}^M \mathcal{P}(X^{(m)})_i. \quad (20)$$

A channel-wise residue vector is

$$R_i = \left(\hat{\mathbf{B}}_5 - \mathbb{E}_{\text{ind}} \mathbf{B}_5, \hat{\mathbf{B}}_6 - \mathbb{E}_{\text{ind}} \mathbf{B}_6, \hat{\mathbf{B}}_7 - \mathbb{E}_{\text{ind}} \mathbf{B}_7, \hat{\mathbf{S}}_8 - \mathbb{E}_{\text{ind}} \mathbf{S}_8, \hat{\mathbf{S}}_9 - \mathbb{E}_{\text{ind}} \mathbf{S}_9 \right)_i. \quad (21)$$

Definition 3 (Poole coherence). *A simple global correlation statistic is*

$$\mathcal{C}_{\mathcal{P}} = \sum_{i \in V} \|R_i\|_2^2. \quad (22)$$

A large value of $\mathcal{C}_{\mathcal{P}}$ indicates that local Poole channel rates are not explained by the product of observed local marginals. It does not by itself prove quantum entanglement or a particular physical noise model. It is a local structured-correlation detector.

6 Defect cardinality logic

Around a stable Poole sheet, the channel-resolved birth rule becomes exact-cardinality logic. Let S be a one-cell-thick coordinate sheet in the $z = 0$ plane. A perfect sheet is an exact Poole fixed point: active sheet cells have $N = 8$, and inactive cells immediately above or below the sheet have $N = 9$, which is above the birth window.

Let D be a set of missing cells in the sheet. For a normal-layer candidate cell directly above or below a sheet location $u \in (\mathbb{Z}/L\mathbb{Z})^2$, define the defect-footprint count

$$r_D(u) = \sum_{\eta \in \{-1,0,1\}^2} \mathbf{1}\{u + \eta \in D\}. \quad (23)$$

This counts missing cells from the 3×3 sheet footprint below the candidate. For the first update, assuming no pre-existing active cells in the adjacent normal layer, the candidate's active neighbor count is

$$N = 9 - r_D(u). \quad (24)$$

Therefore

$$\mathbf{B}_7 \iff r_D = 2, \quad \mathbf{B}_6 \iff r_D = 3, \quad \mathbf{B}_5 \iff r_D = 4. \quad (25)$$

Proposition 3 (Channel-cardinality equivalence). *For a normal-layer candidate cell adjacent to a defective sheet, under the first-update sheet approximation,*

$$\mathbf{B}_k = 1 \iff r_D = 9 - k, \quad k \in \{5, 6, 7\}.$$

6.1 Biased exact-cardinality detectors

Let $A_1, \dots, A_n \in \{0, 1\}$ be input defects placed at selected footprint positions, and let $b \in \{0, 1, \dots, 9 - n\}$ be the number of fixed bias defects in the same 3×3 footprint. Define

$$s = \sum_{j=1}^n A_j, \quad r = s + b. \quad (26)$$

By Eq. (25), the birth channels detect exact cardinalities:

$$\mathbf{B}_7 = \mathbf{1}\{s = 2 - b\}, \quad \mathbf{B}_6 = \mathbf{1}\{s = 3 - b\}, \quad \mathbf{B}_5 = \mathbf{1}\{s = 4 - b\}. \quad (27)$$

Thus a biased footprint is a small exact-count circuit.

7 Raw biased-footprint Boolean gates

The exact-cardinality channels above require channel-resolved readout. There is also a lower-level fact: the collapsed raw birth event itself is already a useful Boolean gate.

For an inactive normal-layer candidate adjacent to a defective sheet, the raw birth output is

$$Y = B_5 \vee B_6 \vee B_7. \quad (28)$$

By Eq. (25), this is equivalent to

$$Y = \mathbf{1}\{2 \leq r \leq 4\}. \quad (29)$$

If $A_1, \dots, A_n \in \{0, 1\}$ denote input defects and c denotes fixed bias defects in the same footprint, then

$$Y_{n,c}(A_1, \dots, A_n) = \mathbf{1} \left\{ 2 \leq c + \sum_{j=1}^n A_j \leq 4 \right\}. \quad (30)$$

Theorem 1 (Biased defect-footprint logic). *Under the first-update sheet-footprint assumptions, for two inputs A, B ,*

$$Y_{2,0}(A, B) = A \wedge B, \quad (31)$$

$$Y_{2,1}(A, B) = A \vee B, \quad (32)$$

$$Y_{2,3}(A, B) = \neg(A \wedge B), \quad (33)$$

$$Y_{2,4}(A, B) = \neg(A \vee B). \quad (34)$$

Thus the raw Poole defect-footprint birth operator contains functionally complete Boolean primitives.

Proof. If r cells are missing from the 3×3 sheet footprint, then the inactive normal-layer candidate sees $N = 9 - r$ active neighbors. The Poole birth condition $5 \leq N \leq 7$ is equivalent to $2 \leq r \leq 4$. With $r = c + \sum_j A_j$, Eq. (30) follows. For $n = 2$, substitute $c = 0, 1, 3, 4$. $\square$

The same construction gives one-input identity and inversion gates:

$$Y_{1,1}(A) = A, \quad Y_{1,4}(A) = \neg A. \quad (35)$$

Channel-resolved readout is not needed for NAND or NOR; it becomes necessary for arithmetic compression, where one must distinguish whether the birth arose from B_5 , B_6 , or B_7 .

8 Half-adder and full-adder

Let $A, B \in \{0, 1\}$ be two input defects, and set one fixed bias defect $b = 1$. Then

$$r = A + B + 1.$$

From Eq. (27),

$$B_7 = \mathbf{1}\{A + B = 1\} = A \oplus B, \quad (36)$$

$$B_6 = \mathbf{1}\{A + B = 2\} = A \wedge B. \quad (37)$$

Thus SUM = B_7 and CARRY = B_6 .

For a full-adder, let $A, B, C_{\text{in}} \in \{0, 1\}$ and again set $b = 1$. If

$$s = A + B + C_{\text{in}},$$

then

$$B_7 = \mathbf{1}\{s = 1\}, \quad B_6 = \mathbf{1}\{s = 2\}, \quad B_5 = \mathbf{1}\{s = 3\}. \quad (38)$$

The full-adder sum bit is parity, true for $s = 1$ or $s = 3$, and the carry bit is true for $s \geq 2$. Hence

$$\boxed{\text{SUM} = B_7 \vee B_5, \quad \text{CARRY} = B_6 \vee B_5.} \quad (39)$$

Remark 2 (What is exact, and what remains engineering). *The raw biased-footprint theorem and the channel-resolved full-adder theorem are exact local first-update statements, but they do not yet make an autonomous one-state Poole computer. Routing, fanout, timing, reset, gate isolation, and channel demultiplexing remain engineering and verification problems. If only the collapsed output $\mathcal{P}(x)_i \in \{0, 1\}$ is observed, the full-adder footprint collapses to $A \vee B \vee C_{\text{in}}$, not the full-adder output.*

9 Application to qubit syndrome fields

Surface-code quantum computing represents quantum information using many physical qubits and repeated stabilizer measurements; classical decoders infer corrections from syndrome data [5, 6, 7]. PooleQ/P does not replace these decoders. Instead, it can be defined as a local feature layer on binary syndrome spacetime.

Let

$$S(x, y, t) \in \{0, 1\}$$

be a syndrome-event history. Treating (x, y, t) as a three-dimensional lattice, define

$$\Phi(S)(x, y, t) = (B_5, B_6, B_7, S_5, \dots, S_9, O_{10+}, \psi)(x, y, t). \quad (40)$$

This maps a syndrome history into local features describing event-chain thickness, over-support, local crowding, and hysteretic survival windows. A decoder can then be augmented as

$$\hat{E} = D_\theta(S, \Phi(S)), \quad (41)$$

where D_θ may be a matching, union-find, neural, or hybrid decoder. The claim is not that PooleQ/P is a complete decoder by itself. The claim is that it supplies interpretable local geometry features for measured syndrome data.

10 v5.1 scope note: static Q/P capture versus dynamic Defect Calculus

The v5.1 amendment kept PooleQ/P separate from the Defect Calculus paper. Static first-step Q/P capture is not the same as multi-step response capture. Multi-step planar relaxation and 3D void response belong to the Defect Calculus empirical atlas unless a future dynamic Q/P extension explicitly defines the model, measurement channel, and time window.

This boundary is kept in v5.5. The new benchmark below is a measurement-side structured-correlation benchmark. It does not replace the static footprint theorems or the separate dynamic Defect Calculus program.

11 v5 amendment: rectangle transducer boundary

For an $a \times b$ rectangular planar puncture with $a, b \geq 2$, the Defect Calculus v9 paper gives the raw response

$$B_0(a, b) = 4a + 4b + 12, \quad (42)$$

and the default prime-sharpened response

$$B_\phi(a, b) = 4a + 4b - 4. \quad (43)$$

In perimeter form, with $P = 2a + 2b$,

$$B_0 = 2P + 12, \quad B_\phi = 2P - 4. \quad (44)$$

Thus the defect response is a perimeter-like capture channel.

Theorem 2 (Raw Q/P rectangle reconstruction). *For a raw rectangular puncture, area $A = ab$ and raw response B_0 recover the unordered side pair $\{a, b\}$ by*

$$s = \frac{B_0 - 12}{4} = a + b, \quad (45)$$

$$\Delta = s^2 - 4A, \quad (46)$$

$$\{a, b\} = \left\{ \frac{s - \sqrt{\Delta}}{2}, \frac{s + \sqrt{\Delta}}{2} \right\}. \quad (47)$$

Theorem 3 (Prime Q/P rectangle reconstruction). *For a default prime rectangular puncture, area A and prime response B_ϕ recover the unordered side pair by*

$$s = \frac{B_\phi + 4}{4} = a + b, \quad (48)$$

with the same discriminant formula.

Remark 3 (Corrected reconstruction boundary). *Birth count alone is not a complete rectangle reconstruction invariant. It determines $a + b$ but not ab . The corrected claim is therefore: area plus response reconstructs rectangle dimensions up to exchange. This is a prior-dependent rectangle theorem, not a claim that global counts reconstruct arbitrary masks.*

12 v5.4/v5.5 amendment: null-calibrated and decomposed capture

This amendment turns the earlier correlation-residue idea into a direct empirical benchmark for measured binary fields. The target is not private quantum-state recovery. The target is structured correlation in already-measured binary data.

Let

$$X^{(1)}, \dots, X^{(M)} \in \{0, 1\}^V$$

be repeated measured binary fields. For example, these may be p-bit samples, measured qubit readout strings, or syndrome-event histories. Define the empirical channel-rate vector

$$\hat{\mu}_\Phi = \frac{1}{M} \sum_{m=1}^M \text{rate}_V[\Phi(X^{(m)})]. \quad (49)$$

12.1 Same-density IID and shuffled controls

Let

$$\hat{p} = \frac{1}{M|V|} \sum_{m=1}^M \sum_{i \in V} X_i^{(m)}$$

be the observed active density. The homogeneous same-density IID null is

$$X_i^{\text{null}} \sim \text{Bernoulli}(\hat{p}), \quad i \in V,$$

independent over sites. In the synthetic benchmark, all classes are also forced to have exactly the same active count. This prevents the score from separating classes merely because one class has more active cells.

For each structured field X , define a shuffled control X^{shuf} by randomly permuting the voxel values. Then

$$\sum_i X_i^{\text{shuf}} = \sum_i X_i,$$

but local geometry is destroyed. If a score is high for X and collapses for X^{shuf} , the score is detecting local arrangement rather than active density.

12.2 Raw and robust capture scores

For a selected channel set $\mathcal{K}$, let μ_0 and σ_0 be the empirical same-density IID mean and standard deviation of the channel-rate vector. The raw null-calibrated capture score is

$$Z_{\mathcal{K}}(X) = \left\| \frac{\hat{\mu}_{\Phi_{\mathcal{K}}}(X) - \mu_0}{\sigma_0 + \varepsilon} \right\|_2. \quad (50)$$

Because rare channels such as S_9 and O_{10+} may have near-zero IID variance in sparse fields, v5.5 continues to report a softened robust score

$$R_{\mathcal{K}}(X) = \left(\sum_{k \in \mathcal{K}} \text{asinh}^2 \left[\frac{1}{\lambda} \frac{\hat{\mu}_{\Phi_k}(X) - \mu_{0,k}}{\sigma_{0,k} + \varepsilon} \right] \right)^{1/2}. \quad (51)$$

In the v5.5 run, $\lambda = 5$. The raw score remains useful as a detector. The robust score is safer for comparison and plotting when a structured class activates channels that are nearly absent under the null.

12.3 Decomposed PooleQ/P score groups

The v5.5 amendment decomposes capture into three interpretable components:

$$\mathcal{K}_B = \{B_5, B_6, B_7\}, \quad (52)$$

$$\mathcal{K}_H = \{S_8, S_9, O_{10+}\}, \quad (53)$$

$$\mathcal{K}_{\psi} = \{\psi_{\text{abs}}\}. \quad (54)$$

Here $\mathcal{K}_B$ measures birth-window or boundary geometry, $\mathcal{K}_H$ measures high-support geometry, and $\mathcal{K}_{\psi}$ measures support-strain geometry. The non-redundant combined set is

$$\mathcal{K}_C = \mathcal{K}_B \cup \mathcal{K}_H \cup \mathcal{K}_{\psi}. \quad (55)$$

The collapsed update rate P_{active} is still reported as a descriptive statistic, but it is excluded from $\mathcal{K}_C$ because it overlaps with the individual birth and survival channels.

For non-null classes, geometry shares are defined from the robust components by

$$s_g = \frac{R_g^2}{R_B^2 + R_H^2 + R_\psi^2 + \varepsilon}, \quad g \in \{B, H, \psi\}. \quad (56)$$

Classes with $R_C < 2$ are labeled null-like before interpreting shares, because a percentage decomposition of a near-zero signal is not meaningful.

13 v5.3 shuffled-control benchmark

The v5.3 Colab-safe benchmark used $L = 28$, $p = 0.06$, $M = 50$ samples per class, seed 13, and exact active count 1317 per field. It established the same-density shuffled-control pattern: structured line, chain, burst, and sheet ensembles had high capture scores, while their exact same-density shuffled controls returned to the IID baseline.

Class	Active density	Poole active	Raw $Z_{\mathcal{P}}$	Robust $R_{\mathcal{P}}$
IID null	0.059995	0.017413	1.71	0.34
Lines	0.059995	0.066732	87.94	5.88
Lines shuffled	0.059995	0.017239	1.67	0.33
Random-walk chains	0.059995	0.081722	215.05	8.36
Walks shuffled	0.059995	0.017360	1.90	0.38
Branching chains	0.059995	0.076872	141.35	7.01
Branches shuffled	0.059995	0.017408	1.67	0.33
Compact bursts	0.059995	0.060024	555.14	9.50
Bursts shuffled	0.059995	0.017602	1.89	0.37
Sheets	0.059995	0.075381	657.61	10.04
Sheets shuffled	0.059995	0.017513	1.82	0.36

Table 1: v5.3 same-density shuffled-control benchmark. Every class has the same active density to six decimal places. Structured classes have high capture scores; shuffled controls return to the IID baseline.

This result justified moving from a single score to the decomposed v5.5 score groups above.

14 v5.5 decomposed benchmark results

The v5.5 run used $L = 28$, $p = 0.06$, $M = 50$, seed 23, and exact active count 1317 per field. The verification report passed all checks: same-density generation, active-count audit, shuffle audit, clean robust separation, and structured-beats-shuffled controls. The maximum combined robust score among IID and shuffled controls was 1.0959, while the minimum combined robust score among structured samples was 4.5899.

Class	Active	R_B	R_H	R_ψ	R_C	Signature
IID null	0.059995	0.21	0.02	0.16	0.28	null-like
Straight lines	0.059995	3.49	2.19	3.28	5.28	mixed geometry
Lines shuffled	0.059995	0.24	0.03	0.17	0.31	null-like
Random-walk chains	0.059995	3.62	5.63	4.15	7.88	high-support-led
Walks shuffled	0.059995	0.23	0.02	0.19	0.31	null-like
Branching chains	0.059995	3.87	3.66	3.88	6.61	mixed geometry
Branches shuffled	0.059995	0.21	0.02	0.14	0.27	null-like
Compact bursts	0.059995	3.12	7.06	5.00	9.19	high-support-led
Bursts shuffled	0.059995	0.23	0.02	0.16	0.29	null-like
Sheets	0.059995	3.50	7.40	5.31	9.76	high-support-led
Sheets shuffled	0.059995	0.22	0.04	0.16	0.29	null-like

Table 2: v5.5 decomposed robust capture scores. R_B uses $\{B_5, B_6, B_7\}$, R_H uses $\{S_8, S_9, O_{10+}\}$, R_ψ uses ψ_{abs} , and R_C uses their non-redundant union. The collapsed update rate P_{active} is reported in the result files but is not included in R_C .

Structured class	R_C	Shuffled R_C	Drop	Signature
Straight lines	5.28	0.31	4.97	mixed geometry
Random-walk chains	7.88	0.31	7.57	high-support-led
Branching chains	6.61	0.27	6.34	mixed geometry
Compact bursts	9.19	0.29	8.90	high-support-led
Sheets	9.76	0.29	9.47	high-support-led

Table 3: Paired same-density shuffled-control results from v5.5. Shuffling preserves active count but destroys local geometry. Every structured class remains high while its shuffled control collapses to the null-like range.

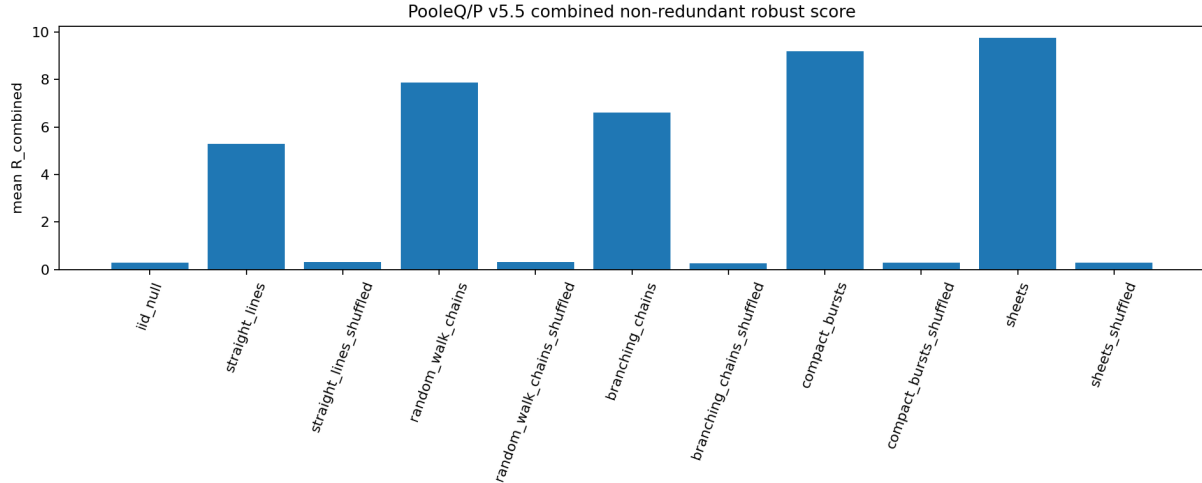


Figure 1: Combined non-redundant robust score R_C for v5.5. Structured fields separate from IID and same-density shuffled controls.

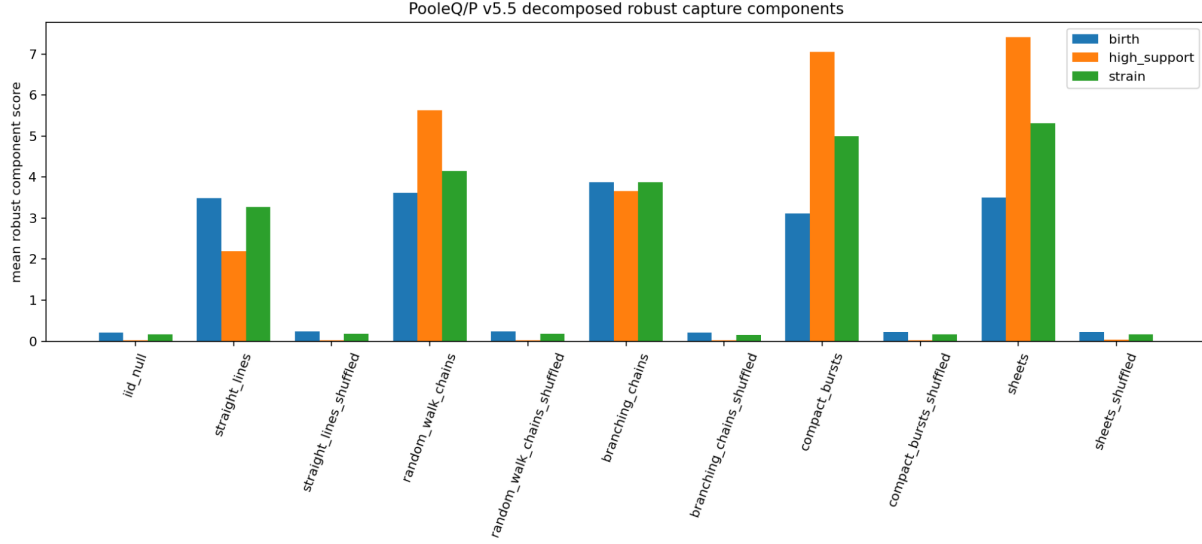


Figure 2: Decomposed robust components. Straight lines and branching chains are mixed; random-walk chains, compact bursts, and sheets are high-support-led.

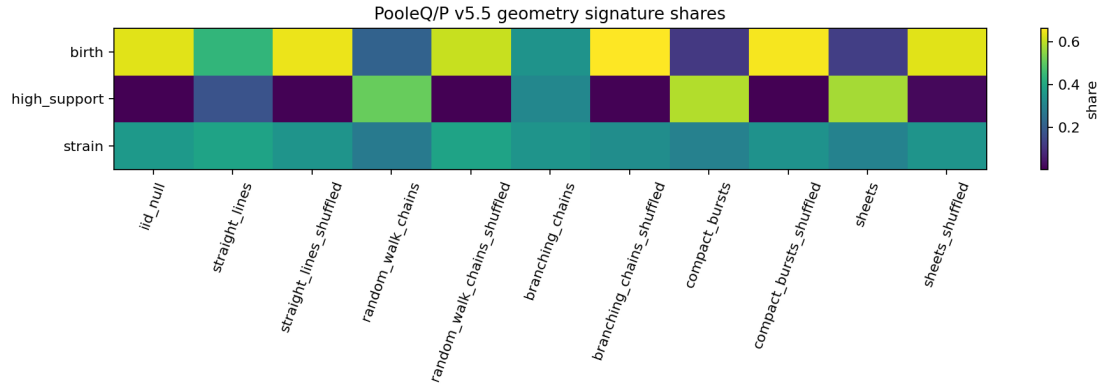


Figure 3: Geometry signature shares. Shares should be interpreted only for non-null classes; IID and shuffled controls are first labeled null-like because their total robust score is below threshold.

14.1 Single-channel interpretation

The strongest single-channel effects agree with the class geometry. Values in Table 4 are mean null-calibrated channel effects and should be treated as diagnostics rather than physical sigma claims.

Class	Top mean channel effects, largest first
Straight lines	ψ_{abs} : 66.8; B_6 : 26.1; B_7 : 23.7; S_8 : 15.6; B_5 : 8.2
Random-walk chains	ψ_{abs} : 159.2; S_8 : 82.6; O_{10+} : 64.1; S_9 : 52.8; B_7 : 33.1
Branching chains	ψ_{abs} : 122.1; B_7 : 40.0; S_8 : 31.9; B_6 : 25.0; O_{10+} : 17.4
Compact bursts	ψ_{abs} : 370.8; O_{10+} : 358.8; S_9 : 91.0; S_8 : 83.4; B_7 : 26.4
Sheets	ψ_{abs} : 508.9; S_8 : 328.9; O_{10+} : 147.7; S_9 : 119.7; B_6 : 28.6

Table 4: Top single-channel effects from the v5.5 benchmark.

The resulting geometry signatures are intuitive. Straight lines are mixed, with birth-window and strain components both active. Branching chains are also mixed, consistent with split local support. Random-walk chains, compact bursts, and sheets are high-support-led, consistent with local bunching, dense support, and over-support channels.

15 v5.6 interpretation: Poole geometry spectrometer

The v5.5 decomposition suggests a cleaner identity for PooleQ/P. The transform should not be described only as a detector that answers whether a field is structured. It is better described as a local-geometry spectrometer: it maps a measured binary field into a low-dimensional signature of the local geometry that produced the channel spectrum.

Definition 4 (Poole geometry signature vector). *For a measured binary field X , define*

$$\mathcal{S}_{\mathcal{P}}(X) = (R_B(X), R_H(X), R_{\psi}(X)), \quad (57)$$

where R_B is the robust birth-window score over $\{B_5, B_6, B_7\}$, R_H is the robust high-support score over $\{S_8, S_9, O_{10+}\}$, and R_{ψ} is the robust strain score over $\{\psi_{\text{abs}}\}$.

The combined score

$$R_C(X) = R_{\mathcal{K}_C}(X)$$

answers whether the field is non-null relative to the selected same-density null model. The direction of $\mathcal{S}_{\mathcal{P}}(X)$ answers what kind of local Poole geometry carries the non-null signal.

Definition 5 (Normalized Poole geometry spectrum). *For non-null fields, define the normalized spectrum*

$$\Sigma_{\mathcal{P}}(X) = \frac{(R_B^2, R_H^2, R_{\psi}^2)}{R_B^2 + R_H^2 + R_{\psi}^2 + \varepsilon}. \quad (58)$$

The entries of $\Sigma_{\mathcal{P}}$ estimate the fractional birth-window, high-support, and strain contributions to the robust non-null signal.

Proposition 4 (Geometry-spectrometer principle). *Let X and X' be measured binary-field ensembles with the same active density. If their PooleQ/P channel-rate vectors differ after same-density null calibration, then they may separate either by magnitude, through R_C , or by direction, through $\Sigma_{\mathcal{P}}$. Thus PooleQ/P can distinguish fields that have the same low-order active density but different local organization, such as chain-like, branch-like, burst-like, or sheet-like support.*

Interpretive proof. The score R_C is computed from the non-redundant channel set $\mathcal{K}_C = \mathcal{K}_B \cup \mathcal{K}_H \cup \mathcal{K}_{\psi}$. If two fields have the same active density, density alone cannot explain a stable separation

from an exact same-density shuffled control. A high R_C therefore indicates non-null local channel structure relative to the null. The decomposition into R_B , R_H , and R_ψ records which disjoint channel groups account for that non-null structure. Normalizing their squared robust components gives Eq. (58), which removes overall magnitude and keeps the direction of the signal in Poole geometry space. $\square$

In the v5.5 run, this interpretation gives the following diagnostic map. IID and shuffled controls are null-like because $R_C < 2$. Straight lines and branching chains are mixed-geometry cases. Random-walk chains, compact bursts, and sheets are high-support-led cases. These labels are not universal physical categories; they are benchmark-calibrated names for regions of the synthetic Poole geometry space.

Class	R_C	Birth share	High-support share	Strain share	Spectrometer readout
IID null	0.28	0.64	0.01	0.36	null-like
Straight lines	5.28	0.44	0.17	0.39	mixed geometry
Random-walk chains	7.88	0.21	0.51	0.28	high-support-led
Branching chains	6.61	0.34	0.31	0.35	mixed geometry
Compact bursts	9.19	0.11	0.59	0.30	high-support-led
Sheets	9.76	0.13	0.58	0.30	high-support-led

Table 5: v5.6 spectrometer interpretation of the v5.5 decomposed benchmark. Shares are interpreted only after the field is non-null by R_C ; the IID row is shown for completeness but remains null-like.

The practical consequence is that PooleQ/P can be tested as a first-pass diagnostic layer. A future v5.6 verifier can use $\mathcal{S}_P(X)$ or $\Sigma_P(X)$ for a simple nearest-centroid geometry classifier. Such a classifier would not establish quantum decoding superiority by itself, but it would test whether PooleQ/P signatures can consistently separate null-like, chain-like, branch-like, burst-like, and sheet-like measured-field geometries under matched controls.

16 Interpretation and updated research program

The v5.6 interpretation supports the following updated claim:

PooleQ/P maps measured binary fields into an interpretable local-geometry signature space. Under exact active-density matching, structured fields remain separated from IID and same-density shuffled controls, while the vector (R_B, R_H, R_ψ) identifies whether the non-null signal is birth-window-led, high-support-led, strain-led, or mixed.

This result does *not* prove that PooleQ/P captures unknown quantum states. It does *not* prove quantum decoder superiority. It does *not* prove physical quantum advantage. It does justify a focused measurement-side claim: PooleQ/P is an interpretable local-geometry feature transform for p-bit fields, readout histories, and syndrome-event data.

The updated PooleQ/P program should now proceed in seven concrete steps.

1. **Geometry-signature classifier.** Build a simple nearest-centroid classifier in (R_B, R_H, R_ψ) or Σ_P space to test whether unknown synthetic measured fields can be labeled as null-like, chain-like, branch-like, burst-like, or sheet-like under matched controls.
2. **Paper-locked v5.5/v5.6 verifier.** Keep the v5.5 Colab verifier as the canonical reproduction script for this amended paper.

3. **Syndrome-like benchmark hardening.** Add temporal error chains, open boundary conditions, erasures, correlated bursts, and measurement-noise flips.
4. **Matched-null controls.** Keep exact active-count shuffling as a required control. Add site-wise marginal nulls when data are spatially inhomogeneous.
5. **Decoder augmentation test.** Compare a baseline classifier or decoder against the same model augmented with PooleQ/P decomposed features.
6. **PooleQ/p sampler.** Implement Eq. (17) and compare IID, Ising-like, and Poole-biased p-bit fields.
7. **Footprint-gate engineering verifier.** Continue checking raw AND, OR, NAND, NOR, and channel-resolved full-adder footprints, including routing, reset, fanout, and interference radius.

17 Conclusion

PooleQ/P Capture is a channel-resolved transform for binary stochastic and measured quantum-derived data. Its core mathematical components are

$$X \mapsto \Phi(X) = (B_5, B_6, B_7, S_5, \dots, S_9, O_{10+}, \psi), \quad (59)$$

$$q_i = (1 - p_i)C_i^{5:7} + p_iC_i^{5:9}, \quad (60)$$

$$B_7 = \text{Exact}_1, \quad B_6 = \text{Exact}_2, \quad B_5 = \text{Exact}_3 \quad (61)$$

for biased defect footprints, plus the decomposed null-calibrated metrics R_B , R_H , R_ψ , and R_C , and the geometry signature vector $\mathcal{S}_P = (R_B, R_H, R_\psi)$.

The amended v5.6 contribution is not a new overclaim. It is an interpretability hardening of the PooleQ/P capture idea into a geometry-spectrometer framing. The v5.5 experiment shows that PooleQ/P detects structured local correlation beyond active density alone, because structured classes remain high while exact same-density shuffled controls collapse to the IID baseline. The v5.6 interpretation adds that the decomposed vector $\mathcal{S}_P$ identifies the direction of the signal in local geometry space: birth-window geometry, high-support geometry, strain geometry, or a mixed combination. This makes PooleQ/P more suitable to present as an interpretable measurement-side feature layer and diagnostic preprocessor for p-bit fields and quantum syndrome histories.

A Machine-checkable footprint logic

The biased raw-gate and full-adder results can be checked by the following minimal Python logic:

```
from itertools import product

def raw_gate(bits, c):
    r = c + sum(bits)
    return int(2 <= r <= 4)

# Two-input raw gates
for A, B in product([0, 1], repeat=2):
    assert raw_gate([A, B], 0) == (A & B)
    assert raw_gate([A, B], 1) == (A | B)
    assert raw_gate([A, B], 3) == (1 - (A & B))
```

```

    assert raw_gate([A, B], 4) == (1 - (A | B))

# Channel-resolved full-adder with one bias defect
for A, B, C in product([0, 1], repeat=3):
    s = A + B + C
    r = s + 1
    B7 = int(r == 2)
    B6 = int(r == 3)
    B5 = int(r == 4)
    SUM = B7 or B5
    CARRY = B6 or B5
    assert SUM == ((A + B + C) % 2)
    assert CARRY == int(A + B + C >= 2)

```

B Glossary

B_k Birth channel: inactive center with exactly k active neighbors.

S_k Survival channel: active center with exactly k active neighbors.

ψ Signed support strain; negative for under-support and positive for over-support.

O_{10+} Active cell with at least ten active neighbors; interpreted as overbulk/crowding.

PooleQ/P Channel-resolved Poole features applied to p-bit data or measured qubit-derived classical data.

Raw biased footprint gate The collapsed birth/no-birth output $Y_{n,c} = \mathbf{1}\{2 \leq c + \sum_j A_j \leq 4\}$.

Channel demultiplexing Any mechanism that converts B_5, B_6, B_7 labels into separate outputs rather than one collapsed live cell.

Null-calibrated capture A comparison of measured Poole channel rates against same-density IID or shuffled controls.

R_B Robust birth-window score using $\{B_5, B_6, B_7\}$.

R_H Robust high-support score using $\{S_8, S_9, O_{10+}\}$.

R_ψ Robust strain score using ψ_{abs} .

R_C Combined non-redundant robust score using birth, high-support, and strain groups; P_{active} is excluded to avoid channel redundancy.

S_P Poole geometry signature vector (R_B, R_H, R_ψ) .

Σ_P Normalized Poole geometry spectrum, using squared component shares of R_B , R_H , and R_ψ .

Geometry spectrometer The v5.6 interpretation of PooleQ/P as a map from measured binary fields into an interpretable local-geometry signature space.

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Appendix C

Appendix C: Full signed reaction-membrane source

Inducible Signed Reaction-Membrane Basins in Poole-Type Cellular Automata

Preparation Beats Forcing: A Benchmark Study of Symmetry, Rule Windows, and Broad-Forcing Degradation

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Abstract

We introduce a benchmarkable theory of signed reaction-interface induction in a three-dimensional Poole-type cellular automaton. The state space is $s_i \in \{-1, 0, +1\}$, and local updates are governed by birth and survival windows on a 26-neighbor Moore lattice, together with an opposite-support collapse threshold k . Although the motivating language is antimatter-inspired, the model is a computational abstraction: it is not a proposal for antimatter production, storage, shielding, or propulsion hardware. The central finding is the *Poole Signed-Membrane Basin*: near-mirrored opposite-sign initialization followed by a short passive warmup reliably induces a thin, high-support, low-imbalance reaction membrane. The compressed lesson is that preparation creates the membrane, rule windows tune it, and broad forcing usually degrades it. Once induced, the membrane is maintained most effectively by passive or gentle local dynamics; broad high-motion forcing consistently degrades performance. A reusable benchmark suite verifies six tasks: induction, maintenance, perturbation stress, survival-window rule ridge, geometry/boundary robustness, and mirror-quality basin size. Across optimization and frozen holdout tests, the canonical $B5-7/S5-9, k=3$ rule is strongly outperformed by a survival ridge centered near $B5-7/S3-11, k=3$ and $B5-7/S4-11, k=3$. The results support a rule-sensitive basin-discovery claim rather than a universal-controller claim: stable signed reaction membranes are not primarily forced into existence; they are induced by symmetry-preserving preparation and preserved by local survival dynamics.

1 Scope and motivation

Matter-antimatter annihilation motivates the language of opposite-signed populations, but macroscopic antimatter propulsion remains far beyond present production and storage capabilities. Modern antimatter experiments produce, decelerate, trap, and transport tiny numbers of antiprotons or anti-atoms using accelerator and cryogenic electromagnetic trap infrastructure. CERN’s AD/ELENA complex supplies low-energy antiprotons for precision experiments, with ELENA reducing antiprotons from 5.3 MeV to about 0.1 MeV and increasing the number trappable by experiments [1]. Antihydrogen has been magnetically confined for 1000 seconds in ALPHA [3], and BASE recently demonstrated transport of a small antiproton sample in a portable cryogenic trap [2]. Propulsion-oriented reviews emphasize that production, collection, storage, and directed use of annihilation products remain major barriers [4].

This paper therefore does not claim to solve antimatter production, storage, or propulsion. Instead, it studies a lower-level computational problem suggested by such systems: how opposite-signed populations form, stabilize, and lose localized reaction fronts under local rules and forcing. The result is a benchmarked computational morphology: an induced signed membrane basin.

2 Model

2.1 Signed lattice

We use a cubic lattice $\Lambda \subset \mathbb{Z}^3$ with a 26-neighbor Moore neighborhood. Each cell has state

$$s_i(t) \in \{-1, 0, +1\}, \quad (1)$$

where $+1$ and -1 denote opposite signed excitations and 0 denotes inactive background. Let N_i^+ and N_i^- be the number of positive and negative neighbors of cell i . A Poole-type rule is written

$$B[a, b]/S[c, d], \quad (2)$$

where an inactive cell is born into a sign when same-sign support lies in $[a, b]$ and exceeds opposite support by a margin, while an active cell survives when same-sign support lies in $[c, d]$.

Opposite-support collapse is controlled by a threshold k . A candidate positive cell is removed if $N_i^- \geq k$, and a candidate negative cell is removed if $N_i^+ \geq k$. The canonical early rule was $B5-7/S5-9, k = 3$; later tests found a stronger induced-membrane ridge near $B5-7/S3-11, k = 3$ and $B5-7/S4-11, k = 3$.

2.2 Reaction-interface measurements

The local signed overlap proxy is

$$q_i = \min(N_i^+ + \mathbf{1}_{s_i=+1}, N_i^- + \mathbf{1}_{s_i=-1}). \quad (3)$$

We define the interface support as

$$Q = \sum_{i \in \Omega} q_i \mathbf{1}_{q_i \geq q_0}, \quad (4)$$

where Ω is the engine/chamber mask and q_0 is a small threshold. The scale-normalized interface density is

$$Q_\rho = \frac{Q}{|\Omega|}. \quad (5)$$

The signed imbalance is

$$\epsilon_\pm = \frac{|N_+ - N_-|}{N_+ + N_- + \varepsilon}. \quad (6)$$

The membrane thickness σ_y is the support-weighted standard deviation of interface support along the signed separation axis y . A high-quality membrane has high Q_ρ , low σ_y , and low $\epsilon_\pm$.

We summarize this morphology with the Poole signed-membrane quality proxy

$$\mathcal{M}_P = \frac{Q_\rho}{1 + \sigma_y} (1 - \epsilon_\pm). \quad (7)$$

This quantity is not a physical energy or thrust law. It is a compact benchmark descriptor: support density multiplied by thinness and signed balance. In this paper, the named basin is the high- $\mathcal{M}_P$ regime reached by symmetry-preserving preparation.

The benchmark also reports a practical score used for model comparison:

$$J_{\text{practical}} = J_{\text{hybrid}} + 0.35J_{\text{pure}} + 0.20J_{\text{balance}} - 0.20\epsilon_\pm, \quad (8)$$

where the J sub-scores combine interface support, collimation proxy, balance, strain penalty, and membrane thinness. Because a single score can hide tradeoffs, all central claims are also reported using Q_ρ , σ_y , imbalance, good-regime rate, and stress behavior.

3 Experimental protocol

3.1 Preparation protocols

The key preparation protocols are:

1. random signed initialization;
2. mirrored signed initialization, in which the negative population is a reflected counterpart of the positive population;
3. mirrored initialization followed by a passive membrane-lock warmup for 10–20 ticks;
4. low-quality mirrored states used to map basin size;
5. preformed sheet controls, which test whether simply placing a sheet is sufficient.

The central induction protocol in the final benchmark is mirrored signed initialization plus a short passive warmup. This protocol is then followed by one of several maintainers: passive membrane-lock, Hybrid Guard, focus-only motion, interface-blind sparse motion, same-move random control, or static broad forcing.

3.2 Benchmark tasks

The attached benchmark artifact implements six paper-facing tasks:

1. induction: random versus mirrored versus mirrored-plus-warmup;
2. maintenance: passive/gentle controllers after induction;
3. perturbation stress: 0%–10% post-induction perturbations;
4. survival-window ridge: rule-window comparisons around the canonical and optimized rules;
5. geometry/boundary robustness: $L = 32, 40, 48$ and zero versus absorbing outlet boundary;
6. mirror-quality basin size: 25%–100% mirror quality.

All tables report means over seeds with bootstrap confidence intervals in the CSV outputs. The attached Python runner reproduces the benchmark.

4 Results

4.1 Mirrored preparation induces the membrane basin

Mirrored signed preparation plus warmup is the strongest causal lever. Table 1 shows that mirrored-plus-warmup protocols produce high $J_{\text{practical}}$, high Q_{ρ} , thin membranes, and 100% good-regime rates, whereas random Hybrid Guard and preformed sheet controls are substantially weaker.

Table 1: Induction benchmark. Mirrored preparation with passive warmup induces the membrane basin; forcing a preformed sheet does not.

Protocol	$J_{\text{practical}}$	Q_{ρ}	σ_y	Good rate
mirror+10 membrane	0.785	0.548	0.000	100.0%
mirror+20 hybrid	0.751	0.522	0.001	100.0%
mirror+10 hybrid	0.745	0.520	0.007	100.0%
mirror+0 hybrid	0.700	0.507	0.080	100.0%
random hybrid	0.492	0.450	0.648	68.8%
preformed preformed sheet hybrid	0.254	0.406	1.713	43.8%

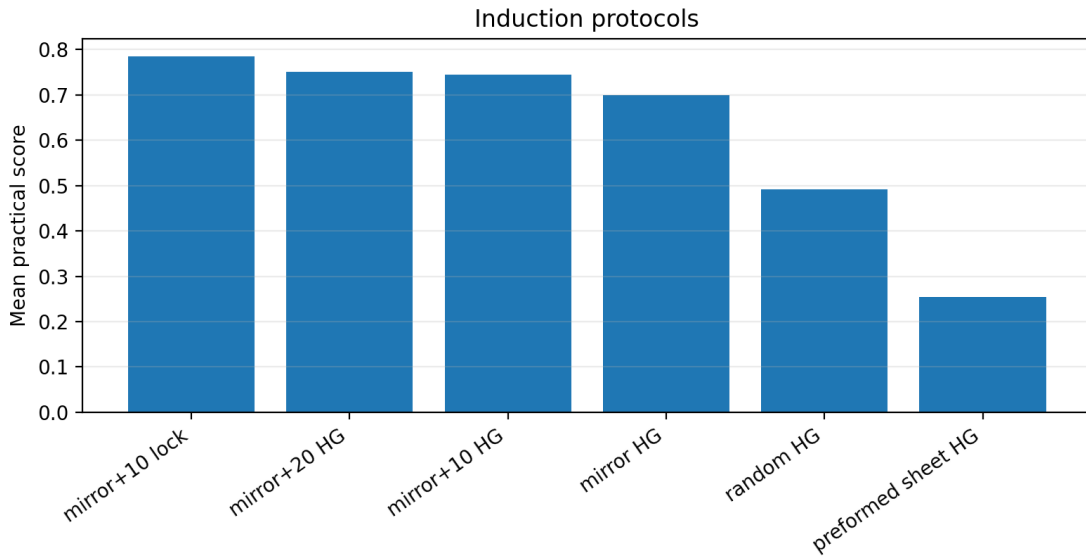


Figure 1: Practical score across induction protocols. The best protocols use mirrored signed preparation with a short passive warmup.

4.2 After induction, passive and gentle dynamics maintain the membrane

A surprising result is that after successful induction, the passive membrane-lock maintainer is usually strongest. Hybrid Guard remains a valid gentle maintainer, but is not uniquely optimal after the basin is induced. Broad static forcing is consistently worse despite much higher motion rates.

Table 2: Maintenance benchmark after induced preparation. Passive membrane-lock and gentle local maintainers preserve the basin; static broad forcing is worse.

Maintainer	$J_{\text{practical}}$	Q_{ρ}	σ_y	$\epsilon_{\pm}$	Moves/tick
maintain membrane	0.785	0.548	0.000	0.003	0.000
maintain random	0.771	0.536	0.006	0.003	0.777
maintain interfaceblind	0.761	0.529	0.000	0.003	0.806
maintain hybrid	0.745	0.520	0.007	0.004	2.079
maintain focus	0.744	0.519	0.007	0.004	2.081
maintain static	0.602	0.485	0.191	0.014	84.233

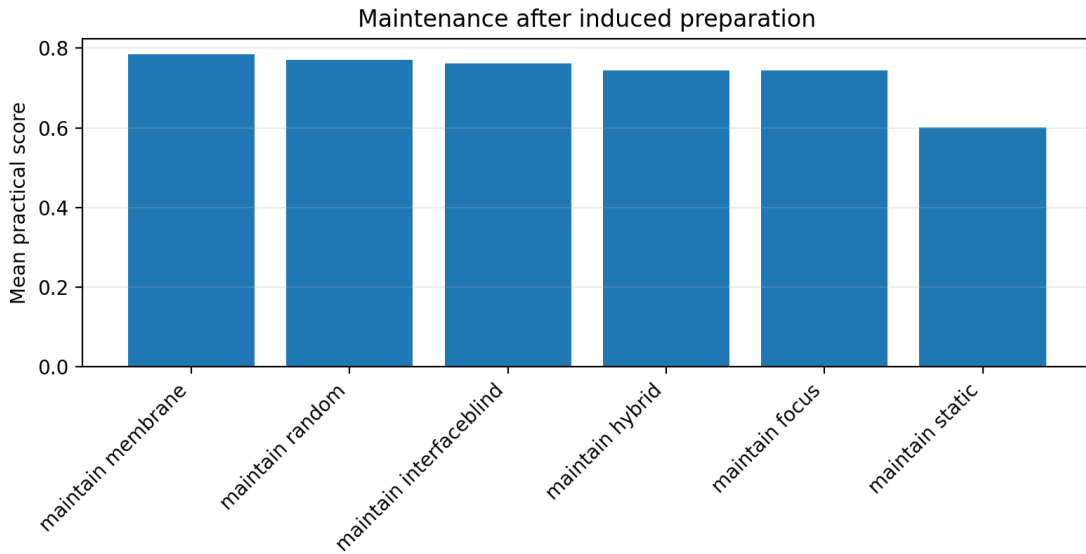


Figure 2: Maintenance comparison after induction. The membrane is most efficiently maintained by passive/gentle dynamics rather than broad static forcing.

4.3 The induced basin is perturbation robust

The best-rule stress tests perturb the prepared membrane after warmup. The induced basin remains in a good regime through perturbations up to 10% under passive and gentle maintainers. Static forcing remains lower-scoring at every perturbation level.

Table 3: Perturbation stress response under the best holdout rule. Entries are mean practical scores.

Controller	0%	1%	3%	5%	10%
membrane lock	0.776	0.766	0.728	0.739	0.707
hybrid guard	0.740	0.734	0.708	0.725	0.692
focus only	0.740	0.731	0.711	0.725	0.688
static nozzle	0.608	0.606	0.613	0.603	0.599

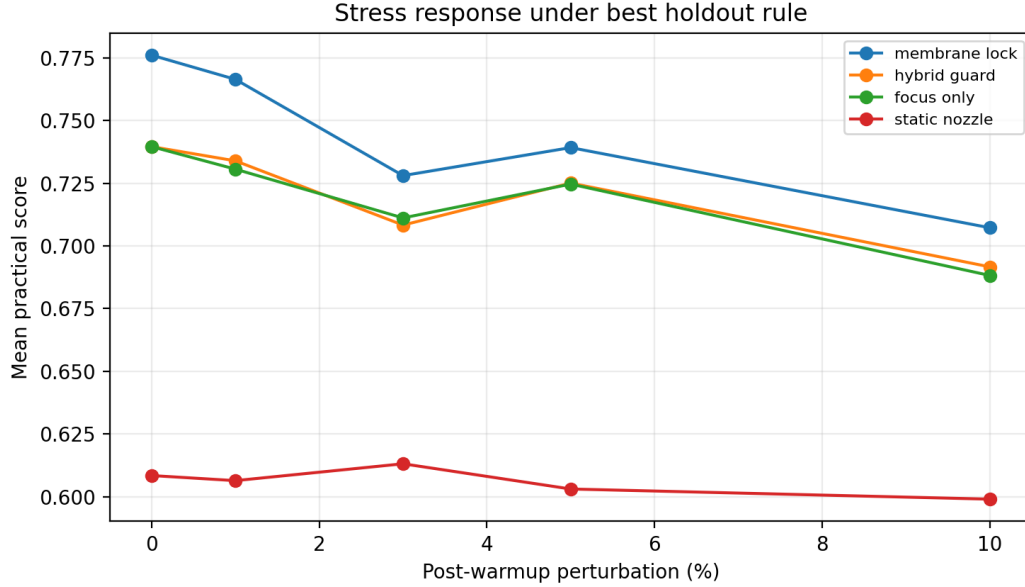


Figure 3: Stress response under the best holdout rule. Passive and gentle maintainers preserve the basin; broad static forcing remains weaker.

4.4 Rule-window ridge

The canonical $B5-7/S5-9, k = 3$ rule is not optimal after induction. Survival-window sweeps show a ridge centered near $S3-11$, $S3-12$, and $S4-11$. The current strongest confirmed holdout rule is $B5-7/S3-11, k = 3$, with $S4-11$ appearing as a close benchmark winner under the reusable benchmark suite. The safest summary is therefore a ridge family rather than a single sacred rule.

Table 4: Frozen holdout rule ridge. The canonical rule is strongly outperformed by a survival ridge near $S3-11/S4-11$.

Rule	Holdout gentle J	Good rate	Q_ρ	σ_y
$B5-7/S3-11, k = 3$	0.754	100.0%	0.527	0.0019
$B5-7/S3-12, k = 3$	0.750	100.0%	0.510	0.0010
$B5-7/S4-11, k = 3$	0.748	100.0%	0.524	0.0073
$B5-7/S5-12, k = 3$	0.745	100.0%	0.506	0.0066
$B5-7/S4-12, k = 3$	0.742	100.0%	0.504	0.0021
$B5-7_S4-13_k3$	0.739	100.0%	0.490	0.0017
$B5-7_S5-13_k3$	0.736	100.0%	0.487	0.0012
$B5-7/S5-11, k = 3$	0.731	100.0%	0.524	0.0382

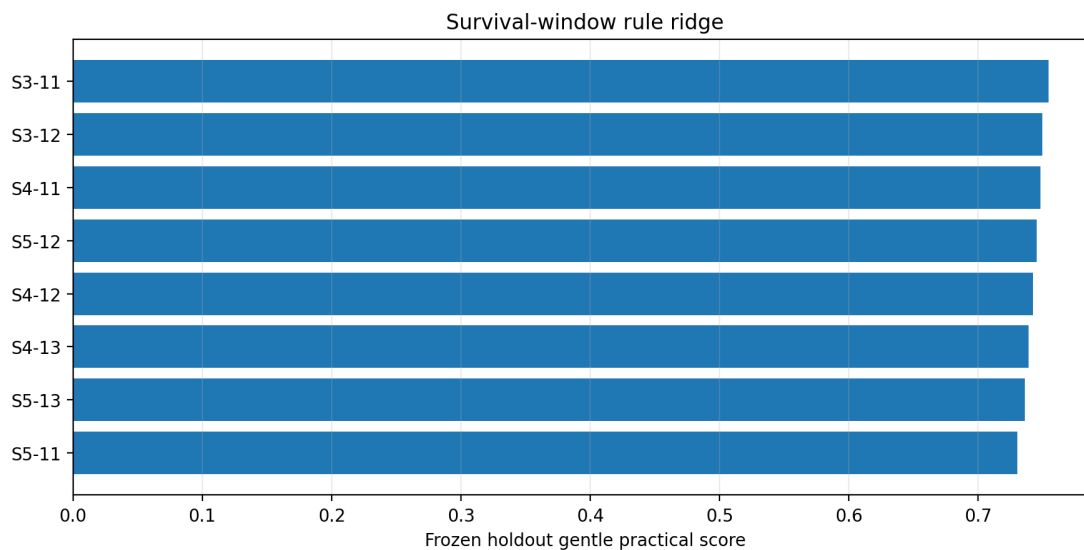


Figure 4: Frozen holdout rule ridge. Higher survival ceilings around 11–12 and lower survival minima around 3–4 produce thinner, more stable induced membranes than the canonical rule.



Figure 5: Representative best-rule membrane slice. The interface is extended in the chamber plane but nearly one-cell thin along the signed separation axis.

4.5 Geometry and mirror-quality robustness

The normalized classifier resolves earlier false negatives at small lattice size. Geometry/boundary tests show that the induced basin persists across $L = 32, 40, 48$ and zero versus absorbing outlet boundaries when judged by normalized interface density. The basin is not purely a single-geometry artifact. Mirror-quality tests show a sharp basin-size threshold: perfect and high-quality mirror preparation are strong, while low mirror quality collapses performance.

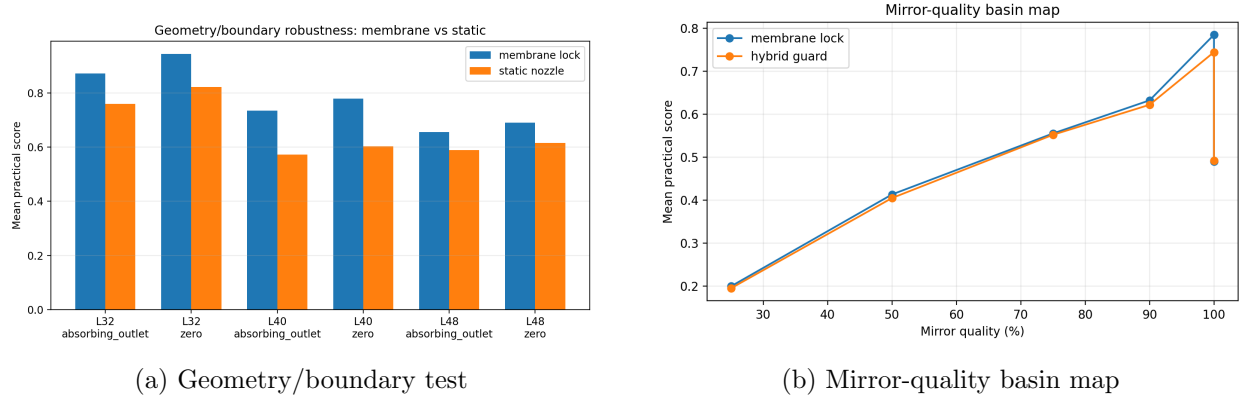


Figure 6: Geometry and basin-size tests. Induced membranes are robust under normalized geometry assessment, but depend strongly on signed mirror quality.

5 Discussion

5.1 What the result is

The core result is not that a particular active controller universally wins. Early tests showed that Hybrid Guard can outperform controls in a naturally sparse regime, but later holdout and matched-preparation tests showed that passive membrane-lock and other gentle local maintainers can be as good or better once the membrane is induced. The stronger result is basin-level: mirrored signed preparation plus short passive warmup induces a high-support, one-cell-thin, low-imbalance membrane that is stable under local dynamics and robust to moderate perturbation. In short: preparation creates the membrane; rules tune it; broad forcing usually destroys or degrades it.

5.2 What the result is not

This model is not a physical antimatter trap or propulsion device. Real antiprotons and antihydrogen are produced, cooled, trapped, and transported with high-energy accelerator and cryogenic electromagnetic apparatus [1, 2, 5]. The model has no electromagnetic fields, plasma kinetics, quantum processes, material wall interactions, radiative transport, or annihilation-product physics. Its contribution is computational morphology: it defines a reproducible signed-interface benchmark and identifies rule/preparation conditions under which a thin reaction membrane forms.

5.3 Interpretation for antimatter-inspired modeling

The result suggests a useful abstraction for future simulations of controlled release: opposite-signed reaction interfaces may be more effectively created by symmetry-preserving preparation than by broad forcing. In the benchmark, static high-motion forcing repeatedly degrades the induced

membrane, while passive and gentle local dynamics preserve it. This mirrors a general design intuition: once a delicate reaction front exists, excessive forcing can thicken, imbalance, or scatter it.

5.4 Limitations

Important limitations remain. First, the model is an abstract cellular automaton. Second, the rule ridge may depend on the chamber geometry and measurement definitions. Third, the practical score is only a composite benchmark metric; therefore all conclusions are supported with additional quantities such as Q_ρ , σ_y , imbalance, and stress behavior. Fourth, the exact physical interpretation of the widened survival ridge is not yet established.

6 Conclusion

We presented a signed Poole-type cellular automaton benchmark for reaction-membrane induction and maintenance. The main finding is an inducible signed membrane basin: near-mirrored opposite-sign initialization followed by short passive warmup produces a high-support, thin, balanced interface. We call this high- $\mathcal{M}_P$ regime the Poole Signed-Membrane Basin. The basin is robust under perturbation and geometry tests when evaluated with normalized interface metrics. The canonical $B5-7/S5-9, k=3$ rule is strongly outperformed by a survival-window ridge near $B5-7/S3-11, k=3$ and $B5-7/S4-11, k=3$. Once induced, the membrane is maintained well by passive or gentle local dynamics, while broad static forcing degrades it. The attached benchmark suite provides a reproducible artifact for future work. The final interpretation is deliberately narrow: this is not an antimatter propulsion design, but a reproducible computational result on how opposite-signed populations can be induced into stable, thin reaction interfaces.

A Benchmark artifact

The package includes `poole_signed_reaction_membrane_benchmark_v1.py`, reduced CSV outputs, and raw result ZIPs. The benchmark certificate reports that induction, stress, geometry, and rule-ridge criteria reproduce the paper claims. The artifact is intended as the canonical reproducibility target for this paper.

B Recommended citation of the benchmark

If reused, cite this package as the Poole Signed Reaction Membrane Benchmark v1 and report: rule label, preparation protocol, maintainer, Q_ρ , σ_y , imbalance, good-regime rate, perturbation rate, and seed set.

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Appendix D

Appendix D: Full bridge core source

Poole Bridge Theorem and Spectral-Basin Benchmark Core

Magnum Opus Build v0.6

Rooke Poole

June 2026

Abstract

This v0.6 bridge core integrates the exact theorem-level connection between the Poole Defect Calculus and channel-resolved PooleQ/P spectrometry, then connects that theorem layer to the signed reaction-membrane basin benchmarks. The exact statement is local: for a planar defect in a stable coordinate sheet, the first-step selector written in defect coordinates (q_D, r_D) is identical to the corresponding PooleQ/P event-channel readout. The empirical layer is deliberately separate: pre-warmup spectral signatures are tested as predictors of later signed membrane quality M_P . The v1.3 rule-ridge benchmark contains 144 cases over B5-7/S5-9/k3, B5-7/S4-11/k3, and B5-7/S3-11/k3. Structured signed preparations remain separated from all tested controls across the ridge: the maximum control M_P is 0.01667, while the minimum structured M_P is 0.06187. Spectral features predict all-case membrane quality better than density and corrected mirror baselines under random holdout and leave-rule-out tests, but structured-only leave-rule-out prediction remains unresolved. The safe conclusion is therefore sharp but bounded: the theorem bridge is exact; the empirical ridge benchmark supports basin-entry generalization across rule space; full cross-rule prediction of fine-grained structured membrane quality remains an open rule-aware problem.

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1 Purpose of this bridge core

This document is the bridge chapter-core for the larger Poole Local-Geometry Program manuscript. It has two roles. First, it states the exact bridge theorem connecting the Poole Defect Calculus to channel-resolved PooleQ/P spectrometry. Second, it records the empirical bridge benchmark that tests whether pre-warmup channel spectra predict signed reaction-membrane basin entry.

The proof layer and benchmark layer are intentionally separated. The defect/channel identity is a theorem about the first update of a planar defect. The spectral-basin predictor is a benchmark claim about finite signed-lattice experiments. The final manuscript should preserve that separation because it makes the result stronger, not weaker.

1.1 Claim lanes

1. **Theorem-level:** exact consequences of the local update rule.
2. **Verifier-level:** facts reproduced by included scripts.
3. **Benchmark-level:** finite experimental results reported with baselines and held-out splits.
4. **Empirical-atlas level:** observed patterns not yet theoremized.
5. **Open-problem level:** unresolved bridge claims or future theorem targets.

2 The exact defect/channel bridge

Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3$ and let the base unsigned Poole update be

$$P(x)_i = \mathbf{1}\{5 \leq N_i(x) \leq 7 + 2x(i)\}, \quad (1)$$

where $N_i(x)$ is the 26-neighbor Moore active count. Equivalently, inactive sites are born in channels B_5, B_6, B_7 , while active sites survive in channels S_5, S_6, S_7, S_8, S_9 .

Let $D \subset (\mathbb{Z}/L\mathbb{Z})^2$ be a defect mask removed from an otherwise stable coordinate sheet $z = c$. Define the open and closed defect counts

$$q_D(u) = \sum_{\eta \in \{-1,0,1\}^2 \setminus \{0\}} \mathbf{1}_{u+\eta \in D}, \quad r_D(u) = \sum_{\eta \in \{-1,0,1\}^2} \mathbf{1}_{u+\eta \in D}. \quad (2)$$

Theorem 2.1 (Defect/channel bridge). *For the first update of a defective coordinate sheet under the base Poole rule, the local event channel is determined by*

$$u \notin D : \quad S_{8-q_D(u)}, \quad (3)$$

$$u \in D : \quad B_{8-q_D(u)}, \quad (4)$$

$$z = c \pm 1 : \quad B_{9-r_D(u)}. \quad (5)$$

Consequently, the planar defect selector written in (q_D, r_D) coordinates and the channel-resolved PooleQ/P readout written in $(B_5, B_6, B_7, S_5, \dots, S_9)$ coordinates contain the same first-step local information.

Proof. All active cells initially lie in the sheet. An active sheet site $u \notin D$ has exactly $8 - q_D(u)$ active in-plane neighbors, so its survival channel is $S_{8-q_D(u)}$. A missing in-plane defect cell $u \in D$ sees the same $8 - q_D(u)$ active planar neighbors but is inactive, so a birth event, when accepted, is in channel $B_{8-q_D(u)}$. A normal-layer candidate above or below u sees the full 3×3 sheet footprint minus the closed defect footprint, hence $9 - r_D(u)$ active neighbors, so its possible birth channel is $B_{9-r_D(u)}$. These three cases exhaust the first-step local response near the defective sheet; cells two or more layers away see zero active neighbors on the first update. $\square$

Remark 2.2 (Boundary of the theorem). *The theorem is exact for the first-step local planar selector under the stated sheet-defect setup. It is not a multi-step relaxation theorem, a non-planar void theorem, or a physical-device claim.*

3 PooleQ/P spectrometry as the bridge language

For a binary field X , the channel-resolved PooleQ/P vector keeps the accepted birth channels, accepted survival channels, high-support channel, and strain diagnostic instead of immediately collapsing to a single live/dead update:

$$\Phi_i(X) = (B_5, B_6, B_7, S_5, S_6, S_7, S_8, S_9, O_{10+}, \psi)_i. \quad (6)$$

The null-calibrated geometry signature is written in compact form as

$$\mathcal{S}_P(X) = (R_B(X), R_H(X), R_\psi(X), R_C(X)), \quad (7)$$

where R_B measures birth-window geometry, R_H high-support geometry, R_ψ support-strain geometry, and R_C aggregate channel contrast. The exact theorem above explains why this readout is the natural bridge language: defect-footprint counts select the same channel groups that PooleQ/P preserves.

4 Signed fields and the spectral-basin feature map

For a signed field $s \in \{-1, 0, +1\}^{\Lambda_L}$ define four binary projections:

$$X_+ = \mathbf{1}_{s=+1}, \quad X_- = \mathbf{1}_{s=-1}, \quad X_{|s|} = \mathbf{1}_{|s|=1}, \quad X_{\text{int}} = \mathbf{1}_{N_i^+ > 0 \wedge N_i^- > 0}. \quad (8)$$

The v1.3 spectral-basin vector uses PooleQ/P signatures of those projections, plus plus/minus spectral match features and corrected mirror diagnostics:

$$\mathcal{B}(s_0) = (\mathcal{S}_P(X_+), \mathcal{S}_P(X_-), \mathcal{S}_P(X_{|s|}), \mathcal{S}_P(X_{\text{int}}), \mu_{\text{mirror}}, \mu_{\text{mirror}}^{\text{shift}}, \epsilon_{\pm}, \Delta_{+/-}). \quad (9)$$

The empirical hypothesis is that $\mathcal{B}(s_0)$ predicts final signed membrane quality $M_P(s_T)$ after passive warmup better than density-only and corrected mirror-only baselines.

5 Benchmark design

The bridge benchmark compares structured signed sheet-pair preparations against controls. The hard-mode v1.3 suite includes aligned sheet pairs, shifted sheet pairs, degraded sheet pairs, wide-gap sheet pairs, patch pairs, mirrored-random controls, same-density random controls, signed-shuffle controls, and single-sign sheet controls.

The rule-ridge run used

$$B5-7/S5-9/k3, \quad B5-7/S4-11/k3, \quad B5-7/S3-11/k3. \quad (10)$$

The target variable is final membrane quality M_P . The benchmark reports R^2 , RMSE, MAE, Spearman rank correlation, and median-threshold AUC for density-only, mirror-only, design-parameter, protocol-label, rule/protocol-label, early-interface, spectral-bridge, and combined-numeric models.

6 First-pass and hard-mode progression

The first bridge run was intentionally easy: structured cases were strongly separated from controls. The hard-mode run added centered mirror correction, protocol-one-hot baselines, structured-only splits, shifted/degraded/wide-gap/patch cases, single-sign controls, and optional rule-ridge tests. The methodological lesson is important: the result should not be presented as a single number. It should be presented as a progression of increasingly hostile controls.

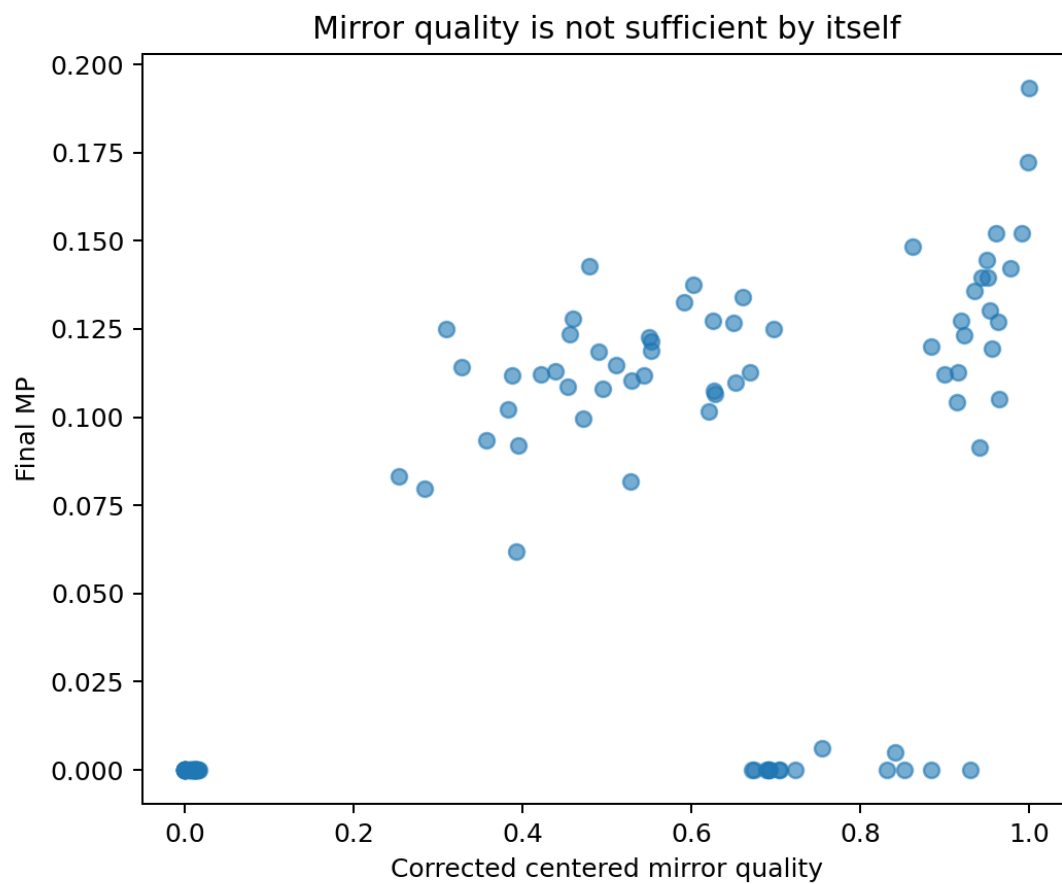


Figure 1: Hard-mode mirror-quality sanity check. High corrected mirror quality is not sufficient for basin entry: mirrored random controls can score high on mirror quality while remaining near zero in final membrane quality.

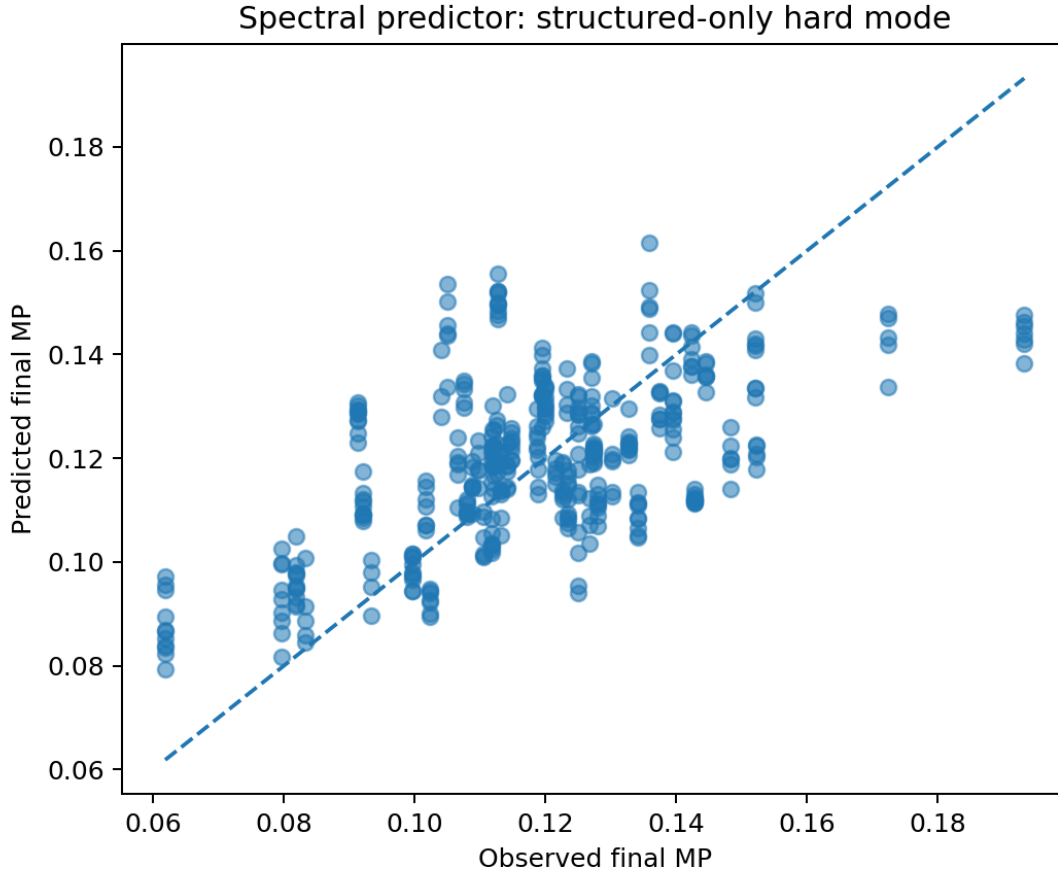


Figure 2: Hard-mode structured-only prediction scatter. This split tests whether spectral features carry signal within the basin-forming family, rather than only separating obvious controls from structured cases.

7 Rule-ridge basin-entry result

The rule-ridge benchmark is the most important bridge stress test in v0.6. It asks whether the basin-entry separation survives across survival windows. It does.

Rule-ridge separation. Across 144 ridge cases, the maximum control M_P was 0.01667, while the minimum structured M_P was 0.06187. The midpoint separating threshold is approximately 0.03927. The weakest structured case is 3.71 times larger than the strongest control case.

Table 1: Rule-ridge protocol summary. Structured sheet-pair protocols enter the signed membrane basin; controls remain near zero even when mirror quality is high.

Protocol	n	Mean MP	Min MP	Max MP	Mean Qrho	Mean mirror
mirrored_random_control	17.0000	0.0031	0.0000	0.0167	0.0038	0.7541
same_density_random	9.0000	0.0000	0.0000	0.0000	0.0000	0.0097
sheet_pair_aligned	20.0000	0.2451	0.0913	0.3131	0.2517	0.9556
sheet_pair_degraded	19.0000	0.1604	0.0921	0.2253	0.1710	0.4927
sheet_pair_shifted	20.0000	0.1919	0.1017	0.2734	0.2047	0.5562
sheet_pair_widegap	12.0000	0.1797	0.1120	0.2569	0.1873	0.9249
sheet_patch_pair	12.0000	0.1451	0.0619	0.2277	0.1654	0.4232
signed_shuffle_control	21.0000	0.00032	0.0000	0.0067	0.000837	0.0108
single_sign_sheet_control	14.0000	0.0000	0.0000	0.0000	0.0000	0.0000

Table 2: Rule-ridge summary by rule and family. The S3-11 and S4-11 survival windows produce about twice the structured mean membrane quality of canonical S5-9 in this benchmark.

Rule	Family	n	Mean MP	Min MP	Max MP	Mean Qrho	Mean mirror
B5-7/S3-11/k3	control	16.0000	0.0016	0.0000	0.0167	0.0024	0.1836
B5-7/S3-11/k3	structured	32.0000	0.2277	0.1164	0.3131	0.2409	0.6981
B5-7/S4-11/k3	control	24.0000	0.0011	0.0000	0.0157	0.0016	0.1711
B5-7/S4-11/k3	structured	24.0000	0.2232	0.1120	0.2989	0.2333	0.6818
B5-7/S5-9/k3	control	21.0000	0.000293	0.0000	0.0061	0.000293	0.2901
B5-7/S5-9/k3	structured	27.0000	0.1127	0.0619	0.1428	0.1223	0.6324

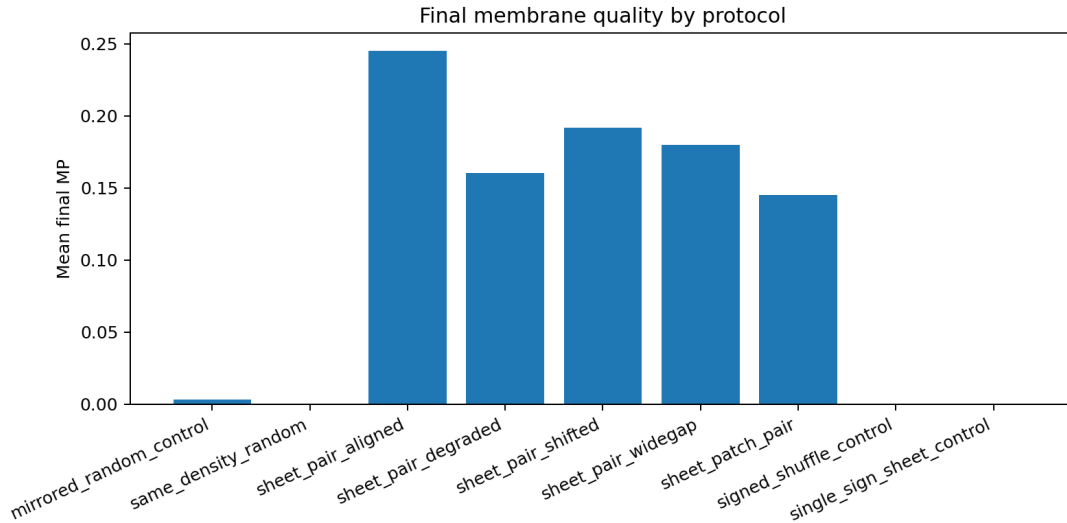


Figure 3: Final membrane quality by protocol in the v1.3 rule-ridge bridge run.

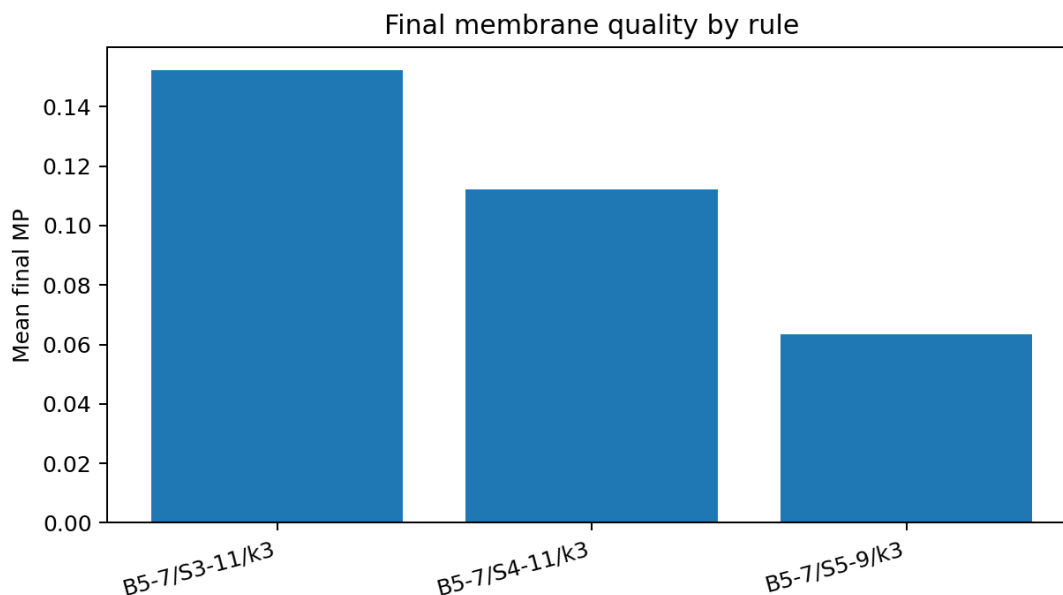


Figure 4: Final membrane quality by rule in the v1.3 rule-ridge bridge run. B5-7/S3-11/k3 and B5-7/S4-11/k3 strongly improve structured membrane quality relative to B5-7/S5-9/k3.

8 Predictor results and boundaries

The spectral bridge has strong all-case signal across the rule ridge, but the strongest statement is basin-entry separation, not perfect fine-grained quality prediction. Spectral features outperform density and corrected mirror baselines in all-case random-holdout and all-case leave-rule-out testing. Protocol/rule-label and combined numeric baselines can outperform spectral-only models in some scopes, which is why the final manuscript should report all baselines rather than only the best-looking result.

Table 3: v1.3 rule-ridge model comparison. Negative structured-only leave-rule-out R2 rows are retained as an explicit limitation: fixed spectral features do not yet solve fine-grained structured-quality prediction across unseen rules.

Scope	Split	Model	R2	RMSE	MAE	Spearman	AUC
all cases	random holdout	density baseline	0.1442	0.0977	0.0829	0.4496	0.7038
all cases	random holdout	mirror baseline corrected	0.4972	0.0743	0.0581	0.7304	0.8648
all cases	random holdout	protocol onehot baseline	0.8140	0.0449	0.0316	0.8485	0.9324
all cases	random holdout	rule protocol onehot baseline	0.8870	0.0351	0.0296	0.9159	0.9743
all cases	random holdout	spectral bridge	0.7684	0.0506	0.0384	0.8399	0.9523
all cases	random holdout	combined numeric	0.8693	0.0372	0.0314	0.9137	0.9823
all cases	leave rule out	density baseline	-0.0155	0.1067	0.0935	0.0800	0.5659
all cases	leave rule out	mirror baseline corrected	0.3341	0.0864	0.0684	0.5520	0.7967
all cases	leave rule out	protocol onehot baseline	0.6737	0.0605	0.0433	0.7656	0.9082
all cases	leave rule out	rule protocol onehot baseline	0.6692	0.0609	0.0447	0.7611	0.9078
all cases	leave rule out	spectral bridge	0.5876	0.0680	0.0505	0.7380	0.9252
all cases	leave rule out	combined numeric	0.6770	0.0602	0.0455	0.7863	0.9279
structured only	random holdout	density baseline	0.0418	0.0623	0.0531	0.3058	0.6103
structured only	random holdout	mirror baseline corrected	0.0381	0.0613	0.0526	0.4310	0.6851
structured only	random holdout	protocol onehot baseline	0.2192	0.0582	0.0503	0.5126	0.7257
structured only	random holdout	rule protocol onehot baseline	0.7633	0.0309	0.0244	0.8576	0.9161
structured only	random holdout	spectral bridge	0.1377	0.0600	0.0491	0.4519	0.7067
structured only	random holdout	combined numeric	0.7430	0.0325	0.0255	0.8585	0.9193
structured only	leave rule out	density baseline	-0.5615	0.0834	0.0729	-0.2011	0.3368
structured only	leave rule out	mirror baseline corrected	-0.4648	0.0808	0.0714	-0.0760	0.4071
structured only	leave rule out	protocol onehot baseline	-0.3843	0.0785	0.0695	-0.0046	0.4512
structured only	leave rule out	rule protocol onehot baseline	-0.4064	0.0792	0.0693	-0.0639	0.4228
structured only	leave rule out	spectral bridge	-0.5239	0.0824	0.0716	-0.0810	0.3920
structured only	leave rule out	combined numeric	-0.1076	0.0703	0.0558	0.3062	0.6318

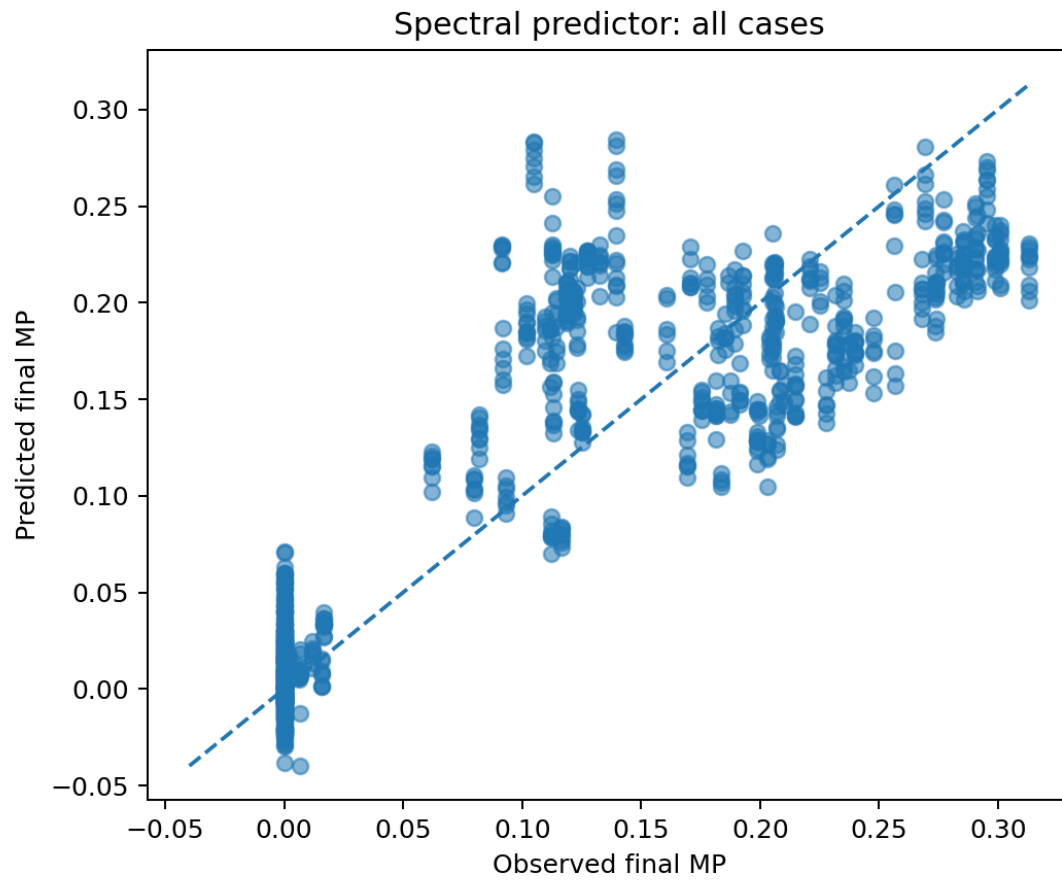


Figure 5: Rule-ridge all-case prediction scatter for the spectral bridge model.

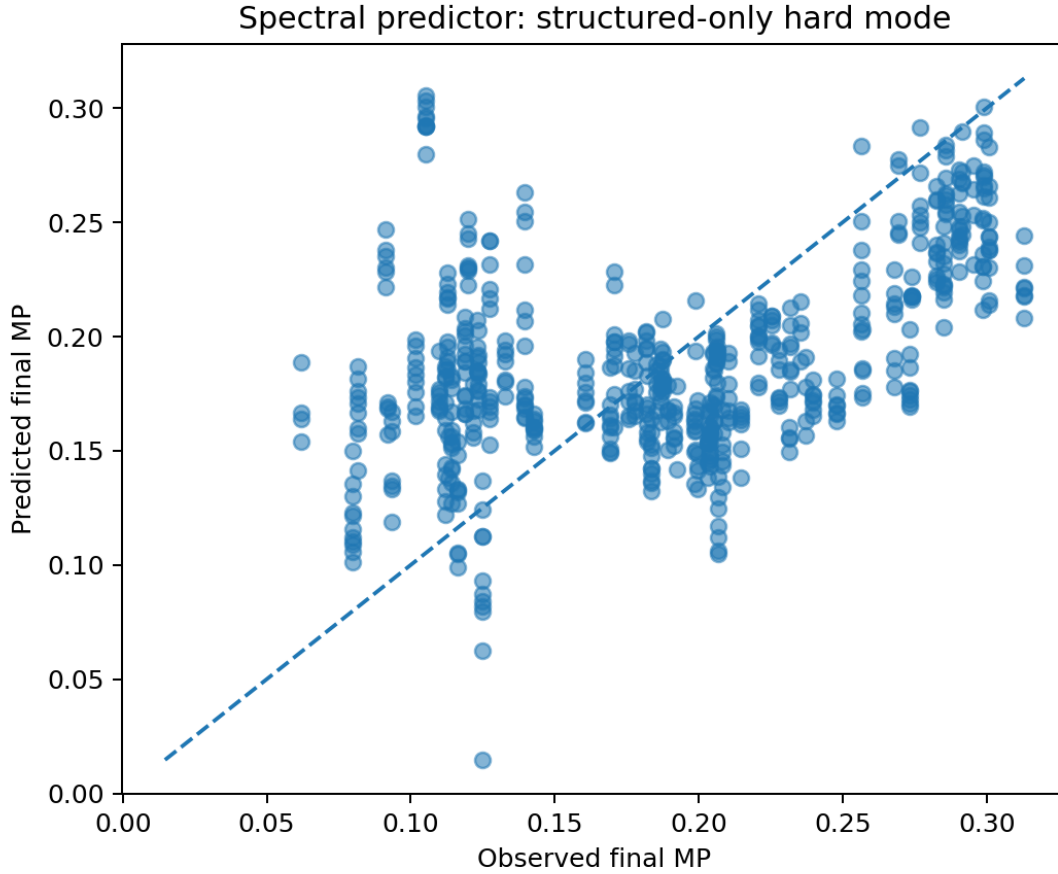


Figure 6: Rule-ridge structured-only prediction scatter. This is the hardest current split and motivates rule-aware v1.4 features.

9 Coefficient interpretation

The coefficient table is not a theorem, but it is useful for interpretation. In the all-case ridge model, support-strain and absolute-channel contrast terms are prominent. In structured-only fitting, plus/minus mismatch, initial imbalance, and birth-window terms carry much of the signal. This is consistent with the bridge story: basin entry is governed by local birth-window geometry, support strain, and signed spectral matching rather than by density or global mirror symmetry alone.

Table 4: Largest standardized spectral coefficients by absolute magnitude for the v1.3 rule-ridge run.

Scope	Feature	Beta
all cases	t0_imbalance	-0.0166
all cases	abs_Rpsi	0.0156
all cases	abs_RC	0.0156
all cases	abs_minus_int_RC	0.0121
all cases	abs_minus_int_Rpsi	0.0121
all cases	pm_RB_absdiff	-0.0096
all cases	abs_int_Rpsi_mean	0.0090
all cases	abs_int_RC_mean	0.0089
all cases	pm_RB_min	0.0067
all cases	t0_active_density_signed	-0.0061
all cases	int_RC	-0.0047
all cases	int_Rpsi	-0.0046
structured only	pm_RB_absdiff	-0.0194
structured only	t0_imbalance	-0.0135
structured only	spectral_coupling_proxy	0.0120
structured only	mirror_quality	0.0120
structured only	pm_RC_absdiff	0.0120
structured only	pm_Rpsi_absdiff	0.0119
structured only	mirror_quality_best_shift	0.0108
structured only	abs_minus_int_RH	-0.0075
structured only	int_RH	0.0074
structured only	mirror_shift_gain	-0.0073
structured only	abs_minus_int_Rpsi	0.0047
structured only	abs_minus_int_RC	0.0046

10 Safe manuscript conclusion

The exact bridge theorem should be stated strongly. The empirical predictor claim should be stated carefully:

The theorem bridge is exact: the first-step planar defect selector and the channel-resolved PooleQ/P readout are the same local information expressed in different coordinates. The empirical ridge benchmark supports basin-entry generalization across rule space: structured signed preparations remain separated from controls under all tested survival windows, and spectral features outperform density and corrected mirror baselines for all-case prediction under random-holdout and leave-rule-out tests. Full structured-only cross-rule prediction of fine-grained membrane quality remains unresolved and should be treated as a rule-aware v1.4 target.

11 Open targets for v1.4

1. Add explicit rule-window features to the spectral bridge, including survival-window width, birth/survival overlap, and collapse-threshold interaction terms.
2. Replace fixed spectral signatures with rule-conditioned signatures $\mathcal{S}_{P,\theta}$ so the channel readout knows which survival window is being tested.
3. Add leave-protocol-out tests to separate geometry-level prediction from protocol naming.
4. Add cross-size testing with $L = 20, 28, 36$ or larger if runtime allows.
5. Define a basin-entry classifier separate from the continuous M_P regressor; the current data suggest the entry boundary is more stable than fine-grained quality prediction.

A Claim ledger

The bridge claim ledger is included as `data/bridge_claim_ledger_v0_6.csv`. The key claims are summarized here.

Table 5: Bridge claim ledger summary.

ID	Title	Statement	Lane	Boundary
B01	Exact defect-channel bridge	The first-step planar defect selector can be written equivalently as local channel events: active sheet sites map to $S_{\{8-q_D\}}$, in-plane defect cells map to $B_{\{8-q_D\}}$, and normal-layer candidates map to $B_{\{9-r_D\}}$.	theorem-level	Exact only for the first update around a coordinate stable sheet under the stated planar-defect setup.
B02	Q/P bridge interpretation	The Defect Calculus selector and the PooleQ/P channel readout are the same first-step local information in different coordinates.	theorem-level / interpretation	Does not imply quantum-state capture or reconstruction. It is a classical feature-map statement.
B03	Spectral-basin hypothesis	A pre-warmup spectral-basin vector computed from X_+ , X_- , $—s—$, and interface projections should predict later signed membrane quality M_P.	benchmark-level hypothesis	Not a theorem. It must be reported with baselines and held-out splits.
B04	Mirror is insufficient	Corrected mirror quality alone does not explain basin entry; mirrored random controls have high mirror quality but near-zero M_P.	benchmark-level	Applies to the tested initialization families and rules.
B05	Structured/control separation across rule ridge	Structured signed preparations beat all tested controls under B5-7/S5-9/k3, B5-7/S4-11/k3, and B5-7/S3-11/k3.	benchmark-level	Finite benchmark result, not a universal guarantee over all rules.
B06	Survival ridge strengthens basin	B5-7/S3-11/k3 and B5-7/S4-11/k3 produce about twice the structured mean M_P of canonical B5-7/S5-9/k3 in the ridge benchmark.	benchmark-level	Specific to the tested protocols, L, warmup, and collapse k.
B07	All-case spectral prediction	The spectral bridge predicts all-case M_P better than density and corrected mirror baselines under random holdout and leave-rule-out ridge tests.	benchmark-level	Protocol/rule one-hot or combined numeric models may exceed spectral-only scores in some scopes.

ID	Title	Statement	Lane	Boundary
B08	Structured-only rule-aware limitation	Structured-only leave-rule-out prediction remains unresolved; no tested fixed feature family generalized well across held-out rules.	open-problem / limitation	Requires rule-aware spectral features or separate theoremization before stronger claims.
B09	Basin-entry vs quality prediction	The bridge is stronger for basin-entry/separation than for fine-grained cross-rule structured quality prediction.	bounded conclusion	This distinction should be explicit in the main manuscript.

B Data and reproducibility manifest

The package contains the v1 first-pass bridge outputs, v1.3 hard-mode outputs, v1.3 rule-ridge outputs, generated figures, compact summary tables, claim ledger, and Colab-ready runner source. The full file manifest is stored as `data_manifest_v0.6.csv`. No source file is required to be hidden from the mathematical record; the final master manuscript can cite the package manifest for reproducibility.

C Integration notes for the Magnum Opus

This v0.6 bridge core should be merged after the Signed Reaction-Membrane Basin Core. The theorem proof belongs near the end of the Defect Calculus/Q/P transition. The benchmark belongs after the signed-membrane metrics and before the global limitation chapter. The structured-only leave-rule-out limitation should remain visible, because it motivates the rule-aware spectral program without weakening the theorem bridge.